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On the Power of Adaptivity for ε-Best Arm Identification in Linear Bandits

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Part I

The power of adaptivity for ε-best arm identification in linear bandits

Overview

This blueprint guides the Lean formalization of the paper *On the Power of Adaptivity for ε-Best Arm Identification in Linear Bandits* by Arnab Maiti, Yunbei Xu and Kevin Jamieson (COLT 2026). The paper studies the minimax sample complexity of ε-best arm identification in linear bandits with Gaussian noise: a compact action set $\mathcal{X} \subset \mathbb{R}^d$ spanning $\mathbb{R}^d$, an unknown reward vector θ, observations $y_t = \langle x_t, \theta\rangle + \eta_t$, and the goal of returning $\hat{x} \in \mathcal{X}$ with $\max_{x\in\mathcal{X}}\langle x - \hat{x}, \theta\rangle \le \varepsilon$ with probability at least $1-\delta$. Its results are:

(i) a non-adaptive fixed-design algorithm with sample complexity $O\big((d\log(1/\delta) + w(\mathcal{X})^2)/\varepsilon^2\big)$, where $w(\mathcal{X})$ is a Gaussian width term of the action set (Theorem **??**);

(ii) matching lower bounds: $\Omega(d\log(1/\delta)/\varepsilon^2)$ for every algorithm, adaptive or not (Theorem **??**), and $\Omega(w(\mathcal{X})^2/\varepsilon^2)$ for non-adaptive fixed designs (Theorem **??**);

(iii) properties of the Gaussian width term and its computation for structured action sets (Chapter **??**);

(iv) structured action sets on which adaptivity yields at most logarithmic improvements (Chapter **??**), through a Median-Elimination based adaptive algorithm and adaptive lower bounds;

(v) an ℓ_2-norm estimation procedure with sample complexity $O(d\log(1/\delta)/\varepsilon^2)$ (Theorem **??**);

(vi) an action set on which adaptivity yields a polynomial-factor improvement over every non-adaptive algorithm (Theorem **??**).

Goals of the formalization. Two goals drive the organization of this document. First, formalize the main results of the paper. Second, build a reusable library, to be contributed to *Lean Machine Learning* (LML) and Mathlib: the objects (linear bandits, PAC identification algorithms, fixed designs, optimal experimental design, Gaussian width, Gaussian concentration, sub-exponential variables, divergence decompositions, Median Elimination) are therefore introduced in the generality in which they are useful, even when the paper only needs a special case, and extra work is done when needed to reach that generality.

Deviations from the paper. The blueprint departs from the paper whenever a different route is simpler to formalize while giving the same (or a stronger) statement; every such point is recorded in a remark next to the statement it concerns. The main ones:

- The rounding of a design distribution to a fixed design of size $O(d)$ is proved by a one-sided barrier argument in the style of Batson–Spielman–Srivastava (Section **??**) instead of the cited rounding procedures; the constant 360 of the paper becomes another absolute constant.

- The Sudakov–Fernique inequality is only used in its "covariance domination" special case, which has a one-line proof by Jensen's inequality (Lemma **??**); the general inequality is only needed for the secondary lower bound $w(\mathcal{X}) \geq \Omega(\sqrt{d \log d})$ (Chapter **??**).

- John's theorem in the proof of the adaptive lower bound is replaced by the Kiefer–Wolfowitz theorem on G-optimal designs, which is needed anyway.

- The Bayesian argument of the non-adaptive lower bound is done without Gaussian conditioning, through an orthogonal decomposition and independence of uncorrelated jointly Gaussian vectors.

- Borell–TIS is obtained from the Maurey–Pisier Gaussian concentration argument, with the constant $c_{\mathrm{G}} = \pi^2/4$ in the exponent instead of 1.

- The large-norm regime of the ℓ_2-norm estimation uses a deterministic basis design instead of random Rademacher rows, which removes the need for random-matrix concentration.

- The lower bounds for structured sets are proved directly for randomized algorithms (no Yao principle) and with a single argument for all block sizes.

- Non-adaptive algorithms are deterministic fixed designs with an arbitrary (possibly randomized) recommendation rule, the class of the paper's Theorem 3; the $\Omega(kd^2/\varepsilon^2)$ lower bound of the polynomial separation (Theorem **??**) is stated for this class, whereas the paper's Theorem 7 says "possibly randomized non-adaptive". Budgets are deterministic (Remark **??**).

- The rank-one Hanson–Wright inequality used for the Rademacher directions is replaced by an MGF bound for the square of a sub-Gaussian variable (Lemma **??**), and the Gaussian Hanson–Wright inequality by diagonalization (Lemma **??**); the paper's tail-to-MGF lemma is not needed.

- The width of the multi-task set (Theorem **??**) is computed in Lean without the Helmert matrices: a basis-free reduction bounds the intrinsic width of a non-spanning set by the expected supremum of a Gaussian vector supported on its span, whose covariance is the pseudo-inverse of a design matrix (Lemma **??**); only the description of the span and the action of the uniform design on it are needed.

Reading guide. Part **??** follows the paper: Chapter **??** sets up the model, Chapter **??** the experimental design tools, Chapters **??** to **??** the fixed-design upper bound and the lower bounds, Chapter **??** the Gaussian width, Chapter **??** the structured sets, Chapter **??** the ℓ_2-norm estimation and Chapter **??** the polynomial separation. Part **??** collects the probabilistic and information-theoretic prerequisites that are not in Mathlib or LML, decomposed into small lemmas: Gaussian vectors and linear Gaussian processes, Gaussian concentration, sub-exponential variables, divergences of algorithm-environment sequences, the

Sudakov–Fernique inequality, Median Elimination and Gaussian posteriors of adaptively collected data.

Conventions. An *action set* is a nonempty compact subset of $\mathbb{R}^d$; the paper's standing assumption $\mathrm{span}(\mathcal{X}) = \mathbb{R}^d$ ("spanning") is stated explicitly in the results that need it (designs, widths, upper bounds), because the structured sets of Chapter **??** are not spanning. Time is 0-indexed, as in LML: an algorithm with budget T plays the actions $x_0, \dots, x_{T-1}$ and sums over $t < T$. All random objects live on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and $\mathbb{E}$ is the expectation under $\mathbb{P}$; when the reward vector matters we write $\mathbb{P}_\theta$, $\mathbb{E}_\theta$. Norms are Euclidean unless stated otherwise, and for a positive definite matrix W we write $\|x\|_W^2 := x^\top W x$.

Status. The formalization proceeds in two phases. Phase 1 (done) states the headline results of the paper in Lean, over definitions written in library generality: each such statement carries a `\lean` tag naming its Lean declaration and is marked as formalized, with its proof left to phase 2; the upper bounds are stated in the paper's existential form (Corollaries **??**, **??**, **??** and **??**), the explicit algorithms being part of the proofs. Each headline statement is also a standalone challenge file for `leanprover/comparator` (directory `comparator/`). Phase 2 will formalize the proofs, following this document.

Blueprint author. Rémy Degenne. Anyone is welcome to contribute!

Chapter 1

Model: linear bandits, identification algorithms and the Gaussian width

This chapter fixes the objects of the paper (Section 1 and the notations of Section 1.1), in the form in which they will be formalized on top of the LML framework of sequential algorithms and environments (`Learning.Algorithm`, `Learning.Environment`, `Learning.IsAlgEnvSeq`, `Learning.stationaryEnv`).

1.1 Action sets and the linear Gaussian environment

Definition 1 (Action set)**.** Maiti2026Power.unitBallLet $d \geq 1$. An *action set* (or arm set) is a nonempty compact subset $\mathcal{X} \subset \mathbb{R}^d$. It is *spanning* if $\operatorname{span}(\mathcal{X}) = \mathbb{R}^d$. The paper assumes spanning throughout; in this blueprint the spanning assumption is stated explicitly in every statement that needs it (the existence of positive definite designs, the Gaussian width, the upper bounds), and the definitions of environments, algorithms, regret and PAC guarantees below make sense for every action set, spanning or not (the structured sets of Chapter **??** are not spanning). We write $\mathbb{B}_d := \{x \in \mathbb{R}^d : \|x\| \leq 1\}$ for the unit Euclidean ball and $\mathbb{S}^{d-1}$ for the unit sphere. For $\theta \in \mathbb{R}^d$ the *value* of an arm x is $\langle x, \theta \rangle$ and an *optimal arm* is $x_\star(\theta) \in \arg\max_{x \in \mathcal{X}} \langle x, \theta \rangle$ (it exists by compactness).

Lean remark. The ambient space is `EuclideanSpace  (Fin d)`. The action type of the LML algorithms is the subtype $\mathcal{X}$ with its Borel σ-algebra, so that algorithms play in $\mathcal{X}$ by typing. Compactness is only used through the existence of maximizers of continuous functions and the finiteness of $\sup_{x \in \mathcal{X}} \|x\|$; the spanning assumption is used through the existence of positive definite design matrices (Lemma **??**) and is a separate hypothesis (`Submodule.span     =  `)

of the statements that need it. Finite action sets are the important special case, and several statements are first proved for finite $\mathcal{X}$ and then extended by density (Lemma **??**).

Definition 2 (Linear Gaussian environment)**.** Learning.LinearBandit.linearGaussianEnv, Learning.LinearBandit.linearGaussianKerneldef:arm$_s etFor\theta \in \mathbb{R}^d$, the *linear Gaussian environment* ν_θ is the stationary environment (LML `stationaryEnv`) with reward kernel $x \mapsto \mathcal{N}(\langle x, \theta \rangle, 1)$, $x \in \mathcal{X}$: at each round t the learner plays $x_t \in \mathcal{X}$ and observes $y_t = \langle x_t, \theta \rangle + \eta_t$ where the η_t are i.i.d. standard Gaussian, independent of the past and of the learner's randomness.

Lemma 3 (Noise representation)**.** *Learning.LinearBandit.noise, Learning.LinearBandit.hasCondDistrib$_s ub_i nn$ be an algorithm-environment sequence for an algorithm with actions in $\mathcal{X}$ and the environment ν_θ, on $(\Omega, \mathbb{P})$. Then $\eta_t := y_t - \langle x_t, \theta \rangle$, $t < T$, are i.i.d. $\mathcal{N}(0,1)$ and, for every t, η_t is independent of the pair $\big((x_s, y_s)_{s<t}, x_t\big)$ (jointly, i.e. of the σ-algebra generated by the history up to $t-1$ together with x_t).*

Proof. def:env The LML statement `IsAlgEnvSeq.hasCondDistrib_feedback` gives that the conditional law of y_t given the history up to $t-1$ and x_t is $\mathcal{N}(\langle x_t, \theta \rangle, 1)$; shifting by the measurable function $\langle x_t, \theta \rangle$ of the conditioning variables gives that the conditional law of η_t is $\mathcal{N}(0,1)$, a constant kernel, which is the claimed independence. Joint independence of the η_t follows by induction on t. $\square$

Lean remark. The formalization expresses the independence statement as a *constant* conditional distribution: ‘`HasCondDistrib (noise  X Y (n + 1)) (history n, X (n + 1)) (Kernel.const _ (gaussianReal 0 1)) P`‘ (and similarly at time 0), which is what the concentration argument below consumes.

Lemma 4 (Azuma–Hoeffding for the noise)**.** $ProbabilityTheory.hasSubgaussianMGF_f un_i d_g aussianReal, Proba$ $m \geq 0$ *and* $\varepsilon \geq 0$,

$$\mathbb{P}\Big(\sum_{s<m} \eta_{n+s} \geq \varepsilon\Big) \leq e^{-\varepsilon^2/(2m)}, \qquad \mathbb{P}\Big(\Big|\sum_{s<m} \eta_{n+s}\Big| \geq \varepsilon\Big) \leq 2e^{-\varepsilon^2/(2m)}.$$

Proof. lem:noise$_r epresentation A centered Gaussian$ N(0,1)$is sub-Gaussian with variance proxy$ 1$(its moment ge$ Since the conditional law of η_{t+1} given $\sigma(H_t, x_{t+1})$ is the constant $\mathcal{N}(0,1)$, the conditional moment generating function of η_{t+1} given this σ-algebra is $e^{t^2/2}$ (Mathlib `HasCondSubgaussianMGF`, established through `condExpKernel`); the noise process is adapted to the filtration $\mathcal{F}_s := \sigma(H_{n+s}, x_{n+s+1})$, so the Azuma–Hoeffding inequality of Mathlib (`measure_sum_ge_le_of_hasCondSubgaussianMGF`) applies to the sum of the noises over the block $[n, n+m)$; the lower tail is the same bound for $-\eta$ and the two-sided bound is the union bound. $\square$

Lean remark. This lemma replaces, in the formalization, the combination of Lemma **??** (i.i.d. Gaussians) and the conditioning step of Lemma **??** in the proof of Lemma **??**: the empirical mean of the observations of a block of rounds in which the same action is repeated concentrates around its mean, for the run of the composed algorithm itself.

1.2 Identification algorithms and PAC guarantees

Definition 5 (Identification algorithms; runs; fixed budget)**.** Learning.IdentAlg, Learning.IdentAlg.stoppingTime, Learning.IdentAlg.IsRun, Learning.IdentAlg.IsFixedBudget, Learning.IdentAlg.fixedBudget, Learning.IdentAlg.fixedBudgetRunMeasure, Learning.IdentAlg.IsFixedBudget = (alg, τ, ρ) of an LML algorithm alg with actions in $\mathcal{X}$ and observations in $\mathbb{R}$ (a sampling rule), a *stopping rule*: a measurable predicate on finite histories, "stop after n rounds when their history is $h = (x_t, y_t)_{t<n}$"; and a *recommendation rule* (output rule) ρ: for each n, a kernel from histories of length n to $\mathcal{X}$ which is a probability measure on every history at which the rule stops.

A *run* of Alg against ν_θ on a probability space $(\Omega, \mathbb{P})$ consists of action and observation processes $(x_t, y_t)_{t\in\mathbb{N}}$ forming an algorithm-environment sequence for alg and ν_θ (LML `IsAlgEnvSeq`), and of a recommendation $\hat{x}$ whose conditional law given the history $H_\tau = (x_t, y_t)_{t<\tau}$ is $\rho(H_\tau)$, where the number of rounds played $\tau := \min\{n :$ the stopping rule fires on $(x_t, y_t)_{t<n}\} \in \mathbb{N} \cup \{\infty\}$ is the hitting time of the stopping rule by the process of histories (Mathlib `hittingAfter`), a stopping time of the history filtration. The law of $(H_\tau, \hat{x})$ is the same for all runs (`Learning.isAlgEnvSeq_unique`); we write $\mathbb{P}_\theta$ for it, and $\mathbb{E}_\theta$ for the corresponding expectation, whenever a statement only depends on this law.

An identification algorithm has *fixed budget* $T \geq 0$ if its stopping rule is "stop after exactly T rounds"; then $H_\tau = H_T = (x_t, y_t)_{t<T}$ and only the kernel ρ on histories of length T matters, so it is determined by the pair (alg, ρ) with ρ a Markov kernel from histories of length T to $\mathcal{X}$. All the algorithms of this blueprint have a fixed budget (Remark **??**); *fixed-confidence* algorithms, which stop adaptively, are covered by the same definition.

Lean remark. The randomness of the recommendation is allowed on purpose (the lower bounds are stated for randomized procedures). Budget $T = 0$ is allowed: the history of length 0 lives in a one-point space (`Fin 0 →   ×  `) and the recommendation kernel is then just a probability measure on $\mathcal{X}$; this degenerate case is needed by the singular-design lemma (Lemma **??**) and its use in Chapter **??**. Histories of length $T \geq 1$ are LML's `Iic (T-1) →   ×  `. A deterministic recommendation is the kernel `Kernel.deterministic`. Stopping times are not modelled: the paper's PAC algorithms use a deterministic budget $T = (H_1 \log(1/\delta) + H_2)/\varepsilon^2$ and all its lower bounds are stated for such budgets; a randomized budget can be handled by padding (Remark **??**).

Lemma 6 (The number of rounds is a stopping time)**.** $Learning.IdentAlg.isStoppingTime_stoppingTimedef : ba$ *be an algorithm-environment sequence on* $(\Omega, \mathbb{P})$ *(for any algorithm and environment) and let* τ *be the number of rounds played by an identification algorithm on it (Definition* **??***). Then* τ *is a stopping time of the history filtration* $(\mathcal{F}_n)_{n\in\mathbb{N}}$*,* $\mathcal{F}_n := \sigma((x_t, y_t)_{t\leq n})$ *(LML* `IsAlgEnvSeq.filtration`*):* $\{\tau \leq n\} \in \mathcal{F}_n$ *for every* n*.*

Proof. τ is the hitting time of the measurable set $\{(n, h) :$ the stopping rule fires on $h\}$

by the process $n \mapsto (n, (x_t, y_t)_{t<n})$, which is adapted to $(\mathcal{F}_n)$ since $(x_t, y_t)_{t<n}$ is a function of $(x_t, y_t)_{t \le n}$; the hitting time of a measurable set by an adapted discrete-time process is a stopping time (Mathlib `Adapted.isStoppingTime_hittingAfter`). □

Definition 7 (Simple regret)**.** Learning.LinearBandit.simpleRegretdef:arm$_s$*etThesimple regretof*$f \in$ $\mathcal{X}$ for the reward vector θ is

$$r(\hat{x}, \theta) := \max_{x \in \mathcal{X}} \langle x, \theta \rangle - \langle \hat{x}, \theta \rangle \ \geq 0.$$

We also set $Z(\theta) := \max_{x,y \in \mathcal{X}} \langle x - y, \theta \rangle$, so that $r(\hat{x}, \theta) \leq Z(\theta)$ for every $\hat{x} \in \mathcal{X}$.

Definition 8 ((ε, δ)-PAC algorithm)**.** Learning.LinearBandit.IsPAC, Learning.IdentAlg.IsPACdef:bai$_a lg, def$: *regretLet*$\varepsilon > 0$ and $\delta \in (0, 1)$. An identification algorithm is (ε, δ)-*PAC on* $\mathcal{X}$ if for every $\theta \in \mathbb{R}^d$ and every run of it against ν_θ on a probability space $(\Omega, \mathbb{P})$,

$$\mathbb{P}\big(r(\hat{x}, \theta) \leq \varepsilon\big) \geq 1 - \delta,$$

that is, $\mathbb{P}_\theta(r(\hat{x}, \theta) \leq \varepsilon) \geq 1 - \delta$ for the law $\mathbb{P}_\theta$ of Definition **??**.

Lean remark (existence of runs and universes). The PAC property quantifies over all runs on all probability spaces; for it to have content, runs must exist. For a fixed-budget algorithm Alg with budget T the canonical run lives on the space $(\mathbb{N} \to \mathcal{X} \times \mathbb{R}) \times \mathcal{X}$ of pairs (trajectory, recommendation), with the law "trajectory measure of the sampling rule, then the recommendation kernel applied to the history of the first T rounds" (`Learning.IdentAlg.fixedBudgetRunMeasure`, `IsFixedBudget.isRun_fixedBudgetRunMeasure`); the output kernels of an identification algorithm are required to be s-finite so that this composition-product is well defined. Since Lean's universes cannot be quantified over inside a statement, `Learning.LinearBandit.IsPAC` quantifies over probability spaces in the universe of the ambient space $E \supseteq \mathcal{X}$, which is the universe of the canonical run; runs are transported along measurable maps of the sample space by `IsRun.comp_hasLaw`.

Lemma 9 (Expected regret of a PAC algorithm under a prior)**.** *ProbabilityTheory.integral*$_l e_a dd_m ul_i ntegral_o f_m$ *PAC with budget* T *and let* π *be a probability measure on* $\mathbb{R}^d$ *such that* $\int Z(\theta)\, \pi(d\theta) < \infty$*. Let* κ *be a Markov kernel from* $\mathbb{R}^d$ *to the space of pairs* $(H_T, \hat{x})$ *such that* $\kappa(\theta) = \mathbb{P}_\theta$ *for* π*-almost every* θ *("run* Alg *against* ν_θ *conditionally on* θ*"), and let* $(\theta, H_T, \hat{x}) \sim \pi \otimes \kappa$*. Then*

$$\mathbb{E}\big[r(\hat{x}, \theta)\big] \leq \varepsilon + \delta \int Z(\theta)\, \pi(d\theta).$$

Proof. def:regret Pathwise, $r \leq \varepsilon + Z(\theta) \mathbf{1}\{r > \varepsilon\}$. Taking expectations under $\pi \otimes \kappa$ (Fubini for the composition-product, Mathlib `MeasureTheory.Measure.integral_compProd`), $\mathbb{E}[Z(\theta)\mathbf{1}\{r > \varepsilon\}] = \int Z(\theta)\, \kappa(\theta)(r > \varepsilon)\, \pi(d\theta) = \int Z(\theta)\, \mathbb{P}_\theta(r > \varepsilon)\, \pi(d\theta) \leq \delta \int Z \, d\pi$.

Lean remark. In Lean the lemma is stated for a prior π on a parameter space Θ, an auxiliary "noise" probability space B and a Markov kernel κ :

$\Theta \times B \to O$, i.e. for the joint law $(\pi \otimes \nu) \otimes \kappa$ of $((\theta, b), o)$; the PAC hypothesis is that for every θ the failure probability under $\nu \otimes \kappa(\theta, \cdot)$ is at most δ, and it also requires $\varepsilon \geq 0$ (the statement is false for $\varepsilon < 0$). In the application Θ carries the standard Gaussian g generating θ, B the noise η, and κ is the recommendation kernel applied to the history. The kernel κ is a hypothesis, not a construction: measurability of $\theta \mapsto \mathbb{P}_\theta$ for LML trajectory measures is not available off the shelf. Both uses of the lemma supply κ explicitly: for a fixed design $\kappa(\theta) = \mathcal{N}(X\theta, I_T) \otimes \rho$ (Lemma **??**), and Chapter **??** builds κ from the likelihood ratio of Lemma **??**. ☐

Remark 10 (Stopping times). An algorithm with a stopping time τ and $\mathbb{E}[\tau] \leq T_0$ can be turned into one with deterministic budget $T = T_0/\delta$ by padding with dummy rounds: by Markov's inequality $\mathbb{P}(\tau > T) \leq \delta$, so an (ε, δ)-PAC algorithm with random budget becomes $(\varepsilon, 2\delta)$-PAC with budget T_0/δ. All lower bounds below therefore transfer to expected budgets up to the factor $1/\delta$, which is what the paper does in Appendix I. This reduction is not part of the formalization plan.

1.3 Non-adaptive fixed designs

Definition 11 (Fixed-design algorithm)**.** Learning.fixedDesignAlg, Learning.IdentAlg.IsFixedDesigndef:bai$_a$*lg* $\mathcal{X}$. The associated *fixed-design (non-adaptive) algorithm* is the identification algorithm whose sampling rule is deterministic and ignores the observations, playing x_t at round t (LML `detAlgorithm` with a history-independent next-action function), together with an arbitrary recommendation kernel ρ from $(y_t)_{t<T} \in \mathbb{R}^T$ to $\mathcal{X}$ (budget $T \geq 0$; for $T = 0$ the design is empty and ρ is a probability measure on $\mathcal{X}$). We write $X \in \mathbb{R}^{T \times d}$ for the matrix with rows $x_t^\top$ and $\Sigma_T := X^\top X = \sum_{t<T} x_t x_t^\top$ for its *design matrix*. Since the actions are deterministic, the history H_T is the image of $y = (y_t)_{t<T}$ under the measurable bijection $y \mapsto ((x_t, y_t))_{t<T}$; we identify $(H_T, \hat{x})$ with $(y, \hat{x})$ and write $\mathbb{P}_\theta$ for the law of either.

Lemma 12 (Law of the observations under a fixed design)**.** *Learning.IsAlgEnvSeq.hasCondDistrib*$_n$*oise*$_f$*inVec*, *the observation vector* $y = (y_t)_{t<T}$ *satisfies* $y = X\theta + \eta$ *with* $\eta \sim \mathcal{N}(0, I_T)$*, and the recommendation is* $\hat{x} \sim \rho(y)$*. In particular the law of* $(y, \hat{x})$ *is* $\mathcal{N}(X\theta, I_T) \otimes \rho$.

Proof. lem:noise$_r$*epresentation, lem : gauss*$_p$*iLemma* **??***withdeterministicactions :* *the*η_t are i.i.d. standard Gaussians, so $\eta \sim \mathcal{N}(0, I_T)$ (Lemma **??**). ☐

Lean remark. The noise $\eta_t = y_t - \langle x_t, \theta \rangle$ is i.i.d. $\mathcal{N}(0, 1)$ for *every* algorithm, not only for fixed designs: by Lemma **??** its conditional law given the past (hence given the past noises) is constant, and Lemma **??** in its sequential form applies (`Learning.IsAlgEnvSeq.hasLaw_noise_finVec`, `iIndepFun_noise`, `hasLaw_toLp_noise_finVec` for the standard Gaussian law on `EuclideanSpace ℝ (Fin T)`). For a fixed design the actions are x_t almost surely (LML `action_detAlgorithm_ae_all_eq`) and $y_t = \langle x_t, \theta \rangle + \eta_t$, which gives the law of y (`hasLaw_feedback_finVec_of_fixedDesign`); the joint law with the recommendation is `Learning.IdentAlg.map_finHistory_out_eq`.

Lemma 13 (Transport of a fixed-design algorithm through a linear map)**.** *Learning.LinearBandit.relabelHist, Learning.LinearBandit.transportKernel, Learning.LinearBandit.fixedDesignTransport, Learning.LinearBandit.fixedDesignHistLaw*$_m ap_r elabelHist, Learning.$ *E and $\mathcal{Y} \subseteq F$ be action sets in two Euclidean spaces, let $L : F \to E$ be linear, let $x_0, \ldots, x_{T-1} \in \mathcal{X}$ be a fixed design with recommendation kernel ρ (from $\mathbb{R}^T$ to $\mathcal{X}$), and let $x'_0, \ldots, x'_{T-1} \in \mathcal{Y}$ satisfy $\langle x'_t, \vartheta \rangle = \langle x_t, L\vartheta \rangle$ for all t and $\vartheta \in F$ (for instance $x'_t = L^* x_t$, when $L^* x_t \in \mathcal{Y}$). Let $\pi : \mathcal{X} \to \mathcal{Y}$ be measurable with $r_{\mathcal{Y}}(\pi(\hat{x}), \vartheta) \leq r_{\mathcal{X}}(\hat{x}, L\vartheta)$ for all $\hat{x} \in \mathcal{X}$ and $\vartheta \in F$. The* transported algorithm *is the fixed-design algorithm on $\mathcal{Y}$ with design x' and recommendation kernel $\pi_* \rho$: draw $\hat{x} \sim \rho(y)$ (with the history relabelled by the actions x_t) and output $\pi(\hat{x})$. If the original algorithm is (ε, δ)-PAC on $\mathcal{X}$, the transported algorithm is (ε, δ)-PAC on $\mathcal{Y}$.*

Proof. lem:fixed$_d esign_l aw, def : regretFix \vartheta \in F$ and put $\theta := L\vartheta$. By Lemma **??**, under ν_ϑ on $\mathcal{Y}$ the observations of the transported algorithm are $y'_t = \langle x'_t, \vartheta \rangle + \eta_t = \langle x_t, \theta \rangle + \eta_t$, so y' has the law $\mathcal{N}(X\theta, I_T)$ of the observation vector y of the original design under ν_θ on $\mathcal{X}$; hence the pair (observations, recommendation) of the transported algorithm has the law of $(y, \pi(\hat{x}))$ for the original algorithm under θ. Therefore $\mathbb{P}_\vartheta\big(r_{\mathcal{Y}}(\pi(\hat{x}), \vartheta) \leq \varepsilon\big) \geq \mathbb{P}_\theta\big(r_{\mathcal{X}}(\hat{x}, \theta) \leq \varepsilon\big) \geq 1 - \delta$. $\square$

Lean remark. `fixedDesignTransport A T x x' h` is `IdentAlg.fixedBudget` of the fixed design x' with the output kernel `transportKernel`, the composition of the relabelling of the history (`relabelHist`, a deterministic kernel), of the output kernel of `A` and of the deterministic kernel of π. The proof is a computation with the explicit laws `fixedDesignPairLaw` of Lemma **??**.

1.4 Design matrices and the Gaussian width

Definition 14 (Design distributions and design matrices)**.** Maiti2026Power.designSet, Maiti2026Power.outerSelf, Maiti2026Power.designMatrix, Maiti2026Power.IsDesign, Maiti2026Power.IsDesign.designMatrix$_m em_d esignSet, Maiti2026Power.inv_s mul_s um_o uterSelf_m em_d esignSet$ $(\lambda_x)_{x \in S}$, $S \subset \mathcal{X}$ finite, $\lambda_x \geq 0$, $\sum_{x \in S} \lambda_x = 1$; we write $\lambda \in \triangle_{\mathcal{X}}$ and $\mathrm{supp}(\lambda) := \{x \in S : \lambda_x > 0\}$. Its *design matrix* is

$$A(\lambda) := \sum_{x \in S} \lambda_x \, x x^\top \in \mathcal{S}^d_+.$$

The *set of design matrices* is $\mathcal{M}(\mathcal{X}) := \{A(\lambda) : \lambda \in \triangle_{\mathcal{X}}\} = \mathrm{conv}\{x x^\top : x \in \mathcal{X}\}$. Finally, for a fixed design $x_0, \ldots, x_{T-1}$ the *empirical design distribution* is $\lambda_T := \frac{1}{T} \sum_{t<T} \delta_{x_t}$, so that $\Sigma_T = T\, A(\lambda_T)$.

Lean remark. A design distribution is a finitely supported function $\mathcal{X} \to [0,1]$ summing to 1 (`Finsupp`, or a `Finset` with a weight function; zero weights are allowed and supp is the set of positive weights). Restricting to finitely supported designs loses nothing: by Carathéodory's theorem (Lemma **??**) every barycenter of $\{x x^\top : x \in \mathcal{X}\}$ under a probability measure on $\mathcal{X}$ is also $A(\lambda)$ for a λ supported on at most $d(d+1)/2 + 1$ points. Finitely supported designs are what the fixed-design construction consumes.

Lemma 15 (Basic properties of the set of design matrices)**.** $Maiti2026Power.convex_designSet, Maiti2026Powe$
and every $A \in \mathcal{M}(\mathcal{X})$ *satisfies* $\mathrm{tr}(A) \le \sup_{x\in\mathcal{X}} \|x\|^2$.

Proof. def:design$_m atrixThemap$x $\mapsto xx^\top$ is continuous, so $\{xx^\top : x \in \mathcal{X}\}$ is compact; the convex hull of a compact subset of a finite-dimensional space is compact (Mathlib `Set.Finite.isCompact_convexHull` / `isCompact_convexHull`). Positive semidefiniteness and the trace bound are preserved by convex combinations. ☐

Lemma 16 (Existence of a positive definite design)**.** $Maiti2026Power.exists_posDef_mem_designSetdef : design_m$
$\triangle_{\mathcal{X}}$ *with* $A(\lambda)$ *positive definite. More precisely, if* $b_1, \dots, b_d \in \mathcal{X}$ *is a basis of* $\mathbb{R}^d$ *(which exists since* $\mathrm{span}(\mathcal{X}) = \mathbb{R}^d$*), then the uniform design on* $\{b_1, \dots, b_d\}$ *has* $A(\lambda) \succ 0$.

Proof. def:design$_m atrixFor$v $\neq 0$, $v^\top A(\lambda)v = \frac{1}{d}\sum_i \langle b_i, v\rangle^2 > 0$ since some $\langle b_i, v\rangle \neq 0$ (the b_i span $\mathbb{R}^d$). ☐

Definition 17 (Gaussian width of a compact set for a matrix)**.** Maiti2026Power.gwMat, ProbabilityTheory.gaussianWidthdef:arm$_s etForanonemptycompactset$K $\subseteq \mathbb{R}^d$ (typically $K = \mathcal{X}$, or a region $K = R \subseteq \mathcal{X}$ in Chapter **??**), a positive definite matrix $A \in \mathcal{S}^d_{++}$ and $g \sim \mathcal{N}(0, I_d)$, the *Gaussian width of* K *for* A is

$$w(K; A) := \mathbb{E}\Big[\sup_{x\in K} \langle x, A^{-1/2} g\rangle\Big] = \mathbb{E}\Big[\sup_{x\in K} \langle A^{-1/2} x, g\rangle\Big].$$

Here $A^{-1/2}$ is the positive definite square root of A^{-1} (Mathlib `CFC.sqrt`). In Lean, since $A^{-1/2} g \sim \mathcal{N}(0, A^{-1})$, $w(K; A)$ is defined as the *Gaussian width of* K *for the measure* $\mathcal{N}(0, A^{-1})$, $\int \sup_{x\in K} \langle x, \xi\rangle\, \mathcal{N}(0, A^{-1})(d\xi)$ (`ProbabilityTheory.gaussianWidth`, defined for any measure), which avoids matrix square roots in the definition; the two expressions agree by Lemma **??**. The lemmas of this section that do not involve the design set $\mathcal{M}(\mathcal{X})$ (Lemmas **??**, **??**, **??** and **??**) are stated and proved for an arbitrary nonempty compact index set K; they are written with $K = \mathcal{X}$ for readability.

Lemma 18 (The width for a matrix is well defined)**.** $ProbabilityTheory.integrable_supportFn_of_isGaussian, Mai$
is measurable, integrable, and $0 \le w(\mathcal{X}; A) \le \sup_{x\in\mathcal{X}} \|A^{-1/2} x\| \cdot \sqrt{d}$*. Moreover* $w(\mathcal{X}; A)$ *only depends on the law of the Gaussian vector* $G := A^{-1/2} g \sim \mathcal{N}(0, A^{-1})$*:* $w(\mathcal{X}; A) = \mathbb{E}[\sup_{x\in\mathcal{X}} \langle x, G\rangle]$ *for any* $G \sim \mathcal{N}(0, A^{-1})$.

Proof. lem:sup$_c ountable_d ense, lem : gauss_n orm_m oments, lem : gauss_l inear_i mageMeasurability :$
$thesupremumcanbetakenoveracountabledensesubsetof$X$(Lemma$ **??**$).Integrabilityandtheupperbound :|\sup_x\langle$
$\sup_x \|A^{-1/2} x\|\, \|g\|$ and $\mathbb{E}\|g\| \le \sqrt{\mathbb{E}\|g\|^2} = \sqrt{d}$ (Lemma **??**). Nonnegativity: for any $x_0 \in \mathcal{X}$, $\sup_x \langle A^{-1/2} x, g\rangle \ge \langle A^{-1/2} x_0, g\rangle$, whose expectation is 0. The last claim is that $A^{-1/2} g \sim \mathcal{N}(0, A^{-1})$ (Lemma **??**) and that the expectation of a measurable function only depends on the law. ☐

Definition 19 (Gaussian width of $\mathcal{X}$)**.** Maiti2026Power.gwdef:width$_m atrix, def :$
$design_m atrix, lem : exists_p osdef_d esignLet$X$bespanning.The$ *Gaussian width* $termoftheactionsetisw(\mathcal{X}) :=$

$\inf\left\{w(\mathcal{X};A) : A \in \mathcal{M}(\mathcal{X}) \cap \mathcal{S}^d_{++}\right\} = \inf_{\lambda \in \triangle_{\mathcal{X}},\ A(\lambda) \succ 0} \mathbb{E}\Big[\max_{x \in \mathcal{X}} \langle x, A(\lambda)^{-1/2} g\rangle\Big]$. The set over which the infimum is taken is nonempty by Lemma **??**, so $0 \le w(\mathcal{X}) < \infty$.

Lemma 20 (Homogeneity)**.** *Maiti2026Power.gwMat$_s$muldef : width$_m$atrix, lem : sqrt$_p$rops(Foranarbitraryn* $\in \mathcal{S}^d_{++}$ *and* $c > 0$*,* $w(\mathcal{X}; cA) = c^{-1/2}\, w(\mathcal{X}; A)$*.*

Proof. def:width$_m$*atrix*(cA)$^{-1/2} = c^{-1/2} A^{-1/2}$ and the supremum is positively homogeneous. ☐

Lemma 21 (Monotonicity in the Loewner order)**.** *Maiti2026Power.gwMat$_a$ntidef : width$_m$atrix, lem : gauss$_c$* $B \in \mathcal{S}^d_{++}$ *and* $A \preceq B$*, then* $w(\mathcal{X}; B) \le w(\mathcal{X}; A)$*.*

Proof. lem:loewner$_i$*nv, lem : gauss$_c$omparison, lem : width$_m$atrix$_w$dByLemma* **??**,B$^{-1} \preceq A^{-1}$. The processes $x \mapsto \langle x, G_B\rangle$ and $x \mapsto \langle x, G_A\rangle$ with $G_B \sim \mathcal{N}(0, B^{-1})$, $G_A \sim \mathcal{N}(0, A^{-1})$ are centered linear Gaussian processes whose covariance matrices are ordered, so Lemma **??** (covariance domination) gives $\mathbb{E}\sup_x \langle x, G_B\rangle \le \mathbb{E}\sup_x \langle x, G_A\rangle$; conclude with Lemma **??**. ☐

Lemma 22 (Invariance under linear isomorphisms)**.** *Maiti2026Power.gwMat$_i$mage, Maiti2026Power.gw$_i$mage* $\in \mathbb{R}^{d\times d}$ *be invertible and* $\mathcal{X}' := M\mathcal{X}$*. Then for every* $A \in \mathcal{S}^d_{++}$*,* $w(\mathcal{X}'; MAM^\top) = w(\mathcal{X}; A)$*, the map* $A \mapsto MAM^\top$ *is a bijection from* $\mathcal{M}(\mathcal{X}) \cap \mathcal{S}^d_{++}$ *onto* $\mathcal{M}(\mathcal{X}') \cap \mathcal{S}^d_{++}$*, and* $w(\mathcal{X}') = w(\mathcal{X})$*.*

Proof. lem:gauss$_l$*aw$_o$f$_c$ov, lem : width$_m$atrix$_w$dTheprocess*x $\mapsto \langle Mx, G'\rangle$ with $G' \sim \mathcal{N}(0, (MAM^\top)^{-1})$ is centered Gaussian with covariance $x^\top M^\top (MAM^\top)^{-1} M y = x^\top A^{-1} y$, the same as $x \mapsto \langle x, G\rangle$, $G \sim \mathcal{N}(0, A^{-1})$. Two centered Gaussian vectors with the same covariance have the same law (Lemma **??**), so the two suprema have the same expectation by Lemma **??**. The design matrix of the pushed-forward design is $A(M\lambda) = \sum \lambda_x (Mx)(Mx)^\top = MA(\lambda)M^\top$, and congruence by an invertible matrix preserves positive definiteness. ☐

Lemma 23 (Width of an empirical design)**.** *Maiti2026Power.gw$_l$e$_s$qrt$_m$ul$_g$wMat$_s$umdef : width, def : design$_n$* $\mathcal{X}$ *with* $\Sigma_T = \sum_{t<T} x_t x_t^\top$ *positive definite. Then* $w(\mathcal{X}; \Sigma_T) \ge w(\mathcal{X})/\sqrt{T}$*.*

Proof. lem:width$_h$*omog, def : width*$\Sigma_T = T A(\lambda_T)$ with $A(\lambda_T) \in \mathcal{M}(\mathcal{X}) \cap \mathcal{S}^d_{++}$, so $w(\mathcal{X}; \Sigma_T) = T^{-1/2} w(\mathcal{X}; A(\lambda_T)) \ge T^{-1/2} w(\mathcal{X})$. ☐

Lemma 24 (Symmetry of the width)**.** *Maiti2026Power.gwMat$_n$eg$_s$et, Maiti2026Power.gwMat$_s$ub$_s$etdef : wid* $\in \mathcal{S}^d_{++}$ *and* $G \sim \mathcal{N}(0, A^{-1})$*,* $\mathbb{E}[\max_{x\in\mathcal{X}} \langle x, -G\rangle] = \mathbb{E}[\max_{x\in\mathcal{X}} \langle x, G\rangle] = w(\mathcal{X}; A)$ *and* $\mathbb{E}[Z(G)] = \mathbb{E}[\max_{x,y\in\mathcal{X}} \langle x - y, G\rangle] = 2\, w(\mathcal{X}; A)$*.*

Proof. lem:gauss$_l$*aw$_o$f$_c$ov*-G $\sim \mathcal{N}(0, A^{-1})$ as well (Lemma **??**), and $\max_{x,y} \langle x - y, G\rangle = \max_x \langle x, G\rangle + \max_y \langle y, -G\rangle$. ☐

1.5 Sequential composition of algorithms

The adaptive algorithms of the paper (Chapters **??**, **??** and **??**) are built by running a first algorithm for T_1 rounds and then a second algorithm whose parameters depend on the first history. The following framework lemma isolates what is needed.

Lemma 25 (Sequential composition)**.** *def:bai$_a$lgLetalg$_1$ be an algorithm with actions in $\mathcal{X}$, $T_1 \geq 1$, and for each history h of length T_1 let* $\mathrm{alg}_2(h)$ *be an algorithm, measurably in h. There is an algorithm* alg *(the* composition*) whose policy coincides with that of* alg_1 *at rounds $t < T_1$ and with that of* $\mathrm{alg}_2(H_{T_1})$ *at rounds $t \geq T_1$ (applied to the history after time T_1). For every environment ν, under the law of the composition against ν: the history H_{T_1} has the law of* alg_1 *against ν, and conditionally on $H_{T_1} = h$ the shifted process $(x_{T_1+s}, y_{T_1+s})_{s\geq 0}$ is an algorithm-environment sequence for* $(\mathrm{alg}_2(h), \nu)$*. Consequently, if $(E_h)_h$ is a family of sets of trajectories, jointly measurable in $(h, \cdot)$, with $\mathbb{P}^{\mathrm{alg}_2(h),\nu}(E_h) \geq 1 - \delta$ for every h in a set of* alg_1*-probability at least $1 - \delta'$, then the composition satisfies $\mathbb{P}\big((x_{T_1+s}, y_{T_1+s})_{s\geq 0} \in E_{H_{T_1}}\big) \geq 1 - \delta - \delta'$.*

Proof. def:bai$_a$$lgDefinethepolicyof$ alg$attime$t$\geq T_1$ from the history h' of length t by splitting $h' = (h, h'')$ with h of length T_1 and applying the policy of $\mathrm{alg}_2(h)$ at time $t - T_1$ to h'' (with the initial action distribution of $\mathrm{alg}_2(h)$ at $t = T_1$); measurability in h' holds by assumption. The trajectory measure of the composition (`Learning.trajMeasure`, an Ionescu–Tulcea construction) is, by construction of the kernels, the composition-product of the law of H_{T_1} under alg_1 with the kernel $h \mapsto$ trajectory measure of $(\mathrm{alg}_2(h), \nu)$ shifted by T_1. The probability bound is then Fubini for the composition-product. □

Lean remark. This lemma is a general statement about the LML framework, to be contributed to LML; it is not formalized. The measurability of $h \mapsto \mathrm{alg}_2(h)$ means that the policy kernels and initial distributions, as functions of $(h, \cdot)$, are kernels; in all uses below $\mathrm{alg}_2(h)$ is a fixed algorithm reparametrized by a measurable function of h (a finite set of candidate arms, or a scale r_0). The adaptive lower bound (Lemma **??**) only needs the following special case, in which the composed algorithm is written down directly and the first phase is a PAC algorithm; Median Elimination and the region algorithm of Chapter **??** are *phased algorithms* (Lemma **??**), for which the composition is replaced by a round-by-round analysis using the independence of the future noise from the past.

Lemma 26 (Algorithms agreeing on a prefix; transfer of PAC guarantees)**.** *Learning.Algorithm.AgreeUntil, Learning.IsAlgEnvSeqUntil.of$_a$$greeUntil$, $Learning.IsAlgEnvSeq.isAlgEnvSe$ alg'$betwoalgorithmswiththesameinitialactiondistributionandthesamepoliciesattimest$ $< T_1$ (they* agree until time T_1*). Then every algorithm-environment sequence for* alg *(in any environment ν) is an algorithm-environment sequence for* alg' *until time T_1 (LML* ***`IsAlgEnvSeqUntil`****), and the law of its history H_{T_1+1} is the law of the history of* alg' *against ν. Consequently, let* $\mathsf{Alg} = (\mathrm{alg}', \rho)$ *be an identification algorithm with budget $T_1 + 1$, let $(x_t, y_t)_t$ be an algorithm-environment* *sequence for* alg$'$ *until time* T_1 *on* $(\Omega, \mathbb{P})$ *and let* $\hat{x}$ *be a random variable with conditional law* $\rho(H_{T_1+1})$ *given* H_{T_1+1}*. Then* $(H_{T_1+1}, \hat{x})$ *has the law* $\mathbb{P}_\theta$ *of Definition* **??***, and if* Alg *is* δ*-PAC for a measurable goodness predicate, then* $\mathbb{P}(\hat{x}$ *is good*$) \geq 1 - \delta$.

Proof. def:bai$_alg, def: pacThefirstassertionisimmediatefromthedefinitions, sincetheconditionsof$ `IsAlgEn` only involve the policies at times $t < T_1$. The law of the history is then the image of the trajectory measure of (alg', ν) (`Learning.eq_trajMeasure_map_frestrictLe_of_isAlgEnvSeqUntil`). For the second assertion, the law of $(H_{T_1+1}, \hat{x})$ is the composition-product of the law of H_{T_1+1} with ρ, and so is the law of (history, recommendation) under the canonical run of Alg (`Learning.IdentAlg.fixedBudgetRunMeasure`); the PAC property applied to the canonical run gives the bound. □

Lemma 27 (Seeded algorithms and the seed representation of a run)**.** *Learning.SeededAlg, Learning.SeededAlg.toAlgorithm, Learning.SeededAlg.policy*$_a$*pply, Learning.noiseEnv, Learning on* U *(the law of the seeds), a measurable first action* $a_0 : U \to \mathcal{X}$ *and measurable maps* $\mathrm{next}_n : \mathcal{X}^{n+1} \times \mathbb{R}^{n+1} \times U \to \mathcal{X}$*: at every round a fresh seed* $u_n \sim \mu_U$ *is drawn, independently of everything else, and the action* $x_n = \mathrm{next}_{n-1}(H_{n-1}, u_n)$ *is played* $(x_0 = a_0(u_0))$*. It is an algorithm in the sense of Definition* **??***, whose policy at round* n *is the kernel* $h \mapsto \mu_U \circ \mathrm{next}_{n-1}(h, \cdot)^{-1}$*. A* noise environment *answers the action* x *with* $F(x, e)$ *for a fresh noise* $e \sim \mu_E$*; the linear Gaussian environment* ν_θ *is the case* $F(x, e) = \langle x, \theta \rangle + e$*,* $\mu_E = \mathcal{N}(0, 1)$*.*

Seed representation. *On the product space* $(U \times E)^{\mathbb{N}}$ *with the product measure* $\mathbb{P}_{\mathrm{seed}} := (\mu_U \otimes \mu_E)^{\otimes \mathbb{N}}$ *define recursively* $x_0 := a_0(u_0)$*,* $y_n := F(x_n, e_n)$ *and* $x_{n+1} := \mathrm{next}_n(H_n, u_{n+1})$*. Then* $(x_n, y_n)_n$ *is an algorithm-environment sequence for the seeded algorithm and the noise environment. Since the law of the history of an algorithm-environment sequence is determined by the algorithm and the environment (by induction on the rounds, from the conditional laws of the steps), the history* H_T *of* every *run of the seeded algorithm in the noise environment has the law of the recursively defined history under* $\mathbb{P}_{\mathrm{seed}}$*. Consequently, for the fixed-budget identification algorithm with budget* T *and a deterministic output* $g(H_T)$*: if for every* θ*,* $\mathbb{P}_{\mathrm{seed}}\big(g(H_T)$ *is good for* $\theta\big) \geq 1 - \delta$*, then the algorithm is* δ*-PAC.*

Proof. def:bai$_alg, def: pac, def: envTheconditionallawofthecoordinate(\mathrm{u}_{n+1}, e_{n+1})$ given the coordinates $0, \ldots, n$ is the constant $\mu_U \otimes \mu_E$ (independence of the coordinates of a product measure), and H_n is a measurable function of these coordinates. Hence the conditional law of $x_{n+1} = \mathrm{next}_n(H_n, u_{n+1})$ given H_n is the policy kernel, and the conditional law of $y_{n+1} = F(x_{n+1}, e_{n+1})$ given (H_n, x_{n+1}) is the feedback kernel $\mu_E \circ F(x_{n+1}, \cdot)^{-1}$: these are the defining properties of an algorithm-environment sequence. The uniqueness of the history law is the induction $H_{n+1} \sim \mathrm{law}(H_n) \otimes (\text{step kernel at } n)$. The PAC statement follows since the output of the fixed-budget algorithm is almost surely $g(H_T)$. □

Lean remark. This framework replaces Lemma **??** for the randomized algorithms of Chapter **??**: the Rademacher directions are the seeds, and the analysis of an adaptive composition is a computation on the product space of seeds and noises, where independence between the rounds of different phases is the product structure. The deterministic phased algorithms of Lemma **??** are the case of a trivial seed space.

Lemma 28 (Phased algorithms)**.** *Learning.PhasedAlg, Learning.PhasedAlg.state,* $Learning.PhasedAlg.state_congr, Learning.PhasedAlg.toAlgorithm, Learning.PhasedAlg.toIdentAlg, Learnin$
1, an initial state s_0*, measurable designs* $\mathrm{act}_\ell : S \to \mathcal{X}^{n_\ell}$ *and measurable updates* $\mathrm{upd}_\ell : S \times \mathbb{R}^{n_\ell} \to S$*: during phase* ℓ *(the rounds* $\tau_\ell \le t < \tau_\ell + n_\ell$*,* $\tau_\ell := \sum_{k<\ell} n_k$*) it plays the design* $\mathrm{act}_\ell(s_\ell)$ *of the state* s_ℓ *at the start of the phase, and at the end of the phase it sets* $s_{\ell+1} := \mathrm{upd}_\ell(s_\ell, y^{(\ell)})$ *where* $y^{(\ell)}$ *are the observations of the phase. It is a deterministic algorithm:* s_ℓ *is a measurable function of the observations before time* τ_ℓ*. With a budget of* L *phases and a measurable output* $\mathrm{out} : S \to \mathcal{O}$ *of the final state, it is a fixed-budget identification algorithm with budget* τ_L*. Prepending a fixed-design phase whose observations set the state (*`PhasedAlg.cons`*) gives a phased algorithm.*

Along any run in the linear Gaussian bandit ν_θ*: the actions of phase* ℓ *are* $\mathrm{act}_\ell(s_\ell)$*, the observations are* $\langle \mathrm{act}_\ell(s_\ell)_j, \theta\rangle + \eta_j^{(\ell)}$ *where the noise* $\eta^{(\ell)} \sim \mathcal{N}(0, I_{n_\ell})$ *is independent of* s_ℓ*, and* $s_{\ell+1} = \mathrm{upd}_\ell(s_\ell, \langle \mathrm{act}_\ell(s_\ell), \theta\rangle + \eta^{(\ell)})$*. Consequently, if* $G_0, \dots, G_L \subseteq S$ *are measurable sets of* good *states with* $s_0 \in G_0$ *and, for every* $\ell < L$ *and every* $s \in G_\ell$*,* $\mathbb{P}_{\eta\sim\mathcal{N}(0,I_{n_\ell})}\big(\mathrm{upd}_\ell(s, \langle \mathrm{act}_\ell(s), \theta\rangle + \eta) \in G_{\ell+1}\big) \ge 1 - \delta_\ell$*, then* $\mathbb{P}(s_L \in G_L) \ge 1 - \sum_{\ell<L} \delta_\ell$*; and if moreover* $\mathrm{out}(s)$ *is good for every* $s \in G_L$*, the identification algorithm is* δ*-PAC for* $\delta \ge \sum_{\ell<L} \delta_\ell$ *(for every* θ*, with good sets depending on* θ*).*

Proof. lem:noise$_a zuma, lem : noise_r epresentation, lem : prefix_t ransferThestateatthestartofphase\ell$ only depends on the observations before the phase (`state_congr`), so the phased algorithm is the deterministic algorithm whose next action at time t in phase ℓ is $\mathrm{act}_\ell(s_\ell)_{t-\tau_\ell}$ (`toAlgorithm`), and along a run the actions are almost surely these (`ae_action_start_add_eq`, from the deterministic policy). The noise $\eta_t := y_t - \langle x_t, \theta\rangle$ satisfies: conditionally on the history up to time $t-1$ and x_t, $\eta_t \sim \mathcal{N}(0,1)$ (Lemma **??**). Hence, for every m, the noises $(\eta_{m+j})_{j<n}$ are i.i.d. $\mathcal{N}(0,1)$ conditionally on the history of the first m rounds (`hasCondDistrib_noise_window`; abstractly, a sequence W_k whose conditional law given $(Z, W_0, \dots, W_{k-1})$ is a constant ν for every k has the constant conditional law $\nu^{\otimes n}$ given Z, `hasCondDistrib_pi_of_hasCondDistrib_` by induction on n with the product structure of the joint law). Since s_ℓ is a function of the history of the first τ_ℓ rounds, the noise of phase ℓ is independent of s_ℓ with law $\mathcal{N}(0, I_{n_\ell})$ (`hasCondDistrib_phaseNoise`), and the observations of the phase are the means plus this noise. The per-phase step is Fubini for the product law of $(s_\ell, \eta^{(\ell)})$: for any measurable $G \subseteq S$ and $G' \subseteq S \times \mathbb{R}^n$ with $\mathbb{P}_\eta((s,\eta) \in G') \ge 1 - \delta$ for all $s \in G$, $\mathbb{P}((s_\ell, \eta^{(\ell)}) \in G') \ge \mathbb{P}(s_\ell \in G) - \delta$ (`HasCondDistrib.measureReal_sub_le`), which gives $\mathbb{P}(s_{\ell+1} \in G_{\ell+1}) \ge \mathbb{P}(s_\ell \in G_\ell) - \delta_\ell$ (`measureReal_stateProc_mem_sub_le`); induction over the phases gives the union bound (`one_sub_sum_le_measureReal_stateProc_mem`). The output of a run of the identification algorithm is almost surely $\mathrm{out}(s_L)$ (the output kernel is deterministic), which gives the PAC guarantee (`isPAC_toIdentAlg`). □

Chapter 2

Optimal experimental design and rounding

This chapter collects the deterministic (linear-algebraic and convex-analytic) tools on which the fixed-design algorithm of Chapter **??** and the lower bounds of Chapters **??** and **??** rest. It proves:

- the facts about the Loewner order, eigenvalues and matrix square roots that are used throughout (Section **??**); Mathlib already provides the order (`Matrix.instPartialOrder`, scoped in `MatrixOrder`), positive (semi)definiteness (`Matrix.PosDef`, `Matrix.PosSemidef`), the spectral theorem (`Matrix.IsHermitian.spectral_theorem`) and the square root through the continuous functional calculus (`CFC.sqrt`);
- the existence of a G-optimal design, i.e. the "D-optimal implies G-optimal" half of the Kiefer–Wolfowitz theorem (Section **??**). This replaces the use of John's theorem in the paper (Appendix B.2) and gives the lemma stated without proof in the paper's Appendix B.1;
- Carathéodory's theorem for design matrices, and the continuity and coercivity of $A \mapsto w(\mathcal{X}; A)$, which give a minimizer of the Gaussian width (Section **??**);
- a rounding theorem (Section **??**): every design distribution λ can be rounded, for every budget $T \geq 4d$, to a multiset of T points of its support whose design matrix dominates $(T/4)A(\lambda)$ in the Loewner order. This replaces the rounding procedures cited by the paper [**?**, **?**]; the proof is the one-sided barrier argument of Batson, Spielman and Srivastava [**?**] with unit weights.

Nothing in this chapter is probabilistic except the two lemmas on the width, which only use the Gaussian facts of Chapter **??**. Since action sets (Definition **??**) are not assumed to span $\mathbb{R}^d$, every statement of this chapter that relies on the existence of a positive definite design (Lemma **??**) or on the width $w(\mathcal{X})$

(Definition **??**) carries an explicit hypothesis "$\mathcal{X}$ is spanning"; the linear-algebra facts, Carathéodory's theorem, the continuity of the width and the rounding theorem hold for every action set.

Notation. $\mathcal{S}^d$ is the space of symmetric real $d \times d$ matrices, $\mathcal{S}^d_+ \subset \mathcal{S}^d$ the positive semidefinite ones and $\mathcal{S}^d_{++} \subset \mathcal{S}^d_+$ the positive definite ones. For $A, B \in \mathcal{S}^d$ we write $A \preceq B$ if $B - A \in \mathcal{S}^d_+$ and $A \prec B$ if $B - A \in \mathcal{S}^d_{++}$ (the *Loewner order*). For $A \in \mathcal{S}^d$ we write $\lambda_1(A) \le \cdots \le \lambda_d(A)$ for its eigenvalues with multiplicity, $\lambda_{\min}(A) := \lambda_1(A)$, $\lambda_{\max}(A) := \lambda_d(A)$, and $\|A\|_{\mathrm{op}}$ for the operator norm, $\|A\|_F$ for the Frobenius norm. For $W \in \mathcal{S}^d_{++}$, $\|x\|^2_W := x^\top W x$.

2.1 The Loewner order and matrix square roots

Lemma 29 (Loewner order and quadratic forms)**.** $Matrix.dotProduct_mulVec_le_of_le, Matrix.le_iff_dotProduct_mu$
$B \in \mathcal{S}^d$. Then $A \preceq B$ if and only if $x^\top A x \le x^\top B x$ for every $x \in \mathbb{R}^d$, and $A \prec B$ if and only if $x^\top A x < x^\top B x$ for every $x \neq 0$. In particular $\preceq$ is a partial order on $\mathcal{S}^d$ compatible with addition and with multiplication by nonnegative scalars, and $A \preceq B$ together with $B \preceq A$ implies $A = B$.

Proof. By definition $B - A \in \mathcal{S}^d_+$ means that $x^\top (B - A)x \ge 0$ for all x, which is the first claim; the second is the definition of $\mathcal{S}^d_{++}$. Compatibility with addition and nonnegative scalars is immediate on quadratic forms. Antisymmetry: if $x^\top (B-A)x = 0$ for all x then the symmetric matrix $B-A$ vanishes (polarization: $2\,x^\top (B - A)y = (x + y)^\top (B - A)(x + y) - x^\top (B - A)x - y^\top (B - A)y = 0$ for all x, y). □

Lean remark. Mathlib's `Matrix.PosSemidef` is defined through `Matrix.posSemidef_iff_dotProduct_mu` (Hermitian plus nonnegative quadratic form), and the order instance is `Matrix.le_iff : A  B  (B - A).PosSemidef` in `Mathlib/Analysis/Matrix/Order.lean` (`open scoped MatrixOrder`). Over $\mathbb{R}$ the Hermitian condition is symmetry. Antisymmetry is part of the `PartialOrder` instance.

Lemma 30 (Congruence preserves the Loewner order)**.** $Matrix.mul_mul_conjTranspose_le_of_le, Matrix.conjTran$
$B \in \mathcal{S}^d$ with $A \preceq B$ and let $M \in \mathbb{R}^{k \times d}$. Then $MAM^\top \preceq MBM^\top$ in $\mathcal{S}^k$; in particular (take $A = 0$) congruence by an arbitrary matrix M maps positive semidefinite matrices to positive semidefinite matrices. If moreover $k = d$, M is invertible and $A \prec B$, then $MAM^\top \prec MBM^\top$; in particular congruence by an invertible matrix maps $\mathcal{S}^d_{++}$ into $\mathcal{S}^d_{++}$.

Proof. lem:loewner$_q$uadraticFor$z \in \mathbb{R}^k$, $z^\top M(B - A)M^\top z = (M^\top z)^\top (B - A)(M^\top z) \ge 0$ by Lemma **??**; with $A = 0$ this says that $MBM^\top$ is positive semidefinite whenever B is, for every M. If M is invertible and $z \neq 0$ then $M^\top z \neq 0$, so the quantity is > 0 when $A \prec B$. □

Lean remark. `Matrix.PosSemidef.mul_mul_conjTranspose_same` and `Matrix.PosDef.mul_mul_conjTranspose_same` (the latter needs injectivity of $M^\top$, i.e. `Function.Injective M.vecMul`) applied to $B - A$.

Lemma 31 (Eigenvalues and quadratic forms)**.** $Matrix.IsHermitian.trace_cfc, Matrix.IsHermitian.trace_inv_su$
$\in \mathcal{S}^d$. *There is an orthonormal basis* $u_1, \ldots, u_d$ *of* $\mathbb{R}^d$ *with* $Au_j = \lambda_j(A)u_j$*, i.e.* $A = U \operatorname{diag}(\lambda_1(A), \ldots, \lambda_d(A))U^\top = \sum_j \lambda_j(A)u_j u_j^\top$ *with* $U = [u_1 \cdots u_d]$ *orthogonal. Consequently:*

(i) $\lambda_{\min}(A)\|x\|^2 \le x^\top Ax \le \lambda_{\max}(A)\|x\|^2$ *for all* x*, with equality on the left at* $x = u_1$ *and on the right at* $x = u_d$*; hence* $\lambda_{\min}(A) = \min_{\|x\|=1} x^\top Ax$*,* $A \in \mathcal{S}^d_+$ *iff* $\lambda_{\min}(A) \ge 0$*,* $A \in \mathcal{S}^d_{++}$ *iff* $\lambda_{\min}(A) > 0$*, and for* $c \in \mathbb{R}$*,* $cI \preceq A$ *iff* $\lambda_{\min}(A) \ge c$*.*

(ii) If $A \preceq B$ *then* $\lambda_{\min}(A) \le \lambda_{\min}(B)$*; in particular* $\lambda_{\min}(A + P) \ge \lambda_{\min}(A)$ *for* $P \in \mathcal{S}^d_+$*.*

(iii) $\|A\|_{\mathrm{op}} = \max_j|\lambda_j(A)|$*, which equals* $\lambda_{\max}(A)$ *when* $A \in \mathcal{S}^d_+$*.*

(iv) $\operatorname{tr} A = \sum_j \lambda_j(A)$*,* $\det A = \prod_j \lambda_j(A)$*,* $\operatorname{tr}(A^2) = \sum_j \lambda_j(A)^2 \le d \max_j \lambda_j(A)^2$*, and for every* $c \in \mathbb{R}$ *the matrix* $A - cI$ *has the same eigenvectors with eigenvalues* $\lambda_j(A) - c$*.*

(v) If $A \in \mathcal{S}^d_{++}$ *then* $A^{-1} = U \operatorname{diag}(1/\lambda_j(A))U^\top \in \mathcal{S}^d_{++}$*,* $A^{-2} = U \operatorname{diag}(1/\lambda_j(A)^2)U^\top$*,* $\operatorname{tr}(A^{-1}) = \sum_j 1/\lambda_j(A)$*,* $\operatorname{tr}(A^{-2}) = \sum_j 1/\lambda_j(A)^2$ *and* $\|A^{-1}\|_{\mathrm{op}} = 1/\lambda_{\min}(A)$*.*

Proof. lem:loewner$_quadraticTheexistenceoftheorthonormaleigenbasisisthespectraltheoremforrealsymmetr$
$= \sum_j \langle x, u_j\rangle u_j$ gives $x^\top Ax = \sum_j \lambda_j\langle x, u_j\rangle^2$ and $\|x\|^2 = \sum_j\langle x, u_j\rangle^2$, from which (i) follows by bounding each λ_j by $\lambda_{\min}$ and $\lambda_{\max}$. (ii) is (i) combined with Lemma **??**: $\lambda_{\min}(B) = \min_{\|x\|=1} x^\top Bx \ge \min_{\|x\|=1} x^\top Ax$. (iii): $\|Ax\|^2 = \sum_j \lambda_j^2\langle x, u_j\rangle^2 \le \max_j \lambda_j^2\|x\|^2$ with equality at the corresponding u_j. (iv): trace and determinant are invariant under conjugation by U; $A^2 = U \operatorname{diag}(\lambda_j^2)U^\top$; $A - cI = U \operatorname{diag}(\lambda_j - c)U^\top$. (v): $U \operatorname{diag}(1/\lambda_j)U^\top$ is a two-sided inverse of A since $U^\top U = I$, and the remaining identities are (iii) and (iv) applied to A^{-1}. □

Lean remark. `Matrix.IsHermitian.spectral_theorem`, `Matrix.IsHermitian.eigenvalues`, `Matrix.IsHermitian.eigenvectorUnitary`, `Matrix.IsHermitian.det_eq_prod_eigenvalues`, `Matrix.IsHermitian.trace_eq_sum_eigenvalues` (in `Mathlib/Analysis/Matrix/Spectrum.lean`); `Matrix.PosDef.eigenvalues_pos`, `Matrix.PosSemidef.eigenvalues_nonneg`. Mathlib's eigenvalues are indexed by the index type of the matrix (not sorted), so $\lambda_{\min}$ is `Finset.univ.inf'` of the eigenvalue function; statement (i) is the only place where sorting matters and it is best stated as "$\min_j \lambda_j\|x\|^2 \le x^\top Ax$". The identity $x^\top Ax = \sum_j \lambda_j\langle x, u_j\rangle^2$ is the coordinate form of the spectral theorem in the eigenbasis (`OrthonormalBasis` of `EuclideanSpace`). The formalization never sorts the eigenvalues: the condition "$\lambda_{\min}(A) > c$" is encoded as $\forall x \in \operatorname{spec}_{\mathbb{R}}(A),\ c < x$ (the real spectrum is the range of the eigenvalues, `Matrix.IsHermitian.spectrum_real_eq_range_eigenvalues`), which is equivalent to $A - cI \in \mathcal{S}^d_{++}$ (`Matrix.IsHermitian.posDef_sub_smul_one_iff`), and the traces of functions of A are computed through the continuous functional calculus: $\operatorname{tr}(\texttt{cfc}\ f\ A) = \sum_j f(\lambda_j)$ (`Matrix.IsHermitian.trace_cfc`), so that $\operatorname{tr}((A - cI)^{-1}) = \sum_j(\lambda_j - c)^{-1}$ and $\operatorname{tr}((A - cI)^{-2}) = \sum_j(\lambda_j - c)^{-2}$ for c outside the spectrum. This covers the parts of (i), (ii), (iv), (v) used by the rounding argument; the operator-norm statements (iii) and the bound $\mathrm{tr}(A^2) \le d\lambda_{\max}^2$ are not formalized.

Lemma 32 (Square roots of positive definite matrices)**.** *Matrix.PosDef.sqrt,* $Matrix.PosSemidef.sqrt_mul_sqrt, Matrix.PosDef.sqrt_mul_sqrt_inv, Matrix.PosDef.sqrt_inv_mul_sqrt, Matrix.Po$ $\in \mathcal{S}_{++}^d$ *with spectral decomposition* $A = U\,\mathrm{diag}(\lambda_j)U^\top$*. The matrix* $A^{1/2} := U\,\mathrm{diag}(\sqrt{\lambda_j})U^\top$ *is the unique* $S \in \mathcal{S}_+^d$ *with* $S^2 = A$ *(for* $A \in \mathcal{S}_+^d$ *the same formula defines the unique positive semidefinite square root). Moreover:*

(i) $A^{1/2} \in \mathcal{S}_{++}^d$*, it commutes with* A*, its eigenvalues are* $\sqrt{\lambda_j(A)}$ *and* $\lambda_{\min}(A^{1/2}) = \sqrt{\lambda_{\min}(A)}$*;*

(ii) $A^{-1/2} := (A^{1/2})^{-1} = (A^{-1})^{1/2} = U\,\mathrm{diag}(\lambda_j^{-1/2})U^\top \in \mathcal{S}_{++}^d$*,* $(A^{-1/2})^2 = A^{-1}$*,* $\|A^{-1/2}\|_{\mathrm{op}} = \lambda_{\min}(A)^{-1/2}$*;*

(iii) $A^{1/2}A^{-1/2} = A^{-1/2}A^{1/2} = I$ *and* $A^{-1/2}AA^{-1/2} = I$*,* $A^{1/2}A^{-1}A^{1/2} = I$*;*

(iv) $x^\top Ax = \|A^{1/2}x\|^2$ *and* $x^\top A^{-1}x = \|A^{-1/2}x\|^2$ *for all* x*; in particular* $\|x\|_{A^{-1}} = \|A^{-1/2}x\|$*;*

(v) $\langle u, v\rangle = \langle A^{1/2}u, A^{-1/2}v\rangle$ *for all* $u, v \in \mathbb{R}^d$*;*

(vi) for $c > 0$*,* $(cA)^{1/2} = \sqrt{c}\,A^{1/2}$ *and* $(cA)^{-1/2} = c^{-1/2}A^{-1/2}$*; and* $I^{1/2} = I$*.*

Proof. lem:eigen$_q uadratic_b ounds \mathrm{S}_0 := U\,\mathrm{diag}(\sqrt{\lambda_j})U^\top$ is symmetric with $S_0^2 = U\,\mathrm{diag}(\lambda_j)U^\top = A$ and is positive definite by Lemma **??**(i) since $\sqrt{\lambda_j} > 0$. Uniqueness: if $S \in \mathcal{S}_+^d$ and $S^2 = A$, then S commutes with $A = S^2$, hence preserves the eigenspaces of A; on the eigenspace of A for λ, S is a positive semidefinite operator whose square is $\lambda\,\mathrm{id}$, hence (diagonalizing S there) equals $\sqrt{\lambda}\,\mathrm{id}$; so $S = S_0$. (i) is read off the decomposition. (ii): $U\,\mathrm{diag}(\lambda_j^{-1/2})U^\top$ is the inverse of $A^{1/2}$ and the positive square root of $A^{-1} = U\,\mathrm{diag}(\lambda_j^{-1})U^\top$ (Lemma **??**(v)), and its largest eigenvalue is $\lambda_{\min}(A)^{-1/2}$. (iii) follows from (ii). (iv): $x^\top Ax = x^\top A^{1/2}A^{1/2}x = \|A^{1/2}x\|^2$ since $A^{1/2}$ is symmetric, and likewise for A^{-1}. (v): $\langle A^{1/2}u, A^{-1/2}v\rangle = u^\top A^{1/2}A^{-1/2}v = \langle u, v\rangle$ by (iii). (vi): $cA = U\,\mathrm{diag}(c\lambda_j)U^\top$ and $\sqrt{c\lambda_j} = \sqrt{c}\sqrt{\lambda_j}$. ☐

Lean remark. Mathlib defines the square root through the continuous functional calculus: `CFC.sqrt` (`Mathlib/Analysis/SpecialFunctions/ContinuousFunctionalCalculus/Rpow/Ba` with `CFC.sqrt_mul_sqrt_self`, `CFC.sqrt_nonneg`, `CFC.sqrt_unique`, `CFC.sqrt_eq_iff`, and for matrices `Matrix.PosSemidef.inv_sqrt : (CFC.sqrt A) ¹ = CFC.sqrt A ¹` and `Matrix.PosSemidef.det_sqrt` in `Mathlib/Analysis/Matrix/Order.lean`. The explicit spectral formula is available through `Matrix.IsHermitian.cfc` and `Matrix.IsHermitian.cfc_eq` (`Mathlib/Analysis/Matrix/HermitianFunctionalCalculus.lean`), which state that `cfc f A` equals $U\,\mathrm{diag}(f(\lambda_j))U^\top$; this is the bridge between the abstract `CFC.sqrt` and the concrete computations of this chapter. Since Lemma **??** is the only place where the definition of $A^{1/2}$ is unfolded, the rest of the blueprint can use whichever definition is convenient. Note that `CFC.sqrt`

`A = 0` for A not positive semidefinite; all uses below are for $A \in \mathcal{S}^d_{++}$. The formalization uses `CFC.sqrt` and proves the items (i), (iii), (iv), (v) directly from `CFC.sqrt_mul_sqrt_self` and `Matrix.PosSemidef.inv_sqrt`, without the spectral formula; the eigenvalue statements of (i)–(ii) and the scaling rule (vi) are not formalized yet (the latter is proved inline where it is used, in `Maiti2026Power.gwMat_smul`).

Lemma 33 (Inversion reverses the Loewner order)**.** $Matrix.PosDef.of_le, Matrix.PosDef.inv_le_inv lem : loewne$ $\in \mathcal{S}^d_{++}$ *and* $B \in \mathcal{S}^d$ *with* $A \preceq B$*. Then* $B \in \mathcal{S}^d_{++}$ *and* $B^{-1} \preceq A^{-1}$*.*

Proof. lem:loewner$_congruence, lem : loewner_quadratic, lem : eigen_quadratic_bounds, lem : sqrt_props$B $\in \mathcal{S}^d_{++}$: for $x \neq 0$, $x^\top Bx \geq x^\top Ax > 0$ (Lemma **??**). Let $C := A^{-1/2}BA^{-1/2} \in \mathcal{S}^d$. Congruence of $A \preceq B$ by $M = A^{-1/2}$ (Lemma **??**) gives $I = A^{-1/2}AA^{-1/2} \preceq C$ (Lemma **??**(iii)). By Lemma **??**(i) applied to $C - I$, every eigenvalue μ_j of C satisfies $\mu_j \geq 1$; in particular $C \in \mathcal{S}^d_{++}$. Write $C = V \operatorname{diag}(\mu_j)V^\top$ with V orthogonal. Then $I - C^{-1} = V \operatorname{diag}(1-1/\mu_j)V^\top$ has nonnegative eigenvalues $1 - 1/\mu_j \geq 0$ (Lemma **??**(v)), so $C^{-1} \preceq I$. Now $C^{-1} = A^{1/2}B^{-1}A^{1/2}$ (indeed $A^{-1/2}BA^{-1/2} \cdot A^{1/2}B^{-1}A^{1/2} = I$ by Lemma **??**(iii)), and congruence of $C^{-1} \preceq I$ by $A^{-1/2}$ gives $B^{-1} = A^{-1/2}C^{-1}A^{-1/2} \preceq A^{-1/2}IA^{-1/2} = A^{-1}$. □

Lean remark. With the C*-algebra structure on `Matrix (Fin d) (Fin d)` (scoped instances of `Mathlib/Analysis/CStarAlgebra/Matrix.lean`, operator norm), this is `CStarAlgebra.inv_le_inv` in `Mathlib/Analysis/CStarAlgebra/ContinuousFunctionalCa` stated for units a, b with $0 \leq a \leq b$. The proof above is the elementary route in case the instance plumbing (the order on matrices versus the C*-algebra order, `Matrix.instStarOrderedRing`) turns out to be painful.

Lemma 34 (Scalar domination and inverses)**.** $Matrix.PosDef.inv_le_inv_smul, Matrix.PosDef.dotProduct_inv_m$ > 0*,* $A \in \mathcal{S}^d_{++}$ *and* $B \in \mathcal{S}^d$ *with* $cA \preceq B$*. Then* $B \in \mathcal{S}^d_{++}$*,* $B^{-1} \preceq c^{-1}A^{-1}$*, and* $\|x\|^2_{B^{-1}} \leq c^{-1}\|x\|^2_{A^{-1}}$ *for every* $x \in \mathbb{R}^d$*.*

Proof. lem:loewner$_inv, lem : loewner_quadratic$cA $\in \mathcal{S}^d_{++}$, so Lemma **??** gives $B \in \mathcal{S}^d_{++}$ and $B^{-1} \preceq (cA)^{-1} = c^{-1}A^{-1}$; the quadratic-form inequality is Lemma **??**. □

Lemma 35 (Trace of the inverse: AM–HM)**.** $Matrix.PosDef.trace_inv_eq_sum_inv_eigenvalues, Matrix.PosDef.ca$ $\in \mathcal{S}^d_{++}$*,* $\operatorname{tr}(A^{-1}) \geq d^2/\operatorname{tr}(A)$*.*

Proof. lem:eigen$_quadratic_boundsByLemma$ **??**$(iv)--(v)$,$\operatorname{tr} A = \sum_j \lambda_j$ and $\operatorname{tr}(A^{-1}) = \sum_j 1/\lambda_j$ with all $\lambda_j > 0$. By the Cauchy–Schwarz inequality, $d^2 = \big(\sum_j \sqrt{\lambda_j} \cdot \lambda_j^{-1/2}\big)^2 \leq \sum_j \lambda_j \cdot \sum_j \lambda_j^{-1}$. □

2.2 Kiefer–Wolfowitz: existence of a G-optimal design

Recall (Definition **??**) that design distributions $\lambda \in \triangle_{\mathcal{X}}$ are finitely supported and that $\mathcal{M}(\mathcal{X}) = \{A(\lambda)\} = \operatorname{conv}\{xx^\top : x \in \mathcal{X}\}$. The Kiefer–Wolfowitz theorem [**?**] (see also [**?**, Chapter 9] and [**?**, Chapter 21]) states that a design maximizes $\det A(\lambda)$ if and only if it minimizes $\max_{x\in\mathcal{X}}\|x\|^2_{A(\lambda)^{-1}}$, and that the minimal value is d. We only need the implication "D-optimal $\Rightarrow$ G-optimal" together with the value d.

Definition 36 (G-value of a matrix)**.** Maiti2026Power.gValuedef:arm$_set, lem : eigen_quadratic_boundsFor$A $\in \mathcal{S}^d_{++}$ the *G-value* of A on $\mathcal{X}$ is

$$g_{\mathcal{X}}(A) := \max_{x\in\mathcal{X}} x^\top A^{-1}x = \max_{x\in\mathcal{X}}\|x\|^2_{A^{-1}}.$$

The maximum exists since $x \mapsto x^\top A^{-1}x$ is continuous and $\mathcal{X}$ is compact and nonempty.

Lemma 37 (The G-value of a design matrix is at least d)**.** *Maiti2026Power.sum$_m$ul$_d$otProduct$_i$nv$_m$ulVec, Mai*
$\triangle_{\mathcal{X}}$ with $A(\lambda) \in \mathcal{S}^d_{++}$. Then

$$\sum_{x\in\operatorname{supp}\lambda} \lambda_x\, x^\top A(\lambda)^{-1}x = d,$$

hence $g_{\mathcal{X}}(A(\lambda)) \geq \max_{x\in\operatorname{supp}\lambda} x^\top A(\lambda)^{-1}x \geq d$. If $g_{\mathcal{X}}(A(\lambda)) = d$, then $x^\top A(\lambda)^{-1}x = d$ for every $x \in \operatorname{supp}\lambda$.

Proof. def:g$_v alue, def : design_m atrixWrite$A := A($\lambda$). For each x, $x^\top A^{-1}x = \operatorname{tr}(A^{-1}xx^\top)$ (cyclicity of the trace). By linearity of the trace, $\sum_x \lambda_x x^\top A^{-1}x = \operatorname{tr}\left(A^{-1}\sum_x \lambda_x xx^\top\right) = \operatorname{tr}(A^{-1}A) = \operatorname{tr}(I_d) = d$. Since the λ_x are nonnegative and sum to one, a weighted average equal to d forces the maximum over the support to be $\geq d$, and $g_{\mathcal{X}}(A) \geq$ that maximum because $\operatorname{supp}\lambda \subset \mathcal{X}$. If $g_{\mathcal{X}}(A) = d$, all terms $x^\top A^{-1}x$, $x \in \operatorname{supp}\lambda$, are $\leq d$ and their λ-average is d, so each equals d (a nonnegative combination $\sum_x \lambda_x(d - x^\top A^{-1}x) = 0$ of nonnegative terms with $\lambda_x > 0$ vanishes termwise). □

Definition 38 (D-optimal design matrix)**.** def:design$_m$*atrixAD-optimal design matrixisamatrix*A$_D \in$ $\mathcal{M}(\mathcal{X})$ maximizing det over $\mathcal{M}(\mathcal{X})$: $\det A_D = \max_{A\in\mathcal{M}(\mathcal{X})} \det A$.

Lemma 39 (Existence and positivity of a D-optimal design)**.** *Maiti2026Power.exists$_i$sMaxOn$_d$et$_d$esignSet, M*
exists. Assume moreover that $\mathcal{X}$ is spanning. Then $\det A_D > 0$; consequently $A_D \in \mathcal{S}^d_{++}$.

Proof. lem:designset$_b asic, lem : exists_p osdef_d esign, lem : eigen_q uadratic_b ounds$M(X)$iscompactandnonempty$
is continuous (a polynomial in the entries), so a maximizer exists. If $\mathcal{X}$ is spanning, Lemma **??** gives $A' \in \mathcal{M}(\mathcal{X}) \cap \mathcal{S}^d_{++}$, and $\det A' = \prod_j \lambda_j(A') > 0$ (Lemma **??**(i),(iv)); hence $\det A_D \geq \det A' > 0$. Finally $A_D \in \mathcal{S}^d_+$ (Lemma **??**), so its eigenvalues are ≥ 0 with product $\det A_D > 0$, so they are all > 0 and $A_D \in \mathcal{S}^d_{++}$. □

Lean remark. `IsCompact.exists_isMaxOn` with `Continuous.matrix_det`;
the last step is `Matrix.PosSemidef.posDef_iff_det_ne_zero` (`Mathlib/Analysis/Matrix/Order.lean`).

Lemma 40 (Determinant along a rank-one mixture)**.** $Matrix.det_a dd_v ecMulVec, Maiti2026Power.det_s mul_a dd_s$
$\in \mathbb{R}^{d\times d}$ *be invertible and* $u, v \in \mathbb{R}^d$*. Then (matrix determinant lemma)*

$$\det(A + uv^\top) = \det A\,\big(1 + v^\top A^{-1}u\big).$$

Consequently, for every $x \in \mathbb{R}^d$ *and* $t \neq 1$*,*

$$\det\big((1-t)A + t\,xx^\top\big) = (1-t)^d \det A\left(1 + \frac{t}{1-t}\,x^\top A^{-1}x\right).$$

Proof. $A+uv^\top = A\,(I+(A^{-1}u)v^\top)$ and $\det(I+wv^\top) = 1+v^\top w$ (the Weinstein–Aronszajn identity $\det(I+CD) = \det(I+DC)$ with $C = w$ a column and $D = v^\top$ a row; Mathlib's `Matrix.det_one_add_replicateCol_mul_replicateRow`). For the second identity apply the first to the invertible matrix $(1-t)A$, whose determinant is $(1-t)^d \det A$ and whose inverse is $(1-t)^{-1}A^{-1}$, with $u = t\,x$ and $v = x$. □

Lemma 41 (Sign of the derivative at a one-sided maximum)**.** $HasDerivAt.nonpos_o f_e ventually_l e_n hdsGTLetf$ *:* $R \to \mathbb{R}$ *be differentiable at* a *and assume that* $f(t) \le f(a)$ *for all* t *in a right neighborhood of* a*. Then* $f'(a) \le 0$*.*

Proof. The difference quotients $(f(a+t) - f(a))/t$ are ≤ 0 for small $t > 0$ and converge to $f'(a)$ as $t \to 0^+$ (`HasDerivAt.tendsto_slope_zero_right`). □

Lean remark. The outline proves the Kiefer–Wolfowitz theorem through the directional derivative $\frac{d}{dt}\log\det(A+tM)|_{t=0} = \operatorname{tr}(A^{-1}M)$ for a symmetric direction M, whose proof goes through the spectral theorem. Since D-optimality is only tested along the segments from A_D to the rank-one matrices $xx^\top$, the derivative of $t \mapsto \det((1-t)A_D + t\,xx^\top)$ at $t = 0$ is available in closed form from Lemma **??**, and the formalization follows this elementary route.

Theorem 42 (Kiefer–Wolfowitz: existence of a G-optimal design)**.** *Maiti2026Power.IsGOptimalDesign,*
$Maiti2026Power.exists_i sGOptimalDesign, Maiti2026Power.IsGOptimalDesign.gValue_e q, Maiti2026Power.$
$\triangle_{\mathcal{X}}$ *(finitely supported, as every design distribution) such that* $A_G := A(\lambda_G) \in \mathcal{S}^d_{++}$ *and* $g_{\mathcal{X}}(A_G) = d$*, i.e.* $\max_{x\in\mathcal{X}}\|x\|^2_{A_G^{-1}} = d$*; in particular* A_G *minimizes* $g_{\mathcal{X}}$ *over* $\mathcal{M}(\mathcal{X}) \cap \mathcal{S}^d_{++}$ *(Lemma* **??***). Moreover every* $x \in \operatorname{supp}\lambda_G$ *satisfies* $x^\top A_G^{-1}x = d$*. (A numerical bound on* $|\operatorname{supp}\lambda_G|$ *is available but deliberately not part of the statement; see Remark* **??***.)*

Proof. lem:exists$_{do}ptimal, def: design_m atrix, lem: caratheodory_d esign, lem: designset_b asic, lem: det_r ank_o ne_m ix, lem: deriv_n onpos_r ight_m ax, lem: g_g e_d Let \mathrm{A}_D$ be a D-optimal design matrix; since $\mathcal{X}$ is spanning, $A_D \in \mathcal{S}^d_{++}$ (Lemma **??**). By definition of $\mathcal{M}(\mathcal{X})$ (Definition **??**) there is a finitely supported $\lambda_G \in \triangle_{\mathcal{X}}$ with $A(\lambda_G) = A_D$ (Lemma **??** shows that it can even be chosen with $|\operatorname{supp}\lambda_G| \le d^2+1$, which we do not need); write $A_G := A_D$.

Fix $x \in \mathcal{X}$ and set $g := x^\top A_G^{-1}x$. For $t \in [0,1]$, $B := xx^\top = A(\delta_x) \in \mathcal{M}(\mathcal{X})$ and $(1-t)A_G + tB \in \mathcal{M}(\mathcal{X})$ by convexity (Lemma **??**), so by D-optimality $\det((1-t)A_G + t\,xx^\top) \le \det A_G$. By Lemma **??** and $\det A_G > 0$, this says that

$$\varphi(t) := (1-t)^d\Big(1 + \frac{t}{1-t}\,g\Big) \le 1 = \varphi(0) \qquad \text{for all } t \in (0,1).$$

The function φ is differentiable at 0 with $\varphi'(0) = -d + g$ (product rule: $\frac{d}{dt}(1-t)^d|_0 = -d$ and $\frac{d}{dt}\frac{t}{1-t}|_0 = 1$), so Lemma **??** gives $g - d \le 0$, i.e. $x^\top A_G^{-1} x \le d$. As $x \in \mathcal{X}$ was arbitrary, $g_\mathcal{X}(A_G) \le d$. Lemma **??** gives $g_\mathcal{X}(A_G) \ge d$, hence $g_\mathcal{X}(A_G) = d$, and the same lemma gives $x^\top A_G^{-1} x = d$ on the support. □

Lean remark. The formalization packages the conclusion as the predicate `Maiti2026Power.IsGOptimalDesign    w`: w is a design distribution on $\mathcal{X}$ (`Maiti2026Power.IsDesign`), its design matrix is positive definite and $x^\top A(w)^{-1} x \le d$ for all $x \in \mathcal{X}$; `Maiti2026Power.exists_isGOptimalDesign` is the existence statement, and the value $g_\mathcal{X}(A_G) = d$ and the support property are the lemmas `IsGOptimalDesign.gValue_eq` and `IsGOptimalDesign.dotProduct_inv_mulVec_eq_card`. The theorem needs $d \ge 1$ (the type   nonempty), as does Lemma **??**.

Remark 43 (Support size of the G-optimal design). The proof of Theorem **??** may produce λ_G through Lemma **??**, hence with $|\operatorname{supp} \lambda_G| \le d^2 + 1$; the paper (Appendix B.1) states the existence of a G-optimal design with $|\operatorname{supp} \lambda_*| \le d(d+1)/2$ (the sharper bound uses Carathéodory's theorem in the space of symmetric matrices and that the optimal matrices lie on the boundary of the hull). This numerical bound is deliberately kept out of the statements of Theorem **??** and Lemma **??**: only the finiteness of the support is ever used. In Chapter **??**, λ_G replaces the John ellipsoid of the paper: Corollary **??** provides exactly the normalized points and the isotropy identity that the paper extracts from John's theorem.

Corollary 44 (Whitened arms have norm at most $\sqrt{d}$)**.** *Maiti2026Power.IsGOptimalDesign.norm$_s$qrt$_i$nv$_l$e, M*
be as in Theorem **??***. Every* $x \in \mathcal{X}$ *satisfies* $\|A_G^{-1/2} x\|^2 = x^\top A_G^{-1} x \le d$.

Proof. thm:kiefer$_wolfowitz, lem: sqrt_props \|A_G^{-1/2} x\|^2 = x^\top A_G^{-1} x$ by Lemma **??**(iv), and $x^\top A_G^{-1} x \le g_\mathcal{X}(A_G) = d$. □

Corollary 45 (Normalized G-optimal design)**.** *Maiti2026Power.normalizeMat, Maiti2026Power.IsGOptimalDesign.norm$_n$ormalizeMat$_e$q$_o$ne, Maiti2026Power.IsGOptimalDesign.designM*
A_G *be as in Theorem* **??** *and set* $M := d^{-1/2} A_G^{-1/2}$, $v_x := Mx = A_G^{-1/2} x / \sqrt{d}$ *for* $x \in \operatorname{supp} \lambda_G$, *and* $\mathcal{X}' := M\mathcal{X} = \{Mx : x \in \mathcal{X}\}$. *Then:*

(i) $\|v_x\| = 1$ *for every* $x \in \operatorname{supp} \lambda_G$*;*

(ii) $\sum_{x \in \operatorname{supp} \lambda_G} \lambda_{G,x}\, v_x v_x^\top = I_d/d$*;*

(iii) $\mathcal{X}' \subset \mathbb{B}_d$, *and* $v_x \in \mathcal{X}'$ *for every* $x \in \operatorname{supp} \lambda_G$*;* $\mathcal{X}'$ *is compact and spans* $\mathbb{R}^d$.

Proof. thm:kiefer$_wolfowitz, cor: g_optimal_norm_bound, lem: sqrt_props (i): \|v_x\|^2 = x^\top A_G^{-1} x/d = 1$ by Lemma **??**(iv) and the support property of Theorem **??**. (ii): $\sum_x \lambda_{G,x} v_x v_x^\top = \frac{1}{d} A_G^{-1/2} \big(\sum_x \lambda_{G,x} x x^\top\big) A_G^{-1/2} = \frac{1}{d} A_G^{-1/2} A_G A_G^{-1/2} = I_d/d$ (Lemma **??**(iii)). (iii): for $x \in \mathcal{X}$, $\|Mx\|^2 = x^\top A_G^{-1} x/d \le 1$ by Corollary **??**; $v_x = Mx$ with $x \in \operatorname{supp} \lambda_G \subset \mathcal{X}$; M is invertible, so $M\mathcal{X}$ is compact and spanning. □

2.3 Carathéodory, continuity and attainment of the width

Lemma 46 (Carathéodory for design matrices)**.** *Maiti2026Power.exists$_i$sDesign$_o$f$_m$em$_d$esignSet, Maiti2026F* $\in \mathcal{M}(\mathcal{X})$ *is of the form* $A = A(\lambda)$ *for a design distribution* $\lambda \in \triangle_{\mathcal{X}}$ *with* $|\operatorname{supp}\lambda| \le d^2 + 1$.

Proof. def:design$_m$*atrix*M(X) $= \operatorname{conv} S$ with $S := \{xx^\top : x \in \mathcal{X}\} \subset \mathbb{R}^{d\times d}$, a real vector space of dimension d^2. By Carathéodory's theorem, every point of $\operatorname{conv} S$ is a convex combination of an affinely independent family of points of S, which has at most d^2+1 elements: $A = \sum_{i=1}^k \alpha_i s_i$ with $k \le d^2+1$, $\alpha_i > 0$, $\sum_i \alpha_i = 1$, $s_i \in S$ pairwise distinct. Choose $x_i \in \mathcal{X}$ with $s_i = x_i x_i^\top$; the x_i are pairwise distinct since the s_i are. Define $\lambda_{x_i} := \alpha_i$; then $\lambda \in \triangle_{\mathcal{X}}$, $A(\lambda) = A$ and $|\operatorname{supp}\lambda| = k \le d^2+1$. ☐

Lean remark. `eq_pos_convex_span_of_mem_convexHull` (`Mathlib/Analysis/Convex/Caratheodory.l` writes a point of the hull as a positive combination of an affinely independent family, and `AffineIndependent.card_le_finrank_succ` bounds its cardinality by `finrank` + 1. Applying it in the space of symmetric matrices would give the bound $d(d+1)/2+1$ (its `finrank` is not in Mathlib and would have to be computed from the basis of matrices E_{ii} and $E_{ij} + E_{ji}$, $i < j$); the formalization applies it in the full matrix space `Matrix` , of dimension d^2, which is sufficient for every use in this blueprint (the support size of λ_G and λ_1 only needs to be finite). Design distributions are encoded as finitely supported functions `EuclideanSpace` → (`Finsupp`) with the predicate `Maiti2026Power.IsDesign` (support in $\mathcal{X}$, nonnegative, summing to one), and $A(\lambda)$ is the linear map `Maiti2026Power.designMatrix`; the equality of $\mathcal{M}(\mathcal{X})$ with the set of design matrices of design distributions is `Maiti2026Power.mem_designSet_iff`.

Lemma 47 (Local Lipschitz continuity of the square root)**.** *lem:sqrt$_p$rops, lem : eigen$_q$uadratic$_b$oundsForB, B'* $\in \mathcal{S}^d_{++}$,

$$\|B^{1/2} - B'^{1/2}\|_{\mathrm{op}} \le \|B^{1/2} - B'^{1/2}\|_F \le \frac{d\,\|B - B'\|_{\mathrm{op}}}{\sqrt{\lambda_{\min}(B)} + \sqrt{\lambda_{\min}(B')}}.$$

Consequently $B \mapsto B^{1/2}$ *is continuous on* $\mathcal{S}^d_{++}$*, and* $A \mapsto A^{-1/2}$ *is continuous on* $\mathcal{S}^d_{++}$.

Proof. lem:sqrt$_p$*rops, lem : eigen$_q$uadratic$_b$oundsLet*X := B$^{1/2} - B'^{1/2}$. Then $B^{1/2}X + XB'^{1/2} = B - B'$ (expand, using $(B^{1/2})^2 = B$ and $(B'^{1/2})^2 = B'$). Let $(u_i, a_i)_i$ be an orthonormal eigenbasis of $B^{1/2}$ and $(v_j, b_j)_j$ one of $B'^{1/2}$; by Lemma **??**(i), $a_i \ge \alpha := \sqrt{\lambda_{\min}(B)}$ and $b_j \ge \beta := \sqrt{\lambda_{\min}(B')}$. Since $B^{1/2}$ is symmetric, $u_i^\top B^{1/2} = a_i u_i^\top$, hence $u_i^\top(B-B')v_j = u_i^\top B^{1/2} X v_j + u_i^\top X B'^{1/2} v_j = (a_i + b_j)\, u_i^\top X v_j$, so $|u_i^\top X v_j| \le \|B - B'\|_{\mathrm{op}}/(\alpha+\beta)$. The Frobenius norm is invariant under orthogonal changes of basis, so $\|X\|_F^2 = \sum_{i,j}(u_i^\top X v_j)^2 \le d^2\|B - B'\|_{\mathrm{op}}^2/(\alpha+\beta)^2$. The operator norm is bounded by the Frobenius norm. Continuity at $B_0 \in \mathcal{S}^d_{++}$: the bound with $B' = B_0$ gives $\|B^{1/2} - B_0^{1/2}\|_{\mathrm{op}} \le$

$d\,\|B-B_0\|_{\mathrm{op}}/\sqrt{\lambda_{\min}(B_0)}$ for every $B \in \mathcal{S}^d_{++}$, with a constant independent of B. Finally $A \mapsto A^{-1}$ is continuous on the (open) set of invertible matrices (entries of $A^{-1} = \mathrm{adj}(A)/\det A$ are rational functions with nonvanishing denominator), maps $\mathcal{S}^d_{++}$ to $\mathcal{S}^d_{++}$, and $A^{-1/2} = (A^{-1})^{1/2}$ (Lemma **??**(ii)). $\square$

Lean remark. This lemma is not in the outline; it is stated locally because the continuity of `CFC.sqrt` in its argument is not readily available in Mathlib for matrices, whereas the Sylvester-equation argument above is elementary. Continuity of the inverse is `continuousAt_matrix_inv` (`Mathlib/Topology/Instances/Matrix.lean`) or `NormedRing.inverse_continuousAt`. All norms on the finite-dimensional space of matrices are equivalent, so the choice of norm in the statement is immaterial for continuity.

Lemma 48 (Continuity of the width in the matrix)**.** *def:width$_m$atrix, lem : width$_m$atrix$_w$d, lem : design$_s$qrt$_c$ontinuous, lem : gauss$_n$orm$_m$omentsLetR* $:= \sup_{x\in\mathcal{X}}\|x\|$. *For* $A, A' \in \mathcal{S}^d_{++}$,

$$|w(\mathcal{X};A) - w(\mathcal{X};A')| \le R\sqrt{d}\,\|A^{-1/2} - A'^{-1/2}\|_{\mathrm{op}}.$$

Consequently $A \mapsto w(\mathcal{X};A)$ *is continuous on* $\mathcal{S}^d_{++}$.

Proof. lem:width$_m$*atrix$_w$d, lem : design$_s$qrt$_c$ontinuous, lem : gauss$_n$orm$_m$omentsForeveryg* $\in \mathbb{R}^d$, two suprema of functions on $\mathcal{X}$ differ by at most the supremum of the difference: $|\sup_x\langle A^{-1/2}x, g\rangle - \sup_x\langle A'^{-1/2}x, g\rangle| \le \sup_x|\langle (A^{-1/2} - A'^{-1/2})x, g\rangle| \le R\,\|A^{-1/2}-A'^{-1/2}\|_{\mathrm{op}}\|g\|$ (Cauchy–Schwarz). Both suprema are integrable (Lemma **??**); taking expectations and using $\mathbb{E}\|g\| \le \sqrt{d}$ (Lemma **??**) gives the inequality. Continuity follows from the continuity of $A \mapsto A^{-1/2}$ on $\mathcal{S}^d_{++}$ (Lemma **??**). $\square$

Lemma 49 (Coercivity of the width on design matrices)**.** *def:arm$_s$et, def : width$_m$atrix, def : design$_m$atrix, lem : sqrt$_p$rops, lem : gauss$_1$d$_m$oments, lem : gauss$_l$inear$_i$mageLetXbespanningandlet* $(A_n)_n$ *be a sequence in* $\mathcal{M}(\mathcal{X}) \cap \mathcal{S}^d_{++}$ *converging to a singular matrix* A_∞*. Then* $w(\mathcal{X};A_n) \to +\infty$*. More precisely, for every* $A \in \mathcal{S}^d_{++}$*, every unit vector* u *and all* $x, x' \in \mathcal{X}$,

$$w(\mathcal{X};A) \;\ge\; \frac{\|A^{-1/2}(x-x')\|}{\sqrt{2\pi}} \;\ge\; \frac{|\langle u, x-x'\rangle|}{\sqrt{2\pi}\,\sqrt{u^\top A u}},$$

and if $A_n \in \mathcal{M}(\mathcal{X})$ *with* $u^\top A_n u \to 0$ *then there are* $x, x' \in \mathcal{X}$ *with* $\langle u, x-x'\rangle \ne 0$.

Proof. lem:sqrt$_p$*rops, lem : gauss$_1$d$_m$oments, lem : gauss$_l$inear$_i$mage, lem : width$_m$atrix$_w$d, lem : designset$_b$asicLower bound.Let*a $:= \mathrm{A}^{-1/2}x$, $b := A^{-1/2}x'$. For all g, $\sup_{z\in\mathcal{X}}\langle A^{-1/2}z, g\rangle \ge \max(\langle a, g\rangle, \langle b, g\rangle) = \frac{1}{2}\langle a+b, g\rangle + \frac{1}{2}|\langle a-b, g\rangle|$. Taking expectations, $\mathbb{E}\langle a+b, g\rangle = 0$ and $\langle a-b, g\rangle \sim \mathcal{N}(0, \|a-b\|^2)$ (Lemma **??**) has $\mathbb{E}|\langle a-b, g\rangle| = \|a-b\|\sqrt{2/\pi}$ (Lemma **??**), so $w(\mathcal{X};A) \ge \|a-b\|/\sqrt{2\pi} = \|A^{-1/2}(x-x')\|/\sqrt{2\pi}$. Next, by Lemma **??**(v) and Cauchy–Schwarz, $|\langle u, x - x'\rangle| = |\langle A^{1/2}u, A^{-1/2}(x-x')\rangle| \le \|A^{1/2}u\|\,\|A^{-1/2}(x-x')\|$ with $\|A^{1/2}u\|^2 = u^\top A u$ (Lemma **??**(iv)).

A separating pair exists. Suppose $A_n = A(\lambda^{(n)}) \in \mathcal{M}(\mathcal{X})$ with $u^\top A_n u \to 0$, and suppose for contradiction that $\langle u, x - x' \rangle = 0$ for all $x, x' \in \mathcal{X}$, i.e. $\langle u, x \rangle = c$ for all $x \in \mathcal{X}$ and some constant c. Then $u^\top A_n u = \sum_x \lambda_x^{(n)} \langle u, x \rangle^2 = c^2$ for all n, so $c = 0$, so $u \perp \mathcal{X}$ and hence $u \perp \operatorname{span}(\mathcal{X}) = \mathbb{R}^d$ (this is where the spanning hypothesis is used), contradicting $\|u\| = 1$.

Divergence. $A_\infty \in \mathcal{S}_+^d$ since $\mathcal{S}_+^d$ is closed and $A_n \in \mathcal{S}_+^d$, and A_∞ is singular, so there is a unit vector u with $A_\infty u = 0$; then $u^\top A_n u \to u^\top A_\infty u = 0$, and $u^\top A_n u > 0$ for each n. Pick x, x' as in the previous paragraph. The lower bound gives $w(\mathcal{X}; A_n) \geq |\langle u, x - x' \rangle| / (\sqrt{2\pi}\sqrt{u^\top A_n u}) \to +\infty$. □

Remark 50. The restriction to $A_n \in \mathcal{M}(\mathcal{X})$ (rather than arbitrary $A_n \in \mathcal{S}_{++}^d$, as the outline of the chapter first suggested) is necessary: for $\mathcal{X} = \{e_1 + e_2, e_1 - e_2\} \subset \mathbb{R}^2$ and $A_n = \operatorname{diag}(1/n, 1)$, one has $w(\mathcal{X}; A_n) = \mathbb{E}[\sqrt{n} g_1 + |g_2|] = \sqrt{2/\pi}$ for all n, although $A_n \to \operatorname{diag}(0,1)$ is singular. The point is that a design matrix built from $\mathcal{X}$ cannot degenerate in a direction u along which $\mathcal{X}$ is not constant.

Lemma 51 (The width is attained on a positive definite design)**.** *def:arm$_s$et, $def:width, def:design_matrix, lem:designset_basic, lem:width_matrix_continuous, lem:width_coerciveLetXbespanning.ThereisA_1 \in \mathcal{M}(\mathcal{X}) \cap \mathcal{S}_{++}^d$ with $w(\mathcal{X}; A_1) = w(\mathcal{X})$, i.e. a (finitely supported) design distribution $\lambda_1 \in \triangle_{\mathcal{X}}$ with $A_1 = A(\lambda_1) \in \mathcal{S}_{++}^d$ and $w(\mathcal{X}; A(\lambda_1)) = w(\mathcal{X})$.*

Proof. def:width, def:design$_matrix, lem:designset_basic, lem:width_matrix_continuous, lem:width_coercive, lem:exists_posdef_designSince$Xisspanning$, theinfimumdefining$w(X)$isoveranonemptyset(L$ $\mathcal{M}(\mathcal{X}) \cap \mathcal{S}_{++}^d$ with $w(\mathcal{X}; A_n) \to w(\mathcal{X})$. By compactness of $\mathcal{M}(\mathcal{X})$ (Lemma **??**) a subsequence A_{n_k} converges to some $A_\infty \in \mathcal{M}(\mathcal{X})$. If A_∞ were singular, Lemma **??** would give $w(\mathcal{X}; A_{n_k}) \to +\infty$, contradicting convergence to the finite value $w(\mathcal{X})$. Hence $A_\infty \in \mathcal{S}_{++}^d$ ($A_\infty \in \mathcal{S}_+^d$ and nonsingular), and by continuity (Lemma **??**) $w(\mathcal{X}; A_\infty) = \lim_k w(\mathcal{X}; A_{n_k}) = w(\mathcal{X})$. Take $A_1 := A_\infty$; the representation $A_1 = A(\lambda_1)$ with a finitely supported $\lambda_1 \in \triangle_{\mathcal{X}}$ is the definition of $\mathcal{M}(\mathcal{X})$ (Definition **??**). □

Lean remark. Not formalized: the fixed-design algorithm of Chapter **??** is built in Lean from an *approximate* minimizer of the width (any A_1 with $w(\mathcal{X}; A_1)^2 \leq w(\mathcal{X})^2 + (\log 2)/4$, which exists by definition of the infimum), so the attainment is not needed for Theorem **??**.

Lean remark. The argument is a standard "minimize a continuous coercive function on a compact set": it can be organized as follows. The set $K_c := \{A \in \mathcal{M}(\mathcal{X}) : w(\mathcal{X}; A) \leq c\}$ for $c > w(\mathcal{X})$ is nonempty, contained in $\mathcal{S}_{++}^d$ (by coercivity and closedness of $\mathcal{M}$), and closed in $\mathcal{M}(\mathcal{X})$ (sequential closedness: a limit of elements of K_c is nonsingular by coercivity, and then the width is continuous there), hence compact, and a minimizer of the continuous function $w(\mathcal{X}; \cdot)$ on K_c (`IsCompact.exists_isMinOn`) is a global minimizer. Subsequence extraction is `IsCompact.tendsto_subseq`. As for Theorem **??**, Lemma **??** shows that λ_1 can be chosen with $|\operatorname{supp} \lambda_1| \leq d(d+1)/2 + 1$; this numerical bound is kept out of the statement (Remark **??**).

2.4 Rounding a design distribution to a fixed design

Throughout this section, $\lambda \in \triangle_{\mathcal{X}}$ is a design distribution with $A := A(\lambda) \in \mathcal{S}^d_{++}$, support $\operatorname{supp}\lambda = \{v_1, \dots, v_m\}$ and weights $p_i := \lambda_{v_i} > 0$, $\sum_{i=1}^m p_i = 1$. The *whitened* vectors are $u_i := A^{-1/2}v_i$, so that (Lemma **??**(iii))

$$\sum_{i=1}^{m} p_i\, u_i u_i^\top = A^{-1/2}\Big(\sum_i p_i v_i v_i^\top\Big)A^{-1/2} = A^{-1/2}AA^{-1/2} = I_d.$$

The goal is Theorem **??**: for every $T \geq 4d$ one can pick T points of the support (with repetitions) whose empirical design matrix is at least $(T/4)A$. This is the lower-barrier half of the argument of [**?**], with all added rank-one matrices given weight one (no upper barrier is needed since only a lower bound on Σ_T is used).

Definition 52 (Lower barrier potential)**.** Maiti2026Power.barrier, Maiti2026Power.barrier$_{e}q_{s}um, Maiti2026Po$
$\in \mathcal{S}^d$ and $l \in \mathbb{R}$ with $l < \lambda_{\min}(B)$, so that $B - lI \in \mathcal{S}^d_{++}$, the *lower barrier potential* is

$$\Phi_l(B) := \operatorname{tr}\big((B - lI)^{-1}\big) = \sum_{j=1}^{d} \frac{1}{\lambda_j(B) - l}.$$

The second expression is Lemma **??**(iv)–(v). Note that $\Phi_l(B) \geq 1/(\lambda_{\min}(B) - l)$ (keep only the smallest eigenvalue), that $\Phi_l(B)$ is increasing in l, and that $\Phi_{-d}(0) = \operatorname{tr}((dI)^{-1}) = 1$.

Lemma 53 (Sherman–Morrison, trace form)**.** $Matrix.inv_{a}dd_{v}ecMulVec, Matrix.trace_{i}nv_{a}dd_{v}ecMulVec_{s}elfle$
$\in \mathcal{S}^d_{++}$ *and* $u \in \mathbb{R}^d$*. Then* $M + uu^\top \in \mathcal{S}^d_{++}$*,* $1 + u^\top M^{-1}u > 0$*,*

$$(M+uu^\top)^{-1} = M^{-1} - \frac{M^{-1}uu^\top M^{-1}}{1 + u^\top M^{-1}u}, \qquad \operatorname{tr}\big((M+uu^\top)^{-1}\big) = \operatorname{tr}(M^{-1}) - \frac{u^\top M^{-2}u}{1 + u^\top M^{-1}u}.$$

Proof. lem:loewner$_{q}uadratic, lem : eigen_{q}uadratic_{b}ounds$M + uu$^\top \in \mathcal{S}^d_{++}$ since $x^\top(M + uu^\top)x = x^\top Mx + \langle u, x\rangle^2 > 0$ for $x \neq 0$; and $u^\top M^{-1}u \geq 0$ since $M^{-1} \in \mathcal{S}^d_{++}$ (Lemma **??**(v)). Set $s := u^\top M^{-1}u$ and $N := M^{-1} - M^{-1}uu^\top M^{-1}/(1+s)$. Then

$$(M+uu^\top)N = I - \frac{uu^\top M^{-1}}{1+s} + uu^\top M^{-1} - \frac{u\,(u^\top M^{-1}u)\,u^\top M^{-1}}{1+s} = I + uu^\top M^{-1}\Big(1 - \frac{1}{1+s} - \frac{s}{1+s}\Big) = I,$$

so $N = (M + uu^\top)^{-1}$ (a right inverse of a square matrix is the inverse). Taking traces, $\operatorname{tr}(M^{-1}uu^\top M^{-1}) = \operatorname{tr}(u^\top M^{-1}M^{-1}u) = u^\top M^{-2}u$ by cyclicity. □

Lean remark. This is the special case $U = u$ (a $d \times 1$ matrix), $V = u^\top$, $C = (1)$ of the Woodbury identity `Matrix.add_mul_mul_inv_eq_sub` in `Mathlib/LinearAlgebra/Matrix/NonsingularInverse.lean`; the direct verification above is probably shorter than instantiating it (`Matrix.inv_eq_right_inv` turns the one-sided computation $(M + uu^\top)N = I$ into $(M + uu^\top)^{-1} = N$). The rank-one matrix $uu^\top$ is `Matrix.vecMulVec u u`.

Lemma 54 (One barrier step)**.** *Maiti2026Power.barrierScore, Maiti2026Power.barrier*$_l t_b arrier_a dd_h alf, Maiti2$ $\in \mathbb{R}$, $l' := l + 1/2$, *and* $B \in \mathcal{S}^d$ *with* $\lambda_{\min}(B) > l'$. *Set* $M := B - l'I \in \mathcal{S}^d_{++}$, $\Delta := \Phi_{l'}(B) - \Phi_l(B)$, *and for* $u \in \mathbb{R}^d$

$$L_B(u) := \frac{u^\top M^{-2} u}{\Delta} - u^\top M^{-1} u.$$

Then $\Delta > 0$, *and if* $L_B(u) \geq 1$ *then* $\lambda_{\min}(B + uu^\top) > l'$ *and* $\Phi_{l'}(B + uu^\top) \leq \Phi_l(B)$.

Proof. def:barrier$_potential, lem : sherman_morrison_trace, lem : eigen_quadratic_bounds\Delta >$ 0: with $\lambda_j := \lambda_j(B)$, $\Phi_{l'}(B) - \Phi_l(B) = \sum_j \left(\frac{1}{\lambda_j - l'} - \frac{1}{\lambda_j - l}\right)$ and each term is positive since $0 < \lambda_j - l' < \lambda_j - l$. Next, $uu^\top \in \mathcal{S}^d_+$, so $\lambda_{\min}(B + uu^\top) \geq \lambda_{\min}(B) > l'$ (Lemma **??**(ii)), and $\Phi_{l'}(B + uu^\top)$ is defined. Since $(B + uu^\top) - l'I = M + uu^\top$, Lemma **??** gives

$$\Phi_{l'}(B + uu^\top) = \mathrm{tr}(M^{-1}) - \frac{u^\top M^{-2} u}{1 + u^\top M^{-1} u} = \Phi_{l'}(B) - \frac{u^\top M^{-2} u}{1 + u^\top M^{-1} u}.$$

The hypothesis $L_B(u) \geq 1$ reads $u^\top M^{-2} u/\Delta \geq 1 + u^\top M^{-1} u$; multiplying by $\Delta/(1 + u^\top M^{-1} u) > 0$ gives $u^\top M^{-2} u/(1 + u^\top M^{-1} u) \geq \Delta$, hence $\Phi_{l'}(B + uu^\top) \leq \Phi_{l'}(B) - \Delta = \Phi_l(B)$. ☐

Lemma 55 (Averaging the barrier criterion)**.** *Maiti2026Power.barrier*$_a veragedef : barrier_potential, lem : barr$ $\mathbb{R}^d$ *and* $p_1, \ldots, p_m \geq 0$ *with* $\sum_i p_i = 1$ *and* $\sum_i p_i u_i u_i^\top = I_d$. *Let* $l \in \mathbb{R}$, $l' := l + 1/2$, *and* $B \in \mathcal{S}^d$ *with* $\lambda_{\min}(B) > l$ *and* $\Phi_l(B) \leq 1$. *Then* $\lambda_{\min}(B) \geq l + 1 > l'$, *and with* L_B *as in Lemma* **??**,

$$\sum_{i=1}^m p_i \, L_B(u_i) \ \geq\ 2 - \Phi_l(B) \ \geq\ 1;$$

in particular there is an index i *with* $L_B(u_i) \geq 1$.

Proof. def:barrier$_potential, lem : barrier_step, lem : eigen_quadratic_bounds Since 1/(\lambda_{\min}(B) -$ $l) \leq \Phi_l(B) \leq 1$ and $\lambda_{\min}(B) - l > 0$, we get $\lambda_{\min}(B) - l \geq 1$, so $\lambda_{\min}(B) \geq l + 1 > l + 1/2 = l'$ and Lemma **??** applies: $M = B - l'I \in \mathcal{S}^d_{++}$ and $\Delta > 0$.

Let $\lambda_j := \lambda_j(B)$, $a_j := \lambda_j - l \geq 1$ and $b_j := a_j - 1/2 = \lambda_j - l' \geq 1/2$, so that M has eigenvalues b_j (Lemma **??**(iv)). Define

$$P := \mathrm{tr}(M^{-2}) = \sum_j \frac{1}{b_j^2}, \qquad Q := \sum_j \frac{1}{a_j b_j} > 0, \qquad \Phi_l(B) = \sum_j \frac{1}{a_j}, \qquad \Phi_{l'}(B) = \mathrm{tr}(M^{-1}) = \sum_j \frac{1}{b_j}.$$

By linearity of the trace and $\sum_i p_i u_i u_i^\top = I$: $\sum_i p_i \, u_i^\top M^{-2} u_i = \mathrm{tr}(M^{-2} \sum_i p_i u_i u_i^\top) = P$ and $\sum_i p_i \, u_i^\top M^{-1} u_i = \mathrm{tr}(M^{-1}) = \Phi_{l'}(B)$. Hence $\sum_i p_i L_B(u_i) = P/\Delta - \Phi_{l'}(B)$. Now

$$\Delta = \sum_j \left(\frac{1}{b_j} - \frac{1}{a_j}\right) = \sum_j \frac{a_j - b_j}{a_j b_j} = \frac{Q}{2}, \qquad \Phi_{l'}(B) = \Phi_l(B) + \Delta = \Phi_l(B) + \frac{Q}{2},$$

so $\sum_i p_i L_B(u_i) = 2P/Q - \Phi_l(B) - Q/2$, and the claim $\sum_i p_i L_B(u_i) \geq 2 - \Phi_l(B)$ is equivalent to $2P/Q - Q/2 \geq 2$, i.e. (multiplying by $Q/2 > 0$) to $P - Q \geq Q^2/4$. We have $P - Q = \sum_j \left(\frac{1}{b_j^2} - \frac{1}{a_j b_j}\right) = \sum_j \frac{a_j - b_j}{a_j b_j^2} = \frac{1}{2}\sum_j \frac{1}{a_j b_j^2} \geq 0$, and by Cauchy–Schwarz,

$$Q = \sum_j \frac{1}{\sqrt{a_j}\, b_j} \cdot \frac{1}{\sqrt{a_j}} \leq \Big(\sum_j \frac{1}{a_j b_j^2}\Big)^{1/2} \Big(\sum_j \frac{1}{a_j}\Big)^{1/2} = \sqrt{2(P-Q)}\,\sqrt{\Phi_l(B)} \leq \sqrt{2(P-Q)},$$

using $\Phi_l(B) \leq 1$. Thus $Q^2 \leq 2(P-Q)$ and $Q^2/4 \leq (P-Q)/2 \leq P - Q$. Finally $2 - \Phi_l(B) \geq 1$ because $\Phi_l(B) \leq 1$, and a p-weighted average of the numbers $L_B(u_i)$ being ≥ 1 forces $\max_i L_B(u_i) \geq 1$. $\square$

Theorem 56 (Rounding to a fixed design of any size $T \geq 4d$)**.** $Maiti2026Power.exists_rounding def : design_m atr$ $\triangle_{\mathcal{X}}$ *with* $A := A(\lambda) \in \mathcal{S}^d_{++}$ *and let* $T \geq 4d$ *be an integer. There exist* $x_0, \ldots, x_{T-1} \in \operatorname{supp}\lambda$ *(with repetitions allowed) such that*

$$\Sigma_T := \sum_{t<T} x_t x_t^\top \ \succeq\ \frac{T}{4} A(\lambda).$$

Proof. lem:barrier$_a$$verage, lem : barrier_s tep, lem : loewner_c ongruence, lem : eigen_q uadratic_b ounds, lem : sqrt_p rops, def : barrier_p otentialUsethenotationofthebeginningofthesection$:v$_1$, p_i, $u_i = A^{-1/2} v_i$ with $\sum_i p_i u_i u_i^\top = I$. We construct by induction on $t \leq T$ indices $i_0, \ldots, i_{t-1} \in \{1, \ldots, m\}$ such that $B_t := \sum_{s<t} u_{i_s} u_{i_s}^\top$ and $l_t := -d + t/2$ satisfy the invariant

$$\lambda_{\min}(B_t) > l_t \qquad \text{and} \qquad \Phi_{l_t}(B_t) \leq 1.$$

Base case $t = 0$: $B_0 = 0$ has $\lambda_{\min}(B_0) = 0 > -d = l_0$ and $\Phi_{-d}(0) = \operatorname{tr}((dI)^{-1}) = d \cdot \frac{1}{d} = 1$. *Step:* given the invariant at t, Lemma **??** (with $l = l_t$, $l' = l_t + 1/2 = l_{t+1}$, $B = B_t$) provides an index i_t with $L_{B_t}(u_{i_t}) \geq 1$, and Lemma **??** then gives, for $B_{t+1} = B_t + u_{i_t} u_{i_t}^\top$, $\lambda_{\min}(B_{t+1}) > l_{t+1}$ and $\Phi_{l_{t+1}}(B_{t+1}) \leq \Phi_{l_t}(B_t) \leq 1$.

After T steps, $\lambda_{\min}(B_T) > l_T = -d + T/2 \geq T/4$, where the last inequality is $T/4 \geq d$, i.e. $T \geq 4d$. By Lemma **??**(i), $B_T \succeq (T/4)I$. Set $x_t := v_{i_t} \in \operatorname{supp}\lambda$. Since $v_i = A^{1/2} u_i$ (Lemma **??**(iii)),

$$\sum_{t<T} x_t x_t^\top = \sum_{t<T} A^{1/2} u_{i_t} u_{i_t}^\top A^{1/2} = A^{1/2} B_T A^{1/2} \ \succeq\ A^{1/2}\Big(\frac{T}{4} I\Big) A^{1/2} = \frac{T}{4} A$$

by congruence with $M = A^{1/2}$ (Lemma **??**) and $A^{1/2} A^{1/2} = A$. $\square$

Lean remark. The proof is an existence statement proved by induction on t: "for every t there is a family `x : Fin t → E` of points of the support whose B_t satisfies the invariant" (`Maiti2026Power.exists_rounding`); the greedy choice at each step is the point provided by Lemma **??** (`Maiti2026Power.barrier_average`, stated for a design distribution with design matrix I, here the whitened distribution $\lambda \circ (A^{-1/2})^{-1}$ of `IsDesign.mapDomain`), and the family is extended with `Fin.snoc`. The invariant "$\lambda_{\min}(B_t) > l_t$" is the spectral condition $\forall \mu \in \operatorname{spec}_{\mathbb{R}}(B_t),\ l_t < \mu$ (see the Lean remark after Lemma **??**), and the eigenvalue inequality at the heart of Lemma **??** is isolated as `barrier_sum_ineq`. The design matrix Σ_T is $\sum_{t<T}$ `outerSelf (x t)` for `x : Fin T → E`. Note that the theorem produces a design of length exactly T with points in the support of λ, which is what Definition **??** consumes.

Corollary 57 (Fixed design with controlled inverse and width)**.** $Maiti2026Power.posDef_of_smul_le, Maiti2026P$ *be the design it provides and* $\Sigma_T := \sum_{t<T} x_t x_t^\top$. *Then* $\Sigma_T \in \mathcal{S}^d_{++}$,

$$\|x\|^2_{\Sigma_T^{-1}} \leq \frac{4}{T}\,\|x\|^2_{A(\lambda)^{-1}} \quad \textit{for all } x \in \mathbb{R}^d, \qquad \textit{and} \qquad w(\mathcal{X};\Sigma_T) \leq \frac{2}{\sqrt{T}}\,w(\mathcal{X};A(\lambda)).$$

Proof. thm:rounding, lem:loewner$_scalar, lem : width_mono, lem : width_homogTheorem$ **??**$gives$(T/4)A $\preceq \Sigma_T$ with $A = A(\lambda) \in \mathcal{S}^d_{++}$. Lemma **??** with $c = T/4$ yields $\Sigma_T \in \mathcal{S}^d_{++}$ and $\|x\|^2_{\Sigma_T^{-1}} \leq (4/T)\|x\|^2_{A^{-1}}$. For the width, Lemma **??** applied to $(T/4)A \preceq \Sigma_T$ (both in $\mathcal{S}^d_{++}$) gives $w(\mathcal{X};\Sigma_T) \leq w(\mathcal{X};(T/4)A) = (T/4)^{-1/2}w(\mathcal{X};A) = (2/\sqrt{T})w(\mathcal{X};A)$ by Lemma **??**. ☐

Lemma 58 (Naive rounding by ceilings)**.** *def:design$_matrix, lem : loewner_quadraticLet\lambda \in$* $\triangle_{\mathcal{X}}$ *with support* $\{v_1,\dots,v_m\}$ *and weights* $p_i > 0$, *and let* $T \geq 2m$ *be an integer. Set* $\kappa_i := \lceil (T-m)p_i \rceil \in \mathbb{N}$. *Then* $\sum_i \kappa_i \leq T$, *and any sequence* $x_0,\dots,x_{T-1}$ *consisting of* κ_i *copies of* v_i *for each* i, *padded with* $T - \sum_i \kappa_i$ *further copies of* v_1, *satisfies*

$$\sum_{t<T} x_t x_t^\top \ \succeq\ \sum_i \kappa_i v_i v_i^\top \ \succeq\ (T-m)\,A(\lambda) \ \succeq\ \frac{T}{2}\,A(\lambda).$$

Proof. def:design$_matrix, lem : loewner_quadraticSince\lceil y \rceil < y+1$, $\sum_i \kappa_i < (T-m)\sum_i p_i + m = T$, so $\sum_i \kappa_i \leq T$ (integers) and the padding is well defined. Each $v_i v_i^\top \in \mathcal{S}^d_+$, so adding the padding copies only increases the sum in the Loewner order (Lemma **??**). Next, $\sum_i \kappa_i v_i v_i^\top - (T-m)A(\lambda) = \sum_i (\kappa_i - (T-m)p_i) v_i v_i^\top$ is a combination of positive semidefinite matrices with nonnegative coefficients $\kappa_i - (T-m)p_i \geq 0$, hence positive semidefinite. Finally $T - m \geq T/2$ because $T \geq 2m$, and $A(\lambda) \in \mathcal{S}^d_+$, so $(T-m)A(\lambda) \succeq (T/2)A(\lambda)$. ☐

Remark 59. Lemma **??** is a one-line substitute for Theorem **??** with a better constant ($T/2$ instead of $T/4$), but it requires $T \geq 2m$ where $m = |\operatorname{supp}\lambda|$ is only known to be finite (at most $d(d+1)/2+1$ when λ is produced by Lemma **??**); it therefore only gives fixed designs with budget $T = \Omega(d^2)$, which is not enough for the $d\log(1/\delta)/\varepsilon^2$ term of Theorem **??** when $\log(1/\delta) \ll d$. Theorem **??** removes this restriction. Either lemma may be used in a first version of the formalization of Chapter **??**, at the price of a weaker theorem in the naive case.

Chapter 3

The non-adaptive fixed-design algorithm

This chapter proves Theorem 1 of the paper (Section 2.1 and Appendix B.1 there): a non-adaptive fixed-design algorithm with budget $T = \lceil 600(w(\mathcal{X})^2 + d\log(2/\delta))/\varepsilon^2\rceil$ is (ε, δ)-PAC on every spanning action set $\mathcal{X}$ (Theorem **??**); the spanning hypothesis is what makes the width $w(\mathcal{X})$ and the G-optimal design exist. The design is obtained by rounding (Theorem **??**) the mixture of a width-minimizing design (Lemma **??**) and a G-optimal design (Theorem **??**); the estimator is least squares; the recommendation is an (approximate) maximizer of the estimated value. The probabilistic input is the law of the least-squares error (Chapter **??**) and the Borell–TIS inequality for linear Gaussian processes in the form of Corollary **??** (Chapter **??**). Compared with the paper: the constant 360 becomes 600 because of the concentration constant $c_{\mathrm{G}} = \pi^2/4$ of Theorem **??** and because the recommendation is only an $\varepsilon/4$-approximate maximizer (a measurable exact argmax selector on a compact set is not available without extra work; Lemma **??**); the rounding step uses Theorem **??** instead of the cited rounding procedure. The chapter ends with a direct treatment of the unit ball (Lemma **??**), which needs none of the width machinery and is used in Chapter **??**.

3.1 The mixed design and its rounding

Definition 60 (Mixed design)**.** Maiti2026Power.mixDesign, Maiti2026Power.designMatrix$_{m}ixDesign, Maiti2$
$\triangle_{\mathcal{X}}$ with $A_1 := A(\lambda_1) \in \mathcal{S}^d_{++}$ and $w(\mathcal{X}; A_1) = w(\mathcal{X})$ (Lemma **??**), and let $\lambda_G \in \triangle_{\mathcal{X}}$ with $A_G := A(\lambda_G) \in \mathcal{S}^d_{++}$ and $g_{\mathcal{X}}(A_G) = d$ (Theorem **??**). The *mixed design distribution* is

$$\lambda_0 := \tfrac{1}{2}\lambda_1 + \tfrac{1}{2}\lambda_G \in \triangle_{\mathcal{X}}, \qquad A_0 := A(\lambda_0) = \tfrac{1}{2}A_1 + \tfrac{1}{2}A_G \in \mathcal{S}^d_{++},$$

where λ_0 is supported on $\operatorname{supp}\lambda_1 \cup \operatorname{supp}\lambda_G$ (finite) with weights $\lambda_{0,x} = \frac{1}{2}\lambda_{1,x} + \frac{1}{2}\lambda_{G,x}$ (a weight being 0 outside the corresponding support).

Lemma 61 (Bounds for the mixed design)**.** $Maiti2026Power.smul_{d}esignMatrix_{l}e_{d}esignMatrix_{m}ixDesign_{l}ef$
$\in \mathcal{X}$, $\|x\|^2_{A_0^{-1}} \le 2d$, *and* $w(\mathcal{X}; A_0) \le \sqrt{2}\, w(\mathcal{X})$.

Proof. lem:loewner$_{s}calar, cor : g_{o}ptimal_{n}orm_{b}ound, lem : width_{m}ono, lem : width_{h}omog, lem : loewner_{q}uadraticSince1\frac{}{2A_1 \in \mathcal{S}_+^d}$, $\frac{1}{2}A_G \preceq A_0$ (Lemma **??**), and Lemma **??** with $c = 1/2$ gives $\|x\|^2_{A_0^{-1}} \le 2\|x\|^2_{A_G^{-1}} \le 2d$ for $x \in \mathcal{X}$, by Corollary **??**. Similarly $\frac{1}{2}A_1 \preceq A_0$ with $\frac{1}{2}A_1, A_0 \in \mathcal{S}^d_{++}$, so by monotonicity of the width (Lemma **??**) and homogeneity (Lemma **??** with $c = 1/2$, $c^{-1/2} = \sqrt{2}$), $w(\mathcal{X}; A_0) \le w(\mathcal{X}; \frac{1}{2}A_1) = \sqrt{2}\, w(\mathcal{X}; A_1) = \sqrt{2}\, w(\mathcal{X})$. □

Lean remark. The mixture `Maiti2026Power.mixDesign w w` and the two bounds are formalized for an arbitrary design distribution λ_1 with $A(\lambda_1) \succ 0$ (the width bound reads $w(\mathcal{X}; A_0) \le \sqrt{2}\, w(\mathcal{X}; A(\lambda_1))$); the choice of a width-minimizing λ_1 (Lemma **??**) is made only when the algorithm is assembled.

3.2 Least squares and the difference process

Definition 62 (Least-squares estimator)**.** Maiti2026Power.leastSquares, Maiti2026Power.designRows, Maiti2026Power.lsMatrix, Maiti2026Power.leastSquares$_{e}q_{t}oEuclideanLindef : fixed_{d}esignForafixeddesignx$
with $\Sigma_T = X^\top X \in \mathcal{S}^d_{++}$ and observations $y \in \mathbb{R}^T$, the *least-squares estimator* is
$$\hat{\theta} := \Sigma_T^{-1} X^\top y = \Sigma_T^{-1} \sum_{t<T} y_t x_t \in \mathbb{R}^d,$$
a linear (hence measurable) function of y.

Lemma 63 (Law of the least-squares error)**.** $Maiti2026Power.leastSquares_{i}nner_{a}dd, Maiti2026Power.lsMatri$
$\hat{\theta} - \theta = \Sigma_T^{-1} X^\top \eta \sim \mathcal{N}(0, \Sigma_T^{-1})$, *where* $\eta \sim \mathcal{N}(0, I_T)$ *is the noise vector. In particular* $\hat{\theta} - \theta \overset{d}{=} \Sigma_T^{-1/2} g$ *with* $g \sim \mathcal{N}(0, I_d)$.

Proof. lem:fixed$_{d}esign_{l}aw, lem : gauss_{l}inear_{i}mage, lem : gauss_{l}aw_{o}f_{c}ov, lem : sqrt_{p}ropsByLemma$ **??**,y = X$\theta + \eta$ with $\eta \sim \mathcal{N}(0, I_T)$. Then $\hat{\theta} = \Sigma_T^{-1} X^\top X\theta + \Sigma_T^{-1} X^\top \eta = \theta + \Sigma_T^{-1} X^\top \eta$. Lemma **??** with the matrix $M := \Sigma_T^{-1} X^\top \in \mathbb{R}^{d \times T}$ gives $M\eta \sim \mathcal{N}(0, MM^\top)$ with $MM^\top = \Sigma_T^{-1} X^\top X \Sigma_T^{-1} = \Sigma_T^{-1}$ (using that Σ_T^{-1} is symmetric). The last claim is Lemma **??** together with $(\Sigma_T^{-1/2})^2 = \Sigma_T^{-1}$ (Lemma **??**(ii)). □

Lean remark. The formalized statement is $\hat{\theta} \sim \mathcal{N}(\theta, \Sigma_T^{-1})$ (`Maiti2026Power.hasLaw_leastSquares_of_f` for the run of the fixed design in the environment ν_θ and any T with $\Sigma_T \succ 0$), obtained from $\hat{\theta} = \theta + \Sigma_T^{-1} X^\top \eta$ (`leastSquares_inner_add`), the standard Gaussian law of η (Lemma **??**) and `multivariateGaussian_zero_map_toEuclideanLin` with $MM^\top = \Sigma_T^{-1}$ (`lsMatrix_mul_transpose`); the representation $\Sigma_T^{-1/2} g$ is not needed.

Lemma 64 (Measurable approximate argmax selector)**.** *$exists_{m}easurable_{o}f_{f}orall_{e}xists_{m}em, exists_{m}easurabl$ $\mathbb{R}^d$ be compact and nonempty and let $\gamma > 0$. There is a Borel measurable map $s_\gamma : \mathbb{R}^d \to \mathcal{X}$ such that for every $v \in \mathbb{R}^d$,*

$$\langle s_\gamma(v), v\rangle \;\; \geq \;\; \max_{x\in\mathcal{X}} \langle x, v\rangle - \gamma.$$

If $\mathcal{X}$ is finite, there is a measurable $s_0 : \mathbb{R}^d \to \mathcal{X}$ with $\langle s_0(v), v\rangle = \max_{x\in\mathcal{X}}\langle x, v\rangle$ for all v (an exact argmax selector).

Proof. lem:sup$_{c}ountable_{d}enseLet$M(v) := $\max_{x\in\mathcal{X}}\langle x, v\rangle$; it is continuous in v (Lipschitz with constant $R := \sup_{x\in\mathcal{X}}\|x\|$, Lemma **??**). Let $(q_n)_{n\in\mathbb{N}}$ be a sequence that is dense in $\mathcal{X}$ ($\mathcal{X}$ is a separable metric space). For each n the set $U_n := \{v \in \mathbb{R}^d : \langle q_n, v\rangle > M(v) - \gamma\}$ is open, as a strict inequality between continuous functions. The U_n cover $\mathbb{R}^d$: given v, let $x^\star \in \mathcal{X}$ attain $M(v)$ and pick n with $\|q_n - x^\star\| < \gamma/(1+\|v\|)$; then $\langle q_n, v\rangle \geq \langle x^\star, v\rangle - \|q_n - x^\star\|\|v\| > M(v) - \gamma$. Define $n(v) := \min\{n : v \in U_n\}$ and $s_\gamma(v) := q_{n(v)}$. Then $s_\gamma(v) \in \mathcal{X}$ and $\langle s_\gamma(v), v\rangle > M(v) - \gamma$ by construction. Measurability: s_γ takes countably many values and $\{v : s_\gamma(v) = q_n\} \supseteq \{v : n(v) = n\} = U_n \setminus \bigcup_{k<n} U_k$; more precisely $s_\gamma^{-1}(\{q\}) = \bigcup_{n: q_n = q}\{n(\cdot) = n\}$ is a countable union of Borel sets, so every preimage of a subset of the countable range is Borel.

For finite $\mathcal{X} = \{q_1, \ldots, q_m\}$ the same construction with $\gamma = 0$ works: the sets $F_n := \{v : \langle q_n, v\rangle = M(v)\}$ are closed (hence Borel), cover $\mathbb{R}^d$ since the maximum is attained, and $s_0(v) := q_{\min\{n: v\in F_n\}}$ is measurable for the same reason. ☐

Lean remark. $n(v)$ is `Nat.find` applied to the covering property; the measurability of a map with countable range whose fibers are measurable is `measurable_to_countable'` (`Mathlib/MeasureTheory/MeasurableSpace/Constructions.lean`), and `TopologicalSpace.exists_countable_dense` gives the dense sequence (turn the countable dense set into a sequence with `Set.Countable.exists_eq_range` after handling emptiness). The codomain is the subtype $\mathcal{X}$ with the Borel σ-algebra of the subspace topology, as in Definition **??**. The recommendation kernel of Definition **??** is then `Kernel.deterministic` of the measurable map $y \mapsto s_\gamma(\hat\theta(y))$. The paper uses an exact argmax; we use $\gamma = \varepsilon/4$ and absorb the loss in the constant (Theorem **??**).

Lemma 65 (Regret is controlled by the difference process)**.** *$Maiti2026Power.simpleRegret_{l}e_{s}upportFn_{a}dd_{s}uppo$ $\mathbb{R}^d$, $\gamma \geq 0$, and $\hat x \in \mathcal{X}$ with $\langle \hat x, \hat\theta\rangle \geq \max_{x\in\mathcal{X}}\langle x, \hat\theta\rangle - \gamma$. Then*

$$r(\hat x, \theta) \leq D + \gamma, \qquad D := \max_{x,x'\in\mathcal{X}} \langle x - x', \hat\theta - \theta\rangle.$$

Proof. def:regret Let $x^\star \in \arg\max_{x\in\mathcal{X}}\langle x, \theta\rangle$. Then

$$r(\hat x, \theta) = \langle x^\star, \theta\rangle - \langle \hat x, \theta\rangle = \langle x^\star - \hat x, \theta - \hat\theta\rangle + \big(\langle x^\star, \hat\theta\rangle - \langle \hat x, \hat\theta\rangle\big).$$

The second bracket is at most $\max_x \langle x, \hat\theta\rangle - \langle \hat x, \hat\theta\rangle \leq \gamma$, and the first term equals $\langle \hat x - x^\star, \hat\theta - \theta\rangle \leq D$ (take $x = \hat x$, $x' = x^\star$). ☐

Lean remark. The formalized bound is $r(\hat{x}, \theta) \le \sup_{x \in \mathcal{X}} \langle x, \theta - \hat{\theta} \rangle + \sup_{x \in \mathcal{X}} \langle x, \hat{\theta} - \theta \rangle + \gamma$ (the two support functions add up to D), for a bounded set $\mathcal{X}$ and without choosing a maximizer $x^\star$: it follows from the subadditivity of the support function $\sup_x \langle x, \theta \rangle \le \sup_x \langle x, \theta - \hat{\theta} \rangle + \sup_x \langle x, \hat{\theta} \rangle$.

Lemma 66 (Concentration of the difference process)**.** *Maiti2026Power.diffSup, Maiti2026Power.diffSup*$_e q_s upportFn_a dd_n eg, Maiti2026Power.continuous_d iffSup, Maiti2026Power.integral$ $\mathcal{S}^d_{++}$*,* $\Delta \sim \mathcal{N}(0, \Sigma^{-1})$*,* $D := \max_{x, x' \in \mathcal{X}} \langle x - x', \Delta \rangle$ *and* $\sigma^2 := \max_{x \in \mathcal{X}} \|x\|^2_{\Sigma^{-1}}$*. Then* $\mathbb{E}D = 2\, w(\mathcal{X}; \Sigma)$*, and for every* $\delta \in (0, 1)$*,*

$$\mathbb{P}\Big(D \le 2\, w(\mathcal{X}; \Sigma) + 2\sigma \sqrt{2 c_{\mathrm{G}} \log(1/\delta)}\Big) \ge 1 - \delta.$$

Proof. lem:width$_s ymm, lem : width_m atrix_w d, cor : sup_b ound_s ymmetric, cor : sup_b ound_w hp, lem : sqrt_p rops Theexpectation$:D $= \max_x \langle x, \Delta \rangle + \max_{x'} \langle x', -\Delta \rangle$ and both maxima have expectation $w(\mathcal{X}; \Sigma)$ by Lemma **??** (the width only depends on the law $\mathcal{N}(0, \Sigma^{-1})$ of Δ, Lemma **??**). The deviation bound is Corollary **??** applied to the centered Gaussian vector $G = \Delta$ with covariance $S = \Sigma^{-1}$ and index set $\mathcal{X}$: there $\sigma^2 = \sup_{x \in \mathcal{X}} x^\top S x = \max_x \|x\|^2_{\Sigma^{-1}}$, and the statement is exactly $\mathbb{P}(D \le \mathbb{E}D + 2\sigma\sqrt{2 c_{\mathrm{G}} \log(1/\delta)}) \ge 1 - \delta$. (Equivalently, apply Corollary **??** to the compact index set $\mathcal{X} - \mathcal{X} = \{x - x' : x, x' \in \mathcal{X}\}$, whose variance proxy is $\max_{x, x'} \|x - x'\|^2_{\Sigma^{-1}} \le (2\sigma)^2$ by the triangle inequality for the norm $\|\cdot\|_{\Sigma^{-1}} = \|\Sigma^{-1/2} \cdot\|$, Lemma **??**(iv).) ☐

3.3 The algorithm and its guarantee

Definition 67 (The fixed-design identification algorithm)**.** Maiti2026Power.lsRecommend, Maiti2026Power.measurable$_l sRecommend, Maiti2026Power.fixedDesignIdentAlg, Maiti2026Power.isFixed$ $(0, 1]$ and $\delta \in (0, 1)$, and set

$$T := \left\lceil \frac{600\,\big(w(\mathcal{X})^2 + d \log(2/\delta)\big)}{\varepsilon^2} \right\rceil.$$

Then $T \ge 4d$, as required by Theorem **??**: indeed $\log(2/\delta) > \log 2 > 0.69$ and $\varepsilon \le 1$, so $T \ge 600 \cdot 0.69\, d / \varepsilon^2 \ge 414\, d \ge 4d$. The *fixed-design identification algorithm* $\mathsf{Alg}_{\mathrm{fd}}(\varepsilon, \delta)$ is the fixed-design algorithm (Definition **??**) with

(i) design $x_0, \ldots, x_{T-1} \in \operatorname{supp} \lambda_0$ given by Theorem **??** applied to the mixed design distribution λ_0 of Definition **??** with budget T; its design matrix Σ_T is positive definite (Corollary **??**);

(ii) estimator $\hat{\theta} = \Sigma_T^{-1} X^\top y$ (Definition **??**);

(iii) recommendation $\hat{x} := s_{\varepsilon/4}(\hat{\theta})$, with $s_{\varepsilon/4}$ the measurable $\varepsilon/4$-approximate argmax selector of Lemma **??**; the recommendation kernel is the Dirac kernel at the measurable map $y \mapsto s_{\varepsilon/4}(\Sigma_T^{-1} X^\top y)$.

Remark 68 (The range of ε and the constant). The definition is restricted to $\varepsilon \in (0,1]$ (the regime of the paper), which matches Theorem **??** and makes the requirement $T \geq 4d$ of Theorem **??** automatic. For $\varepsilon > 1$ one can define the algorithm in the same way with $T := \max\big(\lceil 600(w(\mathcal{X})^2 + d\log(2/\delta))/\varepsilon^2\rceil, 4d\big)$; the proof of Theorem **??** goes through unchanged, since it only uses $T \geq 4d$ and $T \geq 600(w(\mathcal{X})^2 + d\log(2/\delta))/\varepsilon^2$. If $\mathcal{X}$ is finite, the exact selector s_0 can be used in (iii); then $D \leq \varepsilon$ suffices in the proof below, and since that proof gives $D^2 \leq 320\,(w(\mathcal{X})^2 + d\log(2/\delta))/T$ on the good event, the constant 600 can be reduced to 320. The paper's constant is 360.

Theorem 69 (Fixed-design upper bound (Theorem 1 of the paper))**.** *$Maiti2026Power.isPAC_fixedDesignIdent$ $(0,1]$ and $\delta \in (0,1)$. The algorithm $\mathsf{Alg}_{\mathrm{fd}}(\varepsilon,\delta)$ of Definition **??** is (ε,δ)-PAC on $\mathcal{X}$, with budget*

$$T = \left\lceil \frac{600\,(w(\mathcal{X})^2 + d\log(2/\delta))}{\varepsilon^2} \right\rceil \leq \frac{600\,(w(\mathcal{X})^2 + d\log(2/\delta))}{\varepsilon^2} + 1.$$

Corollary 70 (Theorem 1 of the paper, existential form)**.** *$Maiti2026Power.exists_isFixedDesign_isPACthm : u$ $(0,1]$ and $\delta \in (0,1)$. There exist a budget $T \leq 600\,(w(\mathcal{X})^2 + d\log(2/\delta))/\varepsilon^2 + 1$ and a fixed-design identification algorithm with budget T (Definition **??**) that is (ε,δ)-PAC on $\mathcal{X}$.*

Proof. thm:upper Take $\mathsf{Alg}_{\mathrm{fd}}(\varepsilon,\delta)$ of Theorem **??**, which is a fixed-design algorithm with the required budget. ☐

Proof. cor:fixed$_d esign_f rom_d istribution, lem : mixed_d design_b ounds, lem : least_s quares_l aw, lem :$ $difference_p rocess_c oncentration, lem : regret_l e_d ifference_p rocess, lem : approx_a rgmax_s elector, def :$ $pacFix\theta \in \mathbb{R}^d$ and write $L := \log(1/\delta) > 0$, $w := w(\mathcal{X})$.

Design quantities. By Corollary **??** (applied to λ_0, $A_0 = A(\lambda_0)$) and Lemma **??**, for every $x \in \mathcal{X}$,

$$\|x\|^2_{\Sigma_T^{-1}} \leq \frac{4}{T}\|x\|^2_{A_0^{-1}} \leq \frac{8d}{T}, \qquad w(\mathcal{X};\Sigma_T) \leq \frac{2}{\sqrt{T}} w(\mathcal{X};A_0) \leq \frac{2\sqrt{2}\,w}{\sqrt{T}}.$$

Hence $\sigma_T^2 := \max_{x\in\mathcal{X}} \|x\|^2_{\Sigma_T^{-1}} \leq 8d/T$.

Concentration. By Lemma **??**, $\Delta := \hat{\theta} - \theta \sim \mathcal{N}(0, \Sigma_T^{-1})$ under $\mathbb{P}_\theta$. Let $D := \max_{x,x'\in\mathcal{X}} \langle x - x', \Delta\rangle$, a measurable function of y. Lemma **??** gives an event E with $\mathbb{P}_\theta(E) \geq 1 - \delta$ on which

$$D \leq 2\,w(\mathcal{X};\Sigma_T) + 2\sigma_T\sqrt{2c_{\mathrm{G}}L} \leq \frac{4\sqrt{2}\,w}{\sqrt{T}} + 2\sqrt{\frac{8d}{T}}\sqrt{2c_{\mathrm{G}}L} = \frac{4\sqrt{2}\,w + 8\sqrt{c_{\mathrm{G}}dL}}{\sqrt{T}}.$$

Arithmetic. Using $(a+b)^2 \leq 2a^2 + 2b^2$, $(4\sqrt{2})^2 = 32$, $8^2 = 64$,

$$\big(4\sqrt{2}\,w + 8\sqrt{c_{\mathrm{G}}dL}\big)^2 \leq 64\,w^2 + 128\,c_{\mathrm{G}}\,dL \leq 64\,w^2 + 320\,dL \leq 320\,\big(w^2 + d\log(2/\delta)\big),$$

where $128c_{\mathrm{G}} = 32\pi^2 < 32 \cdot 9.87 < 320$ and $L = \log(1/\delta) \leq \log(2/\delta)$. Since $T \geq 600(w^2 + d\log(2/\delta))/\varepsilon^2$, on E,

$$D^2 \leq \frac{320\,(w^2 + d\log(2/\delta))}{T} \leq \frac{320}{600}\,\varepsilon^2 = \frac{8}{15}\varepsilon^2 \leq \frac{9}{16}\varepsilon^2,$$

because $8/15 = 0.5\overline{3} \leq 0.5625 = 9/16$; hence $D \leq 3\varepsilon/4$ on E.

Regret. The recommendation satisfies $\langle \hat{x}, \hat{\theta} \rangle \geq \max_x \langle x, \hat{\theta} \rangle - \varepsilon/4$ (Lemma **??**), so by Lemma **??** with $\gamma = \varepsilon/4$, on E: $r(\hat{x}, \theta) \leq D + \varepsilon/4 \leq 3\varepsilon/4 + \varepsilon/4 = \varepsilon$. Thus $\mathbb{P}_\theta(r(\hat{x}, \theta) \leq \varepsilon) \geq \mathbb{P}_\theta(E) \geq 1 - \delta$, for every θ, which is Definition **??**. The bound $\lceil u \rceil \leq u + 1$ gives the budget bound. □

Lean remark. The formalized algorithm `Maiti2026Power.fixedDesignIdentAlg T x s` takes the design x and the selector s as parameters, and `Maiti2026Power.isPAC_fixedDesignIdentAlg` is the guarantee in the general form "if $\Sigma_T \succ 0$, $\|x\|^2_{\Sigma_T^{-1}} \leq \sigma^2$ on $\mathcal{X}$, s is an $\varepsilon/4$-selector and $2w(\mathcal{X}; \Sigma_T) + 2\sigma\sqrt{2c_{\mathrm{G}} \log(1/\delta)} \leq 3\varepsilon/4$, then the algorithm is (ε, δ)-PAC"; the choice of the design and the arithmetic of the constants are done in the proof of Corollary **??** (`Maiti2026Power.exists_isFixedDesign_isPAC`). Instead of a width-minimizing design λ_1 (Lemma **??**, not formalized) the formal proof uses an approximate minimizer with $w(\mathcal{X}; A_1)^2 \leq w(\mathcal{X})^2 + (\log 2)/4$, which the slack $337.5 - 320 = 17.5$ of the arithmetic absorbs. The case $d = 0$ is treated separately (every recommendation has zero regret). The theorem is an existence statement whose witness is built from non-constructive choices: λ_1 (Lemma **??**), λ_G (Theorem **??**), the rounded design (Theorem **??**) and the selector (Lemma **??**); each is obtained by `Classical.choose` and Definition **??** packages them into a `Learning.detAlgorithm` with a history-independent policy plus a deterministic recommendation kernel. The good event E is defined in terms of y only, so the PAC probability is computed under the law $\mathcal{N}(X\theta, I_T)$ of y (Lemma **??**) and the law of $\Delta = \Sigma_T^{-1} X^\top (y - X\theta)$; no conditioning is needed. The constant $c_{\mathrm{G}} = \pi^2/4$ enters only through the numerical inequality $32\pi^2 < 320$, i.e. $\pi^2 < 10$; the crude bound `Real.pi_lt_four` is not enough ($16 \not< 10$), but `Real.pi_lt_d2` ($\pi < 3.15$) gives $\pi^2 < 9.93 < 10$.

Remark 71 (Asymptotic form). In the asymptotic form of the paper's Theorem 1: for every spanning action set $\mathcal{X}$, every $\varepsilon \in (0, 1]$ and every $\delta \in (0, 1/2]$ there is a non-adaptive (ε, δ)-PAC algorithm with budget $O\big((d \log(1/\delta) + w(\mathcal{X})^2)/\varepsilon^2\big)$, since $\log(2/\delta) \leq 2 \log(1/\delta)$ for $\delta \leq 1/2$. This asymptotic form is not meant to be formalized; Theorem **??** is the explicit statement.

Remark 72 (Comparison with the paper). The paper's algorithm and analysis (Section 2.1) are the same, with three differences: (a) the paper's rounding lemma (Appendix B.1, cited from [**?**, **?**]) requires $T \geq 180d$ and yields $\tau(A_T) \leq 2\tau(A(\lambda_0))$, whereas Theorem **??** yields the Loewner bound $\Sigma_T \succeq (T/4)A_0$ for $T \geq 4d$, which implies the same kind of control with the factor 4 instead of 2; (b) the paper uses the Borell–TIS inequality with constant 1 in the exponent, we have $c_{\mathrm{G}} = \pi^2/4$; (c) the paper's recommendation is an exact maximizer. Items (a)–(c) explain the change of constant from 360 to 600. The dependence on ε, δ, d and $w(\mathcal{X})$ is unchanged.

3.4 The unit ball: a direct algorithm

Lemma 73 (Identification on the unit ball by a basis design)**.** *Maiti2026Power.lnNoise*$_r$*eal*$_n$*orm*$_l$*nDelta*$_g$*t*$_l$*e, M* $\mathbb{B}_d$*,* $\varepsilon > 0$*,* $\delta \in (0,1)$*, and*

$$n := \left\lceil \frac{8d + 48\log(1/\delta)}{\varepsilon^2} \right\rceil, \qquad T := d\,n.$$

Consider the fixed design that plays each standard basis vector $e_1, \dots, e_d$ *exactly* n *times (*$x_t := e_{1+\lfloor t/n \rfloor}$ *for* $t < T$*), the estimator* $\hat\theta \in \mathbb{R}^d$ *whose* i*-th coordinate is the empirical mean of the* n *observations made at* e_i *(this is the least-squares estimator, since* $\Sigma_T = nI_d$*), and the recommendation* $\hat x := \hat\theta/\|\hat\theta\|$ *if* $\hat\theta \neq 0$ *and* $\hat x := e_1$ *otherwise. This fixed-design algorithm with budget* T *is* (ε,δ)*-PAC on* $\mathbb{B}_d$*, and* $T \le (8d^2 + 48d\log(1/\delta))/\varepsilon^2 + d$*.*

Proof. lem:least$_s$quares$_l$aw, lem : fixed$_d$esign$_l$aw, lem : noise$_r$epresentation, lem : gauss$_p$i, lem : gauss$_l$aw$_o$f$_c$ov, lem : chi$_s$quare$_t$ail, def : regretLaw of the error.$\Sigma_T = \sum_{t<T} x_t x_t^\top = n\sum_i e_i e_i^\top = nI_d \in \mathcal{S}^d_{++}$ and $\Sigma_T^{-1}X^\top y$ has i-th coordinate $\frac{1}{n}\sum_{t:x_t=e_i} y_t$, so $\hat\theta$ is the least-squares estimator and Lemma **??** gives $\Delta := \hat\theta - \theta \sim \mathcal{N}(0, I_d/n)$; by Lemma **??**, $\Delta \overset{d}{=} g/\sqrt{n}$ with $g \sim \mathcal{N}(0, I_d)$, so $\|\Delta\|^2 \overset{d}{=} \|g\|^2/n$. (Directly: by Lemmas **??** and **??**, $\hat\theta_i = \theta_i + \frac{1}{n}\sum_{t:x_t=e_i} \eta_t$ with the η_t i.i.d. standard Gaussian, and the d block averages are independent $\mathcal{N}(0,1/n)$ variables, Lemma **??**.)

Good event. By Lemma **??**, $\mathbb{P}(\|g\|^2 \ge 2d + 12\log(1/\delta)) \le \delta$. Let $E := \{\|\Delta\|^2 < (2d + 12\log(1/\delta))/n\}$, so $\mathbb{P}_\theta(E) \ge 1-\delta$. On E, by the choice of n,

$$\|\Delta\|^2 < \frac{2d + 12\log(1/\delta)}{n} \le \frac{(2d + 12\log(1/\delta))\,\varepsilon^2}{8d + 48\log(1/\delta)} = \frac{\varepsilon^2}{4},$$

so $\|\Delta\| \le \varepsilon/2$.

Regret on the good event. On $\mathbb{B}_d$, $\max_{x\in\mathbb{B}_d}\langle x, \theta\rangle = \|\theta\|$ (Cauchy–Schwarz, attained at $\theta/\|\theta\|$ or at any point if $\theta = 0$), so $r(\hat x, \theta) = \|\theta\| - \langle \hat x, \theta\rangle$. If $\hat\theta \neq 0$:

$$\langle \hat x, \theta\rangle = \langle \hat x, \hat\theta\rangle - \langle \hat x, \Delta\rangle = \|\hat\theta\| - \langle \hat x, \Delta\rangle \ge \|\hat\theta\| - \|\Delta\| \ge \|\theta\| - 2\|\Delta\|,$$

using $\|\hat x\| = 1$ and $\|\hat\theta\| \ge \|\theta\| - \|\Delta\|$; hence $r(\hat x, \theta) \le 2\|\Delta\| \le \varepsilon$. If $\hat\theta = 0$: $\|\theta\| = \|\Delta\| \le \varepsilon/2$ and $r(\hat x, \theta) = \|\theta\| - \langle e_1, \theta\rangle \le 2\|\theta\| \le \varepsilon$. In both cases $r(\hat x, \theta) \le \varepsilon$ on E, so $\mathbb{P}_\theta(r(\hat x, \theta) \le \varepsilon) \ge 1 - \delta$ for every θ.

Budget. $T = dn \le d\big((8d + 48\log(1/\delta))/\varepsilon^2 + 1\big)$. □

Lean remark. The two steps of the proof are formalized on the noise space $(\mathbb{R}^n)^d$ with the law $\mathcal{N}(0,1)^{\otimes dn}$ of the noises of the basis rounds: the chi-square tail bound $\mathbb{P}(\|\Delta\| > c) \le \delta$ whenever $2d + 12\log(1/\delta) \le nc^2$ (`lnNoise_real_norm_lnDelta_gt_le`, with Δ the vector of noise averages) and the deterministic regret bound $\|\vartheta\| - \langle v/\|v\|, \vartheta\rangle \le 2\|v - \vartheta\|$ (`norm_sub_inner_normalizeBall_le`, with $v/\|v\| := 0$ when $v = 0$). The fixed-design algorithm itself is not formalized on its own: it is the second phase of the block algorithm of Definition **??**, where the above bounds are used with $(\varepsilon/2, \delta/2)$.

Remark 74. Lemma **??** is a special case of Theorem **??** ($\mathbb{B}_d$ is spanning, so the theorem applies; the lemma itself uses no spanning hypothesis): since $w(\mathbb{B}_d) \le d$ (Proposition **??**), Theorem **??** gives a budget of order $(d^2 + d\log(1/\delta))/\varepsilon^2$ on the ball, the same order as here. The direct argument is recorded because it needs neither the width nor the rounding machinery, only the χ^2 tail bound, and because Chapter **??** composes it with the norm estimation algorithm on each block of the separation instance. The map $y \mapsto \hat{x}$ is measurable as it is continuous on $\{\hat{\theta} \neq 0\}$ and constant on the closed set $\{\hat{\theta} = 0\}$ (`Measurable.piecewise`).

Chapter 4

Lower bound for non-adaptive fixed designs

This chapter proves Theorem 3 of the paper (Section 2.2 and Appendices B.3 and B.4): on a spanning action set $\mathcal{X}$, every non-adaptive fixed-design algorithm that is (ε, δ)-PAC with $\delta \leq 1/10$ uses a budget $T \geq w(\mathcal{X})^2/(70\,\varepsilon^2)$ (Theorem **??**). The argument is Bayesian. Under the prior $\theta \sim \mathcal{N}(0, \tau^2 \Sigma_T^{-1})$, the expected simple regret of *any* recommendation rule is at least a constant times $w(\mathcal{X}; \Sigma_T)$ (Lemma **??**), while the expected regret of an (ε, δ)-PAC algorithm is at most $\varepsilon + \delta\,\mathbb{E}Z(\theta)$ (Lemma **??**). Comparing the two bounds and using $w(\mathcal{X}; \Sigma_T) \geq w(\mathcal{X})/\sqrt{T}$ (Lemma **??**) gives the theorem. Singular design matrices are handled separately (Lemma **??**).

From the prerequisite chapters we use the linear theory of Gaussian vectors of Chapter **??**: linear images (Lemma **??**), joint Gaussianity of linear functionals of a Gaussian vector (Lemma **??**), independence of uncorrelated jointly Gaussian vectors (Lemma **??**, Mathlib `HasGaussianLaw.indepFun_of_covariance_eval`) and the vanishing of $\mathbb{E}\langle f(V), W\rangle$ for independent V, W with W centered (Lemma **??**).

What is new with respect to the paper. The paper uses the Gaussian posterior mean $\mathbb{E}[\theta \mid y] = c\,\Sigma_T^{-1} X^\top y$; conditional expectations of Gaussian vectors are not convenient to formalize, so this step is replaced by the orthogonal decomposition $\theta = W + c\,\Sigma_T^{-1} X^\top y$ with W independent of y (Lemma **??**), which yields the same identity for $\mathbb{E}\langle \hat{x}, \theta\rangle$ (Lemma **??**). The recommendation is an arbitrary Markov kernel, so all statements hold for randomized recommendation rules. The singular case (Appendix B.4 of the paper, stated there for finite $\mathcal{X}$) is proved for every compact spanning $\mathcal{X}$ and every recommendation kernel.

4.1 The Bayesian model and the orthogonal decomposition

Throughout this chapter $x_0, \dots, x_{T-1} \in \mathcal{X}$ is a fixed design of length T, $X \in \mathbb{R}^{T\times d}$ is the matrix with rows $x_t^\top$, $\Sigma_T = X^\top X$ is its design matrix (Definition **??**), and ρ is a recommendation kernel from $\mathbb{R}^T$ to $\mathcal{X}$. In Sections **??** and **??** we assume $T \geq 1$ and $\Sigma_T \succ 0$.

Definition 75 (Gaussian prior on the reward vector)**.** Maiti2026Power.bayesParam, Maiti2026Power.bayesObs, Maiti2026Power.bayesHist, Maiti2026Power.bayesKernel, Maiti2026Power.bayesJoint, Maiti2026Power.map$_b ayesParam_s tdGaussian, Maiti2026Power.map_b ayesHist_s$
0. In the *Bayesian model with parameter* τ, the reward vector θ is a random vector with law $\pi := \mathcal{N}(0, \tau^2\Sigma_T^{-1})$, the noise $\eta \sim \mathcal{N}(0, I_T)$ is independent of θ, the observation vector is $y := X\theta + \eta \in \mathbb{R}^T$ and the recommendation is $\hat{x} \sim \rho(y)$. That is, the joint law of $(\theta, \eta, \hat{x})$ is

$$\big(\mathcal{N}(0, \tau^2\Sigma_T^{-1}) \otimes \mathcal{N}(0, I_T)\big) \otimes \big((\theta, \eta) \mapsto \rho(X\theta + \eta)\big).$$

Equivalently, the joint law of $(\theta, y, \hat{x})$ is $\pi \otimes \kappa$ for the Markov kernel $\kappa(\theta) := \mathcal{N}(X\theta, I_T) \otimes \rho$, and by Lemma **??** $\kappa(\theta)$ is the law $\mathbb{P}_\theta$ of $(y, \hat{x})$ under the fixed-design algorithm run against ν_θ (recall from Definition **??** that $(y, \hat{x})$ is identified with $(H_T, \hat{x})$). This is the setting of Lemma **??**. We write $\mathbb{E}$ for the expectation under this joint law.

Lean remark. In Lean the Bayesian model is built directly on a standard Gaussian pair (g, η) of $\mathbb{R}^d \times \mathbb{R}^T$ (`stdGaussian`$\times$`stdGaussian`): the reward vector is $\theta = \tau\Sigma_T^{-1/2} g$ (`bayesParam`, whose law is $\mathcal{N}(0, \tau^2\Sigma_T^{-1})$ by `map_bayesParam_stdGaussian`), the observations are $y_t = \langle x_t, \theta\rangle + \eta_t$ (`bayesObs`), the history is $((x_t, y_t))_{t<T}$ (`bayesHist`) and the recommendation is drawn from the output rule of the algorithm applied to that history (`bayesKernel`); the joint law of $((g, \eta), \hat{x})$ is `bayesJoint`. Conditionally on g, the law of the history is the fixed-design law of Lemma **??** with reward vector θ (`map_bayesHist_stdGaussian`), which is how the PAC property enters (`measureReal_lt_simpleRegret_bayes_le`).

Lean remark. A randomized recommendation rule $\hat{x} = f(y, U)$ using an internal seed $U \sim \mu$ independent of (θ, η), as in the paper, is the same thing as the Markov kernel $\rho(y) := f(y, \cdot)_{\#}\mu$; only the kernel form is used. The Bayesian model should be set up so that (θ, η) is visibly a linear image of a standard Gaussian vector of $\mathbb{R}^{d+T}$: take $(g, g') \sim \mathcal{N}(0, I_{d+T})$ (`stdGaussian`) and $\theta := \tau\,\Sigma_T^{-1/2} g$, $\eta := g'$; then every linear functional of (θ, η) has `HasGaussianLaw`, which is what Lemma **??** needs. The law of θ is Mathlib's `multivariateGaussian` with mean 0 and covariance $\tau^2\Sigma_T^{-1}$ by Lemma **??**.

Lemma 76 (Orthogonal decomposition of the reward vector)**.** *Maiti2026Power.bayesW , Maiti2026Power.bayesW , Maiti2026Power.bayesY , Maiti2026Power.bayesParam$_s ub_s mul_l eastSquares, Maiti$*
let

$$c := \frac{\tau^2}{1+\tau^2} \in (0,1), \qquad W := \theta - c\,\Sigma_T^{-1} X^\top y \in \mathbb{R}^d.$$

Then (W, y) is a centered jointly Gaussian vector of $\mathbb{R}^d \times \mathbb{R}^T$ with $\operatorname{Cov}(W, y) = \mathbb{E}[Wy^\top] = 0 \in \mathbb{R}^{d\times T}$. *Consequently W and y are independent, and W is integrable with $\mathbb{E}W = 0$.*

Proof. lem:gauss$_pi, lem : gauss_linear_image, lem : gauss_law_of_cov, lem : gauss_joint_linear, lem : gauss_uncorrelated_indep, lem : gauss_norm_momentsSince \Sigma_T^{-1}X^\top X = I_d$,

$$W = \theta - c\,\Sigma_T^{-1}X^\top(X\theta + \eta) = (1-c)\,\theta - c\,\Sigma_T^{-1}X^\top\eta.$$

Joint Gaussianity. Let $V := (\theta, \eta) \in \mathbb{R}^{d+T}$. Let $(g, g') \sim \mathcal{N}(0, I_{d+T})$; its two blocks $g \sim \mathcal{N}(0, I_d)$ and $g' \sim \mathcal{N}(0, I_T)$ are independent (Lemma **??**), and $\tau\Sigma_T^{-1/2} g \sim \mathcal{N}(0, \tau^2\Sigma_T^{-1})$ (Lemmas **??** and **??**). The law of a pair of independent random vectors is the product of the marginals, so $V \overset{d}{=} D\,(g, g')$ with $D := \operatorname{diag}(\tau\Sigma_T^{-1/2}, I_T)$, and hence $V \sim \mathcal{N}(0, S)$ with $S := DD^\top = \operatorname{diag}(\tau^2\Sigma_T^{-1}, I_T)$ (Lemma **??**). Now $W = L_1 V$ and $y = L_2 V$ for the linear maps

$$L_1 := \begin{bmatrix} (1-c)I_d & -c\,\Sigma_T^{-1}X^\top \end{bmatrix} \in \mathbb{R}^{d\times(d+T)}, \qquad L_2 := \begin{bmatrix} X & I_T \end{bmatrix} \in \mathbb{R}^{T\times(d+T)},$$

so by Lemma **??** the pair (W, y) is centered jointly Gaussian with cross-covariance $L_1 S L_2^\top$.

Cross-covariance. Using the block structure of S,

$$L_1 S L_2^\top = (1-c)\,\tau^2\Sigma_T^{-1}X^\top \;-\; c\,\Sigma_T^{-1}X^\top I_T = \big((1-c)\tau^2 - c\big)\,\Sigma_T^{-1}X^\top = 0,$$

because $(1-c)\tau^2 = \tau^2/(1+\tau^2) = c$. (Equivalently, in the form of the outline: $\operatorname{Cov}(\theta, y) = \tau^2\Sigma_T^{-1}X^\top$, $\operatorname{Cov}(y, y) = \tau^2 X\Sigma_T^{-1}X^\top + I_T$, and $\Sigma_T^{-1}X^\top(\tau^2 X\Sigma_T^{-1}X^\top + I_T) = (1+\tau^2)\Sigma_T^{-1}X^\top$, so that $\operatorname{Cov}(W, y) = \tau^2\Sigma_T^{-1}X^\top - c(1+\tau^2)\Sigma_T^{-1}X^\top = 0$.)

Independence and integrability. Uncorrelated jointly Gaussian vectors are independent (Lemma **??**). A Gaussian vector has finite second moments (Lemma **??**), hence is integrable, and it is centered. □

Lemma 77 (Bayes identity for the value of the recommendation)**.** *$Maiti2026Power.integral_inner_bayesParamC$ and with $c = \tau^2/(1+\tau^2)$, the random variables $\langle\hat{x}, \theta\rangle$ and $\langle\hat{x}, \Sigma_T^{-1}X^\top y\rangle$ are integrable and*

$$\mathbb{E}\langle\hat{x}, \theta\rangle = c\,\mathbb{E}\langle\hat{x}, \Sigma_T^{-1}X^\top y\rangle.$$

Proof. lem:posterior$_decomposition, lem : indep_integral_zero, lem : gauss_norm_momentsLet$R $:= \sup_{x\in\mathcal{X}}\|x\| < \infty$ ($\mathcal{X}$ is compact). Integrability: $|\langle\hat{x}, \theta\rangle| \le R\|\theta\|$ and $|\langle\hat{x}, \Sigma_T^{-1}X^\top y\rangle| \le R\|\Sigma_T^{-1}X^\top y\|$, and Gaussian vectors are integrable (Lemma **??**). By Lemma **??**, $\theta = W + c\,\Sigma_T^{-1}X^\top y$, so

$$\langle\hat{x}, \theta\rangle = \langle\hat{x}, W\rangle + c\,\langle\hat{x}, \Sigma_T^{-1}X^\top y\rangle,$$

and it remains to prove $\mathbb{E}\langle\hat{x}, W\rangle = 0$. Let $\bar{\rho}(y) := \int_{\mathcal{X}} x\,\rho(y, dx) \in \mathbb{R}^d$; it is a measurable function of y with $\|\bar{\rho}(y)\| \le R$. Since, given (θ, η), the recommendation has law $\rho(y)$ and W is a function of (θ, η), Fubini for the composition-product gives $\mathbb{E}\langle\hat{x}, W\rangle = \mathbb{E}\langle\bar{\rho}(y), W\rangle$. Now W is integrable and centered, W is independent of y (Lemma **??**) and $\bar{\rho}$ is bounded measurable, so Lemma **??** (with $V = y$ and $f = \bar{\rho}$) gives $\mathbb{E}\langle\bar{\rho}(y), W\rangle = 0$. □

Lean remark. The seed of a randomized rule never appears: integrating the kernel ρ first reduces the claim to the deterministic bounded function $\bar\rho$ of y, and the only independence needed is $W \perp y$, which comes from Mathlib's `HasGaussianLaw.indepFun_of_covariance_eval` through Lemma **??**. No conditional expectation is used anywhere in this chapter.

4.2 Lower bound on the Bayesian expected regret

Lemma 78 (Bayesian lower bound on the expected simple regret)**.** *Maiti2026Power.le$_i$ntegral$_s$impleRegret$_b$ay* 0*, for every recommendation kernel* ρ*,* $r(\hat x, \theta)$ *is integrable and*

$$\mathbb{E}\big[r(\hat x, \theta)\big] \;\geq\; \frac{\tau(1-\tau)}{1+\tau^2}\, w(\mathcal{X}; \Sigma_T).$$

Proof. lem:bayes$_r$*egret$_i$dentity, lem : width$_m$atrix$_w$d, lem : gauss$_l$inear$_i$mage, lem : gauss$_l$aw$_o$f$_c$ovWrite*w$_T := w(\mathcal{X}; \Sigma_T)$ and $c := \tau^2/(1+\tau^2)$.

(i) *Decomposition.* $r(\hat x, \theta) = \max_{x\in\mathcal{X}}\langle x, \theta\rangle - \langle \hat x, \theta\rangle$. Both terms are integrable: the first by Lemma **??** (applied below) and the second by Lemma **??**. Hence $\mathbb{E}[r(\hat x, \theta)] = \mathbb{E}\max_x\langle x, \theta\rangle - \mathbb{E}\langle \hat x, \theta\rangle$.

(ii) *The prior term.* $G := \tau^{-1}\theta \sim \mathcal{N}(0, \Sigma_T^{-1})$ (Lemma **??**), and $\max_x\langle x, \theta\rangle = \tau \max_x\langle x, G\rangle$ since $\tau > 0$. By Lemma **??** (the width only depends on the law of G), $\mathbb{E}\max_x\langle x, \theta\rangle = \tau\, w_T$.

(iii) *The recommendation term.* By Lemma **??** and $\Sigma_T^{-1}X^\top y = \Sigma_T^{-1}X^\top X\theta + \Sigma_T^{-1}X^\top\eta = \theta + \Sigma_T^{-1}X^\top\eta$,

$$\mathbb{E}\langle \hat x, \theta\rangle = c\,\mathbb{E}\langle \hat x, \theta + \Sigma_T^{-1}X^\top\eta\rangle.$$

Pointwise, $\langle \hat x, v\rangle \leq \max_{x\in\mathcal{X}}\langle x, v\rangle$ for every $v \in \mathbb{R}^d$ because $\hat x \in \mathcal{X}$, and $\max_x\langle x, a+b\rangle \leq \max_x\langle x, a\rangle + \max_x\langle x, b\rangle$. Therefore

$$\mathbb{E}\langle \hat x, \theta\rangle \leq c\Big(\mathbb{E}\max_x\langle x, \theta\rangle + \mathbb{E}\max_x\langle x, \Sigma_T^{-1}X^\top\eta\rangle\Big).$$

(iv) *The noise term.* $\Sigma_T^{-1}X^\top\eta$ is a linear image of $\eta \sim \mathcal{N}(0, I_T)$, so $\Sigma_T^{-1}X^\top\eta \sim \mathcal{N}(0, \Sigma_T^{-1}X^\top X\Sigma_T^{-1}) = \mathcal{N}(0, \Sigma_T^{-1})$ (Lemmas **??** and **??**), and Lemma **??** gives $\mathbb{E}\max_x\langle x, \Sigma_T^{-1}X^\top\eta\rangle = w_T$.

(v) *Conclusion.* Combining,

$$\mathbb{E}[r(\hat x, \theta)] \geq \tau w_T - c(\tau w_T + w_T) = \Big(\tau - \frac{\tau^2(1+\tau)}{1+\tau^2}\Big) w_T = \frac{\tau + \tau^3 - \tau^2 - \tau^3}{1+\tau^2}\, w_T = \frac{\tau(1-\tau)}{1+\tau^2}\, w_T.$$

□

Lemma 79 (Optimized constant)**.** $Maiti2026Power.bayesC_identitylem : bayes_regret_lowerWith\tau :=$ $\sqrt{2}-1$*, for every recommendation kernel* ρ*,*

$$\mathbb{E}\big[r(\hat{x},\theta)\big] \geq \frac{\sqrt{2}-1}{2}\, w(\mathcal{X};\Sigma_T) \geq 0.207\, w(\mathcal{X};\Sigma_T).$$

Proof. lem:bayes$_regret_lowerFor\tau = \sqrt{2}-1$: $\tau^2 = 3-2\sqrt{2}$, so $1+\tau^2 = 4-2\sqrt{2} = 2(2-\sqrt{2})$, and $1-\tau = 2-\sqrt{2}$. Hence $\frac{\tau(1-\tau)}{1+\tau^2} = \frac{(\sqrt{2}-1)(2-\sqrt{2})}{2(2-\sqrt{2})} = \frac{\sqrt{2}-1}{2}$. Finally $1.414^2 = 1.999396 \leq 2$ gives $\sqrt{2} \geq 1.414$ and $(\sqrt{2}-1)/2 \geq 0.207$. Apply Lemma **??** (the width is nonnegative by Lemma **??**). (In Lean only the algebraic identity $\tau - c(\tau+1) = (\sqrt{2}-1)/2$ is isolated, `bayesC_identity`; the numerical bound 0.207 is used inline in the proof of Theorem **??**.) $\square$

Remark 80 (Choice of τ)*.* The function $\tau \mapsto \tau(1-\tau)/(1+\tau^2)$ has derivative $(1-2\tau-\tau^2)/(1+\tau^2)^2$, which vanishes on $(0,1)$ exactly at $\tau = \sqrt{2}-1$; this is the maximizer, as stated in the paper. Only the value at $\sqrt{2}-1$ is used.

4.3 Singular designs

Lemma 81 (Singular designs fail)**.** $Maiti2026Power.exists_singular_instances, Maiti2026Power.exists_singular$ $\mathcal{X}$ *be a fixed design of length* $T \geq 0$ *(Definition* **??***) whose design matrix* $\Sigma_T = \sum_{t<T} x_t x_t^\top$ *is singular; if* $T = 0$ *(so that* $\Sigma_0 = 0$*), assume moreover that* $0 \in \mathcal{X}$ *(more generally, it suffices that every direction* v *orthogonal to the design,* $\langle x_t, v\rangle = 0$ *for all* t*, is orthogonal to some point of* $\mathcal{X}$*; for* $T \geq 1$ *this holds with the point* x_0*). Let* $\varepsilon > 0$ *and let* ρ *be any recommendation kernel from* $\mathbb{R}^T$ *to* $\mathcal{X}$*. Then there are* $\theta_+, \theta_- \in \mathbb{R}^d$ *such that*

(i) the laws of $(y,\hat{x})$ *under* θ_+ *and under* θ_- *coincide:* $\mathbb{P}_{\theta_+} = \mathbb{P}_{\theta_-} = \mathcal{N}(0, I_T) \otimes \rho$*;*

(ii) $r(\hat{x},\theta_+) + r(\hat{x},\theta_-) = 4\varepsilon$ *for every* $\hat{x} \in \mathcal{X}$*.*

Consequently there is $\theta \in \{\theta_+,\theta_-\}$ *with* $\mathbb{P}_\theta\big(r(\hat{x},\theta) > \varepsilon\big) \geq 1/2$*; in particular the fixed-design algorithm is not* (ε,δ)*-PAC for any* $\delta < 1/2$*.*

Proof. lem:fixed$_design_law, def : arm_set Step 1: a direction invisible to the design but not to \mathcal{X}. We claim there$ $\in \mathbb{R}^d$ such that $\langle x_t, v\rangle = 0$ for all $t < T$ and $x \mapsto \langle x, v\rangle$ is not constant on $\mathcal{X}$. If $T \geq 1$: since Σ_T is singular there is $v \neq 0$ with $\Sigma_T v = 0$, and then $0 = v^\top \Sigma_T v = \sum_{t<T} \langle x_t, v\rangle^2$ forces $\langle x_t, v\rangle = 0$ for every t. If $\langle \cdot, v\rangle$ were constant on $\mathcal{X}$, its value would be $\langle x_0, v\rangle = 0$, so $v \perp \mathcal{X}$, hence $v \perp \operatorname{span}(\mathcal{X}) = \mathbb{R}^d$ (spanning) and $v = 0$, a contradiction. (This also shows that $\mathcal{X}$ has at least two points when $T \geq 1$.) In general: the singularity of Σ_T gives $v \neq 0$ with $\langle x_t, v\rangle = 0$ for all t (for $T = 0$ any $v \neq 0$ works, as $d \geq 1$), and by assumption $\langle y_0, v\rangle = 0$ for some $y_0 \in \mathcal{X}$ ($y_0 = x_0$ if $T \geq 1$, $y_0 = 0$ if $0 \in \mathcal{X}$); if $\langle \cdot, v\rangle$ were constant on $\mathcal{X}$ its value would be $\langle y_0, v\rangle = 0$, and we conclude as before.

Step 2: the two instances. Let $a := \max_{x\in\mathcal{X}} \langle x, v\rangle$ and $b := \min_{x\in\mathcal{X}} \langle x, v\rangle$ (compactness); by Step 1, $a > b$. Put $\alpha := 4\varepsilon/(a-b) > 0$ and $\theta_\pm := \pm\alpha v$.

Since $\langle x_t, v\rangle = 0$ for all t, $X\theta_+ = X\theta_- = 0$, so by Lemma **??** the law of $(y, \hat{x})$ is $\mathcal{N}(0, I_T) \otimes \rho$ under both instances. This is (i).

Step 3: the regrets. For every $\hat{x} \in \mathcal{X}$,

$$r(\hat{x}, \theta_+) = \max_x \alpha\langle x, v\rangle - \alpha\langle \hat{x}, v\rangle = \alpha\big(a - \langle \hat{x}, v\rangle\big), \qquad r(\hat{x}, \theta_-) = \max_x \big(-\alpha\langle x, v\rangle\big) + \alpha\langle \hat{x}, v\rangle = \alpha\big(\langle \hat{x}, v\rangle - b\big),$$

so $r(\hat{x}, \theta_+) + r(\hat{x}, \theta_-) = \alpha(a - b) = 4\varepsilon$. This is (ii).

Step 4: conclusion. By (ii), for every $\hat{x} \in \mathcal{X}$ at least one of $r(\hat{x}, \theta_+)$, $r(\hat{x}, \theta_-)$ is $\geq 2\varepsilon > \varepsilon$. With $\mu := \mathcal{N}(0, I_T) \otimes \rho$ and $E_\pm := \{(y, x) \in \mathbb{R}^T \times \mathcal{X} : r(x, \theta_\pm) > \varepsilon\}$ (measurable, since $x \mapsto r(x, \theta)$ is continuous), $E_+ \cup E_- = \mathbb{R}^T \times \mathcal{X}$, hence $\mu(E_+) + \mu(E_-) \geq 1$ and one of them is at least $1/2$. By (i), $\mathbb{P}_{\theta_\pm}(r(\hat{x}, \theta_\pm) > \varepsilon) = \mu(E_\pm)$. □

Remark 82 (The case $T = 0$). Budget $T = 0$ is a legitimate fixed design (Definitions **??** and **??**): the design is empty, $\Sigma_0 = 0$ is singular and ρ is a probability measure on $\mathcal{X}$. For $T = 0$ the additional hypothesis ($0 \in \mathcal{X}$, or a direction condition) cannot be dropped: if $\mathcal{X} = \{x_0\}$ with $x_0 \neq 0$ (a spanning action set when $d = 1$), then every recommendation equals x_0 and the regret is identically 0, so the length-0 design is (ε, δ)-PAC. For $T \geq 1$ no such assumption is needed. This is harmless for Theorem **??**, which assumes $T \geq 1$; its conclusion also holds for $T = 0$, since either $\mathcal{X}$ has two points and no length-0 design is (ε, δ)-PAC with $\delta \leq 1/10$, or $\mathcal{X} = \{x_0\}$ and $w(\{x_0\}) = 0$ (as $\mathbb{E}\langle x_0, A^{-1/2} g\rangle = 0$).

Lemma 83 (A PAC design is nonsingular)**.** *Maiti2026Power.posDef$_o f_i$sFixedDesign$_o f_i$sPAC$_o f_f$orall, Maiti2 0 and $\delta < 1/2$. If a fixed-design algorithm with design $x_0, \ldots, x_{T-1} \in \mathcal{X}$ is (ε, δ)-PAC on $\mathcal{X}$, and either $T \geq 1$ or $0 \in \mathcal{X}$ (more generally, every direction orthogonal to the design is orthogonal to some point of $\mathcal{X}$), then $\Sigma_T \succ 0$.*

Proof. lem:singular$_d$*esign$_f$ails, lem : fixed$_d$esign$_l$awOtherwiseLemma* **??***gives*$\theta_\pm$ under which $(y, \hat{x})$ has the same law and $r(\hat{x}, \theta_+) + r(\hat{x}, \theta_-) = 4\varepsilon$ pointwise, so that $\mathbb{P}_{\theta_+}(r \leq \varepsilon) + \mathbb{P}_{\theta_-}(r \leq \varepsilon) \leq 1$, while the PAC property gives $2(1-\delta) > 1$. □

Lemma 84 (The Bayesian step)**.** *Maiti2026Power.gwMat$_l$e$_o f_i$sFixedDesign$_o f_i$sPACdef : fixed$_d$esign, def : 0 and $\delta \in (0, 1/10]$. If a fixed-design algorithm with design $x_0, \ldots, x_{T-1} \in \mathcal{X}$ and $\Sigma_T \succ 0$ is (ε, δ)-PAC on $\mathcal{X}$, then*

$$0.12\, w(\mathcal{X}; \Sigma_T) \leq \varepsilon.$$

Proof. def:bayes$_p$*rior, lem : fixed$_d$esign$_l$aw, lem : pac$_e$xpected$_r$egret, lem : width$_s$ymm, lem : width$_m$atrix$_w$d, lem : gauss$_l$inear$_i$mage, lem : bayes$_r$egret$_l$ower$_o$ptWrite*w$_T$:= $w(\mathcal{X}; \Sigma_T) \geq 0$.

Step 1: upper bound on the Bayesian regret. Let $\tau := \sqrt{2} - 1$ and consider the Bayesian model of Definition **??** with prior $\pi = \mathcal{N}(0, \tau^2 \Sigma_T^{-1})$. Its joint law is $\pi \otimes \kappa$ with the Markov kernel $\kappa(\theta) = \mathcal{N}(X\theta, I_T) \otimes \rho$ (a Gaussian kernel whose mean depends linearly on θ, composed with ρ), and $\kappa(\theta) = \mathbb{P}_\theta$ for every θ by Lemma **??**, with $(H_T, \hat{x})$ identified with $(y, \hat{x})$ as in Definition **??**. Hence Lemma **??** applies with this κ, provided $\int Z\, d\pi < \infty$. With $G := \tau^{-1}\theta \sim$

$\mathcal{N}(0, \Sigma_T^{-1})$ (Lemma **??**) and $Z(\theta) = \tau Z(G)$, Lemma **??** gives $\int Z\, d\pi = \mathbb{E}Z(\theta) = \tau\, \mathbb{E}Z(G) = 2\tau w_T < \infty$. Hence

$$\mathbb{E}\big[r(\hat{x}, \theta)\big] \leq \varepsilon + \delta \cdot 2\tau\, w_T.$$

Step 2: lower bound on the Bayesian regret. By Lemma **??**, $\mathbb{E}[r(\hat{x}, \theta)] \geq 0.207\, w_T$.

Step 3: arithmetic. Combining, $(0.207 - 2\tau\delta)\, w_T \leq \varepsilon$. Since $1.4143^2 = 2.00024 \geq 2$ we have $\tau = \sqrt{2} - 1 \leq 0.4143$, so with $\delta \leq 1/10$, $2\tau\delta \leq 0.08286 \leq 0.083$, and $0.207 - 0.083 = 0.124 \geq 0.12$. Therefore $0.12\, w_T \leq \varepsilon$. ☐

4.4 The lower bound

Theorem 85 (Lower bound for non-adaptive fixed designs)**.** *Maiti2026Power.le$_b$udget$_o$f$_i$sFixedDesign$_o$f$_i$sPA* 0 *and* $\delta \in (0, 1/10]$. *If a fixed-design algorithm with budget* $T \geq 1$ *(design* $x_0, \ldots, x_{T-1} \in \mathcal{X}$ *and any recommendation kernel* ρ*) is* (ε, δ)*-PAC on* $\mathcal{X}$*, then*

$$T \;\geq\; \frac{w(\mathcal{X})^2}{70\, \varepsilon^2}.$$

Proof. lem:lower$_n$*onadaptive$_p$osdef, lem : lower$_n$onadaptive$_b$ayes$_s$tep, lem : width$_e$mpiricalStep 1:* Σ_T *is pos* ≥ 1, $\mathcal{X}$ spanning and $\delta \leq 1/10 < 1/2$). Write $w_T := w(\mathcal{X}; \Sigma_T) \geq 0$.

Step 2: the Bayesian step. Lemma **??** gives $0.12\, w_T \leq \varepsilon$.

Step 3: from w_T to $w(\mathcal{X})$. By Lemma **??**, $w_T \geq w(\mathcal{X})/\sqrt{T}$, so $0.12\, w(\mathcal{X}) \leq \varepsilon\sqrt{T}$; squaring (both sides are nonnegative), $0.0144\, w(\mathcal{X})^2 \leq \varepsilon^2 T$. Since $0.0144 \cdot 70 = 1.008 \geq 1$, this gives $T \geq w(\mathcal{X})^2/(70\, \varepsilon^2)$. ☐

Corollary 86 (The paper's form of the lower bound)**.** *thm:lower$_n$onadaptiveLet$\varepsilon >$ 0 and* $0 \leq H_1 \leq H_2$. *Suppose that for every* $\delta \in (0, 1)$ *there is a fixed-design algorithm that is* (ε, δ)*-PAC on* $\mathcal{X}$ *with budget* $T_\delta \leq (H_1 \log(1/\delta) + H_2)/\varepsilon^2$. *Then*

$$H_2 \;\geq\; \frac{w(\mathcal{X})^2}{70\,(\log 10 + 1)}.$$

Proof. thm:lower$_n$*onadaptiveOnly*$\delta = 1/10$ is used. Theorem **??** gives $T_{1/10} \geq w(\mathcal{X})^2/(70\varepsilon^2)$, while $T_{1/10} \leq (H_1 \log 10 + H_2)/\varepsilon^2 \leq H_2(\log 10 + 1)/\varepsilon^2$ because $H_1 \leq H_2$. Hence $H_2(\log 10 + 1) \geq w(\mathcal{X})^2/70$. ☐

Remark 87 (Comparison with the paper)*.* The paper obtains $H_2 \geq 0.0144\, w(\mathcal{X})^2/(\log 10 + 1)$; we lose slightly by rounding 0.0144 down to 1/70. The statement of Theorem **??** is for a single algorithm and a single δ, which is the form used in Chapter **??**. The lower bound $\Omega(d \log(1/\delta)/\varepsilon^2)$ that the paper states alongside Theorem 3 is the adaptive lower bound of Chapter **??**, which applies in particular to fixed designs.

Chapter 5

Lower bound for adaptive algorithms

This chapter proves Theorem 2 of the paper (Appendix B.2, Theorem *main* and Lemma *baseline* there): for $d \geq 2$ and $\delta < 1/16$, every (ε, δ)-PAC identification algorithm on a spanning action set $\mathcal{X}$, adaptive or not, with a randomized recommendation or not, uses a budget $n \geq c_{\mathrm{ad}}\, d \log(1/\delta)/\varepsilon^2$ with the explicit constant $c_{\mathrm{ad}} = 1/(11556 + 5760\sqrt{2}) \geq 1/20000$ (Theorem **??**). The proof has three steps. First, a linear change of coordinates (Lemma **??**) reduces to a *normalized instance* $\mathcal{X}' \subseteq \mathbb{B}_d$ containing unit vectors $v_1, \ldots, v_m$ with $\sum_j \lambda_j v_j v_j^\top = I_d/d$ (Definition **??**). Second, a two-point argument gives the baseline $n \geq c_0 \log(1/(4\delta))/\varepsilon^2$ (Lemma **??**). Third, a PAC algorithm is turned into a test between $\theta = 0$ and the mixture of the $\theta^{(j)} = 3\varepsilon v_j$ (Lemma **??**), and the mixture's KL to $\mathbb{P}_0$ is at most $9N\varepsilon^2/(2d)$ (Lemma **??**), which forces $N \geq (2d/(9\varepsilon^2)) \log(1/(8\delta))$ (Lemma **??**).

From the prerequisite chapters we use the divergence decomposition for algorithm-environment sequences (Theorem **??**, Corollary **??**), the fact that a common recommendation kernel does not change the KL (Lemma **??**), the Bretagnolle–Huber inequality (Lemma **??**) and the convexity of the KL in its second argument for finite mixtures (Lemma **??**), all from Chapter **??** (Mathlib `ProbabilityTheory.klDiv`; see [**?**, Chapters 14 and 15]), together with Gaussian mean concentration (Lemma **??**, in the form of Lemma **??** for the formalization) and the sequential composition of algorithms (Lemma **??**; the formalization only needs the special case of Lemma **??**).

What is new with respect to the paper. The paper normalizes the instance with the Löwner ellipsoid of $\mathrm{conv}(\mathcal{X} \cup (-\mathcal{X}))$ and John's theorem [**?**]; we use instead the Kiefer–Wolfowitz theorem on G-optimal designs (Theorem **??**, [**?**, **?**]), which is needed anyway for the upper bound and gives exactly the structure the argument uses (Remark **??**). All constants are explicit, the recommendation is a Markov kernel, and the test built from the algorithm is an honest sequential composition in the sense of Lemma **??**. Throughout, $d \geq 2$; the spanning hypothesis on $\mathcal{X}$ enters only through Theorem **??** and is stated where it is needed.

5.1 Linear transformations and the normalized instance

Lemma 88 (Linear transformation of an instance)**.** *def:arm$_s$et, $def : env, def : bai_a lg, def : regret, def : pacLetM \in \mathbb{R}^{d\times d}$ be invertible and $\mathcal{X}' := M\mathcal{X} = \{Mx : x \in \mathcal{X}\}$. Then $\mathcal{X}'$ is an action set (nonempty and compact), and it is spanning if $\mathcal{X}$ is. For every identification algorithm* $\mathsf{Alg} = (\mathrm{alg}, \rho)$ *on $\mathcal{X}$ with budget n there is an identification algorithm* $\mathsf{Alg}' = (\mathrm{alg}', \rho')$ *on $\mathcal{X}'$ with budget n, the* transformed algorithm*, with the following property. For $\theta \in \mathbb{R}^d$ put $\theta' := M^{-\top}\theta$ and let $\Phi\big((x_t, y_t)_{t<n}, \hat{x}\big) := \big((Mx_t, y_t)_{t<n}, M\hat{x}\big)$. Then the law $\mathbb{P}'_{\theta'}$ of $(H'_n, \hat{x}')$ for* Alg' *against the environment $\nu'_{\theta'}$ on $\mathcal{X}'$ is the image of $\mathbb{P}_\theta$ by Φ:*

$$\mathbb{P}'_{\theta'} = \Phi_\# \mathbb{P}_\theta.$$

Moreover $\langle Mx, \theta' \rangle = \langle x, \theta \rangle$ for all $x \in \mathbb{R}^d$, hence the simple regrets satisfy $r'(M\hat{x}, \theta') = r(\hat{x}, \theta)$ (with r' the simple regret on $\mathcal{X}'$), and Alg' *is (ε, δ)-PAC on $\mathcal{X}'$ if and only if* Alg *is (ε, δ)-PAC on $\mathcal{X}$.*

Proof. def:env, def:bai$_a lg, def : pac, lem : noise_r epresentation\mathcal{X}'$ *is an action set.*X'*isthenonemptyimageofac*
$\mathbb{R}^d$ then $\mathrm{span}(M\mathcal{X}) = M\ \mathrm{span}(\mathcal{X}) = M\mathbb{R}^d = \mathbb{R}^d$ since M is invertible.

Values. $\langle Mx, M^{-\top}\theta \rangle = x^\top M^\top M^{-\top}\theta = \langle x, \theta \rangle$. Hence $\max_{x' \in \mathcal{X}'} \langle x', \theta' \rangle = \max_{x \in \mathcal{X}} \langle Mx, \theta' \rangle = \max_{x \in \mathcal{X}} \langle x, \theta \rangle$ and $\langle M\hat{x}, \theta' \rangle = \langle \hat{x}, \theta \rangle$, so $r'(M\hat{x}, \theta') = r(\hat{x}, \theta)$.

The transformed algorithm. Let $\phi := (x \mapsto Mx)$, a homeomorphism from $\mathcal{X}$ onto $\mathcal{X}'$, and let Φ_t be the induced bijection of histories of length t, $\Phi_t((x_s, y_s)_{s<t}) := ((Mx_s, y_s)_{s<t})$. Define alg' by pushing forward the initial action distribution of alg by ϕ, and, at time $t \geq 1$, given a history h' of length t on $\mathcal{X}'$, by playing the image by ϕ of the action drawn by the policy of alg given $\Phi_t^{-1}(h')$. Define $\rho'(h') := \phi_\#\, \rho(\Phi_n^{-1}(h'))$. These are Markov kernels since ϕ and Φ_t^{-1} are measurable.

Identification of the law. Let $(H_n, \hat{x})$ have law $\mathbb{P}_\theta$, i.e. be an algorithm-environment sequence for $(\mathrm{alg}, \nu_\theta)$ followed by $\hat{x} \sim \rho(H_n)$. We check that $\Phi(H_n, \hat{x})$ is an algorithm-environment sequence for $(\mathrm{alg}', \nu'_{\theta'})$ followed by ρ'. Conditionally on the past $\Phi_t(H_t)$, the action Mx_t has the image by ϕ of the conditional law of x_t given H_t, which is the policy of alg at $\Phi_t^{-1}\Phi_t(H_t) = H_t$, i.e. the policy of alg' at $\Phi_t(H_t)$. Conditionally on the past and on Mx_t, the observation y_t has law $\mathcal{N}(\langle x_t, \theta \rangle, 1) = \mathcal{N}(\langle Mx_t, \theta' \rangle, 1)$, which is the reward kernel of $\nu'_{\theta'}$ evaluated at Mx_t (Definition **??**). Finally, conditionally on $\Phi_n(H_n)$, $M\hat{x}$ has law $\phi_\#\rho(H_n) = \rho'(\Phi_n(H_n))$. By uniqueness of the law of an algorithm-environment sequence (`Learning.isAlgEnvSeq_unique`, Definition **??**), $\Phi_\#\mathbb{P}_\theta = \mathbb{P}'_{\theta'}$.

PAC equivalence. For every θ, $\mathbb{P}'_{\theta'}(r'(\hat{x}', \theta') \leq \varepsilon) = \mathbb{P}_\theta(r'(M\hat{x}, \theta') \leq \varepsilon) = \mathbb{P}_\theta(r(\hat{x}, \theta) \leq \varepsilon)$. Since $\theta \mapsto M^{-\top}\theta$ is a bijection of $\mathbb{R}^d$, the PAC condition for all θ' on $\mathcal{X}'$ is equivalent to the PAC condition for all θ on $\mathcal{X}$. ☐

Lean remark. The formalization does *not* transform the algorithm: it works in the original coordinates and only uses the map M inside the analysis. Indeed every quantity of Sections **??** and **??** is of the form $\langle Mx, u\rangle$ for an action $x \in \mathcal{X}$ and a fixed vector u, and $\langle Mx, u\rangle = \langle x, Mu\rangle$ since M is symmetric (`Maiti2026Power.inner_toEuclideanCLM_normalizeMat`); the reward vectors of the normalized instance are therefore replaced by their images Mu on the original instance (e.g. $\theta_\pm = \pm\frac{4\varepsilon}{\varrho}Mu$ in Definition **??**). This lemma is thus not needed for Theorem **??** and is not formalized; the transport of a run along a measurable map of the *sample space* (as opposed to the action type) is `Learning.IdentAlg.IsRun.comp_hasLaw`.

Definition 89 (Normalized instance)**.** Maiti2026Power.normalizeMat, Maiti2026Power.IsGOptimalDesign.norr
$\triangle_{\mathcal{X}}$ be a G-optimal design with $A_G := A(\lambda_G) \succ 0$ (Theorem **??**), with support $\{x_1, \dots, x_m\}$ ($m \ge 1$ distinct points of $\mathcal{X}$) and weights $\lambda_j := \lambda_{G,x_j} > 0$, $\sum_{j=1}^m \lambda_j = 1$. Set

$$M := d^{-1/2} A_G^{-1/2}, \qquad \mathcal{X}' := M\mathcal{X}, \qquad v_j := Mx_j \quad (1 \le j \le m).$$

By Corollaries **??** and **??**: $\mathcal{X}' \subseteq \mathbb{B}_d$ is a spanning action set (Lemma **??**), $v_j \in \mathcal{X}'$, $\|v_j\| = 1$ for every j, and

$$\sum_{j=1}^{m} \lambda_j\, v_j v_j^\top = \frac{1}{d} I_d.$$

In particular $m \ge d \ge 2$ (the left-hand side has rank at most m). We let J be a random index in $\{1, \dots, m\}$ with $\mathbb{P}(J = j) = \lambda_j$, so that $\mathbb{E}[v_J v_J^\top] = I_d/d$. The statements of Sections **??** and **??** are about algorithms on the normalized instance $\mathcal{X}'$; there we write $\mathbb{P}_\theta$, $\mathbb{E}_\theta$ for the law of $(H, \hat{x})$ of the algorithm under consideration against the environment ν'_θ of $\mathcal{X}'$ (Definition **??**), and r for the simple regret on $\mathcal{X}'$.

Remark 90 (Kiefer–Wolfowitz instead of John)*.* The paper obtains points $u_j \in \mathcal{X} \cap \mathbb{S}^{d-1}$ (after normalization by the Löwner ellipsoid of $\mathrm{conv}(\mathcal{X} \cup (-\mathcal{X}))$) and weights $c_j > 0$ with $\sum_j c_j u_j u_j^\top = I_d$, $\sum_j c_j = d$, by John's theorem applied to the polar body. Definition **??** provides the same data with $c_j = d\lambda_j$: after the linear map $M = d^{-1/2} A_G^{-1/2}$ the support points of the G-optimal design are unit vectors whose weighted second-moment matrix is I_d/d, and $\mathcal{X}' \subseteq \mathbb{B}_d$ because $\|A_G^{-1/2} x\|^2 \le d$ for all $x \in \mathcal{X}$. Only these three facts are used below.

5.2 The baseline lower bound

Lemma 91 (A well-separated pair of unit arms)**.** *Maiti2026Power.IsGOptimalDesign.sum*$_m$*ul*$_i$*nner*$_n$*ormalizeM*
$\sum_{j=1}^m \lambda_j \langle v_1, v_j\rangle^2 = \frac{1}{d}$. Consequently there is $k \in \{1, \dots, m\}$ with $\langle v_1, v_k\rangle \le 1/\sqrt{d} \le 1/\sqrt{2}$. Moreover, every k with $\langle v_1, v_k\rangle \le 1/\sqrt{2}$ satisfies $k \ne 1$ and

$$\|v_1 - v_k\|^2 = 2 - 2\langle v_1, v_k\rangle \ge 2 - \sqrt{2} > 0.$$

Proof. def:normalized$_instance\sum_j \lambda_j \langle v_1, v_j\rangle^2 = v_1^\top\big(\sum_j \lambda_j v_j v_j^\top\big)v_1 = v_1^\top (I_d/d)v_1 =$ $\|v_1\|^2/d = 1/d$. A weighted average with weights $\lambda_j > 0$, $\sum_j \lambda_j = 1$, is at least its smallest term, so some k has $\langle v_1, v_k\rangle^2 \le 1/d$, hence $\langle v_1, v_k\rangle \le |\langle v_1, v_k\rangle| \le 1/\sqrt{d}$, and $1/\sqrt{d} \le 1/\sqrt{2}$ because $d \ge 2$. Now let k be any index with $\langle v_1, v_k\rangle \le 1/\sqrt{2}$. Since $\langle v_1, v_1\rangle = 1 > 1/\sqrt{2}$, $k \ne 1$. Finally $\|v_1 - v_k\|^2 = \|v_1\|^2 + \|v_k\|^2 - 2\langle v_1, v_k\rangle = 2 - 2\langle v_1, v_k\rangle \ge 2 - \sqrt{2}$. ☐

Definition 92 (The two-point instance)**.** def:normalized$_instance, lem : separated_pairInthesettingofDefiniti$
0 and let $k \in \{1, \dots, m\}$ be an index with $\langle v_1, v_k\rangle \le 1/\sqrt{2}$, which exists by Lemma **??**; by the same lemma $k \ne 1$, and the distance $\varrho := \|v_1 - v_k\|$ satisfies $\varrho^2 \ge 2 - \sqrt{2} > 0$. Define

$$u := \frac{v_1 - v_k}{\varrho} \in \mathbb{S}^{d-1}, \qquad s := \frac{\langle v_1, u\rangle + \langle v_k, u\rangle}{2}, \qquad \theta_+ := \frac{4\varepsilon}{\varrho}\, u, \qquad \theta_- := -\frac{4\varepsilon}{\varrho}\, u.$$

(The letter ϱ is used for this distance to avoid a clash with the recommendation kernel ρ.) Since $\langle v_1 - v_k, u\rangle = \|v_1 - v_k\|^2/\varrho = \varrho$, we have $\langle v_1, u\rangle = s + \varrho/2$ and $\langle v_k, u\rangle = s - \varrho/2$.

Lemma 93 (Two-point instances and the induced test)**.** *def:lower$_adaptive_two_point_instance, def : regret, def : pacLet\varepsilon > 0$ and let ϱ, u, s, $\theta_\pm$ be as in Definition* **??***. For every $\hat{x} \in \mathcal{X}'$:*

(i) if $r(\hat{x}, \theta_+) \le \varepsilon$ then $\langle \hat{x}, u\rangle \ge s + \varrho/4$;

(ii) if $r(\hat{x}, \theta_-) \le \varepsilon$ then $\langle \hat{x}, u\rangle \le s - \varrho/4$.

Consequently, if an identification algorithm on $\mathcal{X}'$ with budget n is (ε, δ)-PAC, then the $\{0,1\}$-valued measurable function $\hat{H} := \mathbf{1}\{\langle \hat{x}, u\rangle \ge s\}$ of the recommendation satisfies

$$\mathbb{P}_{\theta_+}(\hat{H} = 0) \le \delta, \qquad \mathbb{P}_{\theta_-}(\hat{H} = 1) \le \delta.$$

Proof. def:lower$_adaptive_two_point_instance, def : regret, def : pacRecallfromDefinition$ **??**$that\langle v_1, u\rangle = s + \varrho/2$ and $\langle v_k, u\rangle = s - \varrho/2$.

(i) $r(\hat{x}, \theta_+) \le \varepsilon$ means $\langle \hat{x}, \theta_+\rangle \ge \max_{x \in \mathcal{X}'} \langle x, \theta_+\rangle - \varepsilon \ge \langle v_1, \theta_+\rangle - \varepsilon$ (as $v_1 \in \mathcal{X}'$), i.e. $\frac{4\varepsilon}{\varrho}\langle \hat{x}, u\rangle \ge \frac{4\varepsilon}{\varrho}\big(s + \frac{\varrho}{2}\big) - \varepsilon$. Multiplying by $\varrho/(4\varepsilon) > 0$: $\langle \hat{x}, u\rangle \ge s + \varrho/2 - \varrho/4 = s + \varrho/4$.

(ii) Similarly $\langle \hat{x}, \theta_-\rangle \ge \langle v_k, \theta_-\rangle - \varepsilon$ reads $-\frac{4\varepsilon}{\varrho}\langle \hat{x}, u\rangle \ge -\frac{4\varepsilon}{\varrho}\big(s - \frac{\varrho}{2}\big) - \varepsilon$, i.e. $\langle \hat{x}, u\rangle \le s - \varrho/2 + \varrho/4 = s - \varrho/4$.

For the last claim: by (i), $\{\hat{H} = 0\} = \{\langle \hat{x}, u\rangle < s\} \subseteq \{r(\hat{x}, \theta_+) > \varepsilon\}$, whose $\mathbb{P}_{\theta_+}$-probability is at most δ by the PAC property; by (ii), $\{\hat{H} = 1\} = \{\langle \hat{x}, u\rangle \ge s\} \subseteq \{r(\hat{x}, \theta_-) > \varepsilon\}$, of $\mathbb{P}_{\theta_-}$-probability at most δ. ☐

Lean remark. $\hat{H}$ is the composition of the recommendation kernel ρ with the measurable map $x \mapsto \mathbf{1}\{\langle x, u\rangle \ge s\}$ from $\mathcal{X}'$ to `Bool`; the pair (sampling rule, this composed kernel) is a *test* in the sense of Definition **??** below. The events

$\{\hat H = 0\}$, $\{\hat H = 1\}$ are events of $(H_n, \hat x)$, as required by Lemma ??. In the formalization the two implications (i), (ii) and the error bounds are established inside the proof of Lemma ?? (`Maiti2026Power.baseline_le_budget_of_isPAC`), in the original coordinates (the set $B := \{x : \langle Mx, u\rangle \geq s\}$ of recommendations), and the error probabilities are those of the canonical runs of Definition ?? (`IdentAlg.IsFixedBudget.isRun_fixedBudgetRunMeasure`).

Lemma 94 (KL divergence between the two-point instances)**.** *Learning.LinearBandit.IsRun.klDiv*$_m ap_o ut_l e_o f_s q$ *0 and let* ϱ, $\theta_\pm$ *be as in Definition ??. For every identification algorithm on the normalized instance* $\mathcal{X}' \subseteq \mathbb{B}_d$ *(Definition ??) with budget* n *(any sampling rule, any recommendation kernel),*

$$\mathrm{KL}\big(\mathbb{P}_{\theta_+} \,\big\|\, \mathbb{P}_{\theta_-}\big) \leq \frac{32\, n\, \varepsilon^2}{\varrho^2}.$$

Proof. lem:kl$_d ata_p rocessing_r ecommendation, cor: divergence_d ecomposition_g aussian, def: normalized_i nstanceByLemma$??$, theKLbetweenthelawsof$ $(\mathrm{H}_n, \hat x)$ equals the KL between the laws of H_n (the recommendation kernel is the same under both instances). By Corollary ?? with $\theta_+ - \theta_- = \frac{8\varepsilon}{\varrho} u$,

$$\mathrm{KL}\big(\mathbb{P}_{\theta_+} \big\| \mathbb{P}_{\theta_-}\big) = \frac{1}{2} \sum_{t<n} \mathbb{E}_{\theta_+}\big[\langle x_t, \theta_+ - \theta_-\rangle^2\big] = \frac{32\varepsilon^2}{\varrho^2} \sum_{t<n} \mathbb{E}_{\theta_+}\big[\langle x_t, u\rangle^2\big] \leq \frac{32\, n\, \varepsilon^2}{\varrho^2},$$

using $|\langle x_t, u\rangle| \leq \|x_t\| \|u\| \leq 1$ since $x_t \in \mathcal{X}' \subseteq \mathbb{B}_d$ (Definition ??) and $\|u\| = 1$. ☐

Lemma 95 (Baseline lower bound)**.** *Maiti2026Power.baseline*$_l e_b udget_o f_i sPACdef: normalized_i nstance, def$ *0 and* $\delta \in (0, 1/4)$*. Every* (ε, δ)*-PAC identification algorithm on* $\mathcal{X}'$ *(equivalently, by the Lean remark after Lemma ??, on the spanning action set* $\mathcal{X}$ *itself,* $d \geq 2$*) with budget* n *satisfies*

$$n \;\geq\; \frac{2 - \sqrt{2}}{32\, \varepsilon^2} \log \frac{1}{4\delta}.$$

In particular this holds for $\delta < 1/16$*, the range of Theorem ??.*

Proof. def:lower$_a daptive_t wo_p oint_i nstance, lem: two_p oint_t est, lem: kl_t wo_i nstances, lem: bretagnolle_h uber, lem: separated_p airFix$k,$\varrho$, u, s and $\theta_\pm$ as in Definition ?? (the index k exists by Lemma ??). Let $\hat H$ be the test of Lemma ?? and $A := \{\hat H = 0\}$, an event of $(H_n, \hat x)$. The Bretagnolle–Huber inequality (Lemma ??) applied to $\mu = \mathbb{P}_{\theta_+}$, $\nu = \mathbb{P}_{\theta_-}$ gives

$$\mathbb{P}_{\theta_+}(\hat H = 0) + \mathbb{P}_{\theta_-}(\hat H = 1) \geq \frac{1}{2} \exp\big(-\mathrm{KL}(\mathbb{P}_{\theta_+} \| \mathbb{P}_{\theta_-})\big).$$

The left-hand side is at most 2δ by Lemma ??, so $\exp(-\mathrm{KL}) \leq 4\delta$, i.e. $\mathrm{KL}(\mathbb{P}_{\theta_+} \| \mathbb{P}_{\theta_-}) \geq \log \frac{1}{4\delta}$, which is positive since $\delta < 1/4$. With Lemma ??, $\frac{32 n \varepsilon^2}{\varrho^2} \geq \log \frac{1}{4\delta}$, i.e. $n \geq \frac{\varrho^2}{32\varepsilon^2} \log \frac{1}{4\delta}$, and $\varrho^2 \geq 2 - \sqrt{2}$ (Lemma ??) together with $\log \frac{1}{4\delta} > 0$ gives the claim. ☐

5.3 The mixture testing problem

Definition 96 (Tests and the mixture testing problem)**.** Maiti2026Power.mixtureParam, Maiti2026Power.designWeight, Maiti2026Power.histLaw, Maiti2026Power.mixtureHistLaw, Maiti2026Power.IsDesign.isProbabilityMeasure$_{m}ixtureHistLaw, Maiti2026Power.IsGOptimalDesign.inner$ 0 and, for $1 \leq j \leq m$, $\theta^{(j)} := 3\varepsilon\, v_j$. A *test with budget* $N \geq 1$ on $\mathcal{X}'$ is a pair $\mathsf{Test} = (\mathrm{alg}, \kappa)$ of an LML algorithm alg with actions in $\mathcal{X}'$ and observations in $\mathbb{R}$, and a Markov kernel κ from histories of length N to $\{0,1\}$; run against ν'_θ it produces H_N and the *decision* $\hat{H} \sim \kappa(H_N)$, and we write $\mathbb{P}^{\mathrm{test}}_\theta$ for the law of $(H_N, \hat{H})$ (a probability measure on the product of the history space with $\{0,1\}$). A test is thus an identification algorithm in the sense of Definition **??** whose recommendation type is $\{0,1\}$ instead of $\mathcal{X}'$, and $\mathbb{P}^{\mathrm{test}}_\theta$ is its law in the sense of that definition; when the test is obtained from an identification algorithm (alg, ρ) with budget N by composing ρ with a measurable map $f : \mathcal{X}' \to \{0,1\}$ (as in Lemma **??**), $\mathbb{P}^{\mathrm{test}}_\theta$ is the image of the law $\mathbb{P}_\theta$ of $(H_N, \hat{x})$ under $(h, x) \mapsto (h, f(x))$. The two hypotheses are

$$H_0 :\ \theta = 0, \qquad H_1 :\ \theta = \theta^{(J)} \text{ with } J \sim \lambda_G,$$

with the corresponding laws

$$\mathbb{P}_0 := \mathbb{P}^{\mathrm{test}}_0 \quad \text{and} \quad \mathbb{P}_1 := \sum_{j=1}^{m} \lambda_j\, \mathbb{P}^{\mathrm{test}}_{\theta^{(j)}},$$

the latter being the finite mixture (with the weights λ_j of Definition **??**) of the laws of $(H_N, \hat{H})$; it is a probability measure. We write $\mathbb{E}_0$ for the expectation under $\mathbb{P}_0$. The *errors* of the test are $\mathbb{P}_0(\hat{H} = 1)$ and $\mathbb{P}_1(\hat{H} = 0) = \sum_j \lambda_j \mathbb{P}^{\mathrm{test}}_{\theta^{(j)}}(\hat{H} = 0)$.

Lean remark. A test is an identification algorithm whose recommendation type is `Bool` instead of $\mathcal{X}'$; both are instances of "LML algorithm plus a recommendation kernel into a measurable type", and the KL statements of Chapter **??** (Lemma **??**) are stated for an arbitrary recommendation type. The mixture $\mathbb{P}_1$ is the finite weighted sum of measures $\sum_j \lambda_j \bullet \mathbb{P}^{\mathrm{test}}_{\theta^{(j)}}$ (`Finset.sum` of `Measure`s with `ENNReal` weights); it is not the law of an algorithm-environment sequence for a single θ, which is why the KL to it is controlled by convexity (Lemma **??**) rather than by the divergence decomposition directly. In the formalization the alternatives are written in the original coordinates (`Maiti2026Power.mixtureParam`: $\theta^{(j)} = 3\varepsilon M(Mx_j)$, so that $\langle x, \theta^{(j)} \rangle = 3\varepsilon \langle Mx, v_j \rangle$), the laws $\mathbb{P}^{\mathrm{test}}_\theta$ are the laws `Maiti2026Power.histLaw` of the history of length N under the canonical trajectory measure of the test's sampling rule (the decision is a measurable function of this history, so nothing is lost, cf. Lemma **??**), and the mixture is `Maiti2026Power.mixtureHistLaw`.

Lemma 97 (KL divergence to the mixture)**.** $Maiti2026Power.IsGOptimalDesign.sum_{m}ul_{i}nner_{m}ixtureParam_{s}$ $\sum_{j=1}^{m} \lambda_j \,\mathrm{KL}\big(\mathbb{P}_0 \big\| \mathbb{P}^{\mathrm{test}}_{\theta^{(j)}}\big) = \frac{9\varepsilon^2}{2} \sum_{t<N} \mathbb{E}_0\Big[x_t^\top \Big(\sum_{j=1}^{m} \lambda_j v_j v_j^\top\Big) x_t\Big] = \frac{9\varepsilon^2}{2d} \sum_{t<N} \mathbb{E}_0\big[\|x_t\|^2\big] \leq \frac{9N\varepsilon^2}{2d}$.

Proof. lem:kl$_{m}ixture_{c}onvex, lem : kl_{d}ata_{p}rocessing_{r}ecommendation, cor : divergence_{d}ecomposition_{g}aussian$
$normalized_{i}nstance The first inequality is the convexity of \nu \mapsto$ $\mathrm{KL}(\mu\|\nu)$ for the finite mixture $\mathbb{P}_1 = \sum_j \lambda_j \mathbb{P}^{\text{test}}_{\theta^{(j)}}$ (Lemma **??**). For each j, by Lemma **??** the KL between the laws of $(H_N, \hat{H})$ under 0 and $\theta^{(j)}$ equals the KL between the laws of H_N, and by Corollary **??**

$$\mathrm{KL}\big(\mathbb{P}_0 \big\| \mathbb{P}^{\text{test}}_{\theta^{(j)}}\big) = \frac{1}{2} \sum_{t<N} \mathbb{E}_0\big[\langle x_t, 0 - \theta^{(j)} \rangle^2\big] = \frac{9\varepsilon^2}{2} \sum_{t<N} \mathbb{E}_0\big[\langle x_t, v_j \rangle^2\big].$$

Note that the expectation is under $\mathbb{P}_0$ for every j. Multiplying by λ_j, summing over j and exchanging the finite sum with the expectation, $\sum_j \lambda_j \langle x_t, v_j \rangle^2 = x_t^\top \big(\sum_j \lambda_j v_j v_j^\top\big) x_t = \|x_t\|^2/d$ by Definition **??**, which gives the two equalities. Finally $\|x_t\| \le 1$ since $x_t \in \mathcal{X}' \subseteq \mathbb{B}_d$. ☐

Lemma 98 (Lower bound for the testing problem)**.** *Maiti2026Power.IsGOptimalDesign.exp$_{n}eg_{l}e_{o}f_{m}ixturedef$*
$(0, 1/8)$. *Every test with budget N on $\mathcal{X}'$ such that $\mathbb{P}_0(\hat{H} = 1) \le 2\delta$ and $\mathbb{P}_1(\hat{H} = 0) \le 2\delta$ satisfies*

$$N \ \ge\ \frac{2d}{9\,\varepsilon^2}\, \log \frac{1}{8\delta}.$$

Proof. lem:bretagnolle$_{h}uber, lem : mixture_{k}l Apply the Bretagnolle--Huber inequality (Lemma$ **??**$) to the proba$
$\mathbb{P}_0$, $\nu = \mathbb{P}_1$ and the event $A := \{\hat{H} = 1\}$:

$$4\delta \ge \mathbb{P}_0(\hat{H} = 1) + \mathbb{P}_1(\hat{H} = 0) \ge \frac{1}{2} \exp\big(-\mathrm{KL}(\mathbb{P}_0\|\mathbb{P}_1)\big),$$

so $\mathrm{KL}(\mathbb{P}_0\|\mathbb{P}_1) \ge \log \frac{1}{8\delta} > 0$ (as $\delta < 1/8$). By Lemma **??**, $\frac{9N\varepsilon^2}{2d} \ge \log \frac{1}{8\delta}$. ☐

Lean remark. The formalized statement is the intermediate inequality $\frac{1}{2}\exp(-9N\varepsilon^2/(2d)) \le \alpha + \beta$ for every measurable set E of histories with $\mathbb{P}_0(E) \le \alpha$ and $\mathbb{P}^{\text{test}}_{\theta^{(j)}}(E^c) \le \beta$ for all j (`Maiti2026Power.IsGOptimalDesign.exp_neg_le_of_mixture`); the passage to the bound on N is done in the proof of Theorem **??**.

Lemma 99 (A test built from a PAC algorithm)**.** *Learning.Algorithm.thenRepeat, Learning.IdentAlg.testAlg, Learning.LinearBandit.phaseMean, Learning.IsAlgEnvSeq.isAlgEnvSeqUntil$_{o}f_{t}henR$*
0, $\delta \in (0,1)$ *and let* $\mathsf{Alg} = (\mathrm{alg}, \rho)$ *be an* (ε, δ)*-PAC identification algorithm on* $\mathcal{X}'$ *with budget* n*. Let*

$$n_{\text{est}} := \left\lceil \frac{8 \log(2/\delta)}{\varepsilon^2} \right\rceil, \qquad N := n + n_{\text{est}}.$$

Define the test Test *with budget* N *on* $\mathcal{X}'$ *as follows: for* $t < n$ *play as* alg*; at time* n *draw* $x_n \sim \rho(H_n)$ *(the recommendation of* Alg*); for* $n < t < N$ *play* $x_t := x_n$*; let* $\hat{v} := \frac{1}{n_{\text{est}}} \sum_{s<n_{\text{est}}} y_{n+s}$ *and decide* $\hat{H} := \mathbf{1}\{\hat{v} > \varepsilon\}$*. Then*

$$\mathbb{P}_0(\hat{H} = 1) \le \delta, \qquad \mathbb{P}^{\text{test}}_{\theta^{(j)}}(\hat{H} = 0) \le 2\delta \ \textit{ for every } j,$$

and therefore $\mathbb{P}_1(\hat{H} = 0) \le 2\delta$ *as well.*

Proof. lem:composition, lem:prefix$_{t}ransfer, lem : noise_{r}epresentation, lem : noise_{a}zuma, lem : gauss_{m}ean_{c}oncentration, def : normalized_{i}nstance, def : pac, def : bai_{a}lgConstruction.$Test$isthesequentialcomposition(Lemma$ **??**$)of$ $\mathrm{alg}_1 :=$ alg with $T_1 := n$ and, for a history h of length n, the algorithm $\mathrm{alg}_2(h)$ whose initial action distribution is $\rho(h)$ and whose policy at every later time is deterministic: repeat the first action of the phase. The map $h \mapsto \mathrm{alg}_2(h)$ is measurable because ρ is a kernel. The decision $\hat{H}$ is a deterministic measurable function of H_N, so κ is the deterministic kernel, and $\mathbb{P}_\theta^{\mathrm{test}}$ is the law of $(H_N, \hat{H})$ under the composition run against ν'_θ. Fix $\theta \in \mathbb{R}^d$.

The second phase for a fixed first history. Fix a history h of length n and write $\mathbb{P}^h$ for the law of the algorithm-environment sequence $(x_{n+s}, y_{n+s})_{s<n_{\mathrm{est}}}$ for $(\mathrm{alg}_2(h), \nu'_\theta)$. Under $\mathbb{P}^h$, $x_n \sim \rho(h)$, $x_{n+s} = x_n$ for all s, and by Lemma **??** $y_{n+s} = \langle x_n, \theta\rangle + \eta_{n+s}$ with $(\eta_{n+s})_{s<n_{\mathrm{est}}}$ i.i.d. $\mathcal{N}(0,1)$ independent of x_n. Let

$$\bar{\eta} := \frac{1}{n_{\mathrm{est}}} \sum_{s<n_{\mathrm{est}}} \big(y_{n+s} - \langle x_{n+s}, \theta\rangle\big), \qquad \text{so that} \qquad \hat{v} = \langle x_n, \theta\rangle + \bar{\eta};$$

$\bar{\eta}$ is a measurable function of the second-phase trajectory alone, and Lemma **??** gives $\mathbb{P}^h(|\bar{\eta}| \geq \varepsilon/2) \leq 2\exp(-n_{\mathrm{est}}\varepsilon^2/8) \leq \delta$, because $n_{\mathrm{est}} \geq 8\log(2/\delta)/\varepsilon^2$.

Transfer to the composition. Let $E := \{|\bar{\eta}| < \varepsilon/2\}$, a measurable set of second-phase trajectories, and $E_h := E$ for every h. The family $(E_h)_h$ does not depend on h, so it is jointly measurable in $(h, \cdot)$, and $\mathbb{P}^h(E_h) \geq 1 - \delta$ for *every* h. The last assertion of Lemma **??**, applied with this family and $\delta' = 0$, gives $\mathbb{P}_\theta^{\mathrm{test}}(|\bar{\eta}| < \varepsilon/2) \geq 1 - \delta$. Moreover, by the composition-product structure of Lemma **??** (H_n has the law of alg against ν'_θ, and given $H_n = h$ the first action of the second phase has law $\rho(h)$), the joint law of (H_n, x_n) under the composition is (law of H_n) $\otimes\, \rho$, which is exactly the law $\mathbb{P}_\theta$ of $(H_n, \hat{x})$ of Alg against ν'_θ (Definition **??**); so every statement about $(H_n, \hat{x})$ under Alg holds for (H_n, x_n) under Test.

Under H_0. With $\theta = 0$, $\hat{v} = \bar{\eta}$, so $\{\hat{H} = 1\} = \{\bar{\eta} > \varepsilon\} \subseteq \{|\bar{\eta}| \geq \varepsilon/2\}$ and $\mathbb{P}_0(\hat{H} = 1) \leq \delta$. (The PAC property is not needed here.)

Under $\theta^{(j)}$. Since $v_j \in \mathcal{X}'$ and $\|v_j\| = 1$, $\max_{x\in\mathcal{X}'}\langle x, \theta^{(j)}\rangle \geq \langle v_j, 3\varepsilon v_j\rangle = 3\varepsilon$. By the PAC property of Alg transferred to (H_n, x_n), with $\mathbb{P}_{\theta^{(j)}}^{\mathrm{test}}$-probability at least $1-\delta$, $\langle x_n, \theta^{(j)}\rangle \geq \max_x\langle x, \theta^{(j)}\rangle - \varepsilon \geq 2\varepsilon$. On the intersection of this event with $\{|\bar{\eta}| < \varepsilon/2\}$, which has probability at least $1 - 2\delta$ by the union bound, $\hat{v} \geq 2\varepsilon - \varepsilon/2 = 3\varepsilon/2 > \varepsilon$, i.e. $\hat{H} = 1$. Hence $\mathbb{P}_{\theta^{(j)}}^{\mathrm{test}}(\hat{H} = 0) \leq 2\delta$.

Under H_1. $\mathbb{P}_1(\hat{H} = 0) = \sum_j \lambda_j \mathbb{P}_{\theta^{(j)}}^{\mathrm{test}}(\hat{H} = 0) \leq 2\delta \sum_j \lambda_j = 2\delta$. ☐

Lean remark. The formalization does not use the general composition lemma. The test's sampling rule is the single LML algorithm `Learning.Algorithm.thenRepeat` (`Learning.IdentAlg.testAlg` when ρ is the recommendation rule of Alg): its policy is that of alg at times $t < n$, the kernel ρ applied to the history of the first n rounds at time $t = n$, and the deterministic policy "play the action recorded at index n of the history" afterwards; the recommendation of Alg is thus the action x_n. The three facts used about a run of this algorithm are: it is an algorithm-environment sequence for alg until time $n -$

1 (`IsAlgEnvSeq.isAlgEnvSeqUntil_of_thenRepeat`), x_n has conditional law $\rho(H_n)$ given H_n (`hasCondDistrib_action_succ_of_thenRepeat`), and $x_{n+s} = x_n$ almost surely (`action_add_ae_eq_of_thenRepeat`). The PAC guarantee of Alg is transferred to x_n by Lemma **??** (`IsPAC.le_measureReal_action_succ_of_testAlg`). The concentration of $\bar{\eta}$ is Lemma **??** applied to the run of the test itself (so no conditioning on H_n is needed): $\hat{v} = \langle x_n, \theta\rangle + \bar{\eta}$ almost surely (`phaseMean_ae_eq_of_thenRepeat`, with $\hat{v}$ the empirical mean `Learning.LinearBandit.phaseMean` of the observations of rounds $n, \ldots, N-1$), and the two error bounds are `measureReal_lt_phaseMean_le_of_testAlg` ($\theta = 0$) and `measureReal_phaseMean_le_le_of_testAlg` (an action of value at least 3ε exists), both for runs on any standard Borel probability space. The decision is the event $\{\hat{v} > \varepsilon\}$; no decision kernel is needed because the laws of Definition **??** are laws of histories.

5.4 The adaptive lower bound

Theorem 100 (Lower bound for adaptive algorithms)**.** *$Maiti2026Power.le_budget_of_isPACdef : arm_set, def : p$ ≥ 2, $\varepsilon > 0$ and $\delta \in (0, 1/16)$. Every (ε, δ)-PAC identification algorithm on $\mathcal{X}$ (adaptive or not, with any recommendation kernel) with budget n satisfies*

$$n \;\geq\; \frac{d \log(1/\delta)}{20000\, \varepsilon^2}.$$

More precisely the proof gives $n \geq c_{\mathrm{ad}}\, d \log(1/\delta)/\varepsilon^2$ with $c_{\mathrm{ad}} := \frac{1}{11556+5760\sqrt{2}} \geq \frac{1}{20000}$; the constant c_{ad} is used in the rest of the blueprint.

Proof. lem:transform$_preserves_pac, def : normalized_instance, lem : baseline_lower, lem : test_from_alg, lem : test_lower$ *Step 0: normalization.* Let Alg be (ε, δ)-PAC on $\mathcal{X}$ with budget n. By Lemma **??** with $M = d^{-1/2} A_G^{-1/2}$ (Definition **??**, which uses that $\mathcal{X}$ is spanning), the transformed algorithm Alg' is (ε, δ)-PAC on $\mathcal{X}'$ with the same budget n. We apply the lemmas of Sections **??** and **??** to Alg'. (In the formalization these lemmas are stated in the original coordinates and applied to Alg directly; see the Lean remark after Lemma **??**.)

Step 1: three consequences of $\delta < 1/16$. Write $L := \log(1/\delta) > \log 16 = 4 \log 2$. Then

(a) $\log \frac{1}{4\delta} = L - 2\log 2 \geq L - L/2 = L/2$;

(b) $\log \frac{2}{\delta} = L + \log 2 \leq L + L/4 = \frac{5}{4} L$;

(c) $\log \frac{1}{8\delta} = L - 3\log 2 \geq L - \frac{3}{4} L = L/4$.

Step 2: the baseline. Let $c_0 := (2 - \sqrt{2})/32$. Lemma **??** and (a) give $n \geq c_0 L/(2\varepsilon^2)$, i.e. $L/\varepsilon^2 \leq 2n/c_0$.

Step 3: the budget of the test. By (b) and $n \geq 1$,

$$n_{\mathrm{est}} = \left\lceil \frac{8 \log(2/\delta)}{\varepsilon^2} \right\rceil \leq \frac{8 \log(2/\delta)}{\varepsilon^2} + 1 \leq \frac{10L}{\varepsilon^2} + 1 \leq \frac{20}{c_0}\, n + n,$$

so the test of Lemma **??** built from Alg' has budget $N = n+n_{\text{est}} \le (2+20/c_0)\,n$.

Step 4: the testing lower bound. That test has errors $\mathbb{P}_0(\hat{H}=1) \le \delta \le 2\delta$ and $\mathbb{P}_1(\hat{H}=0) \le 2\delta$ (Lemma **??**), so Lemma **??** (applicable since $\delta < 1/16 < 1/8$) and (c) give

$$N \ge \frac{2d}{9\varepsilon^2}\log\frac{1}{8\delta} \ge \frac{2d}{9\varepsilon^2}\cdot\frac{L}{4} = \frac{dL}{18\,\varepsilon^2}.$$

Step 5: the constant. Combining Steps 3 and 4, $n \ge \frac{N}{2+20/c_0} \ge \frac{dL}{18(2+20/c_0)\,\varepsilon^2}$. Now $20/c_0 = 640/(2-\sqrt{2}) = 320(2+\sqrt{2}) = 640+320\sqrt{2}$, so $18(2+20/c_0) = 18(642+320\sqrt{2}) = 11556+5760\sqrt{2}$, which is c_{ad}^{-1}. Finally $\sqrt{2} \le 1.4143$ gives $5760\sqrt{2} \le 8146.4$ and $11556+5760\sqrt{2} \le 19702.4 \le 20000$, so $c_{\text{ad}} \ge 1/20000$. $\square$

Remark 101 (On the constant and the hypotheses). The paper only states $n = \Omega(d\log(1/\delta)/\varepsilon^2)$; the constant $c_{\text{ad}} \approx 5.1\cdot 10^{-5}$ obtained here is not optimized (the bound $n_{\text{est}} \le (20/c_0+1)n$ of Step 3 is very crude: for the regime of interest, n_{est} is a small multiple of $\log(1/\delta)/\varepsilon^2$ while n is of order $d\log(1/\delta)/\varepsilon^2$). The hypothesis $d \ge 2$ is used only in Lemma **??** (through $1/\sqrt{d} \le 1/\sqrt{2}$), the spanning hypothesis only through Theorem **??** in Definition **??**, and $\delta < 1/16$ only in Step 1 of the proof of Theorem **??**; the baseline lemma holds for $\delta < 1/4$ and the testing lower bound for $\delta < 1/8$. No upper bound on ε is needed. The bound applies in particular to fixed-design algorithms (Definition **??**), which are identification algorithms, and it is the $d\log(1/\delta)/\varepsilon^2$ part of the paper's Theorem 3; the unit-ball case is sharpened in Corollary **??**.

Chapter 6

Properties of the Gaussian width

This chapter studies the Gaussian width term $w(\mathcal{X})$ of Definition **??** (Section 2.3 of the paper and its Appendices on the Gaussian width properties and on the multi-task computations). It proves the three parts of Proposition 4 of the paper as separate statements (Propositions **??**, **??** and **??**), the bounds for the unit ball, the hypercubes and the m-sets, and Theorem 5 (Theorem **??**): a family of finite sets whose width is polynomially smaller than both generic upper bounds. It also introduces the *regionalized* width (Definition **??**) consumed by the adaptive algorithm of Chapter **??**, and the *intrinsic* width $w_{\mathrm{int}}(\mathcal{X})$ of a compact set that does not span $\mathbb{R}^d$ (Definition **??**). The Gaussian width $w(\mathcal{X})$ of Definition **??** is only defined for spanning action sets ($\mathrm{span}(\mathcal{X}) = \mathbb{R}^d$), and so is the G-optimal design A_G; every statement below that involves them carries the spanning hypothesis explicitly. The lemmas on the width for a matrix (Lemmas **??**, **??**, **??** and **??**) hold for an arbitrary nonempty compact index set and are used as such for the regions of a partition.

The inputs are: the G-optimal design of the Kiefer–Wolfowitz theorem (Theorem **??**, Chapter **??**); the expectation of the maximum of Gaussians (Lemma **??**) and Gaussian norm moments (Lemma **??**) of Chapter **??**; for the secondary lower bound $w(\mathcal{X}) \geq \Omega(\sqrt{d \log d})$, the Sudakov–Fernique inequality of Chapter **??**; and for the widths of structured sets, the lower bounds of Chapter **??** combined with the upper bound of Theorem **??**. All constants are explicit. Compared with the paper, the lower bounds for structured sets are stated as explicit inequalities (with admittedly large thresholds on d inherited from the constants of the lower bounds), and the multi-task computation is organized around the Helmert matrices with every linear-algebra step isolated (the Lean proof replaces the explicit basis by a basis-free reduction to a Gaussian vector on the span, see the remarks in Section **??**).

6.1 Upper bounds

Proposition 102 (The width is at most d)**.** *$Maiti2026Power.gw_le_card, Maiti2026Power.IsGOptimalDesign.g$ $\mathbb{R}^d$ be a spanning action set. Then $w(\mathcal{X}) \le d$. More precisely, the G-optimal design matrix A_G of Theorem **??** (which exists since $\mathcal{X}$ is spanning) satisfies $w(\mathcal{X}; A_G) \le d$.*

Proof. cor:g$_optimal_norm_bound, lem: gauss_norm_moments, lem: width_matrix_wdByTheorem$ **??**,$A_G \in \mathcal{M}(\mathcal{X}) \cap \mathcal{S}^d_{++}$, so $w(\mathcal{X}) \le w(\mathcal{X}; A_G)$ by definition of the infimum. For every $g \in \mathbb{R}^d$ and $x \in \mathcal{X}$, the Cauchy–Schwarz inequality and Corollary **??** give $\langle A_G^{-1/2}x, g\rangle \le \|A_G^{-1/2}x\| \, \|g\| \le \sqrt{d}\,\|g\|$, hence $\sup_{x\in\mathcal{X}}\langle A_G^{-1/2}x, g\rangle \le \sqrt{d}\,\|g\|$. Taking expectations (Lemma **??** for integrability) and using $\mathbb{E}\|g\| \le \sqrt{\mathbb{E}\|g\|^2} = \sqrt{d}$ (Lemma **??** with $S = I_d$) yields $w(\mathcal{X}; A_G) \le \sqrt{d} \cdot \sqrt{d} = d$. ☐

Proposition 103 (Width of a finite set)**.** *$Maiti2026Power.gw_le_sqrt_log_ncard, Maiti2026Power.IsGOptimalDe$ m. Then $w(\mathcal{X}) \le \sqrt{2d\log m}$; more precisely $w(\mathcal{X}; A_G) \le \sqrt{2d\log m}$ for the G-optimal design matrix A_G of Theorem **??**.*

Proof. thm:kiefer$_wolfowitz, cor: g_optimal_norm_bound, lem: gauss_max_upper, lem: gauss_linear_imageSince$X$spans$R^d, $m \ge d \ge 1$. If $m = 1$ then $d = 1$, $\mathcal{X} = \{x\}$ and $w(\mathcal{X}; A) = \mathbb{E}\langle A^{-1/2}x, g\rangle = 0 = \sqrt{2d\log 1}$ for every $A \in \mathcal{S}^d_{++}$. Assume $m \ge 2$ and let $u_x := A_G^{-1/2}x$ for $x \in \mathcal{X}$. The family $(Z_x)_{x\in\mathcal{X}}$, $Z_x := \langle u_x, g\rangle$, is a centered Gaussian vector (Lemma **??**) with $\mathrm{Var} Z_x = \|u_x\|^2 \le d$ (Corollary **??**). Lemma **??** with $\sigma^2 = d$ gives $\mathbb{E}\max_{x\in\mathcal{X}} Z_x \le \sqrt{d}\sqrt{2\log m}$, i.e. $w(\mathcal{X}; A_G) \le \sqrt{2d\log m}$, and $w(\mathcal{X}) \le w(\mathcal{X}; A_G)$ as in Proposition **??**. ☐

Definition 104 (Regionalized width)**.** Maiti2026Power.regionalizedGw, Maiti2026Power.gwMat$_le_regionalized$ $\in \mathcal{S}^d_{++}$ and let $\mathcal{R} = (R_1, \dots, R_K)$, $K \ge 1$, be a partition of $\mathcal{X}$ into nonempty compact subsets ($R_i \cap R_j = \emptyset$ for $i \ne j$ and $\bigcup_i R_i = \mathcal{X}$). The *regionalized Gaussian width of $\mathcal{X}$ for A and $\mathcal{R}$* is

$$w(A; \mathcal{X}, \mathcal{R}) := \max_{1\le i\le K} w(R_i; A) = \max_{1\le i\le K} \mathbb{E}\Big[\max_{x\in R_i}\langle x, A^{-1/2}g\rangle\Big], \qquad g \sim \mathcal{N}(0, I_d),$$

where $w(R_i; A)$ is the width for the matrix A of Definition **??** with the nonempty compact index set R_i; it is well defined and nonnegative by Lemma **??** applied to the index set R_i. Since $R_i \subseteq \mathcal{X}$ for every i, the supremum over R_i is at most the supremum over $\mathcal{X}$ pathwise, so $0 \le w(A; \mathcal{X}, \mathcal{R}) \le w(\mathcal{X}; A)$, with equality when $K = 1$.

Lean remark. In Lean the regions are simply a family of sets $\mathcal{R} : \kappa \to \mathcal{P}(\mathbb{R}^d)$ indexed by a finite type κ, and $w(A; \mathcal{R}) := \sup_{i\in\kappa} w(R_i; A)$ (`regionalizedGw`); the action set $\mathcal{X}$ does not appear in the definition, and the partition property (disjointness, covering) is never needed: the statements below only use $R_i \subseteq \mathcal{X}$, and each region nonempty (and finite, with at most p elements). The inner expectation is exactly the width for a matrix of Definition **??** with the index set R_i, which that definition allows (any nonempty compact index set); hence Lemmas **??**, **??** and **??** apply to $w(R_i; A)$ verbatim, and Chapter **??** cites them directly for the regions.

Proposition 105 (Regionalized width of a finite set)**.** *Maiti2026Power.IsGOptimalDesign.regionalizedGw$_l e_s qr$*
= $(R_1, \ldots, R_K)$ a partition of $\mathcal{X}$ with $|R_i| \le p$ for all i, where $p \ge 1$. Then, for the G-optimal design matrix A_G of Theorem ?? (which exists since $\mathcal{X}$ is spanning),

$$w(A_G; \mathcal{X}, \mathcal{R}) \le \sqrt{2d \log p}.$$

Proof. cor:g$_o ptimal_n orm_b ound, lem : gauss_m ax_u pper, lem : gauss_l inear_i mageFix$i.$If |R_i| =$ 1, the expectation $\mathbb{E} \max_{x \in R_i} \langle A_G^{-1/2} x, g\rangle$ is the expectation of a single centered Gaussian, i.e. $0 \le \sqrt{2d \log p}$. Otherwise $2 \le |R_i| \le p$ and, exactly as in the proof of Proposition **??**, the variables $Z_x = \langle A_G^{-1/2} x, g\rangle$, $x \in R_i$, are centered Gaussians with variance at most d (Corollary **??**), so Lemma **??** gives $\mathbb{E} \max_{x \in R_i} Z_x \le \sqrt{2d \log |R_i|} \le \sqrt{2d \log p}$. Take the maximum over i. □

6.2 Lower bounds

Proposition 106 (Width of the unit ball)**.** *Maiti2026Power.gwMat$_u$nitBall, Maiti2026Power.one$_l e_s qrt_d iag_m$*
B_d is a spanning action set (it is compact and contains the standard basis $e_1, \ldots, e_d$), so its Gaussian width $w(\mathbb{B}_d)$ of Definition ?? is defined, and

$$\frac{d}{\sqrt{3}} \le \sqrt{\frac{2}{\pi}}\, d \le w(\mathbb{B}_d) \le d.$$

More precisely, for every $A \in \mathcal{S}_{++}^d$, $w(\mathbb{B}_d; A) \ge \sqrt{2/\pi}\, d / \sqrt{\operatorname{tr} A}$.

Proof. prop:width$_l e_d, lem : designset_b asic, lem : gauss_n orm_l ower, lem : gauss_l inear_i mage, lem :$ $width_m atrix_w dThe upper bound is Proposition$ **??**(B$_d$ being spanning). For every $v \in \mathbb{R}^d$, $\sup_{x \in \mathbb{B}_d} \langle x, v\rangle = \|v\|$ (Cauchy–Schwarz, with equality at $x = v/\|v\|$ when $v \ne 0$), so $w(\mathbb{B}_d; A) = \mathbb{E}\|Y\|$ with $Y := A^{-1/2} g \sim \mathcal{N}(0, A^{-1})$ (Lemma **??**). Fix $A \in \mathcal{S}_{++}^d$ and apply Lemma **??** with the weights $c_i := \sqrt{A_{ii} / \operatorname{tr} A}$ ($\sum_i c_i^2 = 1$): $\mathbb{E}\|Y\| \ge \sqrt{2/\pi} \sum_i \sqrt{A_{ii}} \sqrt{(A^{-1})_{ii}} / \sqrt{\operatorname{tr} A}$. Now $\sqrt{A_{ii}} \sqrt{(A^{-1})_{ii}} \ge 1$ for every i: by Cauchy–Schwarz, $1 = \langle e_i, e_i\rangle = \langle A^{1/2} e_i, A^{-1/2} e_i\rangle \le \|A^{1/2} e_i\| \|A^{-1/2} e_i\| = \sqrt{A_{ii}} \sqrt{(A^{-1})_{ii}}$. Hence $w(\mathbb{B}_d; A) \ge \sqrt{2/\pi}\, d / \sqrt{\operatorname{tr} A}$. For $A \in \mathcal{M}(\mathbb{B}_d)$, Lemma **??** gives $\operatorname{tr} A \le \sup_{x \in \mathbb{B}_d} \|x\|^2 = 1$, so $w(\mathbb{B}_d; A) \ge \sqrt{2/\pi}\, d$ for every admissible A, and the infimum $w(\mathbb{B}_d)$ is at least $\sqrt{2/\pi}\, d \ge d/\sqrt{3}$ (as $\pi \le 6$). □

Lemma 107 (Well-separated points in the whitened set)**.** *$exists_p os_w eight_s q_d ist_g e_o f_i sotropic, exists_s eparated_o$*
≥ 2, let $\mathcal{X}$ be an action set and $\lambda \in \triangle_{\mathcal{X}}$ with $\Sigma := A(\lambda) \succ 0$ (such a λ exists if and only if $\mathcal{X}$ is spanning), and let $\mathcal{T} := \Sigma^{-1/2} \mathcal{X} = \{\Sigma^{-1/2} x : x \in \mathcal{X}\}$ and $m := \lfloor d/2 \rfloor$. There exist $t_1, \ldots, t_m \in \mathcal{T}$ such that $\|t_i - t_j\| \ge \sqrt{d/2}$ for all $i \ne j$.

Proof. lem:sqrt$_p ropsWrite$u$_x := \Sigma^{-1/2} x$ for $x \in \operatorname{supp} \lambda$. Then $\sum_x \lambda_x u_x u_x^\top = \Sigma^{-1/2} A(\lambda) \Sigma^{-1/2} = I_d$ (Lemma **??**). We construct the points greedily. Pick $t_1 \in \mathcal{T}$ arbitrarily. Assume $t_1, \ldots, t_k$ have been chosen for some $1 \le k < m$, let $V_k := \operatorname{span}\{t_1, \ldots, t_k\}$ and let Q_k be the orthogonal projection onto $V_k^\perp$. Then

Q_k is symmetric and idempotent, and $\operatorname{tr} Q_k = \dim V_k^\perp = d - \dim V_k \geq d - k$. Therefore

$$\sum_x \lambda_x \|Q_k u_x\|^2 = \sum_x \lambda_x \, u_x^\top Q_k u_x = \operatorname{tr}\Big(Q_k \sum_x \lambda_x u_x u_x^\top\Big) = \operatorname{tr} Q_k \geq d - k,$$

so (the λ_x being a probability vector) some $x \in \operatorname{supp}\lambda$ has $\|Q_k u_x\|^2 \geq d - k$; set $t_{k+1} := u_x \in \mathcal{T}$. For $i \leq k$ we have $Q_k t_i = 0$ since $t_i \in V_k$, and orthogonal projections do not increase norms, so

$$\|t_{k+1} - t_i\| \geq \|Q_k(t_{k+1} - t_i)\| = \|Q_k t_{k+1}\| \geq \sqrt{d-k} \geq \sqrt{d-(m-1)} \geq \sqrt{d/2},$$

where the last step uses $m - 1 \leq d/2 - 1$. This constructs $t_1, \ldots, t_m$ with the required pairwise separation. □

Lean remark. The lemma is formalized in library generality (`Maiti2026Power/Mathlib/Analysis/InnerP` for a weighted family (w_i, u_i) of a d-dimensional inner product space in *isotropic position*, $\sum_i w_i \langle u_i, v\rangle^2 = \|v\|^2$ for all v ($w_i \geq 0$, $\sum_i w_i = 1$), and every m, there are m points $u_{i_1}, \ldots, u_{i_m}$ with positive weights and $\|u_{i_k} - u_{i_l}\|^2 \geq d + 1 - m$ for $k \neq l$ (`exists_separated_of_isotropic`; no hypothesis on m or d). The greedy step uses an orthonormal basis (f_j) of $V_k^\perp$ and Bessel's inequality $\|u - t_i\|^2 \geq \sum_j \langle u - t_i, f_j\rangle^2 = \sum_j \langle u, f_j\rangle^2$ instead of the projection Q_k. The isotropy of the whitened design, $\sum_x \lambda_x \langle \Sigma^{-1/2} x, v\rangle^2 = \|v\|^2$, is `sum_mul_inner_sqrt_inv_sq`.

Proposition 108 (Lower bound $\sqrt{d \log d}$; secondary)**.** *Maiti2026Power.sqrt$_l$og$_l$e$_g$wdef : widthLet$\mathcal{X} \subset$ $\mathbb{R}^d$ be a spanning action set with $d \geq 4$. Then*

$$w(\mathcal{X}) \;\geq\; \frac{1}{8}\sqrt{d \, \log\lfloor d/2 \rfloor}.$$

Proof. lem:whitened$_s$eparated$_p$oints, thm : sudakov$_f$ernique, lem : gauss$_m$ax$_l$ower, lem : width$_m$atrix$_w$d, lem : gauss$_l$inear$_i$mage, lem : exists$_p$osdef$_d$esignTheconstant1$_{\frac{}{8}}$ is $c_{\max}/2$ with $c_{\max} = \frac{1}{4}$ the constant of Lemma **??**, which holds for every $m \geq 2$ (no threshold); the condition $d \geq 4$ ensures $m := \lfloor d/2 \rfloor \geq 2$. Fix $\lambda \in \triangle_{\mathcal{X}}$ with $\Sigma := A(\lambda) \succ 0$ (it exists since $\mathcal{X}$ is spanning, Lemma **??**); it suffices to lower bound $w(\mathcal{X}; \Sigma) = \mathbb{E} \sup_{t \in \mathcal{T}} \langle t, g\rangle$, $\mathcal{T} = \Sigma^{-1/2}\mathcal{X}$ (Lean: `Maiti2026Power.sqrt_log_le_gwMat`, for every design λ with $A(\lambda) \succ 0$). Take $t_1, \ldots, t_m \in \mathcal{T}$ from Lemma **??**. Set $Z_i := \langle t_i, g\rangle$; $(Z_1, \ldots, Z_m)$ is a centered Gaussian vector (Lemma **??**) with $\mathbb{E}(Z_i - Z_j)^2 = \|t_i - t_j\|^2 \geq d/2$ for $i \neq j$. Let $\xi_1, \ldots, \xi_m$ be i.i.d. standard Gaussians and $Y_i := \frac{\sqrt{d}}{2}\xi_i$, so that $\mathbb{E}(Y_i - Y_j)^2 = \frac{d}{4} \cdot 2 = d/2 \leq \mathbb{E}(Z_i - Z_j)^2$ for $i \neq j$ (and $0 = 0$ for $i = j$). The Sudakov–Fernique inequality (Theorem **??**, applied to the pair (Y, Z) whose increments are ordered as required) gives $\mathbb{E}\max_i Y_i \leq \mathbb{E}\max_i Z_i$. Hence

$$w(\mathcal{X}; \Sigma) \geq \mathbb{E}\max_{i \leq m} Z_i \geq \mathbb{E}\max_{i \leq m} Y_i = \frac{\sqrt{d}}{2}\,\mathbb{E}\max_{i \leq m} \xi_i \geq \frac{\sqrt{d}}{2} \cdot \frac{1}{4}\sqrt{\log m} = \frac{1}{8}\sqrt{d \log m},$$

by Lemma **??** (applicable as $m \geq 2$). Taking the infimum over λ proves the first inequality. For the second, $\lfloor d/2 \rfloor \geq \sqrt{d}$ for $d \geq 6$ (check $d = 6, 7$ directly:

$3 \geq \sqrt{7}$; for $d \geq 8$, $\lfloor d/2 \rfloor \geq (d-1)/2 \geq \sqrt{d}$ because $(\sqrt{d}-1)^2 \geq 2$), so $\log\lfloor d/2 \rfloor \geq \frac{1}{2}\log d$ and $\frac{1}{8}\sqrt{d\log\lfloor d/2 \rfloor} \geq \frac{1}{8}\sqrt{\frac{1}{2}d\log d} = \frac{1}{8\sqrt{2}}\sqrt{d\log d}$. ☐

Corollary 109 (Lower bound $\sqrt{d\log d}$)**.** *prop:width$_l$ower$_s$qrt$_d$logdLetX$\subset \mathbb{R}^d$ be a spanning action set with $d \geq 6$. Then $w(\mathcal{X}) \geq \frac{1}{8\sqrt{2}}\sqrt{d\log d}$.*

Proof. prop:width$_l$ower$_s$qrt$_d$logdFord ≥ 6, $\lfloor d/2 \rfloor \geq (d-1)/2 \geq \sqrt{d}$ (as $(d-1)^2 \geq 4d$ for $d \geq 6$), so $\log\lfloor d/2 \rfloor \geq \frac{1}{2}\log d$ and Proposition **??** gives the claim. ☐

Proposition 110 (Width lower bounds from sample complexity lower bounds)**.** *def:pac, def:width Let $\mathcal{X}$ be a spanning action set, $\delta_0 \in (0,1)$ and $L \geq 0$. Assume that for every $\varepsilon \in (0,1]$, every (ε,δ_0)-PAC identification algorithm on $\mathcal{X}$ with budget $T \geq 1$ satisfies $T \geq L/\varepsilon^2$. Then*

$$w(\mathcal{X})^2 \;\geq\; \frac{L}{600} - d\log\frac{2}{\delta_0}.$$

Proof. thm:upper, def:fixed$_d$esign$_a$lgorithmFix$\varepsilon \in (0,1]$. By Theorem **??** (which requires $\mathcal{X}$ spanning), the fixed-design algorithm of Definition **??** with parameters (ε,δ_0) is (ε,δ_0)-PAC with budget $T(\varepsilon) = \lceil 600(w(\mathcal{X})^2 + d\log(2/\delta_0))/\varepsilon^2 \rceil < 600(w(\mathcal{X})^2 + d\log(2/\delta_0))/\varepsilon^2 + 1$, and $T(\varepsilon) \geq 4d \geq 1$. The hypothesis gives $L/\varepsilon^2 \leq T(\varepsilon)$, hence, multiplying by ε^2, $L < 600(w(\mathcal{X})^2 + d\log(2/\delta_0)) + \varepsilon^2$. Letting $\varepsilon \to 0$ gives $L \leq 600(w(\mathcal{X})^2 + d\log(2/\delta_0))$, which is the claim. ☐

Remark 111. The outline of this chapter allowed an additional -1 on the right-hand side (coming from the ceiling in the budget at $\varepsilon = 1$); letting $\varepsilon \to 0$ removes it, as shown above. The hypothesis of Proposition **??** is exactly what the lower bounds of Section **??** provide: they show that if $1 \leq T \leq L/\varepsilon^2$ then some instance fails with probability at least some $p_0 > \delta_0$, so an (ε,δ_0)-PAC algorithm with budget $T \geq 1$ must have $T > L/\varepsilon^2$. The restriction to budgets $T \geq 1$ is harmless since the algorithm of Theorem **??** has budget at least $4d$.

Corollary 112 (Widths of hypercubes and m-sets)**.** *def:width, def:msets The three action sets below are spanning (see the end of the proof), so their Gaussian widths are defined.*

(i) For $\mathcal{X} = \{-1,1\}^d$: $w(\mathcal{X})^2 \geq d^2/60000 - 3d$; in particular $w(\mathcal{X}) \geq d/350$ for $d \geq 360000$.

(ii) For $\mathcal{X} = \{0,1\}^d$: $w(\mathcal{X})^2 \geq d^2/240000 - 3d$; in particular $w(\mathcal{X}) \geq d/700$ for $d \geq 1500000$.

*(iii) For the m-sets $\mathcal{X} = \{x \in \{0,1\}^d : \sum_i x_i = m\}$ (Definition **??**) with $n := d - m + 1 \geq 20m$: $w(\mathcal{X})^2 \geq mn/(1.5 \cdot 10^6) - 3d$; in particular $w(\mathcal{X}) \geq \sqrt{md}/1800$ for $m \geq 10^7$.*

*Together with Propositions **??** and **??**, this shows $w(\{-1,1\}^d) = \Theta(d)$, $w(\{0,1\}^d) = \Theta(d)$ and, for m-sets with $d - m + 1 \geq 20m$, $\sqrt{md} \lesssim w(\mathcal{X}) \lesssim \sqrt{md\log(ed/m)}$ (using $\log\binom{d}{m} \leq m\log(ed/m)$).*

Proof. prop:width$_{l}ower_{f}rom_{p}ac, thm : lower_{h}ypercube_{p}m, thm : lower_{h}ypercube_{0}1, thm : lower_{m}sets(i) Theorem$ **??** $: for every \varepsilon > 0$, every identification algorithm on $\{-1,1\}^d$ with budget $1 \le T \le d^2/(100\varepsilon^2)$ has an instance on which it fails with probability at least $1/6$. Take $\delta_0 := 1/7 < 1/6$: an (ε,δ_0)-PAC algorithm with budget $T \ge 1$ must have $T > d^2/(100\varepsilon^2)$, so Proposition **??** applies with $L = d^2/100$ and gives $w(\mathcal{X})^2 \ge d^2/60000 - d\log 14 \ge d^2/60000 - 3d$ ($\log 14 < 2.64$). If $d \ge 360000$ then $3d \le d^2/60000 - d^2/122500$ (equivalent to $d \ge 3/(1/60000 - 1/122500) \approx 352800$), so $w(\mathcal{X})^2 \ge d^2/122500$ and $w(\mathcal{X}) \ge d/350$.

(ii) Theorem **??** gives the failure probability $1/6$ for $T \le d^2/(400\varepsilon^2)$; with $\delta_0 = 1/7$ and $L = d^2/400$, $w(\mathcal{X})^2 \ge d^2/240000 - 3d$, and for $d \ge 1500000$ this is at least $d^2/490000$ (equivalent to $d \ge 3/(1/240000 - 1/490000) \approx 1411200$), so $w(\mathcal{X}) \ge d/700$.

(iii) Theorem **??** gives the failure probability $2/9$ for $T \le mn/(2500\varepsilon^2)$; with $\delta_0 = 1/5 < 2/9$ and $L = mn/2500$, $w(\mathcal{X})^2 \ge mn/(1.5\cdot 10^6) - d\log 10 \ge mn/(1.5\cdot 10^6) - 3d$. Since $m \le n/20$ we have $d = n + m - 1 \le 21n/20$, i.e. $n \ge 20d/21$, so $mn/(1.5\cdot 10^6) \ge md/(1.575\cdot 10^6)$. For $m \ge 10^7$, $3d \le md/(1.575\cdot 10^6) - md/(3.15\cdot 10^6)$ (equivalent to $m \ge 9.45\cdot 10^6$), hence $w(\mathcal{X})^2 \ge md/(3.15\cdot 10^6)$ and $w(\mathcal{X}) \ge \sqrt{md}/1775 \ge \sqrt{md}/1800$.

The spanning hypotheses (needed by Proposition **??**, i.e. by Theorem **??**) hold: $\{-1,1\}^d$ and $\{0,1\}^d$ contain a basis, and the m-sets span $\mathbb{R}^d$ for $1 \le m \le d-1$ (Definition **??**), which is implied by $n \ge 20m$. $\square$

Remark 113. The thresholds on d and m in Corollary **??** come from the additive term $d\log(2/\delta_0)$ in the budget of Theorem **??**, which must be dominated by $L/600$; they are not intrinsic. The unit ball is treated directly in Proposition **??**, which is much sharper than what Corollary **??** would give through Proposition **??**.

6.3 Multi-task multi-armed bandits

The multi-task bandit action set below does not span $\mathbb{R}^d$, so the Gaussian width $w(\mathcal{X})$ of Definition **??** is not defined for it. As in the paper, one measures it instead through an isometric identification of its span with a Euclidean space of the right dimension; to avoid overloading the notation $w(\mathcal{X})$, this second notion is called the *intrinsic width* and written $w_{\mathrm{int}}(\mathcal{X})$.

Definition 114 (Intrinsic width of a set spanning a subspace)**.** Maiti2026Power.intrinsicGwdef:width, lem:width$_{i}soLet$X$\subset \mathbb{R}^d$ be an action set (nonempty compact, not necessarily spanning) and let $V := \mathrm{span}(\mathcal{X})$ have dimension $r \ge 1$. Let $U \in \mathbb{R}^{d\times r}$ be any matrix with orthonormal columns ($U^\top U = I_r$) whose range is V. Then $\mathcal{X}_r := U^\top\mathcal{X} = \{U^\top x : x \in \mathcal{X}\} \subset \mathbb{R}^r$ is compact and spans $\mathbb{R}^r$ (the map $z \mapsto U^\top z$ restricted to V is a linear isometry onto $\mathbb{R}^r$ with inverse $u \mapsto Uu$; in particular $x = UU^\top x$ for $x \in \mathcal{X}$), so it is a spanning action set in $\mathbb{R}^r$ to which Definition **??** applies, and we define the *intrinsic width* of $\mathcal{X}$ as

$$w_{\mathrm{int}}(\mathcal{X}) := w(\mathcal{X}_r) = w(U^\top\mathcal{X}).$$

This does not depend on the choice of U: if U' is another such matrix then $M := U^\top U' \in \mathbb{R}^{r \times r}$ is orthogonal ($M^\top M = U'^\top U U^\top U' = U'^\top U' = I_r$ since $UU^\top$ is the orthogonal projection onto V and U' has range V) and $U'^\top x = M^\top U^\top x$ for $x \in V$, so $U'^\top \mathcal{X} = M^\top (U^\top \mathcal{X})$ and $w(U'^\top \mathcal{X}) = w(U^\top \mathcal{X})$ by Lemma **??** (applied to the spanning set $U^\top \mathcal{X} \subset \mathbb{R}^r$ and the invertible matrix $M^\top$). When $V = \mathbb{R}^d$ (i.e. $\mathcal{X}$ is spanning) one has $w_{\mathrm{int}}(\mathcal{X}) = w(\mathcal{X})$, again by Lemma **??**, U being an orthogonal $d \times d$ matrix.

Lean remark. The subspace is `Submodule.span  X`; an orthonormal basis of a finite-dimensional inner product subspace is provided by `stdOrthonormalBasis` or `OrthonormalBasis` of the subspace; the map $x \mapsto U^\top x$ is the coordinate map of this basis composed with the orthogonal projection `orthogonalProjection`, and $u \mapsto Uu$ is the basis' linear isometry. The identification of action sets in $\mathbb{R}^r$ with `EuclideanSpace  (Fin r)` then makes Definition **??** applicable. In Lean, w_{int} is a separate declaration from w (the latter takes the spanning hypothesis as an argument), related by the identity $w_{\mathrm{int}}(\mathcal{X}) = w(\mathcal{X})$ for spanning $\mathcal{X}$.

Definition 115 (Multi-task bandit action set)**.** Maiti2026Power.multitaskSet, Maiti2026Power.oneHot, Maiti2026Power.multitaskSet$_{e}q_{r}ange, Maiti2026Power.ncard_{m}ultitaskSet, Maiti202$ ≥ 1 and $d_1, \ldots, d_m \geq 2$ be integers, $d := \sum_{j \leq m} d_j$, $d_{1:j} := \sum_{s \leq j} d_s$ (with $d_{1:0} = 0$), and let $B_j := \{d_{1:j-1} + 1, \ldots, d_{1:j}\} \subseteq [d]$ be the j-th *block* of coordinates, $|B_j| = d_j$. The *multi-task action set* is

$$\mathcal{X} := \Big\{ x \in \{0,1\}^d : \sum_{i \in B_j} x_i = 1 \text{ for every } j \leq m \Big\}.$$

Thus $|\mathcal{X}| = \prod_j d_j$; an element of $\mathcal{X}$ is determined by the positions $(k_j)_{j \leq m}$ of its ones, $k_j \in B_j$. For $m \geq 2$ the set $\mathcal{X}$ does not span $\mathbb{R}^d$ (Lemma **??**); it is nevertheless an action set in the sense of Definition **??**, and the relevant width is its intrinsic width $w_{\mathrm{int}}(\mathcal{X})$ of Definition **??**. We write $\mathbf{1}_n$ for the all-ones vector of $\mathbb{R}^n$ and e_k for the k-th basis vector.

Lean remark. The natural index type of the coordinates is the sigma type $\Sigma_{j:\mathrm{Fin}\ m}$ Fin (d_j) (block index, position in the block), with $\mathcal{X}$ the image of $\prod_j$ Fin (d_j) under $(k_j)_j \mapsto \sum_j e_{(j,k_j)}$; the block B_j is then the set of indices with first component j. If one prefers Fin d as index type, use the block map b : Fin $d \to$ Fin m and $B_j = b^{-1}(j)$; the offsets $d_{1:j}$ are only needed to translate between the two encodings. All statements below only use the partition $(B_j)_j$ of the coordinates and the sizes d_j. The Lean development uses the sigma type; `oneHot d` is the element with ones at the positions $(k_j)_j = \kappa$, so that $\mathcal{X}$ is the range of `oneHot` (`multitaskSet_eq_range`), $|\mathcal{X}| = \prod_j d_j$, and the support function is $\sup_{x \in \mathcal{X}} \langle x, y \rangle = \sum_j \max_{k \in B_j} y_k$ (`supportFn_multitaskSet`), the identity on which the width computation rests.

Lean remark. The Helmert matrices and the explicit orthonormal basis of Lemma **??** are not formalized: the Lean proof of Proposition **??** avoids any explicit basis of the span through the general reduction `intrinsicGw_le_gaussianWidth` (the Lean form of Lemma **??**), which only needs the span of $\mathcal{X}$ (Lemma **??**, first part), a design matrix Σ of $\mathcal{X}$ (Lemma **??**) and a positive semidefinite matrix S with range in $\operatorname{span}\mathcal{X}$ inverting Σ on $\operatorname{span}\mathcal{X}$; the block-centering maps P_j play the role of $H_{d_j}H_{d_j}^\top$. Definition **??** and Lemma **??** are kept as the paper's route.

Definition 116 (Helmert matrices)**.** For $n \geq 2$, the *Helmert matrix* $H_n \in \mathbb{R}^{n\times(n-1)}$ has entries, for $\ell \in [n]$ and $k \in [n-1]$,

$$(H_n)_{\ell,k} := \begin{cases} \dfrac{1}{\sqrt{k(k+1)}} & \text{if } \ell \leq k, \\ -\dfrac{k}{\sqrt{k(k+1)}} & \text{if } \ell = k+1, \\ 0 & \text{if } \ell \geq k+2. \end{cases}$$

Its k-th column is $h_k := (1,\dots,1,-k,0,\dots,0)^\top/\sqrt{k(k+1)}$ with k ones.

Lemma 117 (Properties of Helmert matrices)**.** *def:helmert For $n \geq 2$: $H_n^\top H_n = I_{n-1}$, $H_n^\top \mathbf{1}_n = 0$, and $H_nH_n^\top = I_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top$. Equivalently, the columns of H_n together with $\mathbf{1}_n/\sqrt{n}$ form an orthonormal basis of $\mathbb{R}^n$, and $H_nH_n^\top$ is the orthogonal projection onto $\mathbf{1}_n^\perp = \{v : \sum_\ell v_\ell = 0\}$.*

Proof. def:helmert $\|h_k\|^2 = (k+k^2)/(k(k+1)) = 1$ and $\langle h_k, \mathbf{1}_n\rangle = (k-k)/\sqrt{k(k+1)} = 0$. For $k < k'$, the entries of $h_{k'}$ on the first $k+1$ coordinates are all equal to $1/\sqrt{k'(k'+1)}$ (because $k+1 \leq k'$), and h_k vanishes beyond coordinate $k+1$, so $\langle h_k, h_{k'}\rangle = \langle h_k, \mathbf{1}_n\rangle/\sqrt{k'(k'+1)} = 0$. Hence $H_n^\top H_n = I_{n-1}$, $H_n^\top \mathbf{1}_n = 0$, and $\{h_1,\dots,h_{n-1},\mathbf{1}_n/\sqrt{n}\}$ is an orthonormal family of n vectors in $\mathbb{R}^n$, i.e. an orthonormal basis. The resolution of the identity for an orthonormal basis, $\sum_k h_kh_k^\top + \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top = I_n$, is the last identity. □

Lemma 118 (An orthonormal basis of the span of the multi-task set)**.** *$Maiti2026Power.mem_span_multitaskSet_i$ $\mathbb{R}^d$ by $\mu_i := 1/d_j$ for $i \in B_j$, $\kappa := \|\mu\|^2 = \sum_{j\leq m} 1/d_j$, $u_0 := \mu/\sqrt{\kappa}$, and for $j \leq m$ the matrix $Q_j \in \mathbb{R}^{d\times(d_j-1)}$ whose rows indexed by B_j form H_{d_j} (row $d_{1:j-1}+\ell$ of Q_j is row ℓ of H_{d_j}) and whose other rows are zero. Let*

$$U := [\,u_0 \;\; Q_1 \;\; \cdots \;\; Q_m\,] \in \mathbb{R}^{d\times r}, \qquad r := 1 + \sum_{j\leq m}(d_j - 1) = d - m + 1.$$

Then $U^\top U = I_r$, $\operatorname{range}(U) = \operatorname{span}(\mathcal{X})$, and $\dim\operatorname{span}(\mathcal{X}) = d-m+1$.

Proof. lem:helmert$_props, def: multitask_set(Lean: only the description$ $\operatorname{span}\mathcal{X} = \{v : \text{all block sums of } v \text{ are equal}\}$ and $\dim\operatorname{span}\mathcal{X} = d-m+1$ are formalized, the latter by rank–nullity for the map $v \mapsto (\sum_{i\in B_j} v_i - \sum_{i\in B_1} v_i)_j$; the matrix U is not needed.) *Orthonormality.* $\|u_0\| = 1$. The columns of Q_j and Q_ℓ have disjoint supports for $j \neq \ell$, so $Q_j^\top Q_\ell = 0$, and $Q_j^\top Q_j = H_{d_j}^\top H_{d_j} = I_{d_j-1}$ (Lemma **??**). Since u_0 is constant on B_j, $u_0^\top Q_j = \frac{1}{d_j\sqrt{\kappa}}\mathbf{1}_{d_j}^\top H_{d_j} = 0$. Hence $U^\top U = I_r$.

Range. Let $R_j := \{v \in \mathbb{R}^d : \operatorname{supp} v \subseteq B_j,\ \sum_{i\in B_j} v_i = 0\}$, a subspace of dimension $d_j - 1$. The columns of Q_j lie in R_j (Lemma **??**: $H_{d_j}^\top \mathbf{1} = 0$) and are $d_j - 1$ orthonormal vectors, so $\operatorname{range}(Q_j) = R_j$, and $\operatorname{range}(U) = \mathbb{R}u_0 \oplus R_1 \oplus \cdots \oplus R_m$ (a direct sum, the summands being pairwise orthogonal). Now $\mu \in \operatorname{span}\mathcal{X}$: $\mu = \frac{1}{|\mathcal{X}|}\sum_{x\in\mathcal{X}} x$, since for $i \in B_j$ exactly $|\mathcal{X}|/d_j$ elements of $\mathcal{X}$ have $x_i = 1$. For $i, i' \in B_j$, $e_i - e_{i'}$ is the difference of two elements of $\mathcal{X}$ that agree outside B_j, hence lies in $\operatorname{span}\mathcal{X}$; these differences span R_j. Therefore $\operatorname{range}(U) \subseteq \operatorname{span}\mathcal{X}$. Conversely, for $x \in \mathcal{X}$ and every j, $\sum_{i\in B_j}(x-\mu)_i = 1 - 1 = 0$, so $x - \mu \in R_1 \oplus \cdots \oplus R_m$ (split $x - \mu$ according to the blocks) and $x \in \operatorname{range}(U)$. Hence $\operatorname{range}(U) = \operatorname{span}\mathcal{X}$ and its dimension is the number r of (orthonormal) columns of U. □

Lemma 119 (Covariance of the uniform design in the intrinsic coordinates)**.** *Maiti2026Power.unifDesignMatrix, Maiti2026Power.unifDesignMatrix$_m em_d esignSet, Maiti2026Power.toEuc$ be the uniform design on $\mathcal{X}$ ($\lambda_x = 1/|\mathcal{X}|$ for every $x \in \mathcal{X}$) and $\Sigma := A(\lambda_u) = \frac{1}{|\mathcal{X}|}\sum_{x\in\mathcal{X}} xx^\top \in \mathbb{R}^{d\times d}$. Then, for $i \in B_j$ and $k \in B_\ell$,*

$$\Sigma_{ik} = \frac{1}{d_j}\mathbf{1}\{i = k\} \quad \textit{if } j = \ell, \qquad \Sigma_{ik} = \frac{1}{d_j d_\ell} \quad \textit{if } j \neq \ell,$$

*and, with U, κ as in Lemma **??**,*

$$\Sigma_r := U^\top \Sigma U = \operatorname{diag}\left(\kappa,\ \tfrac{1}{d_1} I_{d_1-1},\ \ldots,\ \tfrac{1}{d_m} I_{d_m-1}\right) \textit{ (positive definite, } r \times r\textit{)}.$$

Moreover Σ_r is the design matrix of the uniform design on $U^\top\mathcal{X}$: $\Sigma_r = \frac{1}{|\mathcal{X}|}\sum_{x\in\mathcal{X}}(U^\top x)(U^\top x)^\top \in \mathcal{M}(U^\top\mathcal{X})$.

Proof. lem:helmert$_p rops, lem: multitask_b asis(Lean: the formalized statements are the two eigen-relations$ $\Sigma\mu = \kappa\mu$ and $\Sigma v = v/d_j$ for $v \in R_j$, i.e. for $v = P_j w$ the block-centering of a vector w on the block j; they are obtained by splitting the sum over $\prod_j B_j$ according to one coordinate, `sum_pi_split`.) Σ_{ik} is the fraction of $x \in \mathcal{X}$ with $x_i = x_k = 1$. Within a block, two distinct coordinates are never both 1, and $x_i = 1$ for a fraction $1/d_j$ of the elements; across blocks $j \neq \ell$, the positions of the ones in different blocks are chosen independently and uniformly (the set $\mathcal{X}$ is in bijection with $\prod_j B_j$), so the fraction is $1/(d_j d_\ell)$. For $v \in R_j$ (support in B_j, coordinates summing to 0): for $i \in B_j$, $(\Sigma v)_i = \frac{1}{d_j} v_i$, and for $i \in B_\ell$ with $\ell \neq j$, $(\Sigma v)_i = \frac{1}{d_j d_\ell}\sum_{k\in B_j} v_k = 0$; thus $\Sigma v = v/d_j$. For μ: for $i \in B_j$, $(\Sigma\mu)_i = \frac{1}{d_j}\cdot\frac{1}{d_j} + \sum_{\ell\neq j}\sum_{k\in B_\ell}\frac{1}{d_j d_\ell}\cdot\frac{1}{d_\ell} = \frac{1}{d_j}\sum_{\ell\le m}\frac{1}{d_\ell} = \kappa\mu_i$, so $\Sigma\mu = \kappa\mu$ and $\Sigma u_0 = \kappa u_0$. Consequently $u_0^\top \Sigma u_0 = \kappa$, $u_0^\top \Sigma Q_j = \kappa u_0^\top Q_j = 0$, $Q_j^\top \Sigma Q_\ell = \frac{1}{d_\ell} Q_j^\top Q_\ell = \frac{1}{d_\ell}\mathbf{1}\{j = \ell\} I_{d_j - 1}$ (Lemma **??**), which is the block-diagonal form of $U^\top \Sigma U$; it is positive definite since its diagonal entries are positive. The last claim is the identity $U^\top(\sum_x \lambda_x xx^\top)U = \sum_x \lambda_x (U^\top x)(U^\top x)^\top$. □

Lemma 120 (Reduction of the multi-task width to block maxima). *$Maiti2026Power.intrinsicGw_le_gaussianWi$*
$\mathcal{N}(0, I_{d_j-1})$, $j \le m$. Then the intrinsic width of Definition **??** *satisfies*

$$w_{\mathrm{int}}(\mathcal{X}) \;\le\; \sum_{j\le m} \sqrt{d_j}\;\mathbb{E}\Big[\max_{k\le d_j}\big(H_{d_j} g^{(j)}\big)_k\Big].$$

Proof. def:width, lem:width$_m atrix_w d, lem : sqrt_p rops, lem : multitask_d esign_c ov, lem :$
$multitask_b asis, lem : gauss_p i(Lean : thegeneralstatement$`intrinsicGw_le_gaussianWidth`$saysthatforany\Sigma$
$\mathcal{M}(\mathcal{X})$ and any $S \succeq 0$ with range in $V = \operatorname{span}\mathcal{X}$ and $\Sigma S = \mathrm{id}$ on V, one has $w_{\mathrm{int}}(\mathcal{X}) \le \mathbb{E}\sup_{x\in\mathcal{X}}\langle x, G\rangle$ with $G \sim \mathcal{N}(0, S)$: with U the coordinate matrix of the standard orthonormal basis of V, $A := U^\top \Sigma U$ is a positive definite design matrix of $U^\top\mathcal{X}$ with $A^{-1} = U^\top S U$, and $\mathcal{N}(0, U^\top S U)$ is the image of $\mathcal{N}(0, S)$ by $U^\top$ while $\langle U^\top x, U^\top y\rangle = \langle x, y\rangle$ for $x \in V$. For the multi-task set, Σ is the uniform design and $S = L^2$ with $Lv := \kappa^{-3/2}\langle \mu, v\rangle\mu + \sum_j \sqrt{d_j} P_j v$ (`mtL`), whose coordinates in the block j are $\langle w_j, v\rangle + \sqrt{d_j} v_k$ for a fixed vector w_j (`mtL_apply_mk`); the block maxima of Lg, $g \sim \mathcal{N}(0, I_d)$, are then $\langle w_j, g\rangle + \sqrt{d_j}\max_{k\in B_j} g_k$, and the first term has mean zero.) By Definition **??** with the matrix U of Lemma **??**, $w_{\mathrm{int}}(\mathcal{X}) = w(U^\top\mathcal{X})$, where $U^\top\mathcal{X}$ is a spanning action set in $\mathbb{R}^r$, $r = d - m + 1$, so that Definition **??** applies to it; and by Lemma **??**, $\Sigma_r \in \mathcal{M}(U^\top\mathcal{X}) \cap \{$positive definite $r\times r$ matrices$\}$, so

$$w_{\mathrm{int}}(\mathcal{X}) \le w(U^\top\mathcal{X}; \Sigma_r) = \mathbb{E}\Big[\max_{x\in\mathcal{X}}\langle U^\top x, \Sigma_r^{-1/2} g_r\rangle\Big], \qquad g_r \sim \mathcal{N}(0, I_r).$$

The matrix Σ_r is diagonal with positive entries, so $\Sigma_r^{-1/2} = \operatorname{diag}(\kappa^{-1/2}, \sqrt{d_1} I_{d_1-1}, \dots, \sqrt{d_m} I_{d_m-1})$ (this diagonal matrix is positive definite and its square is Σ_r^{-1}; by uniqueness of the positive square root, Lemma **??**, it is $\Sigma_r^{-1/2}$). Write $g_r = (g_0, g^{(1)}, \dots, g^{(m)})$ with $g_0 \sim \mathcal{N}(0,1)$ and $g^{(j)} \sim \mathcal{N}(0, I_{d_j-1})$ independent (Lemma **??**). For $x \in \mathcal{X}$ with ones at positions $k_j \in B_j$, $U^\top x = (\langle u_0, x\rangle, Q_1^\top x, \dots, Q_m^\top x)$ with $\langle u_0, x\rangle = \langle \mu, x\rangle/\sqrt{\kappa} = \kappa/\sqrt{\kappa} = \sqrt{\kappa}$ and $Q_j^\top x = H_{d_j}^\top e_{k_j}$ ($e_{k_j} \in \mathbb{R}^{d_j}$ with the block-local index). Hence

$$\langle U^\top x, \Sigma_r^{-1/2} g_r\rangle = \kappa^{-1/2}\sqrt{\kappa}\, g_0 + \sum_{j\le m}\sqrt{d_j}\langle H_{d_j}^\top e_{k_j}, g^{(j)}\rangle = g_0 + \sum_{j\le m}\sqrt{d_j}\,\big(H_{d_j} g^{(j)}\big)_{k_j}.$$

As x ranges over $\mathcal{X}$, $(k_j)_j$ ranges over all of $\prod_j B_j$ independently across blocks, so the maximum over x is $g_0 + \sum_j \sqrt{d_j}\max_{k\le d_j}(H_{d_j} g^{(j)})_k$. Taking expectations and using $\mathbb{E}g_0 = 0$ gives the claim (each maximum is integrable by Lemma **??**). $\square$

Lemma 121 (Maximum of centered Gaussians). *$Maiti2026Power.integral_iSup_mtL_le, ciSup_const_add_const_muld$*
≥ 2, $Z \sim \mathcal{N}(0, I_n)$, $\bar{Z} := \frac{1}{n}\sum_{k\le n} Z_k$ and $g \sim \mathcal{N}(0, I_{n-1})$. Then

$$\mathbb{E}\Big[\max_{k\le n}(H_n g)_k\Big] = \mathbb{E}\Big[\max_{k\le n}(Z_k - \bar{Z})\Big] = \mathbb{E}\Big[\max_{k\le n} Z_k\Big] \le \sqrt{2\log n}.$$

Proof. lem:gauss$_l inear_i mage, lem : gauss_l aw_o f_c ov, lem : gauss_m ax_u pper, lem :$
$helmert_p rops(Lean : intheformalizationthecenteredmaximumappearsdirectlyas\max_k(g_k-$

$\bar{g}) = \max_k g_k - \bar{g}$ with $\mathbb{E}\bar{g} = 0$, so no change of variables is needed; `integral_iSup_mtL_le` is the statement $\mathbb{E}\max_{k\in B_j}(Lg)_k \le \sqrt{2d_j \log d_j}$.) By Lemma **??**, $H_n g \sim \mathcal{N}(0, H_n H_n^\top) = \mathcal{N}(0, P)$ with $P := I_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^\top$ (Lemma **??**), and $Z - \bar{Z}\mathbf{1}_n = PZ \sim \mathcal{N}(0, PP^\top) = \mathcal{N}(0, P)$ since P is a symmetric projection. Two centered Gaussian vectors with the same covariance have the same law (Lemma **??**), so $\mathbb{E}\max_k (H_n g)_k = \mathbb{E}\max_k (Z_k - \bar{Z})$. Next, $\max_k(Z_k - \bar{Z}) = \max_k Z_k - \bar{Z}$ and $\mathbb{E}\bar{Z} = 0$. Finally $\mathbb{E}\max_k Z_k \le \sqrt{2\log n}$ by Lemma **??** with $\sigma^2 = 1$. ☐

Proposition 122 (Width of the multi-task set)**.** *$Maiti2026Power.intrinsicGw_multitaskSet_ledef : multitask_se$* $\sum_{j\le m}\sqrt{2d_j \log d_j}$. Equivalently, for the matrix U of Lemma **??**, the spanning action set $U^\top \mathcal{X} \subset \mathbb{R}^{d-m+1}$ has Gaussian width $w(U^\top\mathcal{X}) \le \sum_{j\le m}\sqrt{2d_j\log d_j}$.

Proof. lem:multitask$_w$*idth$_r$eduction, lem : centered$_g$aussian$_m$axCombineLemma* **??***withLemma* **??***appliedto* $= \mathrm{d}_j \ge 2$): $\sqrt{d_j}\,\mathbb{E}\max_k (H_{d_j} g^{(j)})_k \le \sqrt{d_j}\sqrt{2\log d_j}$. ☐

Theorem 123 (A finite set with a small width; Theorem 5 of the paper)**.** *$Maiti2026Power.width_separationdef : multitask_set, def : width_subspace, prop : width_le_d, prop : width_finiteL$* ≥ 2 *and consider the multi-task action set $\mathcal{X}$ with $m-1$ blocks of size $d_j = 2$ ($j < m$) and one block of size $d_m = m^2$, so that $d = m^2 + 2m - 2$ and $|\mathcal{X}| = 2^{m-1}m^2$. Write $r := \dim \operatorname{span}(\mathcal{X})$ and let $U \in \mathbb{R}^{d\times r}$ be as in Definition* **??***, so that $U^\top\mathcal{X} \subset \mathbb{R}^r$ is a finite spanning action set with $|U^\top\mathcal{X}| = |\mathcal{X}|$ and $w(U^\top\mathcal{X}) = w_{\mathrm{int}}(\mathcal{X})$. Then:*

(i) $r = \dim\operatorname{span}(\mathcal{X}) = m^2 + m - 1$, *and* $d/2 \le r \le d$;

(ii) $w_{\mathrm{int}}(\mathcal{X}) \le 2(m-1)\sqrt{\log 2} + 2m\sqrt{\log m} \le 3\sqrt{d\log d}$;

(iii) $\sqrt{d\log|\mathcal{X}|} \ge 0.49\, d^{3/4}$, *and also* $\sqrt{r\log|\mathcal{X}|} \ge 0.34\, d^{3/4}$ *for the intrinsic dimension r.*

In particular the generic upper bounds $w(U^\top\mathcal{X}) \le r$ (Proposition **??***) and $w(U^\top\mathcal{X}) \le \sqrt{2r\log|\mathcal{X}|}$ (Proposition* **??***), both applicable to the spanning set $U^\top\mathcal{X} \subset \mathbb{R}^r$, are of order at least $d^{3/4}$, while the intrinsic width $w_{\mathrm{int}}(\mathcal{X}) = w(U^\top\mathcal{X})$ is $O(\sqrt{d\log d})$.*

Proof. prop:width$_m$*ultitask, lem : multitask$_b$asis, prop : width$_l$e$_d$, prop : width$_f$inite(i)Lemma* **??** $:\dim\operatorname{span}\mathcal{X}$ $d - m + 1 = m^2 + m - 1$. It is at most d, and $d - m + 1 \ge d/2$ is equivalent to $d \ge 2m - 2$, which holds.

(ii) Proposition **??**: $w_{\mathrm{int}}(\mathcal{X}) \le (m-1)\sqrt{2\cdot 2\log 2} + \sqrt{2m^2\log(m^2)} = 2(m-1)\sqrt{\log 2} + 2m\sqrt{\log m}$. Since $m^2 \le d \le 2m^2$ (the second inequality is $2m - 2 \le m^2$), we have $m \le \sqrt{d}$ and $\log m \le \frac{1}{2}\log d$; also $\log 2 \le \frac{1}{2}\log d$ because $d \ge 6 \ge 4$. Hence $w_{\mathrm{int}}(\mathcal{X}) \le 2m(\sqrt{\log 2} + \sqrt{\log m}) \le 2\sqrt{d}\cdot 2\sqrt{\frac{1}{2}\log d} = 2\sqrt{2}\sqrt{d\log d} \le 3\sqrt{d\log d}$.

(iii) $\log|\mathcal{X}| = (m-1)\log 2 + 2\log m \ge (m-1)\log 2 \ge \frac{m}{2}\log 2$ for $m \ge 2$, and $m \ge \sqrt{d/2}$ from $d \le 2m^2$; so $\log|\mathcal{X}| \ge \frac{\log 2}{2\sqrt{2}}\sqrt{d} \ge 0.245\sqrt{d}$ and $\sqrt{d\log|\mathcal{X}|} \ge \sqrt{0.245}\, d^{3/4} \ge 0.49\, d^{3/4}$. With $r \ge d/2$, $\sqrt{r\log|\mathcal{X}|} \ge \sqrt{0.1225}\, d^{3/4} \ge 0.34\, d^{3/4}$.

The "in particular" clause: $U^\top\mathcal{X}$ is a finite spanning action set in $\mathbb{R}^r$ with $|U^\top\mathcal{X}| = |\mathcal{X}|$ (the map $x \mapsto U^\top x$ is injective on $\operatorname{span}\mathcal{X}$), so Propositions **??** and **??** apply to it and give $w(U^\top\mathcal{X}) \le r$ and $w(U^\top\mathcal{X}) \le \sqrt{2r\log|\mathcal{X}|}$; the right-hand sides are at least $d/2$ and $0.34\sqrt{2}\,d^{3/4}$ respectively by (i) and (iii), whereas $w(U^\top\mathcal{X}) = w_{\mathrm{int}}(\mathcal{X}) \le 3\sqrt{d\log d}$ by (ii). □

Remark 124. The paper states Theorem 5 as: there is a finite $\mathcal{X}$ with $\dim\operatorname{span}\mathcal{X} = \Theta(d)$, $w(\mathcal{X}) \le O(\sqrt{d\log d})$ (the width being understood as the intrinsic width $w_{\mathrm{int}}(\mathcal{X})$, since $\mathcal{X}$ is not spanning) and $\sqrt{d\log|\mathcal{X}|} \ge \Omega(d^{3/4})$; the explicit form above is what will be formalized. The intuition is that the width of the multi-task set behaves like $\sum_j \sqrt{d_j\log d_j}$, whereas the cardinality bound only sees $\log|\mathcal{X}| = \sum_j \log d_j$ and the dimension bound only sees $\sum_j d_j$; blocks of size 2 are cheap for the width but expensive for the cardinality bound. Section **??** shows a matching lower bound of order $(\sum_j \sqrt{d_j})^2/\varepsilon^2$ on the sample complexity of adaptive algorithms for this set.

Chapter 7

Structured sets with only logarithmic gains from adaptivity

This chapter formalizes Section 2.4 of the paper. Its first part (Section **??**) makes the fixed-design algorithm of Chapter **??** adaptive, following Appendix C of the paper: the finite action set is partitioned into regions, a fixed design identifies a candidate arm in each region, and Median Elimination (Chapter **??**) selects among the candidates. The resulting sample complexity, Theorem 6 of the paper (Theorem **??**), is $O(d\log((|\mathcal{X}|/d)_+/\delta)/\varepsilon^2)$, a $\log d$ improvement over the finite-set bound of Theorem **??** in some regimes. The second part (Section **??**) proves, for adaptive and randomized algorithms, the lower bounds on the δ-independent term of the sample complexity summarized in Table 1 of the paper: multi-task bandits (Theorem **??**), the two hypercubes (Theorems **??** and **??**), the m-sets (Theorem **??**) and the unit ball (Corollary **??**, from Chapter **??**).

Inputs: the G-optimal design and the rounding theorem of Chapter **??**, the least-squares estimator and the concentration of Chapters **??** and **??**, Median Elimination (Theorem **??**, Corollary **??**) and the sequential composition of Lemma **??**; for the lower bounds, Pinsker's inequality and the divergence decomposition of Chapter **??**. Deviations from the paper (decided in the overview): the algorithm only uses the G-optimal design (not the width minimizer), the lower bounds are proved directly for randomized algorithms without Yao's principle (the uniform average over a finite family of instances is a finite sum of probabilities under each instance), and the multi-task lower bound treats all block sizes $d_j \geq 2$ at once by a single Pinsker–Cauchy–Schwarz argument instead of the paper's two cases $d_j \geq 100$ and $d_j < 100$. All constants are explicit.

7.1 An adaptive version of the fixed-design algorithm

Throughout this section $\mathcal{X} \subset \mathbb{R}^d$ is a *finite spanning* action set ($\mathrm{span}(\mathcal{X}) = \mathbb{R}^d$, hence $|\mathcal{X}| \geq d$), $\varepsilon \in (0, 1]$ and $\delta \in (0, 1)$. The spanning hypothesis is what makes the G-optimal design of Theorem **??** available; it is repeated in each statement.

Definition 125 (Region-based adaptive algorithm)**.** Maiti2026Power.regionCandidates, Maiti2026Power.regionAlg, Maiti2026Power.start$_r egionAlgdef : bai_a lg, def : regionalized_w idth, thm : kiefer_v$ $= (\mathrm{R}_1, \ldots, R_K)$, $K \geq 1$, be a partition of $\mathcal{X}$ into nonempty subsets. Let $\lambda_G \in \triangle_{\mathcal{X}}$ and $A_G = A(\lambda_G) \succ 0$ be the G-optimal design of Theorem **??** (which exists since $\mathcal{X}$ is spanning), and let $w_{\mathcal{R}} := w(A_G; \mathcal{X}, \mathcal{R})$ (Definition **??**). Set

$$T_1 := \left\lceil \frac{640 \left(w_{\mathcal{R}}^2 + d \log(4/\delta) \right)}{\varepsilon^2} \right\rceil,$$

which satisfies $T_1 \geq 640 \log 4 \cdot d \geq 4d$. The *region algorithm* with parameters $(\mathcal{R}, \varepsilon, \delta)$ is the sequential composition (Lemma **??**) of the following two phases.

(1) *Fixed design.* Let $x_0, \ldots, x_{T_1 - 1} \in \operatorname{supp} \lambda_G$ be the design given by Theorem **??** applied to λ_G and T_1, so that $\Sigma_{T_1} := \sum_{t<T_1} x_t x_t^\top \succeq \frac{T_1}{4} A_G$. Play it, observe $y \in \mathbb{R}^{T_1}$, and form the least-squares estimator $\hat{\theta} := \Sigma_{T_1}^{-1} X^\top y$ (Definition **??**). For each $i \leq K$ let $x^{(i)} := s_i(\hat{\theta})$, where $s_i : \mathbb{R}^d \to R_i$ is a measurable exact argmax selector for the finite set R_i (Lemma **??** with $\eta = 0$): $\langle x^{(i)}, \hat{\theta} \rangle = \max_{x \in R_i} \langle x, \hat{\theta} \rangle$.

(2) *Median Elimination.* Run Median Elimination (Definition **??**) with parameters $(\varepsilon/2, \delta/2)$ on the K-armed bandit induced by the candidates $x^{(1)}, \ldots, x^{(K)}$ (Corollary **??**: arm a plays $x^{(a)}$); it uses exactly $N_{ME}(K, \varepsilon/2, \delta/2)$ rounds and returns an index $\hat{a}$.

The recommendation is $\hat{x} := x^{(\hat{a})}$. The budget is the deterministic number $T := T_1 + N_{ME}(K, \varepsilon/2, \delta/2)$.

Lean remark. The region algorithm is the phased algorithm (Lemma **??**) `regionAlg`, obtained by prepending to Median Elimination on the state (c, S) (Definition **??**, `PhasedAlg.cons`) a first phase of length T_1 which plays the design x and whose observations y set the state to $(\,(s_i(\hat{\theta}(y)))_{i \leq K},\ \mathrm{Fin}\, K\,)$ (`regionCandidates`); the regions need only cover $\mathcal{X}$ (they need not be disjoint). The fixed design of phase (1) is obtained from the existence statement of Theorem **??** by choice, in Corollary **??**. The recommendation is the surviving candidate (`MedianElim.meOut`) and the budget is $T_1 + N_{\mathrm{ME}}(K, \varepsilon/2, \delta/2)$ (`start_regionAlg`).

Lemma 126 (Width of a region under a scalar Loewner bound)**.** *Maiti2026Power.gwMat$_l e_o f_s mul_l edef : width_($* $\subset \mathbb{R}^d$ *be nonempty and compact (typically a region of a partition of $\mathcal{X}$), $A, B \in \mathcal{S}_{++}^d$ and $c > 0$ with $B \succeq cA$. Then, in the notation of Definition* **??** *with the* *index set R (for $g \sim \mathcal{N}(0, I_d)$, $w(R; A) = \mathbb{E}[\max_{x\in R}\langle x, A^{-1/2}g\rangle]$, well defined by Lemma **??**),*

$$w(R; B) \leq \frac{1}{\sqrt{c}}\, w(R; A), \qquad \textit{i.e.} \qquad \mathbb{E}\Big[\max_{x\in R}\langle x, B^{-1/2}g\rangle\Big] \leq \frac{1}{\sqrt{c}}\, \mathbb{E}\Big[\max_{x\in R}\langle x, A^{-1/2}g\rangle\Big].$$

Proof. lem:width$_mono, lem: width_homog (Formalized for c = T/4, the case of the rounding theorem.) Bothlemm$
$\in \mathcal{S}_{++}^d$ and $cA \preceq B$, Lemma **??** (monotonicity in the Loewner order, index set R) gives $w(R; B) \leq w(R; cA)$, and Lemma **??** (homogeneity, index set R) gives $w(R; cA) = c^{-1/2} w(R; A)$. $\square$

Lemma 127 (The candidate of the optimal region is good)**.** *Maiti2026Power.one$_s$ub$_l$e$_m$easureReal$_r$egionCand*
$\mathbb{R}^d$, let $x_\star \in \arg\max_{x\in\mathcal{X}}\langle x, \theta\rangle$ and let i_* be the index of the region containing $x_\star$. Under ν_θ, phase (1) of the region algorithm satisfies

$$\mathbb{P}_\theta\Big(\langle x^{(i_*)}, \theta\rangle \geq \max_{x\in\mathcal{X}}\langle x, \theta\rangle - \frac{\varepsilon}{2}\Big) \geq 1 - \frac{\delta}{2}.$$

Proof. cor:fixed$_design_from_distribution, cor: g_optimal_norm_bound, lem: least_squares_law, lem:$
$log_gains_region_width_bound, cor: sup_bound_symmetric, lem: width_matrix_wd (In Lean the statement is about the nc$
$the observations are \langle x_t, \theta\rangle + \eta_t$ with $\eta \sim \mathcal{N}(0, I_{T_1})$, and the conclusion holds with probability at least $1 - \delta'$ over η for any design with $\Sigma_{T_1} \succ 0$, $\|x\|^2_{\Sigma_{T_1}^{-1}} \leq \sigma^2$ on $\mathcal{X}$ and $2w(R_i; \Sigma_{T_1}) + 2\sigma\sqrt{2c_{\mathrm{G}}\log(1/\delta')} \leq \varepsilon'$ for all i; the arithmetic below is `region_phase1_arith`.)

Let $\Delta := \hat{\theta} - \theta$ and $D := \max_{x,x'\in R_{i_*}}\langle x - x', \Delta\rangle$.

Deterministic step. Since $x^{(i_*)}$ maximizes $\langle\cdot, \hat{\theta}\rangle$ over $R_{i_*} \ni x_\star$,

$$\langle x^{(i_*)}, \theta\rangle = \langle x^{(i_*)}, \hat{\theta}\rangle - \langle x^{(i_*)}, \Delta\rangle \geq \langle x_\star, \hat{\theta}\rangle - \langle x^{(i_*)}, \Delta\rangle = \langle x_\star, \theta\rangle + \langle x_\star - x^{(i_*)}, \Delta\rangle \geq \langle x_\star, \theta\rangle - D.$$

So it suffices to show $\mathbb{P}_\theta(D \leq \varepsilon/2) \geq 1 - \delta/2$.

Law of Δ. By Corollary **??** (applied to λ_G and the design of phase (1)), $\Sigma_{T_1} \succ 0$ and $\|x\|^2_{\Sigma_{T_1}^{-1}} \leq \frac{4}{T_1}\|x\|^2_{A_G^{-1}} \leq \frac{4d}{T_1}$ for all $x \in \mathcal{X}$ (Corollary **??**). By Lemma **??**, $\Delta \sim \mathcal{N}(0, \Sigma_{T_1}^{-1})$.

Expectation. Since the expectation of the supremum of the process $x \mapsto \langle x, \Delta\rangle$ over the compact index set R_{i_*} only depends on the law of Δ (Lemma **??** with the index set R_{i_*}), $\mathbb{E}\max_{x\in R_{i_*}}\langle x, \Delta\rangle = w(R_{i_*}; \Sigma_{T_1})$, and Lemma **??** with $B = \Sigma_{T_1} \succeq \frac{T_1}{4}A_G$, $c = T_1/4$, gives $w(R_{i_*}; \Sigma_{T_1}) \leq \frac{2}{\sqrt{T_1}}w(R_{i_*}; A_G) \leq \frac{2w_{\mathcal{R}}}{\sqrt{T_1}}$, the last step by Definition **??** ($w_{\mathcal{R}} = \max_i w(R_i; A_G)$).

Concentration. Corollary **??** for the centered Gaussian process $x \mapsto \langle x, \Delta\rangle$ on the finite index set R_{i_*}, with $\sigma^2 := \max_{x\in R_{i_*}}\|x\|^2_{\Sigma_{T_1}^{-1}} \leq 4d/T_1$ and failure probability $\delta/2$: $\mathbb{E}D = 2\mathbb{E}\max_{x\in R_{i_*}}\langle x, \Delta\rangle$ and, with probability at least $1 - \delta/2$,

$$D \leq \mathbb{E}D + 2\sigma\sqrt{2c_{\mathrm{G}}\log(2/\delta)} \leq \frac{4w_{\mathcal{R}}}{\sqrt{T_1}} + \frac{4\sqrt{d}}{\sqrt{T_1}}\sqrt{2c_{\mathrm{G}}\log(2/\delta)} = \frac{4}{\sqrt{T_1}}\Big(w_{\mathcal{R}} + \sqrt{2c_{\mathrm{G}}\, d\log(2/\delta)}\Big).$$

Arithmetic. Using $(a+b)^2 \le 2a^2 + 2b^2$, $2c_{\mathrm{G}} = \pi^2/2 < 5$ and $\log(2/\delta) \le \log(4/\delta)$,

$$\frac{16}{T_1}\Big(w_{\mathcal{R}} + \sqrt{2c_{\mathrm{G}}\, d\log(2/\delta)}\Big)^2 \le \frac{32\,(w_{\mathcal{R}}^2 + 5\,d\log(4/\delta))}{T_1} \le \frac{160\,(w_{\mathcal{R}}^2 + d\log(4/\delta))}{T_1} \le \frac{\varepsilon^2}{4},$$

by the choice $T_1 \ge 640(w_{\mathcal{R}}^2 + d\log(4/\delta))/\varepsilon^2$. Hence $D \le \varepsilon/2$ on the good event. □

Lemma 128 (Covering a finite set by small regions)**.** *Set.exists*$_c$*over*$_n$*card*$_l$*eAfinitesetof* m $\ge d \ge 1$ *elements is covered by* d *nonempty subsets of at most* $\lfloor m/d\rfloor + 1$ *elements each.*

Proof. Enumerate the set as $y_0, \ldots, y_{m-1}$ and let the i-th region be $\{y_k : k \equiv i \pmod d\}$; it is nonempty since $i < d \le m$ and has at most $\lfloor m/d\rfloor + 1$ elements. □

Theorem 129 (Theorem 6 of the paper)**.** *Maiti2026Power.isPAC*$_r$*egionAlg*, *Maiti2026Power.regionGood*, *Ma*
$(0,1]$, $\delta \in (0,1)$.

(i) For every family $\mathcal{R} = (R_1, \ldots, R_K)$ *of nonempty subsets covering* $\mathcal{X}$*, the region algorithm of Definition* **??** *is* (ε,δ)*-PAC on* $\mathcal{X}$*, with budget* $T_1 + N_{ME}(K, \varepsilon/2, \delta/2)$.

(ii) Let $m := |\mathcal{X}|$, $K := d$*, and let* $\mathcal{R}$ *be a family of* d *nonempty subsets covering* $\mathcal{X}$*, of size at most* $p := \lfloor m/d\rfloor + 1$ *(it exists since* $m \ge d$*, Lemma* **??***). Then the budget of the region algorithm satisfies*

$$T \le C_{\mathrm{reg}}\,\frac{d\,\big(\log((m/d)_+) + \log(4/\delta)\big)}{\varepsilon^2}, \qquad C_{\mathrm{reg}} := 1281 + 6\,C_{ME},$$

where $(u)_+ := \max(1,u)$ *and* C_{ME} *is the constant of Lemma* **??***. In particular, for* $\delta \le 1/4$, $T \le 2C_{\mathrm{reg}}\, d\log\big((m/d)_+/\delta\big)/\varepsilon^2$.

Corollary 130 (Theorem 6 of the paper, existential form)**.** *Maiti2026Power.exists*$_i$*sPAC*$_o$*f*$_f$*initethm* : *log*$_g$*ain*
$:= |\mathcal{X}|$, $\varepsilon \in (0,1]$ *and* $\delta \in (0,1)$*. There exist a budget* $T \le C_{\mathrm{reg}}\, d\,(\log((m/d)_+) + \log(4/\delta))/\varepsilon^2$*, with* $C_{\mathrm{reg}} = 1281 + 6C_{ME} = 97281$*, and an identification algorithm with budget* T *that is* (ε,δ)*-PAC on* $\mathcal{X}$*.*

Proof. thm:log$_g$$ains, lem : me_budget, lem : cover_regions, thm : kiefer_wolfowitz, thm : roundingTaketheregionalgorithmofTheorem$ **??**$(ii); itsbudgetisboundedby(ii)with\mathrm{C}_{ME} =$ 16000 from Lemma **??**. (In Lean the case $d = 0$ is handled separately, as in Corollary **??**.) □

Proof. lem:region$_p$$hase1, cor : median_elimination_linear, lem : composition, lem : me_budget, prop : width_partition, lem : phased, lem : cover_regions$**Lean.**$(i)is$`isPAC_regionAlg`$, bythephase-by-phaseanalysisofLemma$ **??**$withthegoodstates$`regionGood` $: afterthefirstphase, thesurvivingsetis$Fin Ka
optimal (Lemma **??**); each round of Median Elimination preserves the goodness (Lemma **??**) with the target value $\max_x\langle x,\theta\rangle - \varepsilon/2$; the final surviving candidate has regret at most $\varepsilon/2 + \sum_\ell \varepsilon_\ell(\varepsilon/2) \le \varepsilon$. The failure probabilities $\delta/2 + \sum_\ell \delta_\ell(\delta/2) \le \delta$ add up over the phases. (ii) is the arithmetic `logGains_budget_arith` with $p = \lfloor m/d\rfloor + 1 \le 2(m/d)_+$.

(i) Fix θ, $x_\star$ and i_* as in Lemma **??**. Let E_1 be the event $\{\langle x^{(i_*)}, \theta\rangle \geq \max_x\langle x, \theta\rangle - \varepsilon/2\}$, which depends on the phase-(1) history only and has probability at least $1-\delta/2$ (Lemma **??**). For every phase-(1) history h (with candidates $x^{(1)}, \dots, x^{(K)}$ determined by h), Corollary **??** gives, for the induced K-armed Gaussian bandit with means $\langle x^{(a)}, \theta\rangle$, that the output $\hat{a}$ satisfies $\langle x^{(\hat{a})}, \theta\rangle \geq \max_{a\leq K}\langle x^{(a)}, \theta\rangle - \varepsilon/2$ with probability at least $1-\delta/2$; call $E_2(h)$ this event of the phase-(2) trajectory. By Lemma **??** (with $\delta' = \delta/2$ for the set of histories E_1 and $\delta/2$ for the conditional bound), the composed algorithm satisfies $\mathbb{P}_\theta(E_1 \cap E_2(H_{T_1})) \geq 1-\delta$. On this event,

$$\langle \hat{x}, \theta\rangle = \langle x^{(\hat{a})}, \theta\rangle \geq \langle x^{(i_*)}, \theta\rangle - \frac{\varepsilon}{2} \geq \max_{x\in\mathcal{X}}\langle x, \theta\rangle - \varepsilon,$$

i.e. $r(\hat{x}, \theta) \leq \varepsilon$. Since θ is arbitrary, the algorithm is (ε, δ)-PAC.

(ii) *Phase (1).* By Proposition **??** with $p = \lfloor m/d\rfloor + 1$, $w_{\mathcal{R}}^2 \leq 2d\log p$. Now $\lfloor m/d\rfloor \geq 1$ since $m \geq d$, so $p \leq 2\lfloor m/d\rfloor \leq 2(m/d)_+$ and $\log p \leq \log((m/d)_+) + \log 2$. Hence, using $2\log 2 = \log 4 \leq \log(4/\delta)$,

$$T_1 \leq \frac{640\big(2d\log((m/d)_+) + 2d\log 2 + d\log(4/\delta)\big)}{\varepsilon^2} + 1 \leq \frac{1280\, d\big(\log((m/d)_+) + \log(4/\delta)\big)}{\varepsilon^2} + 1,$$

and $1 \leq d\log(4/\delta)/\varepsilon^2$ (as $\varepsilon \leq 1$, $\log 4 > 1$), so $T_1 \leq 1281\, d(\log((m/d)_+) + \log(4/\delta))/\varepsilon^2$.

Phase (2). Lemma **??** with $K = d$ and parameters $(\varepsilon/2, \delta/2)$ (admissible: $\varepsilon/2 \leq 1$, $\delta/2 < 1$): $N_{ME}(d, \varepsilon/2, \delta/2) \leq C_{ME}\, d\log(6/\delta)/(\varepsilon/2)^2 = 4C_{ME}\, d\log(6/\delta)/\varepsilon^2$. Since $\log(6/\delta) = \log(4/\delta) + \log(3/2)$ and $\log(3/2) < 0.41 < 0.3\log 4 \leq 0.3\log(4/\delta)$, we get $\log(6/\delta) \leq 1.3\log(4/\delta)$ and $N_{ME}(d, \varepsilon/2, \delta/2) \leq 5.2\, C_{ME}\, d\log(4/\delta)/\varepsilon^2 \leq 6C_{ME}\, d\log(4/\delta)/\varepsilon^2$.

Total. $T = T_1 + N_{ME} \leq (1281 + 6C_{ME})\, d(\log((m/d)_+) + \log(4/\delta))/\varepsilon^2$. For $\delta \leq 1/4$, $\log(4/\delta) \leq 2\log(1/\delta)$ (equivalent to $\delta \leq 1/4$), so $\log((m/d)_+) + \log(4/\delta) \leq 2\log((m/d)_+/\delta)$. □

Remark 131 (On the form of the bound). The paper states the budget as $O(d\log((|\mathcal{X}|/d)_+/\delta)/\varepsilon^2)$. That form cannot hold uniformly in $\delta \in (0,1)$ with a fixed constant (the right-hand side tends to $d\log((|\mathcal{X}|/d)_+)/\varepsilon^2$, possibly 0, as $\delta \to 1$, while any algorithm with the guarantee needs a positive budget), which is why the statement carries the additive $\log(4/\delta)$ and the clean form is given for $\delta \leq 1/4$. When $|\mathcal{X}| \leq d$ (only possible with $|\mathcal{X}| = d$, a basis), $\log((m/d)_+) = 0$ and the bound is $O(d\log(1/\delta)/\varepsilon^2)$, matching the adaptive lower bound Theorem **??**. The improvement over Theorem **??** (whose finite-set form is $O(d\log(|\mathcal{X}|/\delta)/\varepsilon^2)$ by Proposition **??**) is the replacement of $\log|\mathcal{X}|$ by $\log(|\mathcal{X}|/d)$, e.g. a factor $\log d$ when $|\mathcal{X}| = d^{1+o(1)}$. For the multi-armed bandit ($\mathcal{X} = \{e_1, \dots, e_d\}$) this is the classical $\log d$ gain of Median Elimination [**?**].

7.2 Adaptive lower bounds for structured sets

All statements of this section concern an arbitrary identification algorithm Alg with budget $T \geq 1$ on the action set at hand (Definition **??**); both the sampling rule and the recommendation may be randomized. The action sets of this section are nonempty compact sets that need not be spanning (the multi-task set of Definition **??** is not): Definitions **??**, **??**, **??** and **??** and the information-theoretic tools (Corollary **??**, Lemma **??**) are stated for arbitrary action sets in the sense of Definition **??**, so nothing below requires spanning. Lower bounds are obtained by exhibiting a finite family of reward vectors $(\theta_I)_{I\in\mathcal{I}}$ and bounding the *uniform average* $\frac{1}{|\mathcal{I}|}\sum_{I\in\mathcal{I}}\mathbb{E}_{\theta_I}[\cdot]$ of expectations under the individual instances. No minimax (Yao) principle is needed: a bound on the average implies the existence of an instance realizing it.

Lean remark. A family of instances is a function $\theta : \mathcal{I} \to \mathbb{R}^d$ on a `Fintype` $\mathcal{I}$ (in most cases $\mathcal{I} = \mathcal{X}$ itself, or a type of subsets), and the uniform average is `( I, E[ I] f) / Fintype.card` . The conclusion "some I has $\mathbb{P}_{\theta_I}(\cdot) \geq p$" follows from `Finset.exists_le_of_sum_le` (or `Finset.exists_lt_of_sum_lt`) applied to the average.

Lemma 132 (Two-instance Pinsker inequality for identification algorithms)**.** $Learning.LinearBandit.IsRun.klDiv_map_out_le_sum_integral, Learning.LinearBandit.IsRun.abs_measureReal_sub$
≥ 1 *on* $\mathcal{X}$*, and let* $\theta, \theta' \in \mathbb{R}^d$*. Define*

$$\mathrm{KL}(\theta,\theta') := \frac{1}{2}\sum_{t<T}\mathbb{E}_\theta\big[\langle x_t, \theta-\theta'\rangle^2\big].$$

Then for every measurable event E *of* $(H_T, \hat{x})$*,*

$$\mathbb{P}_{\theta'}(E) \leq \mathbb{P}_\theta(E) + \sqrt{\mathrm{KL}(\theta,\theta')/2} \qquad \textit{and} \qquad \mathbb{P}_\theta(E) \leq \mathbb{P}_{\theta'}(E) + \sqrt{\mathrm{KL}(\theta,\theta')/2}.$$

Proof. lem:pinsker, cor:divergence$_decomposition_gaussian, lem : kl_data_processing_recommendationByCorollar$
$itonlyusesthat\sup_{x\in\mathcal{X}}\|x\| < \infty$), the Kullback–Leibler divergence between the laws of H_T under ν_θ and $\nu_{\theta'}$ equals $\mathrm{KL}(\theta,\theta')$, and by Lemma **??** the same holds for the laws $\mathbb{P}_\theta$, $\mathbb{P}_{\theta'}$ of $(H_T,\hat{x})$ (the recommendation kernel is common to both). Pinsker's inequality (Lemma **??**) applied to $\mu = \mathbb{P}_\theta$, $\nu = \mathbb{P}_{\theta'}$ and the set E gives the second inequality; applied to the complement E^c it gives $\mathbb{P}_\theta(E^c) - \mathbb{P}_{\theta'}(E^c) \leq \sqrt{\mathrm{KL}/2}$, i.e. the first one. □

Lean remark. All the events used below depend on the recommendation only, so the Lean statement is for events of $\hat{x}$: `klDiv_map_out_le_sum_integral` bounds the divergence between the laws of the outputs of two runs by $\mathrm{KL}(\theta,\theta')$ (data processing on top of the divergence decomposition), and `abs_measureReal_sub_le_of_sum_le` turns any upper bound K on $\mathrm{KL}(\theta,\theta')$ into $|\mathbb{P}_\theta(E) - \mathbb{P}_{\theta'}(E)| \leq \sqrt{K/2}$, in the two-sided form. `abs_measureReal_sub_le_of_sq_le` is the special case $K = TC/2$ of a uniform bound $\langle x, \theta-\theta'\rangle^2 \leq C$ on $\mathcal{X}$, which is what the hypercubes use; the multi-task and m-set arguments need the data-dependent form, whose K involves the expected number of pulls of a coordinate.

Lemma 133 (Three-point inequality)**.** $Learning.LinearBandit.IsRun.one_sub_le_measureReal_add_of_sq_ledef : bai_a$
$\mathbb{R}^d$ *and* $\kappa \geq 0$ *with* $\mathrm{KL}(\theta^0,\theta^+) \leq \kappa$ *and* $\mathrm{KL}(\theta^0,\theta^-) \leq \kappa$ *(notation of Lemma* **??***).* *Then for every measurable event* E *of* $(H_T, \hat{x})$*,*

$$\mathbb{P}_{\theta^+}(E) + \mathbb{P}_{\theta^-}(E^c) \geq 1 - 2\sqrt{\kappa/2}.$$

Proof. lem:two$_p$oint$_p$inskerLemma **??**$with (\theta,\theta') = (\theta^0,\theta^+)$ and the event E^c gives $\mathbb{P}_{\theta^+}(E^c) \le \mathbb{P}_{\theta^0}(E^c) + \sqrt{\kappa/2}$, i.e. $\mathbb{P}_{\theta^+}(E) \ge \mathbb{P}_{\theta^0}(E) - \sqrt{\kappa/2}$; with (θ^0,θ^-) and the event E it gives $\mathbb{P}_{\theta^-}(E^c) \ge \mathbb{P}_{\theta^0}(E^c) - \sqrt{\kappa/2}$. Sum and use $\mathbb{P}_{\theta^0}(E) + \mathbb{P}_{\theta^0}(E^c) = 1$. ☐

Lemma 134 (From an average regret bound to a failure probability)**.** *Learning.LinearBandit.integral$_s$impleReg*
0 and let $(\theta_I)_{I\in\mathcal{I}}$ be a finite family of reward vectors with $Z(\theta_I) = \max_{x,y\in\mathcal{X}}\langle x - y, \theta_I\rangle = 10\varepsilon_*$ for every I. If

$$\frac{1}{|\mathcal{I}|}\sum_{I\in\mathcal{I}} \mathbb{E}_{\theta_I}\big[r(\hat{x},\theta_I)\big] \ge b\,\varepsilon_* \qquad \textit{for some } b > 1,$$

then there exists $I\in\mathcal{I}$ with $\mathbb{P}_{\theta_I}\big(r(\hat{x},\theta_I) > \varepsilon_*\big) \ge (b-1)/9$. Equivalently, writing $G(\hat{x},\theta) := \langle\hat{x},\theta\rangle - \min_{x\in\mathcal{X}}\langle x,\theta\rangle = Z(\theta) - r(\hat{x},\theta)$ for the gap to the worst arm: if $\frac{1}{|\mathcal{I}|}\sum_I \mathbb{E}_{\theta_I}[G(\hat{x},\theta_I)] \le a\,\varepsilon_*$ with $a < 9$, then some I has $\mathbb{P}_{\theta_I}(r(\hat{x},\theta_I) > \varepsilon_*) \ge (9-a)/9$. In particular, when $\min_x\langle x,\theta_I\rangle = 0$ for all I, $G(\hat{x},\theta_I) = \langle\hat{x},\theta_I\rangle$.

Proof. def:regret For each I, $0 \le r(\hat{x},\theta_I) \le Z(\theta_I) = 10\varepsilon_*$ pathwise, so $r \le \varepsilon_*\mathbf{1}\{r \le \varepsilon_*\} + 10\varepsilon_*\mathbf{1}\{r > \varepsilon_*\} \le \varepsilon_* + 9\varepsilon_*\mathbf{1}\{r > \varepsilon_*\}$ and $\mathbb{E}_{\theta_I}[r] \le \varepsilon_* + 9\varepsilon_*\,p_I$ with $p_I := \mathbb{P}_{\theta_I}(r > \varepsilon_*)$. Averaging over I: $b\varepsilon_* \le \varepsilon_* + 9\varepsilon_*\,\frac{1}{|\mathcal{I}|}\sum_I p_I$, so the average of the p_I is at least $(b-1)/9$ and some p_I is at least that. The second formulation is the first with $b = 10 - a$, since $\mathbb{E}[G] = 10\varepsilon_* - \mathbb{E}[r]$. ☐

Lean remark. What the four lower bounds below actually use is the pathwise inequality of the proof, in the contrapositive direction: if the recommendation of Alg is ε_*-optimal with probability at least $1-\delta$ on the instance θ and $0 \le r(\hat{x},\theta) \le Z$ on $\mathcal{X}$ with $\varepsilon_* \le Z$, then

$$\mathbb{E}_\theta\big[r(\hat{x},\theta)\big] \le \varepsilon_* + (Z - \varepsilon_*)\,\delta,$$

which for $Z = 10\varepsilon_*$ reads $\mathbb{E}_\theta[r] \le \varepsilon_*(1+9\delta)$. This is the Lean statement `integral_simpleRegret_le_of_measureReal_le`. Averaging it over the family and comparing with a lower bound $b\varepsilon_*$ on the average expected regret gives $\delta \ge (b-1)/9$ directly, so no bad instance needs to be exhibited (and `Finset.exists_le_of_sum_le` is not needed).

7.2.1 Multi-task bandits

Definition 135 (Hard instances for the multi-task set)**.** Maiti2026Power.mtSum, Maiti2026Power.mtEps, Maiti2026Power.mtParam, Maiti2026Power.mtParam0, Maiti2026Power.sum$_m tEps, Maiti2026Power.inner_m tParam, Maiti2026Power.supportFn_m tParam, Maiti2$
$taskactionsetofDefinition$ **??**$(blocks$B$_1,\dots,B_m$ of sizes $d_j \ge 2$) and $\varepsilon > 0$. Let

$$S_d := \sum_{s\le m}\sqrt{d_s}, \qquad \varepsilon_j := \frac{\varepsilon\sqrt{d_j}}{S_d}\ (j \le m), \qquad \text{so that} \qquad \sum_{j\le m}\varepsilon_j = \varepsilon.$$

Let $\tilde{\mathcal{X}} := \{\tilde{x} \in \{0,1\}^d : \sum_{i\in B_j}\tilde{x}_i \le 1\ \forall j\} \supseteq \mathcal{X}$. For $\tilde{x} \in \tilde{\mathcal{X}}$ the reward vector $\theta_{\tilde{x}} \in \mathbb{R}^d$ is $\theta_{\tilde{x}}[i] := 10\,\varepsilon_j\,\tilde{x}[i]$ for $i \in B_j$. For $j \le m$ let $\mathcal{X}^{(j)} := \{x \in$

$\tilde{\mathcal{X}} : \sum_{i\in B_\ell} x_i = 1$ for $\ell \neq j$, $\sum_{i\in B_j} x_i = 0\}$ and, for $x \in \mathcal{X}^{(j)}$ and $i \in B_j$, $x^{+i} := x + e_i \in \mathcal{X}$. The map $(x, i) \mapsto x^{+i}$ is a bijection from $\mathcal{X}^{(j)} \times B_j$ onto $\mathcal{X}$, and $|\mathcal{X}^{(j)}| = \prod_{s\neq j} d_s$.

For $x \in \mathcal{X}$ and $\tilde{x} \in \tilde{\mathcal{X}}$, $\langle x, \theta_{\tilde{x}}\rangle = \sum_j 10\varepsilon_j \sum_{i\in B_j} x_i \tilde{x}_i \in [0, 10\varepsilon]$; for $\tilde{x} \in \mathcal{X}$, $\max_{x\in\mathcal{X}}\langle x, \theta_{\tilde{x}}\rangle = 10\varepsilon$ (attained at $x = \tilde{x}$) and $\min_{x\in\mathcal{X}}\langle x, \theta_{\tilde{x}}\rangle = 0$ (choose in every block a position different from that of $\tilde{x}$, possible since $d_j \geq 2$); hence $Z(\theta_{\tilde{x}}) = 10\varepsilon$.

Lean remark. The multi-task set does not span $\mathbb{R}^d$. This is allowed: an action set in the sense of Definition **??** is any nonempty compact set, spanning is a separate hypothesis, and Definitions **??**, **??**, **??** and **??** as well as Lemma **??** apply to $\mathcal{X}$ as they stand, so the lower bound below needs no transport. (Should one prefer to work on a spanning set, Lemma **??** identifies identification algorithms on $\mathcal{X}$ with those on $U^\top\mathcal{X} \subset \mathbb{R}^{d-m+1}$, U as in Definition **??**, through $\theta \mapsto U^\top\theta$; this is what the non-adaptive upper bound of Remark **??** uses.) In the family $(\theta_{\tilde{x}})$, the coordinate $i \in B_j$ of $\hat{x}$ is written $\hat{x}_i \in \{0, 1\}$, and $\hat{x}$ has exactly one nonzero coordinate in each block. In Lean the instances are indexed by the arms themselves, in the form $\kappa : \prod_j B_j$ of Definition **??**: `mtParam`(d, ε, κ) has the value $10\varepsilon_j$ at the position κ_j of the block j, and the alternative `mtParam0`$(d, \varepsilon, \kappa, j)$ is the same vector with the block j zeroed. The relaxed set $\tilde{\mathcal{X}}$ and the sets $\mathcal{X}^{(j)}$ are not needed: the alternative of the block j is a reward vector like any other (it is not required to lie in $\mathcal{X}$), and it depends on κ only through the coordinates $\ell \neq j$ (`mtParam0_update`).

Lemma 136 (Average value of a block)**.** *def:multitask$_i$nstances, def : bai$_a$lgLetAlgbeanidentificationalgorithm taskset X. Fix $j \leq m$ and $x \in \mathcal{X}^{(j)}$. For $i \in B_j$ let $p_i := \mathbb{P}_{\theta_x}(\hat{x}_i = 1)$ and $N_i := \sum_{t<T} x_t[i]$. Then*

(i) $\sum_{i\in B_j} p_i = 1$ *and* $\sum_{i\in B_j} \mathbb{E}_{\theta_x}[N_i] = T$*;*

(ii) $\mathrm{KL}(\theta_x, \theta_{x^{+i}}) = 50\,\varepsilon_j^2\,\mathbb{E}_{\theta_x}[N_i]$ *for every $i \in B_j$ (notation of Lemma* **??***);*

(iii) $\mathbb{P}_{\theta_{x^{+i}}}(\hat{x}_i = 1) \leq p_i + 5\varepsilon_j\sqrt{\mathbb{E}_{\theta_x}[N_i]}$ *for every $i \in B_j$;*

(iv) the block-j value, averaged over the d_j instances x^{+i}, satisfies

$$\frac{1}{d_j}\sum_{i\in B_j} \mathbb{E}_{\theta_{x^{+i}}}\Big[\sum_{i'\in B_j} \hat{x}_{i'}\,\theta_{x^{+i}}[i']\Big] \leq \frac{10\,\varepsilon_j}{d_j}\Big(1 + 5\,\varepsilon_j\sqrt{d_j\,T}\Big).$$

Proof. lem:two$_p$oint$_p$insker, def : *multitask$_i$nstances(i)Bothandevery*x_t belong to $\mathcal{X}$, so they have exactly one coordinate equal to 1 in B_j: $\sum_{i\in B_j} \hat{x}_i = 1$ pathwise, and $\sum_{i\in B_j} N_i = T$ pathwise; take expectations.

(ii) $\theta_x - \theta_{x^{+i}} = -10\varepsilon_j e_i$, so $\langle x_t, \theta_x - \theta_{x^{+i}}\rangle^2 = 100\varepsilon_j^2 x_t[i]^2 = 100\varepsilon_j^2 x_t[i]$ (as $x_t[i] \in \{0, 1\}$), and $\mathrm{KL}(\theta_x, \theta_{x^{+i}}) = \frac{1}{2}\sum_t \mathbb{E}_{\theta_x}[100\varepsilon_j^2 x_t[i]] = 50\varepsilon_j^2\mathbb{E}_{\theta_x}[N_i]$.

(iii) Lemma **??** with $\theta = \theta_x$, $\theta' = \theta_{x^{+i}}$ and $E = \{\hat{x}_i = 1\}$: $\mathbb{P}_{\theta_{x^{+i}}}(E) \leq p_i + \sqrt{25\varepsilon_j^2\mathbb{E}_{\theta_x}[N_i]}$.

(iv) In block j the vector $\theta_{x^{+i}}$ has the single nonzero coordinate $\theta_{x^{+i}}[i] = 10\varepsilon_j$, so $\sum_{i' \in B_j} \hat{x}_{i'} \theta_{x^{+i}}[i'] = 10\varepsilon_j \hat{x}_i$ and its expectation is $10\varepsilon_j \mathbb{P}_{\theta_{x^{+i}}}(\hat{x}_i = 1)$. Summing (iii) over $i \in B_j$ and using (i) and the Cauchy–Schwarz inequality,

$$\sum_{i \in B_j} \mathbb{P}_{\theta_{x^{+i}}}(\hat{x}_i = 1) \leq 1{+}5\varepsilon_j \sum_{i \in B_j} \sqrt{\mathbb{E}_{\theta_x}[N_i]} \leq 1{+}5\varepsilon_j \sqrt{d_j \sum_{i \in B_j} \mathbb{E}_{\theta_x}[N_i]} = 1{+}5\varepsilon_j \sqrt{d_j T}.$$

Multiply by $10\varepsilon_j / d_j$. □

Lean remark. The four parts are carried out inside the proof of Theorem **??** rather than as separate declarations, since they all speak about the same fixed run measures: (i) is the pathwise identity $\sum_{i \in B_j} \hat{x}_i = \sum_{i \in B_j} x_t[i] = 1$ integrated, (ii) is Lemma **??** in its data-dependent form applied to $\theta = \texttt{mtParam0}(d, \varepsilon, \kappa, j)$ and $\theta' = \texttt{mtParam}(d, \varepsilon, \kappa^{j \to i})$, and (iv) is Cauchy–Schwarz (Lemma **??**) over the block.

Lemma 137 (Cauchy–Schwarz for square roots)**.** $Maiti2026Power.sum_sqrt_le_sqrt_card_mul_sumFornonnegativer$ *indexed by a finite set* K*,* $\sum_{k \in K} \sqrt{a_k} \leq \sqrt{|K| \sum_{k \in K} a_k}$*.*

Proof. Cauchy–Schwarz applied to the vectors $(1)_k$ and $(\sqrt{a_k})_k$ gives $\big(\sum_k \sqrt{a_k}\big)^2 \leq |K| \sum_k a_k$; take square roots. □

Lemma 138 (Averaging over one block)**.** $Fintype.sum_sum_updatedef : multitask_setLetj$ $\leq m$ *and let* g *be a real function of the arms* $\kappa \in \prod_\ell B_\ell$ *of the multi-task set. Writing* $\kappa^{j \to i}$ *for* κ *with its* j*-th coordinate replaced by* $i \in B_j$*,*

$$\sum_{\kappa} \sum_{i \in B_j} g\big(\kappa^{j \to i}\big) = d_j \sum_{\kappa} g(\kappa).$$

Proof. The map $(\kappa, i) \mapsto (\kappa^{j \to i}, \kappa_j)$ is an involution of $\big(\prod_\ell B_\ell\big) \times B_j$, so the left-hand side, which is the sum of $g(\kappa^{j \to i})$ over all pairs (κ, i), equals the sum of $g(\kappa)$ over all pairs, namely $d_j \sum_\kappa g(\kappa)$. This replaces the bijection $\mathcal{X}^{(j)} \times B_j \to \mathcal{X}$ of Definition **??**, which would require naming the set $\mathcal{X}^{(j)}$ of arms with an empty block. □

Theorem 139 (Lower bound for multi-task bandits)**.** $Maiti2026Power.le_of_isPAC_multitaskSetdef : multitask$ *taskactionset,*$\varepsilon > 0$*, and let* Alg *be an identification algorithm on* $\mathcal{X}$ *with budget* $T \geq 1$ *satisfying*

$$T \leq \frac{S_d^2}{20000\,\varepsilon^2}, \qquad S_d = \sum_{j \leq m} \sqrt{d_j}.$$

Then $\frac{1}{|\mathcal{X}|} \sum_{\tilde{x} \in \mathcal{X}} \mathbb{E}_{\theta_{\tilde{x}}} \langle \hat{x}, \theta_{\tilde{x}} \rangle \leq 5.36\,\varepsilon$*, and there exists* $\tilde{x} \in \mathcal{X}$ *with* $\mathbb{P}_{\theta_{\tilde{x}}}\big(r(\hat{x}, \theta_{\tilde{x}}) > \varepsilon\big) \geq 0.4$*.*

Corollary 140 (PAC lower bound for multi-task bandits)**.** $Maiti2026Power.multitaskSet_lt_budget_of_isPACthm$ *taskactionset(blocksizes*$d_j \geq 2$*) and* $\varepsilon > 0$*. An* (ε, δ)*-PAC algorithm on* $\mathcal{X}$ *with budget* $T \geq 1$ *and* $\delta < 0.4$ *has* $T > S_d^2 / (20000\,\varepsilon^2)$*.*

Proof. thm:lower$_m ultitaskIf$T $\leq S_d^2/(20000\varepsilon^2)$, Theorem **??** gives an instance $\theta_{\tilde{x}}$ with $\mathbb{P}_{\theta_{\tilde{x}}}(r(\hat{x},\theta_{\tilde{x}}) > \varepsilon) \geq 0.4 > \delta$, contradicting the PAC property. ☐

Proof. lem:multitask$_b lock_a verage, lem : fail_p rob_f rom_e xpected_v alue, def : multitask_i nstances, lem : multitask_b lock_r eindex, lem : sum_s qrt_c sFor \in \mathcal{X}$, $\langle \hat{x}, \theta_{\tilde{x}} \rangle = \sum_{j \leq m} V_j(\tilde{x})$ with $V_j(\tilde{x}) := \sum_{i \in B_j} \hat{x}_i \theta_{\tilde{x}}[i]$. Fix j. Using the bijection $\mathcal{X}^{(j)} \times B_j \to \mathcal{X}$, $(x,i) \mapsto x^{+i}$, and $|\mathcal{X}| = |\mathcal{X}^{(j)}|\, d_j$,

$$\frac{1}{|\mathcal{X}|} \sum_{\tilde{x} \in \mathcal{X}} \mathbb{E}_{\theta_{\tilde{x}}}[V_j(\tilde{x})] = \frac{1}{|\mathcal{X}^{(j)}|} \sum_{x \in \mathcal{X}^{(j)}} \frac{1}{d_j} \sum_{i \in B_j} \mathbb{E}_{\theta_{x^{+i}}}[V_j(x^{+i})] \leq \frac{10\varepsilon_j}{d_j}\Big(1+5\varepsilon_j\sqrt{d_j T}\Big)$$

by Lemma **??**(iv). With the budget assumption,

$$\varepsilon_j \sqrt{d_j T} = \frac{\varepsilon\sqrt{d_j}}{S_d}\sqrt{d_j}\sqrt{T} \leq \frac{\varepsilon\, d_j}{S_d} \cdot \frac{S_d}{\sqrt{20000}\,\varepsilon} = \frac{d_j}{\sqrt{20000}},$$

so, using $d_j \geq 2$ and $50/\sqrt{20000} < 0.36$,

$$\frac{10\varepsilon_j}{d_j}\Big(1 + 5\varepsilon_j\sqrt{d_j T}\Big) \leq \frac{10\varepsilon_j}{d_j} + \frac{50\,\varepsilon_j}{\sqrt{20000}} \leq 5\varepsilon_j + 0.36\,\varepsilon_j = 5.36\,\varepsilon_j.$$

Summing over j and using $\sum_j \varepsilon_j = \varepsilon$ gives the bound $5.36\,\varepsilon$ on the average value. Since $\min_x \langle x, \theta_{\tilde{x}} \rangle = 0$ and $Z(\theta_{\tilde{x}}) = 10\varepsilon$ for $\tilde{x} \in \mathcal{X}$ (Definition **??**), Lemma **??** (gap form, $\varepsilon_* = \varepsilon$, $a = 5.36$) gives an instance with failure probability at least $(9 - 5.36)/9 = 0.4044 \geq 0.4$. The PAC consequence is immediate: an (ε,δ)-PAC algorithm has failure probability at most $\delta < 0.4$ on every instance.

Lean remark. The Lean statement `le_of_isPAC_multitaskSet` is the contrapositive form: an (ε,δ)-PAC algorithm with $T \leq S_d^2/(20000\varepsilon^2)$ has $\delta \geq 2/5$. It is obtained by comparing the upper bound $5.36\,\varepsilon$ on the average value with the lower bound $9\varepsilon(1-\delta)$ that Lemma **??** (PAC form) gives on every instance, using $Z(\theta_{\tilde{x}}) = 10\varepsilon$; the averaging over the block uses Lemma **??**. ☐

Remark 141*.* The paper proves this bound (with the average value 7ε and failure probability 0.1) by splitting into $d_j \geq 100$ and $d_j < 100$ and using deterministic algorithms plus Yao's principle. The argument above, a single Pinsker plus Cauchy–Schwarz step per block, handles all $d_j \geq 2$ uniformly and directly for randomized algorithms; the equality $\varepsilon_j\sqrt{d_j T} = d_j/\sqrt{20000}$ at the maximal budget makes the dependence on d_j cancel exactly. Since $d \leq S_d^2 \leq md$, the lower bound is at least $d/(20000\varepsilon^2)$ and at most $md/(20000\varepsilon^2)$; for $m = 1$ it is the classical $\Omega(K/\varepsilon^2)$ bound for $K = d_1$ arms.

7.2.2 Hypercubes

Both hypercubes of Table 1 are sets of vertices of a box, and the same argument covers them.

Definition 142 (Boxes with two values per coordinate)**.** Maiti2026Power.cubeSet, Maiti2026Power.cubeVec, Maiti2026Power.cubeSet$_eq_range, Maiti2026Power.cubeSet_finite, Maiti2026Power.$ u_+ be two reals. The *box* they define is the action set

$$\mathcal{C}(u) := \{x \in \mathbb{R}^d : x_i \in \{u_-, u_+\} \text{ for all } i \leq d\},$$

a finite — hence compact — set of 2^d points, whose elements are indexed by the sign patterns $s \in \{-,+\}^d$ through $c(s)_i := u_{s_i}$. Both hypercubes are boxes: $\{-1,1\}^d$ is the box of $(u_-, u_+) = (-1,1)$ and $\{0,1\}^d$ that of $(u_-,u_+) = (0,1)$. For a sign pattern s let $\sigma(s) \in \{-1,1\}^d$ be the vector with $\sigma(s)_i = +1$ if $s_i = +$ and -1 otherwise, and let $\sigma(s)^{\setminus j}$ be $\sigma(s)$ with the coordinate j replaced by 0. For $c > 0$,

$$\max_{x\in\mathcal{C}(u)} \langle x, c\,\sigma(s)\rangle = \sum_{i\leq d} \max\big(u_+ c\sigma(s)_i, u_- c\sigma(s)_i\big) = c\sum_{i\leq d} u_{s_i}\sigma(s)_i,$$

and for $x \in \mathcal{C}(u)$ the simple regret is $c(u_+ - u_-)$ times the Hamming distance to the maximizer $c(s)$:

$$r\big(x, c\,\sigma(s)\big) = c\,(u_+ - u_-)\,\#\{i \leq d : x_i \neq u_{s_i}\},$$

so $0 \leq r \leq c(u_+ - u_-)d$ on $\mathcal{C}(u)$ and $Z(c\,\sigma(s)) = c(u_+ - u_-)d$.

Lemma 143 (Lower bound on a box)**.** *Maiti2026Power.le$_of_isPAC_cubeSetdef : cube_set, def : pacLetC(u)beabo$ let $\varepsilon > 0$, let M satisfy $|u_-| \leq M$ and $|u_+| \leq M$, and let $c > 0$ be normalized by $c\,(u_+ - u_-)\,d = 10\varepsilon$. Every (ε,δ)-PAC identification algorithm on $\mathcal{C}(u)$ with budget T satisfying $T\,(cM)^2 \leq 1/4$ has $\delta \geq 1/6$.*

Proof. lem:log$_gains_three_point, lem : fail_prob_from_expected_value, def : cube_setThehardinstancesarethe2^d$ reward vectors $c\,\sigma(s)$, $s \in \{-,+\}^d$.

One coordinate. Fix $j \leq d$ and a sign pattern s, and let s' be s with the sign of the coordinate j flipped. The vector $\theta^0 := c\,\sigma(s)^{\setminus j}$ is the same for s and for s', and $\theta^0 - c\,\sigma(s) = -c\,\sigma(s)_j e_j$, so for every $x \in \mathcal{C}(u)$, $\langle x, \theta^0 - c\,\sigma(s)\rangle^2 = c^2 x_j^2 \leq c^2M^2 =: C$, and likewise with s'. Since $\hat{x} \in \mathcal{C}(u)$ takes only the two values $u_\pm$ in the coordinate j, the event $\{\hat{x}_j \neq u_{s_j}\}$ is $E := \{\hat{x}_j = u_-\}$ when $s_j = +$ and E^c when $s_j = -$, so Lemma **??** gives

$$\mathbb{P}_{c\sigma(s)}(\hat{x}_j \neq u_{s_j}) + \mathbb{P}_{c\sigma(s')}(\hat{x}_j \neq u_{s'_j}) \geq 1 - 2\sqrt{TC/4} \geq \frac{1}{2},$$

using $T(cM)^2 \leq 1/4$.

Averaging. The map $s \mapsto s'$ is an involution of $\{-,+\}^d$, so summing the display over the 2^d sign patterns gives $2\sum_s \mathbb{P}_{c\sigma(s)}(\hat{x}_j \neq u_{s_j}) \geq 2^d/2$: the average over s of $\mathbb{P}_{c\sigma(s)}(\hat{x}_j \neq u_{s_j})$ is at least $1/4$. Summing over $j \leq d$ and using the Hamming form of the regret (Definition **??**),

$$\frac{1}{2^d}\sum_s \mathbb{E}_{c\sigma(s)}\big[r(\hat{x}, c\sigma(s))\big] = c(u_+ - u_-)\sum_{j\leq d}\frac{1}{2^d}\sum_s \mathbb{P}_{c\sigma(s)}(\hat{x}_j \neq u_{s_j}) \geq c(u_+ - u_-)\frac{d}{4} = \frac{10\varepsilon}{4} = 2.5\,\varepsilon.$$

Conclusion. On every instance $0 \leq r \leq Z(c\sigma(s)) = 10\varepsilon$, so Lemma **??** (PAC form) gives $\mathbb{E}_{c\sigma(s)}[r] \leq \varepsilon + 9\varepsilon\delta$. Averaging over s and comparing with 2.5ε yields $2.5 \leq 1 + 9\delta$, i.e. $\delta \geq 1/6$. ☐

Theorem 144 (Lower bound for the hypercube $\{-1,1\}^d$)**.** *def:bai$_a$lg*$, def : pac, def : regret, def : cube_s etLetX=$ *{-1,1}*^{d} *and* $\varepsilon > 0$*. For* $s \in \{-1,1\}^d$ *let* $\theta_s := \frac{5\varepsilon}{d}s$*. Let* Alg *be an identification algorithm on* $\mathcal{X}$ *with budget* $T \geq 1$ *satisfying* $T \leq d^2/(100\,\varepsilon^2)$*. Then* $\frac{1}{2^d}\sum_s \mathbb{E}_{\theta_s}[r(\hat{x},\theta_s)] \geq 2.5\,\varepsilon$ *and there exists* $s \in \{-1,1\}^d$ *with* $\mathbb{P}_{\theta_s}(r(\hat{x},\theta_s) > \varepsilon) \geq 1/6$*.*

Proof. lem:lower$_c ube, lem : fail_p rob_f rom_e xpected_v alue, def : cube_s et$\{-1,1\}^{d} is the box of $(u_-, u_+) = (-1,1)$, with $M = 1$ and $c = 5\varepsilon/d$ (so that $c(u_+ - u_-)d = 10\varepsilon$) and $\theta_s = c\,\sigma(s)$; the budget assumption is $T(cM)^2 = 25\varepsilon^2 T/d^2 \leq 1/4$. The average regret bound is the display of the proof of Lemma **??**, and Lemma **??** with $\varepsilon_* = \varepsilon$ and $b = 2.5$ gives an instance with failure probability at least $1.5/9 = 1/6$. ☐

Corollary 145 (PAC lower bound for the hypercube $\{-1,1\}^d$)**.** *Maiti2026Power.hypercubePM*$_l t_b udget_o f_i sPAC$
0. An (ε,δ)*-PAC algorithm on* $\{-1,1\}^d$ *with budget* $T \geq 1$ *and* $\delta < 1/6$ *has* $T > d^2/(100\,\varepsilon^2)$*.*

Proof. lem:lower$_c ubeIf$T $\leq d^2/(100\varepsilon^2)$, Lemma **??** applied to the box of $(-1,1)$ with $M = 1$ and $c = 5\varepsilon/d$ gives $\delta \geq 1/6$, a contradiction. (For $d = 0$ the bound reads $0 < T$, which holds since $T \geq 1$.) ☐

Theorem 146 (Lower bound for the hypercube $\{0,1\}^d$)**.** *def:bai$_a$lg*$, def : pac, def : regret, def : cube_s etLetX=$ *{0,1}*^{d} *and* $\varepsilon > 0$*. For* $s \in \{-1,1\}^d$ *let* $\theta_s := \frac{10\varepsilon}{d}s$ *and* $\sigma(s) := \mathbf{1}\{s = 1\} \in \{0,1\}^d$ *(coordinatewise). Let* Alg *be an identification algorithm on* $\mathcal{X}$ *with budget* $T \geq 1$ *satisfying* $T \leq d^2/(400\,\varepsilon^2)$*. Then* $\frac{1}{2^d}\sum_s \mathbb{E}_{\theta_s}[r(\hat{x},\theta_s)] \geq 2.5\,\varepsilon$ *and there exists* s *with* $\mathbb{P}_{\theta_s}(r(\hat{x},\theta_s) > \varepsilon) \geq 1/6$*.*

Proof. lem:lower$_c ube, lem : fail_p rob_f rom_e xpected_v alue, def : cube_s et$\{0,1\}^{d} is the box of $(u_-, u_+) = (0,1)$, with $M = 1$ and $c = 10\varepsilon/d$ (so that $c(u_+ - u_-)d = 10\varepsilon$); the budget assumption is $T(cM)^2 = 100\varepsilon^2 T/d^2 \leq 1/4$. The rest is as for Theorem **??**: the vertex maximizing $\langle x, \theta_s\rangle$ is $\mathbf{1}\{s = 1\}$ and the regret is $\frac{10\varepsilon}{d}$ times the Hamming distance to it. ☐

Corollary 147 (PAC lower bound for the hypercube $\{0,1\}^d$)**.** *Maiti2026Power.hypercube01*$_l t_b udget_o f_i sPAClem$
0. An (ε,δ)*-PAC algorithm on* $\{0,1\}^d$ *with budget* $T \geq 1$ *and* $\delta < 1/6$ *has* $T > d^2/(400\,\varepsilon^2)$*.*

Proof. lem:lower$_c ubeAsforCorollary$ **??**$, withc = 10\varepsilon/d.$ ☐

Remark 148. The paper's Appendix on hypercubes only sketches these proofs ("analogous to the $d_j = 2$ case of the multi-task argument, with the expected gap to the worst arm instead of the expected value"). The constant differs between the two hypercubes: for $\{-1,1\}^d$ the instances are at distance $5\varepsilon/d$ from the zeroed alternative in coordinate j, for $\{0,1\}^d$ at distance $10\varepsilon/d$ (the range of $\langle x, \theta\rangle$ over the cube must be 10ε in both cases), so the divergence is four times larger and the admissible budget four times smaller. Both give $H_2 \geq \Omega(d^2/\varepsilon^2)$, matching the upper bound $O((d\log(1/\delta) + d^2)/\varepsilon^2)$ of Theorem **??** since $w \leq d$ (Proposition **??**).

7.2.3 m-sets

Definition 149 (m-sets and their hard instances)**.** Maiti2026Power.mSet, Maiti2026Power.msParam, Maiti2026Power.msVec, Maiti2026Power.msVec$_m em_m Set, Maiti2026Power.mSet_n onempty, Maiti2026Power$
$\leq m \leq d-1$. The *m-set* action set is $\mathcal{X} := \{x \in \{0,1\}^d : \sum_{i \leq d} x_i = m\}$, with $|\mathcal{X}| = \binom{d}{m}$; it spans $\mathbb{R}^d$ (it contains x_S for every m-subset S; $x_S - x_{S'}$ with $S \triangle S' = \{i, i'\}$ gives $e_i - e_{i'}$, and $\sum_S x_S$ is a positive multiple of $\mathbf{1}$; together these span $\mathbb{R}^d$). We write $n := d - m + 1$ and, for $\varepsilon > 0$, $\Delta := 10\varepsilon/m$. For every $S \subseteq [d]$ (of any size) let $\theta^{(S)} := \Delta \, \mathbf{1}_S \in \mathbb{R}^d$ (Δ on S, 0 elsewhere). For a recommendation $\hat{x} \in \mathcal{X}$ let $\hat{S} := \operatorname{supp} \hat{x}$, so $|\hat{S}| = m$ and $\langle \hat{x}, \theta^{(S)} \rangle = \Delta |\hat{S} \cap S|$. If $|S| = m$ and $d \geq 2m$, then $\max_{x \in \mathcal{X}} \langle x, \theta^{(S)} \rangle = m\Delta = 10\varepsilon$, $\min_{x \in \mathcal{X}} \langle x, \theta^{(S)} \rangle = 0$ and $Z(\theta^{(S)}) = 10\varepsilon$.

Lemma 150 (Equivalent sampling of (B, i))**.** *$Finset.sum_powersetCard_succ_sum_mem def : msetsForeveryfunc$*
$i)withB \subseteq [d]$, $|B| = m-1$, $i \notin B$,

$$\frac{1}{\binom{d}{m}} \sum_{|S|=m} \frac{1}{m} \sum_{i \in S} F(S \setminus \{i\}, i) = \frac{1}{\binom{d}{m-1}} \sum_{|B|=m-1} \frac{1}{n} \sum_{i \notin B} F(B, i).$$

In words: drawing S uniformly among m-subsets and then i uniformly in S gives the same law for $(S \setminus \{i\}, i)$ as drawing B uniformly among $(m-1)$-subsets and then i uniformly in $[d] \setminus B$.

Proof. def:msets The map $(S, i) \mapsto (S \setminus \{i\}, i)$ is a bijection from $\{(S, i) : |S| = m,\ i \in S\}$ onto $\{(B, i) : |B| = m-1,\ i \notin B\}$ (inverse $(B, i) \mapsto (B \cup \{i\}, i)$), so both sides are sums of F over the same set of pairs, with the constant weights $1/(m\binom{d}{m})$ and $1/(n\binom{d}{m-1})$ respectively. These are equal: $n\binom{d}{m-1} = (d - m + 1)\frac{d!}{(m-1)!\,(d-m+1)!} = \frac{d!}{(m-1)!\,(d-m)!} = m\binom{d}{m}$. ☐

Lean remark. The Lean statement is the unnormalized identity $\sum_{|S|=m} \sum_{i \in S} F(S \setminus \{i\}, i) = \sum_{|B|=m-1} \sum_{i \notin B} F(B, i)$ (the bijection alone, proved with `Finset.sum_nbij'` on the two Σ-types); the normalization $n\binom{d}{m-1} = m\binom{d}{m}$ is Mathlib's `Nat.choose_succ_right_eq` and is applied once, at the end of the proof of Theorem **??**.

Lemma 151 (Good coordinates under an alternative instance)**.** *def:msets, def:bai$_a$lgLetAlgbeanidentificationalgorithmwithbudgetTonthem$-$setsandletB $\subseteq [d]$ with $|B| = m-1$. For $i \in [d]$ let $N_i := \sum_{t<T} x_t[i]$ and $A_i := \{i \in \hat{S}\}$. Then $\sum_{i \notin B} \mathbb{E}_{\theta^{(B)}}[N_i] \leq mT$ and $\sum_{i \notin B} \mathbb{P}_{\theta^{(B)}}(A_i) \leq m$, and the set*

$$G_B := \left\{ i \in [d] \setminus B : \mathbb{E}_{\theta^{(B)}}[N_i] \leq \frac{4mT}{n} \quad \textit{and} \quad \mathbb{P}_{\theta^{(B)}}(A_i) \leq \frac{4m}{n} \right\}$$

satisfies $|G_B| \geq n/2$.

Proof. def:msets Every x_t and $\hat{x}$ have exactly m ones, so $\sum_{i \leq d} N_i = mT$ and $\sum_{i \leq d} \mathbf{1}_{A_i} = m$ pathwise; restrict to $i \notin B$ and take expectations. Let $G^1 := \{i \notin B : \mathbb{E}N_i > 4mT/n\}$; then $|G^1| \cdot 4mT/n < \sum_{i \in G^1} \mathbb{E}N_i \leq mT$ unless $G^1 = \emptyset$, so $|G^1| < n/4$. Likewise $G^2 := \{i \notin B : \mathbb{P}(A_i) > 4m/n\}$ has $|G^2| < n/4$. Since $|[d] \setminus B| = n$, $|G_B| = n - |G^1 \cup G^2| > n/2$. ☐

Lemma 152 (Pinsker step for m-sets)**.** *def:msets, lem:msets$_good_setIn the setting of Lemma ??, for every i$ $\in G_B$,*

$$\mathrm{KL}(\theta^{(B)}, \theta^{(B\cup\{i\})}) \le \frac{2\Delta^2 mT}{n} \qquad \textit{and} \qquad \mathbb{P}_{\theta^{(B\cup\{i\})}}(i \in \hat{S}) \le \frac{4m}{n} + \Delta\sqrt{\frac{mT}{n}}.$$

Proof. lem:two$_point_pinsker, lem : msets_good_set \theta^{(B)} - \theta^{(B\cup\{i\})} = -\Delta e_i$, so $\mathrm{KL}(\theta^{(B)}, \theta^{(B\cup\{i\})}) = \frac{1}{2}\sum_t \mathbb{E}_{\theta^{(B)}}[\Delta^2 x_t[i]^2] = \frac{\Delta^2}{2}\mathbb{E}_{\theta^{(B)}}[N_i] \le \frac{\Delta^2}{2}\cdot\frac{4mT}{n}$ by Lemma **??**. Lemma **??** with $E = A_i$ gives $\mathbb{P}_{\theta^{(B\cup\{i\})}}(A_i) \le \mathbb{P}_{\theta^{(B)}}(A_i) + \sqrt{\mathrm{KL}/2} \le \frac{4m}{n} + \sqrt{\Delta^2 mT/n}$. ☐

Lean remark. Lemmas **??** and **??** are carried out inside the proof of Theorem **??** rather than as separate declarations, since they speak about the run measures of the fixed algorithm: G_B is a `Finset.filter` of B^c, the counting argument bounds the cardinalities of the two exceptional sets by $n/4$ each from $\sum_i \mathbb{E}N_i = mT$ and $\sum_i \mathbb{P}(A_i) = m$, and the Pinsker step is Lemma **??** in its data-dependent form.

Theorem 153 (Lower bound for m-sets)**.** *Maiti2026Power.le$_of_isPAC_mSetdef : msets, def : pacLetXbethem-$ $= d - m + 1 \ge 20m$, let $\varepsilon > 0$ and let* Alg *be an identification algorithm on $\mathcal{X}$ with budget $T \ge 1$ satisfying*

$$T \le \frac{m\,n}{2500\,\varepsilon^2}.$$

Then $\frac{1}{\binom{d}{m}}\sum_{|S|=m}\mathbb{E}_{\theta^{(S)}}\langle \hat{x}, \theta^{(S)}\rangle \le 7\varepsilon$ and there exists an m-subset S with $\mathbb{P}_{\theta^{(S)}}(r(\hat{x}, \theta^{(S)}) > \varepsilon) \ge 2/9$.

Corollary 154 (PAC lower bound for m-sets)**.** *Maiti2026Power.mSet$_lt_budget_of_isPACthm : lower_msets, def :$ $= d - m + 1 \ge 20m$ and $\varepsilon > 0$. An (ε,δ)-PAC algorithm on $\mathcal{X}$ with budget $T \ge 1$ and $\delta < 2/9$ has $T > mn/(2500\,\varepsilon^2)$.*

Proof. thm:lower$_msetsAsforCorollary$ **??**$, from Theorem$ **??**. ☐

Proof. lem:msets$_good_set, lem : msets_pinsker_step, lem : msets_equivalent_sampling, lem : fail_prob_from_expected_value, def : msetsNotethat$n $\ge 20m$ implies $d = n + m - 1 \ge 21m - 1 \ge 2m$, so Definition **??** gives $Z(\theta^{(S)}) = 10\varepsilon$ and $\min_x\langle x, \theta^{(S)}\rangle = 0$ for every m-subset S.

Constants. $4m/n \le 0.2$, and $\Delta\sqrt{mT/n} \le \frac{10\varepsilon}{m}\sqrt{\frac{m}{n}\cdot\frac{mn}{2500\varepsilon^2}} = \frac{10\varepsilon}{m}\cdot\frac{m}{50\varepsilon} = 0.2$. Hence, by Lemma **??**, $\mathbb{P}_{\theta^{(B\cup\{i\})}}(i \in \hat{S}) \le 0.4$ for every $(m-1)$-subset B and every $i \in G_B$.

Average over $i \notin B$. For fixed B, bounding the terms with $i \in G_B$ by 0.4 and the others by 1, and using $|G_B| \ge n/2$ (Lemma **??**),

$$\frac{1}{n}\sum_{i\notin B}\mathbb{P}_{\theta^{(B\cup\{i\})}}(i \in \hat{S}) \le \frac{1}{n}\Big(0.4\,|G_B| + (n - |G_B|)\Big) \le \frac{1}{n}\Big(0.4\cdot\frac{n}{2} + \frac{n}{2}\Big) = 0.7.$$

Average value. For an m-subset S, $\mathbb{E}_{\theta^{(S)}}\langle \hat{x}, \theta^{(S)}\rangle = \Delta\, \mathbb{E}_{\theta^{(S)}}|\hat{S} \cap S| = \Delta \sum_{i \in S} \mathbb{P}_{\theta^{(S)}}(i \in \hat{S}) = 10\varepsilon \cdot \frac{1}{m} \sum_{i\in S} \mathbb{P}_{\theta^{(S)}}(i \in \hat{S})$. Averaging over S and applying Lemma **??** to $F(B,i) := \mathbb{P}_{\theta^{(B\cup\{i\})}}(i \in \hat{S})$ (note $\theta^{(S)} = \theta^{(B\cup\{i\})}$ for $B = S \setminus \{i\}$),

$$\frac{1}{\binom{d}{m}} \sum_{|S|=m} \mathbb{E}_{\theta^{(S)}}\langle \hat{x}, \theta^{(S)}\rangle = 10\varepsilon\cdot \frac{1}{\binom{d}{m-1}} \sum_{|B|=m-1} \frac{1}{n} \sum_{i \notin B} \mathbb{P}_{\theta^{(B\cup\{i\})}}(i \in \hat{S}) \le 10\varepsilon\cdot 0.7 = 7\varepsilon.$$

Conclusion. Lemma **??** (gap form, $a = 7$) gives an m-subset S with $\mathbb{P}_{\theta^{(S)}}(r > \varepsilon) \ge 2/9$.

Lean remark. The Lean statement `le_of_isPAC_mSet` is the contrapositive form: an (ε,δ)-PAC algorithm with $1 \le T \le mn/(2500\varepsilon^2)$ has $\delta \ge 2/9$. It compares the upper bound $7\varepsilon\binom{d}{m}$ on $\sum_{|S|=m} \mathbb{E}_{\theta^{(S)}}\langle \hat{x}, \theta^{(S)}\rangle$ with the lower bound $9\varepsilon(1-\delta)\binom{d}{m}$ coming from Lemma **??** (PAC form) and $Z(\theta^{(S)}) = 10\varepsilon$. The sizes are handled by $m = k+1$ and $d = k+n$ rather than by truncated subtraction. ☐

Lean remark. The instances are indexed by `Finset (Fin d)` of card m (`Finset.powersetCard m Finset.univ`) and the alternatives by subsets of card $m-1$; $\hat{S}$ is the support of $\hat{x}$ viewed as a `Finset`. Lemma **??** is a reindexing of a double sum along the bijection $(S,i) \mapsto (S\setminus\{i\}, i)$ together with the identity $n\binom{d}{m-1} = m\binom{d}{m}$ (`Nat.choose_succ_right_eq` or `Nat.succ_mul_choose_eq`). Lemma **??** is a counting (Markov) argument on a finite sum.

7.2.4 The unit ball

Corollary 155 (Adaptive lower bound for the unit ball)**.** *Maiti2026Power.unitBall$_l$e$_b$udget$_o$f$_i$sPACdef : pac, c* ≥ 2, $\varepsilon > 0$ *and* $\delta \le 1/500$*. Every* (ε,δ)*-PAC identification algorithm on* $\mathbb{B}_d$ *with budget* T *satisfies* $T \ge d^2/(1000\,\varepsilon^2)$*.*

Proof. cor:ball$_p$ac$_l$owerThisisCorollary ??ofChapter ??, whichreplacesthecitationof [?]inthepaper'sAppend theGaussian−priorargument(Theorem ??(ii))givesaninstanceθ with $\mathbb{E}_\theta[r(\hat{x},\theta)] \ge 3d/(40\sqrt{T})$ and $\|\theta\| \le 10d/\sqrt{T}$, and the PAC property bounds $\mathbb{E}_\theta[r] \le \varepsilon + 2\|\theta\|\delta \le \varepsilon + 20\delta d/\sqrt{T}$; for $\delta \le 1/500$ this gives $(3/40 - 1/25)\, d/\sqrt{T} \le \varepsilon$, i.e. $T \ge (7/200)^2 d^2/\varepsilon^2 = 49d^2/(40000\varepsilon^2) \ge d^2/(1000\varepsilon^2)$. ☐

7.3 Summary of the sample complexities

The non-adaptive upper bound of Theorem **??** requires a spanning action set. For the multi-task set, which is not spanning, it is applied to the spanning set $U^\top \mathcal{X} \subset \mathbb{R}^{d-m+1}$ of Definition **??** and transported back by the following lemma, the analogue for a subspace of the transformation lemma **??** (which handles invertible maps of $\mathbb{R}^d$).

Lemma 156 (Transport of identification algorithms along the span)**.** *def:arm$_s$et, def : bai$_a$lg, def : pac, def : width$_s$ubspaceLetX*$\subset \mathbb{R}^d$ *be an action set with* $V := \mathrm{span}(\mathcal{X})$ *of dimension* $r \ge 1$*, let* $U \in \mathbb{R}^{d\times r}$ *have orthonormal columns with* *range V, and let $\mathcal{X}_r := U^\top \mathcal{X} \subset \mathbb{R}^r$, a spanning action set in $\mathbb{R}^r$ (Definition* **??***). The map $\phi : \mathcal{X}_r \to \mathcal{X}$, $u \mapsto Uu$, is a homeomorphism with inverse $x \mapsto U^\top x$, and for all $u \in \mathbb{R}^r$, $\theta \in \mathbb{R}^d$,*

$$\langle Uu, \theta\rangle = \langle u, U^\top \theta\rangle, \qquad \text{in particular} \qquad \langle U^\top x, U^\top \theta\rangle = \langle x, \theta\rangle \ \text{ for } x \in \mathcal{X},\ \theta \in V.$$

For every identification algorithm $\mathsf{Alg}_r = (\mathrm{alg}_r, \rho_r)$ on $\mathcal{X}_r$ with budget T there is an identification algorithm $\mathsf{Alg} = (\mathrm{alg}, \rho)$ on $\mathcal{X}$ with budget T (the transported algorithm*: play Uu whenever Alg_r plays u, feed it the same observations, recommend $U\hat{x}_r$) such that, for every $\theta \in \mathbb{R}^d$ with $\theta_r := U^\top\theta$, the law of $(H_T, \hat{x})$ under ν_θ is the image of the law of $(H_{T,r}, \hat{x}_r)$ under ν_{θ_r} by $((u_t, y_t)_{t<T}, \hat{x}_r) \mapsto ((Uu_t, y_t)_{t<T}, U\hat{x}_r)$, and the simple regrets agree pathwise: $r_{\mathcal{X}}(U\hat{x}_r, \theta) = r_{\mathcal{X}_r}(\hat{x}_r, \theta_r)$. Consequently Alg is (ε, δ)-PAC on $\mathcal{X}$ if and only if Alg_r is (ε, δ)-PAC on $\mathcal{X}_r$.*

Proof. def:env, def:regret, lem:transform$_p reserves_p acTheidentities$:$\langle Uu, \theta\rangle = u^\top U^\top \theta$, and for $\theta \in V$, $UU^\top\theta = \theta$ ($UU^\top$ is the orthogonal projection onto V), so $\langle U^\top x, U^\top\theta\rangle = \langle x, UU^\top\theta\rangle = \langle x, \theta\rangle$. The map ϕ is continuous with continuous inverse $U^\top$ (as $U^\top U u = u$ and $UU^\top x = x$ for $x \in \mathcal{X} \subset V$), hence a measurable bijection $\mathcal{X}_r \to \mathcal{X}$ with measurable inverse. The reward kernel of ν_θ at Uu is $\mathcal{N}(\langle Uu, \theta\rangle, 1) = \mathcal{N}(\langle u, \theta_r\rangle, 1)$, the reward kernel of ν_{θ_r} at u: the environments correspond under ϕ. The transported algorithm is obtained by mapping the policy kernels of alg_r through ϕ (and the histories back through ϕ^{-1}), exactly as in Lemma **??**, whose proof only uses that $x \mapsto Mx$ is a measurable bijection of action sets under which the environments correspond; the same argument (trajectory measures are characterized by their kernels, Ionescu–Tulcea) gives the image-law statement. For the regrets, $\max_{x\in\mathcal{X}}\langle x, \theta\rangle = \max_{u\in\mathcal{X}_r}\langle Uu, \theta\rangle = \max_{u\in\mathcal{X}_r}\langle u, \theta_r\rangle$ (ϕ is onto $\mathcal{X}$) and $\langle U\hat{x}_r, \theta\rangle = \langle \hat{x}_r, \theta_r\rangle$, so the regrets agree. Finally, $\theta \mapsto U^\top\theta$ is onto $\mathbb{R}^r$ (with right inverse $\theta_r \mapsto U\theta_r$), so "for every θ" on $\mathcal{X}$ and "for every θ_r" on $\mathcal{X}_r$ are the same condition, and the PAC equivalence follows from the equality of the probabilities of the events $\{r \le \varepsilon\}$. ☐

Lean remark. The lemma is a special case of a general "change of action type" statement for LML algorithms along a measurable bijection ϕ with measurable inverse between action types, such that the environments correspond ($\kappa_\theta \circ \phi = \kappa_{\theta_r}$); Lemma **??** and Lemma **??** are two more instances. Only the direction "from $\mathcal{X}_r$ to $\mathcal{X}$" is used (to transport the upper bound of Theorem **??**); the converse direction uses the identity $\langle U^\top x, U^\top\theta\rangle = \langle x, \theta\rangle$ for $\theta \in V$ together with the fact that the environment only sees $\langle x, \theta\rangle$, $x \in \mathcal{X}$, i.e. θ only through its projection $UU^\top\theta \in V$.

Remark 157 (Table 1 of the paper)*.* For the structured sets of this chapter, the minimax sample complexity has the form $H_1 \log(1/\delta) + H_2$ with $H_1 = \Theta(d/\varepsilon^2)$ (upper bound: Theorem **??**; lower bound: Theorem **??**). The chapter establishes the following bounds on H_2; the non-adaptive upper bounds come from Theorem **??** and the width bounds of Chapter **??**, the adaptive lower bounds from Section **??**, so that on each of these sets adaptivity can only improve the sample complexity of the fixed-design algorithm by logarithmic factors.

- Unit ball $\mathbb{B}_d$: non-adaptive upper bound $O((d\log(1/\delta)+d^2)/\varepsilon^2)$ (Proposition **??**, $w(\mathbb{B}_d) \le d$); adaptive lower bound $T \ge d^2/(1000\varepsilon^2)$ for $\delta \le 1/500$ (Corollary **??**).

- Hypercubes $\{-1,1\}^d$ and $\{0,1\}^d$: non-adaptive upper bound $O((d\log(1/\delta)+d^2)/\varepsilon^2)$ (Proposition **??**); adaptive lower bounds $T > d^2/(100\varepsilon^2)$, resp. $T > d^2/(400\varepsilon^2)$, for $\delta < 1/6$ (Theorems **??** and **??**).

- m-sets with $d-m+1 \ge 20m$ (the paper writes $m \le d/21$): non-adaptive upper bound $O((d\log(1/\delta)+md\log(ed/m))/\varepsilon^2)$ (Proposition **??** with $\log\binom{d}{m} \le m\log(ed/m)$); adaptive lower bound $T > m(d-m+1)/(2500\varepsilon^2)$ for $\delta < 2/9$ (Theorem **??**). The gap is the factor $\log(d/m)$.

- Multi-task bandits with block sizes $d_1,\dots,d_m \ge 2$: non-adaptive upper bound $O\big((d\log(1/\delta)+(\sum_j \sqrt{d_j\log d_j})^2)/\varepsilon^2\big)$. Here the multi-task set $\mathcal{X}$ is not spanning, so Theorem **??** is applied to the spanning action set $\mathcal{X}_r := U^\top\mathcal{X} \subset \mathbb{R}^r$, $r = d-m+1 \le d$, with U the orthonormal basis of $\operatorname{span}\mathcal{X}$ of Lemma **??**: it gives a fixed-design (ε,δ)-PAC algorithm on $\mathcal{X}_r$ with budget $\lceil 600(w(\mathcal{X}_r)^2 + r\log(2/\delta))/\varepsilon^2\rceil$, where $w(\mathcal{X}_r) = w_{\mathrm{int}}(\mathcal{X}) \le \sum_j \sqrt{2d_j\log d_j}$ (Definition **??**, Proposition **??**); this algorithm is transported to $\mathcal{X}$ by Lemma **??** (play Uu instead of u; $\langle Uu,\theta\rangle = \langle u, U^\top\theta\rangle$), and the transported algorithm is again a deterministic fixed design on $\mathcal{X}$ with a deterministic recommendation. Adaptive lower bound $T > (\sum_j \sqrt{d_j})^2/(20000\varepsilon^2)$ for $\delta < 0.4$ (Theorem **??**). The gap is at most the factor $\max_j \log d_j$.

For the hypercubes and the m-sets, the adaptive lower bounds are also what yields the width lower bounds of Corollary **??**. The region algorithm of Theorem **??** gives, for the multi-armed bandit $\mathcal{X} = \{e_1,\dots,e_d\}$, the bound $O(d\log(1/\delta)/\varepsilon^2)$ of Median Elimination, and more generally saves a factor $\log d$ over the finite-set bound whenever $|\mathcal{X}| = d^{O(1)}$ and δ is constant.

Chapter 8

Estimation of the ℓ_2-norm of the reward vector

This chapter formalizes Section 3 and Appendix D of the paper (the four subsections on ℓ_2-norm estimation): on the action set $\mathcal{X} = \mathbb{B}_d$ one wants to estimate $r := \|\theta\|$ up to an additive error ε with probability at least $1 - \delta$, using $O(d \log(1/\delta)/\varepsilon^2)$ observations. The target is Theorem 8 of the paper (Theorem **??**). The procedure has three ingredients: a statistic built from random Rademacher directions, each sampled repeatedly, which estimates r^2 with an additive error proportional to $r\varepsilon$ once a constant-factor estimate r_0 of r is available (Section **??**); an adaptive multi-scale test which produces such an r_0, or detects that r is either at most ε or at least $\sqrt{d}$ (Section **??**); and a direct estimator for the large-norm regime $r \geq \sqrt{d}$ (Section **??**). The three are composed into a meta-algorithm in Section **??**. The chapter is fully formalized.

From Chapter **??** we use the sub-exponential calculus: the definition **??**, the tail bound for averages (Corollary **??**), the sub-exponential parameters of squared Gaussians (Lemma **??**), of chi-square variables (Lemma **??**) and of squared Rademacher linear forms (Lemma **??**). From the model chapter we use the seed representation of the runs of a seeded algorithm (Lemma **??**): the internal randomness of the algorithm (the Rademacher directions) and the Gaussian noises are the coordinates of an i.i.d. sequence, and every statistic of the run is a measurable function of finitely many of these coordinates. Compared with the paper, the constants are all explicit (no attempt was made to optimize them), the Hanson–Wright inequality is replaced by Lemma **??**, the conditional argument for the noise term is a Fubini argument over the directions, the large-norm regime uses a deterministic basis design instead of random Rademacher rows, which removes the random-matrix concentration step of the paper, and the sequential composition of the phases is replaced by a deterministic round schedule on the seed space. Throughout the chapter $\varepsilon \in (0, 1]$, $\delta \in (0, 1)$, $d \geq 1$, $\mathcal{X} = \mathbb{B}_d$ and $r := \|\theta\|$.

8.1 Estimation algorithms and the Rademacher statistic

Definition 158 (Norm estimation algorithm)**.** Maiti2026Power.IsAccurateNormEstdef:env, def:bai$_a lgAn estimation\ algorithm\ with\ budget\ T on \mathrm{B}_d$ is a pair (alg, ρ) of an LML algorithm with actions in $\mathbb{B}_d$ and observations in $\mathbb{R}$, and a Markov kernel ρ from histories of length T to $[0, \infty)$. Run against ν_θ (Definition **??**) it produces H_T and the estimate $\hat{r} \sim \rho(H_T)$; we write $\mathbb{P}_\theta$ for the law of $(H_T, \hat{r})$. It is (ε, δ)-*accurate* if for every $\theta \in \mathbb{R}^d$,

$$\mathbb{P}_\theta\big(|\hat{r} - \|\theta\|| \le \varepsilon\big) \ge 1 - \delta.$$

Lean remark. This is Definition **??** with the recommendation type $\mathcal{X}$ replaced by $[0, \infty)$ (or simply $\mathbb{R}$): the LML notion of an algorithm with a recommendation is polymorphic in the recommendation type, so no new framework is needed. All the estimators below are deterministic functions of the history, so ρ is `Kernel.deterministic`.

Definition 159 (Rademacher direction)**.** Maiti2026Power.radDir, Maiti2026Power.radDir$_m em_u nitBall, Maiti$
$:= \sigma/\sqrt{d} \in \mathbb{R}^d$ where $\sigma = (\sigma_1, \dots, \sigma_d)$ has i.i.d. coordinates uniform on $\{-1, +1\}$. We write Rad_d for its law (the image of the uniform measure on $\{-1, +1\}^d$ by $\sigma \mapsto \sigma/\sqrt{d}$). Since $\|x\| = 1$, $x \in \mathbb{S}^{d-1} \subset \mathbb{B}_d$.

Lean remark. The sign vector is encoded by a Boolean vector $u \in \{0, 1\}^d$ with the uniform law (`uniformBoolVec`, the seed of the meta-algorithm), through $\sigma_i := \mathrm{signOf}(u_i) \in \{-1, +1\}$; $\mathrm{radDir}(u) := \sigma/\sqrt{d}$.

Lemma 160 (Inner products with a Rademacher direction)**.** *Maiti2026Power.inner$_r$adDir, $Maiti2026Power.c$*
$= \sigma/\sqrt{d}$ be a Rademacher direction and $\theta \in \mathbb{R}^d$. Then $\langle x, \theta\rangle = \langle \sigma, \theta\rangle/\sqrt{d}$ and $d\,\langle x, \theta\rangle^2 = \langle \sigma, \theta\rangle^2$; in particular $\mathbb{E}\big[\langle x, \theta\rangle^2\big] = \|\theta\|^2/d$.

Proof. lem:rademacher$_f ourth_m oment \mathrm{E}[\sigma_i\sigma_j] = \mathbf{1}\{i = j\}$ by independence and $\mathbb{E}\sigma_i = 0$, $\sigma_i^2 = 1$, so $\mathbb{E}[\sigma\sigma^\top] = I_d$ and $\mathbb{E}[xx^\top] = I_d/d$. Then $\mathbb{E}\langle x, \theta\rangle^2 = \theta^\top \mathbb{E}[xx^\top]\theta = \|\theta\|^2/d$ (this is also the second-moment identity of Lemma **??** divided by d). The last identity is $\langle x, \theta\rangle = \langle \sigma, \theta\rangle/\sqrt{d}$. □

Definition 161 (Rademacher block sampling and the squared statistic)**.** Maiti2026Power.rbMeasure, Maiti2026Power.rbMean, Maiti2026Power.rbStat, Maiti2026Power.rbX, Maiti2026Power.rbW, Maiti2026Power.rbStat$_s ub_s q_e q def: rademacher_d irection, def: env, def: regret Let$s
≥ 1 and $K \ge 1$ be integers. The *Rademacher block sampling rule* $\mathrm{RB}(s, K)$ is the sampling rule with actions in $\mathbb{B}_d$ and budget sK which, for each $k < K$, draws at round ks a fresh Rademacher direction $x^{(k)} = \sigma^{(k)}/\sqrt{d}$ and plays the same action $x^{(k)}$ at rounds $ks + \ell$, $0 \le \ell < s$. Its *block space* is the product space

$$\Omega_{s,K} := \big(\{-1, +1\}^d\big)^K \times \big(\mathbb{R}^s\big)^K \quad \text{with the product law} \quad \mathbb{P}_{s,K} := \mathrm{Unif}\big(\{-1, +1\}^d\big)^{\otimes K} \otimes \big(\mathcal{N}(0, 1)^{\otimes s}\big)^{\otimes K}$$

(the directions of the K blocks and the noises of the sK rounds). For $(\sigma, \eta) \in \Omega_{s,K}$ and a reward vector θ with $r = \|\theta\|$, define for $k < K$

$$x^{(k)} := \frac{\sigma^{(k)}}{\sqrt{d}}, \qquad \mu_k := \langle x^{(k)}, \theta\rangle, \qquad \bar{y}_k := \mu_k + \frac{1}{s}\sum_{\ell<s}\eta_{k,\ell}, \qquad Z_k := d\Big(\bar{y}_k^2 - \frac{1}{s}\Big), \qquad \bar{Z} := \frac{1}{K}\sum_{k<K} Z_k,$$

$$X_k := d\mu_k^2 - r^2, \qquad W_k := Z_k - d\mu_k^2.$$

Then, as an algebraic identity,

$$\bar{Z} - r^2 = \frac{1}{K}\sum_{k<K} X_k + \frac{1}{K}\sum_{k<K} W_k.$$

When $\mathrm{RB}(s, K)$ is run against ν_θ, the observations of block k are $y_{ks+\ell} = \mu_k + \eta_{k,\ell}$ (Lemma **??**), so that $\bar{y}_k$ is the average of the observations of block k and $\bar{Z}$ is a statistic of the history.

Lean remark. $\mathrm{RB}(s, K)$ is not a standalone algorithm in Lean: it appears as the *windows* of the meta-algorithm of Definition **??**, whose seeds at the block starts are the sign vectors $\sigma^{(k)}$ and whose noises are the $\eta_{k,\ell}$ (Lemma **??**). All the probabilistic statements of this chapter about $\bar{Z}$ are statements about the block space $(\Omega_{s,K}, \mathbb{P}_{s,K})$ (`rbMeasure`), and are transported to the run through the window projections of Lemma **??**. The identity $\bar{Z} - r^2 = \frac{1}{K}\sum X_k + \frac{1}{K}\sum W_k$ is a rearrangement of $Z_k = X_k + W_k + r^2$.

Lean remark. The directions are the algorithm's internal randomness: in LML the policy of $\mathrm{RB}(s, K)$ is the kernel which, at rounds t with $s \mid t$, ignores the history and returns the measure Rad_d (a finitely supported measure on the subtype $\mathbb{B}_d$), and at the other rounds returns the Dirac mass at x_{t-1}. The quantities $\bar{y}_k$, Z_k, $\bar{Z}$ are measurable functions of the history of length sK. The identity $\bar{Z} - r^2 = \frac{1}{K}\sum X_k + \frac{1}{K}\sum W_k$ is a rearrangement of $Z_k = X_k + W_k + r^2$.

Lemma 162 (Window projections of the seed space)**.** *Maiti2026Power.nsMeasure, Maiti2026Power.rbProj,* $Maiti2026Power.nsMeasure_map_rbProj, Maiti2026Power.indepFun_prefix_rbProj, M$
$\big(\{-1,+1\}^d \times \mathbb{R}\big)^{\mathbb{N}}$ *with the product law* $\mathbb{P}_{\mathrm{seed}} := \big(\mathrm{Unif}(\{-1,+1\}^d) \otimes \mathcal{N}(0,1)\big)^{\otimes\mathbb{N}}$ *(the seed space of Lemma* **??** *for seeds which are uniform sign vectors and standard Gaussian noises), and write* $\omega_t = (\sigma_t, \eta_t)$ *for its coordinates.*

- *(i) For* $a \in \mathbb{N}$, $s, K \geq 1$, *the* Rademacher window projection $\pi_{a,s,K}(\omega) := \big((\sigma_{a+ks})_{k<K}, (\eta_{a+ks+\ell})_{k<K,\ell<s}\big) \in \Omega_{s,K}$ *has law* $\mathbb{P}_{s,K}$ *(Definition* **??***), and it is independent of the prefix* $(\omega_t)_{t<T}$ *whenever* $T \leq a$.
- *(ii) For* $a, n \in \mathbb{N}$, *the* basis window projection $\pi'_{a,n}(\omega) := (\eta_{a+in+\ell})_{i<d,\ell<n} \in (\mathbb{R}^n)^d$ *has law* $\mathcal{N}(0,1)^{\otimes dn}$, *and it is independent of the prefix* $(\omega_t)_{t<T}$ *whenever* $T \leq a$.
- *(iii) For an observation sequence* $y : \mathbb{N} \to \mathbb{R}$, *let* $\bar{Z}_{a,s,K}(y) := \frac{1}{K}\sum_{k<K} d\big((\frac{1}{s}\sum_{\ell<s} y_{a+ks+\ell})^2 - \frac{1}{s}\big)$ *be the squared statistic of the window of rounds* $[a, a+sK)$. *If* $y_{a+ks+\ell} = \langle\sigma_{a+ks}/\sqrt{d}, \theta\rangle + \eta_{a+ks+\ell}$ *for all* $k < K$, $\ell < s$, *then* $\bar{Z}_{a,s,K}(y) = \bar{Z}\big(\pi_{a,s,K}(\omega)\big)$, *the statistic of Definition* **??** *evaluated at the window projection.*

(iv) *For $a \in \mathbb{N}$, the shifted sequence $(\omega_{a+t})_{t\in\mathbb{N}}$ has law $\mathbb{P}_{\mathrm{seed}}$.*

Proof. lem:seeded, def:squared$_s$*tatisticThecoordinatesofani.i.d. sequenceindexedbyaninjectivefamilyofindi* *theimageof* $\mu^{\otimes\mathbb{N}}$ by $\omega \mapsto (\omega_{f(p)})_{p\in J}$ is $\mu^{\otimes J}$ for injective $f : J \to \mathbb{N}$ (`Measure.infinitePi_map_eval_comp`, from the restriction and reindexing lemmas of infinite product measures; for an infinite index set, as in (iv), the image measure is identified through its finite-dimensional cylinders, `Measure.eq_infinitePi`). The window projections are of this form (the indices $a + ks + \ell$, resp. $a + in + \ell$, are pairwise distinct), which gives the laws after splitting the pairs (σ, η) and currying the index (k, ℓ). The prefix and a window with $T \le a$ use disjoint sets of coordinates, and disjoint families of coordinates of an i.i.d. sequence are independent (`iIndepFun_infinitePi` and `indepFun_finset`). Part (iii) is a rewriting: the block average of the observations is $\mu_k + \frac{1}{s}\sum_\ell \eta_{k,\ell}$. □

Lean remark. The lemma replaces the sequential composition and the conditional independence arguments of the paper: on the seed space, every event of the multi-scale phase (rounds $t < T_1$) is a function of the prefix $(\omega_t)_{t<T_1}$, and every statistic of the second phase is a function of a window projection with $a = T_1$; the two are independent by (i)–(ii).

Proof. lem:noise$_r$*epresentation*, *lem* : *gauss*$_p$*i*, *lem* : *gauss*$_l$*inear*$_i$*mageByLemma* **??***the*η_t, $t < sK$, are i.i.d. $\mathcal{N}(0,1)$ and η_t is independent of (H_t, x_t). Since $x_{ks+\ell} = x^{(k)}$ for $\ell < s$, averaging the observations of block k gives $\bar{y}_k = \mu_k + \zeta_k$. The direction $x^{(k)}$ is drawn at round ks from the constant kernel Rad_d, so it is independent of H_{ks}, hence of $(x^{(j)}, \zeta_j)_{j<k}$; and $(\eta_{ks+\ell})_{\ell<s}$ is independent of $(H_{ks}, x^{(k)})$. An induction on k shows that the law of $\big((x^{(j)}, \zeta_j)\big)_{j\le k}$ is the product law: at each step the conditional law of $(x^{(k)}, (\eta_{ks+\ell})_{\ell<s})$ given the past is the constant kernel $\mathrm{Rad}_d \otimes \mathcal{N}(0,1)^{\otimes s}$ (Lemma **??**). Finally $\zeta_k = \frac{1}{s}\mathbf{1}^\top(\eta_{ks+\ell})_{\ell<s} \sim \mathcal{N}(0, 1/s)$ by Lemma **??**. □

Lean remark. The proof is an induction on the blocks using the step decomposition of the trajectory measure (LML `IsAlgEnvSeq.hasCondDistrib_step`): since the policy kernel at round ks does not depend on the history, the conditional law of the block k given H_{ks} is a constant kernel, which is exactly independence from the past. Alternatively one can view $\mathrm{RB}(s, K)$ as the K-fold sequential composition (Lemma **??**) of the one-block rule $\mathrm{RB}(s, 1)$.

8.2 Additive estimation given a constant-factor estimate

Throughout this section $\mathrm{RB}(s, K)$ is run against ν_θ and the notation of Definition **??** is in force. The decomposition $\bar{Z} - r^2 = \frac{1}{K}\sum_k X_k + \frac{1}{K}\sum_k W_k$ separates the fluctuations of the directions (Term 1, the X_k) from the fluctuations of the noise (Term 2, the W_k).

Lemma 163 (Term 1 is sub-exponential)**.** *Maiti2026Power.hasSubexponentialMGF*$_r bX, Maiti2026Power.iInd$ *the variables* $X_k = d\mu_k^2 - r^2$*,* $k < K$*, are independent and each has* HasSubexponentialMGF X_k $(16r^4)$ $(4r^2)$*. If* $r = 0$ *then* $X_k = 0$ *for all* k*.*

Proof. lem:rademacher$_{s}quare_{s}ubexp, lem : direction_{s}econd_{m}oment, def : squared_{s}tatisticByLemma$ **??**,dμ_k^2 = $\langle \sigma^{(k)}, \theta\rangle^2$ where $\sigma^{(k)}$ is a uniform sign vector, so $X_k = \langle \sigma^{(k)}, \theta\rangle^2 - \|\theta\|^2$, which has the sub-exponential parameters $(16r^4, 4r^2)$ by Lemma **??**. The $\sigma^{(k)}$ are i.i.d. by Lemma **??**, hence so are the X_k. When $\theta = 0$, $\mu_k = 0 = r$. ☐

Lemma 164 (Tail bound for Term 1)**.** *Maiti2026Power.rbMeasure*$_r eal_a vg_r bX_g t_l elem : term1_s ubexpAssumer$ $> 0.Foreveryt \geq 0,$

$$\mathbb{P}_{s,K}\Big(\Big|\frac{1}{K}\sum_{k<K} X_k\Big| > t\Big) \leq 2\exp\Big(-K\min\Big(\frac{t^2}{32r^4}, \frac{t}{8r^2}\Big)\Big).$$

If $r = 0$ *the left-hand side is* 0*.*

Proof. cor:subexp$_{a}verage_{t}ail, lem : term1_{s}ubexpCorollary$ **??**$appliedtotheindependentvariables$X$_k$ with the common parameters $(V, b) = (16r^4, 4r^2)$ (Lemma **??**) gives the bound $2\exp(-K\min(t^2/(2V), t/(2b)))$, and $2V = 32r^4$, $2b = 8r^2$. The case $r = 0$ is $X_k = 0$. ☐

Lemma 165 (Term 2 is conditionally sub-exponential)**.** *Maiti2026Power.rbNoise, Maiti2026Power.hasLaw*$_s qrt_m ul_r bMean, Maiti2026Power.hasSubexponentialMGF_r bW, Maiti2026Power.iI$ *let* $x^{(k)} := \sigma^{(k)}/\sqrt{d}$*,* $\mu_k := \langle x^{(k)}, \theta\rangle$*, and let the noises* $(\eta_{k,\ell})_{k<K,\ell<s}$ *be i.i.d.* $\mathcal{N}(0,1)$ *(the noise marginal* $\mathbb{P}^{\eta}_{s,K} := (\mathcal{N}(0,1)^{\otimes s})^{\otimes K}$ *of the block space). Define* $\bar{y}_k$*,* Z_k *and* $W_k := Z_k - d\mu_k^2$ *as in Definition* **??***. Then* $\sqrt{d}\,\bar{y}_k \sim \mathcal{N}(\sqrt{d}\,\mu_k, d/s)$*, the* W_k *are independent and* HasSubexponentialMGF W_k V_k b *with*

$$V_k := 8\Big(\frac{d^2}{s^2} + \mu_k^2\,\frac{d^2}{s}\Big), \qquad b := \frac{4d}{s}.$$

Proof. lem:gaussian$_{s}quare_{s}ubexp, lem : gauss_{l}inear_{i}mage\sqrt{d}\,\bar{y}_k \sim \mathcal{N}(\sqrt{d}\,\mu_k, d/s)$ (Lemma **??**), and

$$W_k = d\bar{y}_k^2 - \frac{d}{s} - d\mu_k^2 = \big(\sqrt{d}\,\bar{y}_k\big)^2 - \Big((\sqrt{d}\,\mu_k)^2 + \frac{d}{s}\Big),$$

which is $X^2 - (\mu^2 + \sigma^2)$ for $X \sim \mathcal{N}(\mu, \sigma^2)$ with $\mu = \sqrt{d}\,\mu_k$, $\sigma^2 = d/s$. Lemma **??** gives the parameters $(8(\sigma^4 + \mu^2\sigma^2), 4\sigma^2) = (8(d^2/s^2 + \mu_k^2 d^2/s), 4d/s)$. Independence of the W_k follows from that of the ζ_k, each W_k being a function of ζ_k alone. ☐

Lemma 166 (Tail bound for Term 2)**.** *Maiti2026Power.rbVbar, Maiti2026Power.sum*$_r bVbar_e q, Maiti2026Pow$ $\frac{1}{K}\sum_{k<K} V_k = 8\Big(\frac{d^2}{s^2} + \frac{d^2}{s}\cdot\frac{1}{K}\sum_{k<K}\mu_k^2\Big)$, $\qquad b := \frac{4d}{s}$, a function of the directions. Then for every $\tau \in \mathbb{R}$ and $t \geq 0$,

$$\mathbb{P}_{s,K}\Big(\Big|\frac{1}{K}\sum_{k<K} W_k\Big| > t\Big) \leq \mathbb{P}_{s,K}(\bar{V} > \tau) + 2\exp\Big(-K\min\Big(\frac{t^2}{2\tau}, \frac{t}{2b}\Big)\Big).$$

Proof. lem:term2$_{c}onditional, lem : statistic_{s}tructure, cor : subexp_{a}verage_{t}ailByLemma$ **??**$thejointlawofthe$
$\mathcal{N}(0,1/s)^{\otimes K}$, and $\frac{1}{K}\sum_k W_k$ is a measurable function $F(x^{(0:K)}, \zeta_{0:K})$. Split the event on $\{\bar V \le \tau\}$, which depends on the directions only, and apply Fubini:

$$\mathbb{P}_\theta\Big(\Big|\frac{1}{K}\sum_k W_k\Big| > t,\ \bar V \le \tau\Big) = \int \mathbf{1}\{\bar V(x) \le \tau\}\, \mathbb{P}_\zeta\Big(\Big|\frac{1}{K}\sum_k W_k(x,\zeta)\Big| > t\Big)\, \mathrm{Rad}_d^{\otimes K}(dx).$$

For fixed directions x, Lemma **??** and Corollary **??** bound the inner probability by $2\exp(-K\min(t^2/(2\bar V(x)), t/(2b)))$, which is at most $2\exp(-K\min(t^2/(2\tau), t/(2b)))$ when $\bar V(x) \le \tau$. Adding $\mathbb{P}_\theta(\bar V > \tau)$ for the complementary event gives the claim. ☐

Lean remark. The block space is the product $\mathrm{Unif}(\{-1,+1\}^d)^{\otimes K} \otimes (\mathcal{N}(0,1)^{\otimes s})^{\otimes K}$ and the bound is proved by Fubini (`MeasureTheory.Measure.prod`) over the directions, with the inner bound of Lemma **??** for each fixed direction vector. No conditional expectation is needed.

Lemma 167 (Bound on the conditional variance proxy)**.** *Maiti2026Power.rbMeasure$_{r}eal_{r}bVbar_{g}t_{l}elem : term2$*
$8\Big(\frac{d^2}{s^2} + \frac{3dr^2}{2s}\Big)$, *one has* $\mathbb{P}_{s,K}(\bar V > \tau) \le 2\exp(-K/128)$.

Proof. lem:term1$_{b}ound, def : squared_{s}tatisticSince1\frac{}{K\sum_k d\mu_k^2 = r^2 + \frac{1}{K}\sum_k X_k}$, on the event $\{|\frac{1}{K}\sum_k X_k| \le r^2/2\}$ we have $\frac{1}{K}\sum_k \mu_k^2 \le 3r^2/(2d)$ and hence $\bar V \le 8(d^2/s^2 + (d^2/s)\cdot 3r^2/(2d)) = \tau$. If $r = 0$ this event is sure and $\bar V = 8d^2/s^2 \le \tau$. If $r > 0$, Lemma **??** with $t = r^2/2$ gives $\min(t^2/(32r^4), t/(8r^2)) = \min(1/128, 1/16) = 1/128$, so the complementary event has probability at most $2\exp(-K/128)$. ☐

Definition 168 (Additive estimation algorithm; Algorithm 1)**.** Maiti2026Power.addS, Maiti2026Power.addKdef:squared$_{s}tatistic, def : norm_{e}stimatorLet\varepsilon \in (0,1]$, $\delta \in (0,1)$ and $r_0 > 0$ (a scale parameter), and set

$$c_0 := 4, \qquad c_1 := 4096, \qquad s(r_0) := \Big\lceil \frac{c_0 d}{r_0^2} \Big\rceil, \qquad K(r_0) := \Big\lceil \frac{c_1 r_0^2 \log(4/\delta)}{\varepsilon^2} \Big\rceil.$$

Algorithm 1 with parameters $(\varepsilon, \delta, r_0)$, written $\mathrm{AddEst}(\varepsilon, \delta, r_0)$, is the estimation algorithm (Definition **??**) with budget $s(r_0)K(r_0)$ whose sampling rule is the Rademacher block sampling rule $\mathrm{RB}(s(r_0), K(r_0))$ of Definition **??** and whose estimate is the deterministic function of the history

$$\hat r := \sqrt{\max(\bar Z, 0)},$$

where $\bar Z$ is the squared statistic of Definition **??**. When ε and δ are fixed we write $s := s(r_0)$ and $K := K(r_0)$.

Lean remark. Only the parameters $s(r_0), K(r_0)$ (natural numbers computed by `Nat.ceil`) are defined on their own; the algorithm itself is the second phase of the meta-algorithm of Definition **??** for the data-dependent scale $r_0 = t_{j_*}$, and its estimate is a case of `NormEstParam.estimate`. The guarantee below is stated on the block space.

Lean remark. The algorithm is a pair (LML algorithm, deterministic recommendation kernel); its parameters s, K are natural numbers computed from $(\varepsilon, \delta, r_0)$ by `Nat.ceil`. The meta-algorithm of Definition **??** instantiates it with a data-dependent r_0 taking finitely many values.

Theorem 169 (Additive estimation with a constant-factor estimate; Algorithm 1)**.** $Maiti2026Power.rbMeasure_real_addEst_gt_le, Maiti2026Power.rbMeasure_real_sqrt_addEst_gt_le, Maiti2$ *$(0,1]$, $\delta \in (0,1)$, $r_0 \geq \varepsilon$, and let $s = s(r_0)$, $K = K(r_0)$ be the parameters of Algorithm 1 (Definition* **??***, with $c_0 = 4$, $c_1 = 4096$). Then for every θ with $r/2 < r_0 \leq 2r$, on the block space $(\Omega_{s,K}, \mathbb{P}_{s,K})$,*

$$\mathbb{P}_{s,K}\Big(|\bar{Z} - r^2| \leq \frac{r\varepsilon}{2}\Big) \geq 1 - \frac{3\delta}{8}, \qquad \textit{hence} \qquad \mathbb{P}_{s,K}\Big(|\hat{r} - r| \leq \frac{\varepsilon}{2}\Big) \geq 1 - \frac{3\delta}{8} \geq 1 - \frac{\delta}{2}.$$

Moreover, for the scales $r_0 = t_j = 2^j\varepsilon$ with $t_j < 2\sqrt{d}$ (the scales $j < J$ of Definition **??***), the budget satisfies*

$$sK \leq C\,\frac{d\log(4/\delta)}{\varepsilon^2}, \qquad C := 32776.$$

Proof. lem:term1$_bound, lem : term2_bound, lem : vbar_bound, def : squared_statistic, def : additive_algorithm, def : multiscale_algorithmWrite$L $:= \log(4/\delta) \geq \log 4 > 1$ and note $\log(16/\delta) = \log 4 + L \leq 2L$. Since $\varepsilon \leq r_0 \leq 2r$ we have $r > 0$; the hypotheses also give $r_0/2 < r < 2r_0$. By Definition **??**, $\{|\bar{Z} - r^2| > r\varepsilon/2\}$ is contained in the union of $E_1 := \{|\frac{1}{K}\sum X_k| > r\varepsilon/4\}$ and $E_2 := \{|\frac{1}{K}\sum W_k| > r\varepsilon/4\}$.

Term 1. Lemma **??** with $t = r\varepsilon/4$: $t^2/(32r^4) = \varepsilon^2/(512r^2)$ and $t/(8r^2) = \varepsilon/(32r)$. Since $\varepsilon \leq 2r \leq 16r$, the minimum is $\varepsilon^2/(512r^2)$, so $\mathbb{P}(E_1) \leq 2\exp(-K\varepsilon^2/(512r^2))$. This is at most $\delta/8$ as soon as $K \geq 512r^2\log(16/\delta)/\varepsilon^2$, which holds because $r^2 < 4r_0^2$ and $\log(16/\delta) \leq 2L$ give $512r^2\log(16/\delta)/\varepsilon^2 \leq 4096\,r_0^2L/\varepsilon^2 \leq K$.

Variance proxy. Lemma **??**: $\mathbb{P}(\bar{V} > \tau) \leq 2e^{-K/128} \leq \delta/8$ because $K \geq 4096r_0^2L/\varepsilon^2 \geq 4096L \geq 128\log(16/\delta)$ (using $r_0 \geq \varepsilon$).

Term 2. Since $s \geq 4d/r_0^2$ we have $d/s \leq r_0^2/4$, hence $b = 4d/s \leq r_0^2$ and, with $r^2 < 4r_0^2$,

$$\tau = 8\Big(\frac{d^2}{s^2} + \frac{3dr^2}{2s}\Big) \leq 8\Big(\frac{r_0^4}{16} + \frac{3r^2r_0^2}{8}\Big) \leq 8\Big(\frac{r_0^4}{16} + \frac{3r_0^4}{2}\Big) = \frac{25}{2}\,r_0^4.$$

With $t = r\varepsilon/4$ and $r > r_0/2$: $t^2/(2\tau) \geq (r^2\varepsilon^2/16)/(25r_0^4) = r^2\varepsilon^2/(400r_0^4) > \varepsilon^2/(1600r_0^2)$ and $t/(2b) \geq r\varepsilon/(8r_0^2) > \varepsilon/(16r_0)$; since $\varepsilon \leq r_0$, $\varepsilon^2/(1600r_0^2) \leq \varepsilon/(16r_0)$, so the minimum in Lemma **??** is at least $\varepsilon^2/(1600r_0^2)$ and $\mathbb{P}(E_2) \leq \mathbb{P}(\bar{V} > \tau) + 2\exp(-K\varepsilon^2/(1600r_0^2))$. The last term is at most $\delta/8$ as soon as $K \geq 1600r_0^2\log(16/\delta)/\varepsilon^2$, which holds since $1600r_0^2\log(16/\delta)/\varepsilon^2 \leq 3200\,r_0^2L/\varepsilon^2 \leq K$.

Union bound. $\mathbb{P}(|\bar{Z} - r^2| > r\varepsilon/2) \leq \delta/8 + \delta/8 + \delta/8 = 3\delta/8$.

From r^2 to r. On the event $|\bar{Z} - r^2| \leq r\varepsilon/2$, since $r > r_0/2 \geq \varepsilon/2$, $\bar{Z} \geq r(r - \varepsilon/2) > 0$, so $\hat{r} = \sqrt{\bar{Z}}$ and

$$|\hat{r} - r| = \frac{|\bar{Z} - r^2|}{\sqrt{\bar{Z}} + r} \leq \frac{r\varepsilon/2}{r} = \frac{\varepsilon}{2}.$$

Budget. For $r_0 = t_j < 2\sqrt{d}$, $4d/r_0^2 > 1$, so $s \le 4d/r_0^2 + 1 \le 8d/r_0^2 = (8d/\varepsilon^2)\,4^{-j}$; and $K \le 4096\,r_0^2 L/\varepsilon^2 + 1 = 4096 \cdot 4^j L + 1$. Hence $sK \le (8d/\varepsilon^2)(4096\,L + 4^{-j}) \le (8d/\varepsilon^2)(4096\,L+1) \le 8 \cdot 4097\,dL/\varepsilon^2 = 32776\,dL/\varepsilon^2$, using $L \ge 1$. □

Remark 170 (On the constants of Algorithm 1). The paper takes c_0, c_1 "large enough"; the values $c_0 = 4$, $c_1 = 4096$ are what the computation above produces with the sub-exponential parameters $(16r^4, 4r^2)$ of Lemma **??** and the threshold $t = r\varepsilon/4$ of the paper. The proof shows that $|\hat{r} - r| \le \varepsilon/2$, twice as much as needed; using the threshold $t = r\varepsilon/2$ instead would divide c_1 (and C) by four. The three failure events are bounded by $\delta/8$ each, so the total failure probability is $3\delta/8$ rather than the $\delta/2$ claimed by the paper; the extra room is not used.

8.3 A constant-factor estimate by a multi-scale test

Definition 171 (Scale test)**.** Maiti2026Power.testS, Maiti2026Power.testKdef:squared$_s$*tatisticForascale*t > 0*andaconfidencelevel*$\delta' \in (0,1)$, the test $\mathrm{Test}(t, \delta')$ is the algorithm $\mathrm{RB}(s, K)$ with

$$s := \left\lceil \frac{c_0 d}{t^2} \right\rceil, \qquad K := \left\lceil c_1' \log(1/\delta') \right\rceil, \qquad c_0 := 4, \quad c_1' := 5120,$$

followed by the binary output $\hat{H} := H_1$ if $U \ge \frac{3}{2}t^2$ and $\hat{H} := H_0$ otherwise, where $U := \bar{Z}$ is the statistic of Definition **??**. The hypothesis H_0 is "$r \le t$" and H_1 is "$r \ge 2t$"; when $t < r < 2t$ neither output counts as an error. Its budget is sK.

Lean remark. As for Algorithm 1, only the parameters are defined on their own; the tests are the windows of the first phase of the meta-algorithm (Definition **??**), and the error bounds below are stated on the block space.

Lemma 172 (Error of the test under H_0)**.** *Maiti2026Power.testS$_f$acts, Maiti2026Power.testK$_f$acts, Maiti202* $(0, 1/4]$ *and* θ *with* $r \le t$*. Then, on the block space of* $\mathrm{RB}(s, K)$, $\mathbb{P}_{s,K}(U \ge \frac{3}{2}t^2) \le 3\delta'/4$*: the test returns* H_1 *with probability at most* $3\delta'/4$*.*

Proof. lem:term1$_b$*ound, lem : term2$_b$ound, lem : vbar$_b$ound, def : squared$_s$tatisticWrite*$\mathrm{T}_1 := \frac{1}{K}\sum_k X_k$, $T_2 := \frac{1}{K}\sum_k W_k$, so $U = r^2 + T_1 + T_2$. If $|T_1| \le t^2/8$ and $|T_2| \le t^2/8$ then $U \le r^2 + t^2/4 \le 5t^2/4 < 3t^2/2$ and the output is H_0. Hence the error event is contained in $\{|T_1| > t^2/8\} \cup \{|T_2| > t^2/8\}$.

Term 1. If $r = 0$ then $T_1 = 0$. Otherwise Lemma **??** with $a_0 = t^2/8$ gives $a_0^2/(32r^4) = t^4/(2048r^4) \ge 1/2048$ and $a_0/(8r^2) = t^2/(64r^2) \ge 1/64$ since $r \le t$; so $\mathbb{P}(|T_1| > t^2/8) \le 2e^{-K/2048}$.

Term 2. Since $s \ge 4d/t^2$: $d/s \le t^2/4$, $b = 4d/s \le t^2$ and $\tau = 8(d^2/s^2 + 3dr^2/(2s)) \le 8(t^4/16 + 3t^4/8) = 7t^4/2$ (using $r \le t$). With $\nu_0 = t^2/8$: $\nu_0^2/(2\tau) \ge (t^4/64)/(7t^4) = 1/448$ and $\nu_0/(2b) \ge (t^2/8)/(2t^2) = 1/16$. Lemmas **??** and **??** give $\mathbb{P}(|T_2| > t^2/8) \le 2e^{-K/128} + 2e^{-K/448}$.

Total. The error probability is at most $2e^{-K/2048} + 2e^{-K/128} + 2e^{-K/448} \leq 6e^{-K/2048}$. Since $\delta' \leq 1/4$, $\log 8 = \frac{3}{2}\log 4 \leq \frac{3}{2}\log(1/\delta')$, hence $2048\log(8/\delta') \leq 2048 \cdot \frac{5}{2}\log(1/\delta') = 5120\log(1/\delta') \leq K$, and $6e^{-K/2048} \leq 6\delta'/8 = 3\delta'/4$. □

Lemma 173 (Error of the test under H_1)**.** *Maiti2026Power.rbMeasure$_r$eal$_t$est$_h$1$_l$edef : scale$_t$estLet$\delta' \in (0, 1/4]$ and θ with $r \geq 2t$. Then $\mathbb{P}_{s,K}(U < \frac{3}{2}t^2) \leq 3\delta'/4$: the test returns H_0 with probability at most $3\delta'/4$.*

Proof. lem:term1$_b$ound, lem : term2$_b$ound, lem : vbar$_b$ound, def : squared$_s$tatisticWithT_1, T_2 as above, if $|T_1| \leq r^2/8$ and $|T_2| \leq r^2/8$ then $U \geq r^2 - r^2/4 = 3r^2/4 \geq 3t^2$ (as $r^2 \geq 4t^2$) and the output is H_1. Hence the error event is contained in $\{|T_1| > r^2/8\} \cup \{|T_2| > r^2/8\}$; here $r > 0$.

Term 1. Lemma **??** with $a_1 = r^2/8$: $\min(a_1^2/(32r^4), a_1/(8r^2)) = \min(1/2048, 1/64) = 1/2048$, so $\mathbb{P}(|T_1| > r^2/8) \leq 2e^{-K/2048}$.

Term 2. As before $d/s \leq t^2/4 \leq r^2/16$ and $b \leq t^2 \leq r^2/4$, and $\tau \leq 8(t^4/16 + 3t^2r^2/8) \leq 8(r^4/256 + 3r^4/32) = 25r^4/32$. With $\nu_1 = r^2/8$: $\nu_1^2/(2\tau) \geq (r^4/64)/(25r^4/16) = 1/100$ and $\nu_1/(2b) \geq (r^2/8)/(r^2/2) = 1/4$. Lemmas **??** and **??** give $\mathbb{P}(|T_2| > r^2/8) \leq 2e^{-K/128} + 2e^{-K/100}$.

Total. At most $6e^{-K/2048} \leq 3\delta'/4$ exactly as in Lemma **??**. □

Definition 174 (Multi-scale algorithm; Algorithm 2)**.** Maiti2026Power.nsScale, Maiti2026Power.nsDelta, Maiti2026Power.nsS, Maiti2026Power.nsK, Maiti2026Power.nsJ, Maiti2026Power.nsStart, Maiti2026Power.one$_l e_n sJ, Maiti2026Power.nsScale_n sJ_l t, Maiti2026Power.NormE$ $(0, 1]$, $\delta \in (0, 1)$ and set, for $j \in \mathbb{N}$,

$$t_j := 2^j\varepsilon, \qquad \delta_j := \frac{\delta}{2^{j+2}}, \qquad s_j := \left\lceil \frac{4d}{t_j^2} \right\rceil, \qquad K_j := \lceil 5120\log(1/\delta_j) \rceil, \qquad J := \min\{j \in \mathbb{N} : t_j \geq 2\sqrt{d}\}.$$

Since $\varepsilon \leq 1 < 2\sqrt{d}$, $J \geq 1$; and $t_J < 4\sqrt{d}$ by minimality. The *multi-scale algorithm* runs the tests $\mathrm{Test}(t_j, \delta_j)$, $j = 0, \ldots, J-1$, one after the other: test j occupies the rounds $[\mathrm{start}_j, \mathrm{start}_{j+1})$ with $\mathrm{start}_j := \sum_{i<j} s_i K_i$, and its budget is the deterministic number $N_{\mathrm{ms}} := \mathrm{start}_J = \sum_{j<J} s_j K_j$. Its output is the index

$$j_* := \min\left(\{j < J : \mathrm{Test}(t_j, \delta_j) \text{ returns } H_0\} \cup \{J\}\right),$$

i.e. the first scale at which the test returns H_0, or J if all J tests return H_1, and the constant-factor estimate is $r_0 := t_{j_*}$. It satisfies $\varepsilon \leq r_0 \leq t_J < 4\sqrt{d}$ and $r_0 \in \{t_0, \ldots, t_J\}$.

Lean remark. The paper stops the loop at j_* and idles afterwards; since the budget is deterministic anyway, the formalization runs all J tests (the tests after j_* are simply not used by the output), which makes the sampling rule of the first phase non-adaptive. The output j_* is a measurable function of the observations of the rounds $t < N_{\mathrm{ms}}$ (`NormEstParam.jStar`, defined by `Nat.find` from the test outcomes `testH1`), with finitely many values; the multi-scale algorithm is the first phase of the meta-algorithm of Definition **??**.

Lean remark. The loop is a sequential composition of J fixed-length phases (Lemma **??**) whose parameters depend on the past: phase j is $\mathrm{Test}(t_j, \delta_j)$ if no earlier test returned H_0 (a measurable function of the history, computed from the statistics of the earlier blocks) and the constant algorithm playing 0 otherwise. The data-dependent output r_0 is a measurable function of the full history of length N_{ms} with finitely many values. The algorithm is adaptive, but in a very restricted sense: only the choice between "continue" and "idle" depends on the data.

Lemma 175 (Good event of the multi-scale algorithm)**.** *Maiti2026Power.NormEstParam.testOK, Maiti2026Power.NormEstParam.badTest, Maiti2026Power.NormEstParam.nsMeasure*$_real_badTest_le, Maiti202$ *scalealgorithmagainst*ν_θ *on the seed space of Lemma* **??***. For* $j < J$ *let* E_j *be the event that the outcome of test* j *is wrong: it returns* H_1 *while* $r \leq t_j$*, or* H_0 *while* $r \geq 2t_j$*. Then* $\mathbb{P}_{\mathrm{seed}}(E_j) \leq 3\delta_j/4$ *and* $\mathbb{P}_{\mathrm{seed}}\big(\bigcup_{j<J} E_j\big) \leq 3\delta/8$*.*

Proof. lem:statistic$_structure, lem: norm_meta_plan, lem: test_error_h0, lem: test_error_h1, def:$ $multiscale_algorithmByLemma$ **??**$, theobservationsofthewindowoftest$j$arey_{\mathrm{start}_j+ks+\ell} =$ $\langle \sigma_{\mathrm{start}_j+ks}/\sqrt{d}, \theta\rangle + \eta_{\mathrm{start}_j+ks+\ell}$, so by Lemma **??**(iii) the statistic U_j of test j is $\bar{Z}$ evaluated at the window projection $\pi_{\mathrm{start}_j,s_j,K_j}$, whose law is the block law $\mathbb{P}_{s_j,K_j}$ (Lemma **??**(i)). Hence E_j is the preimage under this projection of the error event of the test on the block space, and its probability is at most $3\delta_j/4$ by Lemma **??** (if $r \leq t_j$) or Lemma **??** (if $r \geq 2t_j$); when $t_j < r < 2t_j$, $E_j = \emptyset$. Here $\delta_j \leq \delta/4 < 1/4$. The union bound gives $\sum_{j<J} 3\delta_j/4 \leq \frac{3\delta}{4}\sum_{j\geq 0} 2^{-j-2} = \frac{3\delta}{8}$. $\square$

Theorem 176 (Guarantee of the multi-scale algorithm)**.** *Maiti2026Power.NormEstParam.jStar*$_bounds def: m$ $j < J$*, be the events of Lemma* **??** *and let* G *be the complement of* $\bigcup_{j<J} E_j$ *(all tests are correct), so* $\mathbb{P}_{\mathrm{seed}}(G) \geq 1 - 3\delta/8$ *for every* θ*. On* G *the output* j_* *satisfies one of the following:*

(a) $j_* = 0$ *(that is,* $r_0 = \varepsilon$*) and* $r < 2\varepsilon$*;*

(b) $0 < j_* < J$*,* $r/2 < t_{j_*} \leq 2r$*, and* $\varepsilon \leq t_{j_*} < 2\sqrt{d}$*;*

(c) $j_* = J$ *(that is,* $r_0 = t_J \geq 2\sqrt{d}$*) and* $r \geq \sqrt{d}$*.*

Proof. lem:norm$_multiscale_good_event, def: multiscale_algorithmTheprobabilityboundisLemma$ **??**$.Fixanoutc$ $everytest$j $< \mathrm{J} returns \mathrm{H}_1$ if $r \geq 2t_j$ and H_0 if $r \leq t_j$. Recall $t_{j+1} = 2t_j$, $J \geq 1$ and $t_J \geq 2\sqrt{d}$.

If $j_* = 0$: test 0 returned H_0, which is correct only if $r < 2t_0 = 2\varepsilon$.

If $j_* > 0$: test $j_* - 1$ returned H_1, which is correct only if $r > t_{j_*-1} = t_{j_*}/2$, so $t_{j_*} < 2r$. If moreover $j_* < J$, test j_* returned H_0, so $r < 2t_{j_*}$, i.e. $r/2 < t_{j_*}$; and $\varepsilon \leq t_{j_*} < 2\sqrt{d}$ by the definition of J: this is (b). If $j_* = J$ then $\sqrt{d} \leq t_J/2 = t_{J-1} < r$: this is (c). $\square$

Lemma 177 (Budget of the multi-scale algorithm)**.** *Maiti2026Power.sum*$_pow_le, Maiti2026Power.sum_mul_pow_l$ *:=* $\log(4/\delta)$*, the budget of the multi-scale algorithm satisfies*

$$N_{\mathrm{ms}} = \sum_{j<J} s_j K_j \leq \frac{8d}{\varepsilon^2}\Big(6828\,L + 1578\Big) \leq 63736\,\frac{d\log(4/\delta)}{\varepsilon^2} \leq C'\,\frac{d\log(4/\delta)}{\varepsilon^2}, \qquad C' := 64000.$$

Proof. def:multiscale$_{a}lgorithmFor$j $<$ J,t$_j$ $< 2\sqrt{d}$, so $4d/t_j^2 > 1$ and $s_j \le 4d/t_j^2 + 1 \le 8d/t_j^2 = 8d\,4^{-j}/\varepsilon^2$. Next, $\log(1/\delta_j) = L + j\log 2$, so $K_j \le 5120L + 5120\,j\log 2 + 1 \le 5121\,L + 3549\,j$ using $L \ge \log 4 > 1$ and $5120\log 2 \le 3549$. With $\sum_{j\ge 0} 4^{-j} = 4/3$ and $\sum_{j\ge 0} j4^{-j} = 4/9$,

$$N_{\mathrm{ms}} \le \frac{8d}{\varepsilon^2}\sum_{j\ge 0} 4^{-j}(5121L+3549j) = \frac{8d}{\varepsilon^2}\Big(\frac{4}{3}\,5121\,L+\frac{4}{9}\,3549\Big) \le \frac{8d}{\varepsilon^2}\big(6828\,L+1578\big).$$

Finally $1578 \le 1139\,L$ (as $L \ge \log 4 \ge 1.386$), so $N_{\mathrm{ms}} \le 8\cdot 7967\,dL/\varepsilon^2 \le 64000\,dL/\varepsilon^2$. □

8.4 The large-norm regime

Definition 178 (Large-norm estimator; Algorithm 3)**.** Maiti2026Power.lnN, Maiti2026Power.lnNoise, Maiti2026Power.lnDelta, Maiti2026Power.lnR, Maiti2026Power.lnEstdef:fixed$_{d}esign,$ $norm_{e}stimatorLet\varepsilon \in (0,1]$, $\delta \in (0,1)$ and $n := \lceil 48\log(4/\delta)/\varepsilon^2\rceil$. The *large-norm estimator* is the fixed-design estimation algorithm with budget dn which plays each basis vector $e_i \in \mathbb{B}_d$, $i = 1,\dots,d$, exactly n times (in any fixed order), computes the empirical means $\hat\theta_i := \frac{1}{n}\sum_{t:x_t=e_i} y_t$, and outputs

$$\hat R := \|\hat\theta\|^2 - \frac{d}{n}, \qquad \hat r := \sqrt{\max(\hat R, 0)}.$$

On the *noise space* $(\mathbb{R}^n)^d$ with the law $\mathcal{N}(0,1)^{\otimes dn}$ of the noises $\eta = (\eta_{i,\ell})_{i<d,\ell<n}$ of these rounds, $\hat\theta = \theta + \Delta$ with $\Delta_i := \frac{1}{n}\sum_{\ell<n}\eta_{i,\ell}$, and $\hat R$, $\hat r$ are functions of (θ,η).

Lean remark. The basis design is the second phase of the meta-algorithm of Definition **??** when $j_* = J$ (rounds $T_1 + in + \ell$ play e_i); the guarantee below is stated on the noise space, and transported through the basis window projection of Lemma **??**(ii).

Lemma 179 (Decomposition of the large-norm estimator)**.** *Maiti2026Power.lnR$_{s}ub_{e}q$, $Maiti$2026$Power.hasL$*
$\mathcal{N}(0, I_d/n)$ and

$$\hat R - r^2 = L + Q, \qquad L := 2\langle\theta,\Delta\rangle \sim \mathcal{N}\big(0, 4r^2/n\big), \qquad Q := \|\Delta\|^2 - \frac{d}{n} = \frac{\|g\|^2 - d}{n},$$

where $g := \sqrt{n}\,\Delta \sim \mathcal{N}(0, I_d)$; Q has HasSubexponentialMGF Q $(8d/n^2)$ $(4/n)$.

Proof. lem:gauss$_{p}i, lem : gauss_{l}inear_{i}mage, lem : chi_{s}quare_{s}ubexp, lem : subexp_{c}onst_{m}ulThecoordinate_{i}$ is θ_i plus the average of the n noise variables of the rounds where e_i is played. These d averages are independent $\mathcal{N}(0,1/n)$ (a sum of independent Gaussians is Gaussian, Lemmas **??** and **??**), so $\Delta \sim \mathcal{N}(0, I_d/n)$. Expanding $\|\theta+\Delta\|^2 = r^2 + 2\langle\theta,\Delta\rangle + \|\Delta\|^2$ gives the decomposition; L is a linear image of Δ with variance $4\theta^\top(I/n)\theta$ (Lemma **??**). Finally $\|g\|^2 - d$ has parameters $(8d, 4)$ by Lemma **??**, and scaling by $1/n$ gives $(8d/n^2, 4/n)$ by Lemma **??**. □

Theorem 180 (Guarantee in the large-norm regime)**.** $Maiti2026Power.lnNoise_{r}eal_{l}nEst_{g}t_{l}e, Maiti2026Power$
$(0,1]$, $\delta \in (0,1)$ *and* θ *with* $r \geq \sqrt{d}$*. Then, on the noise space,*

$$\mathbb{P}\big(|\hat{R} - r^2| \geq \varepsilon r\big) \leq \frac{\delta}{2}, \qquad \textit{hence} \qquad \mathbb{P}\big(|\hat{r} - r| \leq \varepsilon\big) \geq 1 - \frac{\delta}{2},$$

and the budget satisfies $n \leq 48\log(4/\delta)/\varepsilon^2 + 1 \leq 49\log(4/\delta)/\varepsilon^2$*, so* $dn \leq 49\, d\log(4/\delta)/\varepsilon^2$*.*

Proof. lem:large$_{n}orm_{d}ecomposition, lem : gauss_{t}ail, lem : subexp_{t}ail, lem : gauss_{l}aw_{o}f_{c}ovWrite$L$_\delta$ $:= \log(4/\delta) > 1$; then $n \geq 48L_\delta/\varepsilon^2 \geq 48$ and $n\varepsilon^2 \geq 48L_\delta$. By Lemma **??**, $\{|\hat{R} - r^2| \geq \varepsilon r\} \subseteq \{|L| \geq \varepsilon r/2\} \cup \{|Q| \geq \varepsilon r/2\}$.

Gaussian term. $L \sim \mathcal{N}(0, 4r^2/n)$ and $-L$ has the same law (Lemma **??**), so by Lemma **??** applied to L and $-L$,

$$\mathbb{P}(|L| \geq \varepsilon r/2) \leq 2\exp\Big(-\frac{(\varepsilon r/2)^2}{2 \cdot 4r^2/n}\Big) = 2\exp\Big(-\frac{n\varepsilon^2}{32}\Big) \leq 2\exp\Big(-\frac{3}{2}L_\delta\Big) = 2\Big(\frac{\delta}{4}\Big)^{3/2} = \frac{\delta^{3/2}}{4} \leq \frac{\delta}{4}.$$

Chi-square term. Lemma **??** (both tails) for Q with parameters $(V, b) = (8d/n^2, 4/n)$ and $u := \varepsilon r/2$: $u^2/(2V) = \varepsilon^2 r^2 n^2/(64d) \geq n^2\varepsilon^2/64$ (as $r^2 \geq d$), and $n^2\varepsilon^2/64 = n(n\varepsilon^2)/64 \geq 48 \cdot 48L_\delta/64 = 36L_\delta$; and $u/(2b) = \varepsilon rn/16 \geq \varepsilon n/16 \geq 48L_\delta/(16\varepsilon) \geq 3L_\delta$ (as $r \geq \sqrt{d} \geq 1$ and $\varepsilon \leq 1$). Hence $\mathbb{P}(|Q| \geq \varepsilon r/2) \leq 2\exp(-3L_\delta) = 2(\delta/4)^3 \leq \delta/4$.

Union bound. $\mathbb{P}(|\hat{R} - r^2| \geq \varepsilon r) \leq \delta/2$.

From r^2 *to* r. On the event $|\hat{R} - r^2| < \varepsilon r$, since $\varepsilon \leq 1 \leq r$, $\hat{R} > r(r - \varepsilon) \geq 0$, so $\hat{r} = \sqrt{\hat{R}}$ and $|\hat{r} - r| = |\hat{R} - r^2|/(\sqrt{\hat{R}} + r) < \varepsilon r/r = \varepsilon$.

Budget. $n \leq 48L_\delta/\varepsilon^2 + 1$ and $1 \leq L_\delta/\varepsilon^2$. ☐

Remark 181 (On the deterministic design and the constant 48)*.* The paper draws n random Rademacher rows and uses a random-matrix concentration bound (Vershynin [**?**], Section 4.7) to control the least-squares covariance $(X^\top X)^{-1}$, together with the Gaussian Hanson–Wright inequality. With the deterministic basis design the covariance of $\hat{\theta}$ is exactly I_d/n, so only the one-dimensional Gaussian tail and the chi-square tail are needed, and no invertibility event has to be handled. The value $n = \lceil 32\log(4/\delta)/\varepsilon^2 \rceil$ (which is what one would first write down, and what the blueprint outline suggested) is not sufficient for the stated failure probability $\delta/2$: it makes the Gaussian term alone equal to $2\exp(-\log(4/\delta)) = \delta/2$, leaving nothing for the chi-square term. With 48 the two terms are $\delta^{3/2}/4$ and $\delta^3/32$, whose sum is at most $\delta/2$ for every $\delta \in (0,1)$. The paper also assumes $\varepsilon \leq r/2$ for the final step; with $\varepsilon \leq 1 \leq \sqrt{d} \leq r$ the weaker $\varepsilon \leq r$ suffices.

8.5 The meta-algorithm

Definition 182 (Meta-algorithm for norm estimation; Algorithm 4)**.** Maiti2026Power.NormEstParam, Maiti2026Power.nsN2, Maiti2026Power.nsBudget, Maiti2026Power.NormEstParam.T, Maiti2026Power.NormEstParam.phaseOf, Maiti2026Power.NormEstParam.rbBlock, Maiti2026Power.NormEstParam.basisRound, Maiti2026Power.NormEstParam.planAction, Maiti2026Power.NormEstParam.nextAction, Maiti2026Power.NormEstParam.alg, Maiti2026Power.NormEstParam.addStat, Maiti2026Power.NormEstParam.lnRStat, Maiti2026Power.NormEstParam.estimate, Maiti2026Power.NormEstParam.outputdef:multiscale$_algorithm, de$ $additive_algorithm, def : large_norm_algorithm, lem : seeded, def : norm_estimatorLet \varepsilon \in$ $(0, 1]$, $\delta \in (0, 1)$, let $T_1 := N_{\mathrm{ms}}$ be the budget of the multi-scale algorithm and

$$N_2 := \max\left(dn,\ \max\{s(t_j)K(t_j) : j < J\}\right),$$

where $s(t) = \lceil 4d/t^2 \rceil$, $K(t) = \lceil 4096\, t^2 \log(4/\delta)/\varepsilon^2 \rceil$ are the parameters of Algorithm 1 with $r_0 = t$ (Definition **??**) and n is that of Definition **??**. The *meta-algorithm* $\mathrm{NormEst}(\varepsilon, \delta)$ is the seeded algorithm (Lemma **??**) with budget $N_{\mathrm{norm}}(\varepsilon, \delta) := T_1 + N_2$ and seeds uniform on $\{-1, +1\}^d$, defined by the following *round schedule*. The first phase is the multi-scale algorithm of Definition **??**: for $t < T_1$, if $t = \mathrm{start}_j + ks + \ell$ with $j < J$, $k < K_j$, $\ell < s_j$, the round t belongs to the *Rademacher block* starting at $b := \mathrm{start}_j + ks$, and the action is $x_t := \sigma_b/\sqrt{d}$ where σ_b is the seed of round b (the action of round b, repeated). The second phase, rounds $T_1 \le t < T_1 + N_2$, depends on the output j_* of the first phase:

(a) if $j_* = 0$: the action is 0 and the estimate is $\hat{r} := \varepsilon$;

(b) if $j_* = J$: the rounds $T_1 + in + \ell$ ($i < d$, $\ell < n$) are *basis rounds* playing e_i, and the estimate is the large-norm estimate $\hat{r} = \sqrt{\max(\hat{R}, 0)}$ of Definition **??** computed from their observations;

(c) if $0 < j_* < J$: the rounds $T_1 + ks + \ell$ ($k < K(t_{j_*})$, $\ell < s(t_{j_*})$) form the Rademacher blocks of Algorithm 1 with $r_0 = t_{j_*}$ (Definition **??**), and the estimate is $\hat{r} = \sqrt{\max(\bar{Z}, 0)}$ computed from their observations.

The remaining rounds of the second phase play the action 0 (their observations are not used). The *plan* $\mathrm{plan}(j_*, t, \sigma)$ is the action of round t as a function of j_* and of the seeds σ at the block starts; as a seeded algorithm, the action of round t is computed from the past actions and observations (which determine j_* once $t \ge T_1$, and the action of the current block start) and the fresh seed of round t.

Lean remark. The parameters are gathered in a structure `NormEstParam` (, and their ranges). The schedule is encoded by two functions of (j_*, t): `rbBlock` returns the start of the Rademacher block of round t (if any) and `basisRound` the basis vector of round t (if any); `planAction` is the plan and `nextAction` its computation from the past, so that the seeded algorithm `alg` has $\mathtt{next}_n(h, u) = \mathrm{nextAction}(n + 1, \mathrm{past}(h), j_*(h), u)$. The estimate is the measurable function `estimate` of the observation sequence (`output` on histories of length N_{norm}), and the identification algorithm is the fixed-budget algorithm with this deterministic output. The paper composes the phases sequentially; here the composition is replaced by the deterministic schedule, and the data-dependence is entirely in j_*.

Lemma 183 (The run of the meta-algorithm follows its plan)**.** *Maiti2026Power.NormEstParam.seedAction*$_{e}q_{p}l$ *against* ν_θ *on the seed space* $(\Omega_{\text{seed}}, \mathbb{P}_{\text{seed}})$ *of Lemma* **??***, with seeds* σ_t *and noises* η_t*, and let* $j_* = j_*(\omega)$ *be the output of the first phase. Then for every round* t *the action is the plan action,* $x_t = \mathrm{plan}(j_*, t, \sigma)$*, and* $y_t = \langle x_t, \theta\rangle + \eta_t$*. Consequently:*

(i) for $j < J$*, the statistic* U_j *of test* j *is* $\bar{Z}$ *evaluated at the window projection* $\pi_{\mathrm{start}_j, s_j, K_j}(\omega)$*;*

(ii) on $\{0 < j_* < J\}$*, the statistic* $\bar{Z}$ *of the second phase is* $\bar{Z}$ *evaluated at* $\pi_{T_1, s(t_{j_*}), K(t_{j_*})}(\omega)$*;*

(iii) on $\{j_* = J\}$*, the statistic* $\hat{R}$ *of the second phase is the large-norm statistic evaluated at the basis window projection* $\pi'_{T_1,n}(\omega)$*;*

(iv) j_* *is a function of the prefix* $(\omega_t)_{t<T_1}$*, and the estimate is a function of the observations of the rounds* $t < N_{\text{norm}}$ *only (so the estimate of the run is the estimate computed from the history* $H_{N_{\text{norm}}}$*).*

Proof. def:meta$_{a}lgorithm, lem: seeded, lem: statistic_{s}tructureStronginductionont.Fort$ $< \mathrm{T}_1$ the schedule does not depend on j_*, and the action computed from the past is the plan action because the action of the block start $b \le t$ is, by the induction hypothesis, $\sigma_b/\sqrt{d}$ (or $t = b$ and the fresh seed is used). For $t \ge T_1$, the value of j_* computed from the past observations $(y_r)_{r<T_1}$ is the true j_* (the first phase is complete), and the same argument applies. The observation identity is the definition of the noise environment. Parts (i)–(iii) follow from Lemma **??**(iii) and the corresponding identities for the basis rounds ($\langle e_i, \theta\rangle = \theta_i$); part (iv) from the fact that j_* only reads the rounds $t < T_1$, each of which only depends on the coordinates ω_r, $r \le t$, and that each statistic of the second phase only reads rounds $t < T_1 + N_2$. □

Lean remark. The second phase is "$\mathrm{alg}_2(h)$" in the notation of Lemma **??**, a measurable function of the first-phase history h through $r_0(h) \in \{t_0, \ldots, t_J\}$; since r_0 takes finitely many values, measurability is a finite case distinction. The estimate $\hat{r}$ is a deterministic measurable function of the whole history with values in $[0, \infty)$, i.e. a recommendation kernel in the sense of Definition **??**. Padding with a constant action keeps the budget deterministic, as required by Definition **??**; the padded rounds are simply not used by the estimate.

Theorem 184 (Norm estimation; Theorem 8 of the paper)**.** *Maiti2026Power.NormEstParam.fail2, Maiti2026Power.NormEstParam.fail*$_{s}$*ubset*, $Maiti2026Power.NormEstParam.nsMeasure_{r}eal_{j}s_{i}nter_{p}reima$ $(0, 1]$ *and* $\delta \in (0, 1)$*. The meta-algorithm* $\mathrm{NormEst}(\varepsilon, \delta)$ *is an* (ε, δ)*-accurate estimation algorithm on* $\mathbb{B}_d$*: for every* $\theta \in \mathbb{R}^d$*,* $\mathbb{P}_\theta(|\hat{r} - \|\theta\|| \le \varepsilon) \ge 1 - 7\delta/8$*. Its budget satisfies*

$$N_{\text{norm}}(\varepsilon, \delta) \le C_{\text{norm}} \frac{d \log(4/\delta)}{\varepsilon^2}, \qquad C_{\text{norm}} := 10^5.$$

Corollary 185 (Theorem 8 of the paper, existential form)**.** *Maiti2026Power.exists*$_{i}$$sAccurateNormEstthm: n$ $(0, 1]$ *and* $\delta \in (0, 1)$*. There exist a budget* $T \le C_{\text{norm}}\, d \log(4/\delta)/\varepsilon^2$ *and an* (ε, δ)*-accurate estimation algorithm on* $\mathbb{B}_d$ *with budget* T*.*

Proof. thm:norm$_{e}stimation, lem : seededTake$NormEst$(\varepsilon, \delta)$ of Theorem **??**; it is $(\varepsilon, 7\delta/8)$-accurate, hence (ε, δ)-accurate. (For $d = 0$ every reward vector is 0 and the constant estimate ε with budget 0 is accurate.) ☐

Proof. thm:multiscale$_{g}uarantee, thm : additive_{e}stimation, thm : large_{n}orm_{g}uarantee, lem : multiscale_{s}amples, lem : norm_{m}eta_{p}lan, lem : statistic_{s}tructure, lem : norm_{m}ultiscale_{g}ood_{e}vent, lem : seededCorrectness.ByLemma$ **??**$itsufficestoprovetheboundontheseedspace, where, byLemma$ **??**$(iv), theestin$ computed from the observation sequence. Let G be the good event of Theorem **??**, $\mathbb{P}_{\mathrm{seed}}(G^c) \leq 3\delta/8$. For $1 \leq j \leq J$ let F_j be the failure event of the second phase at outcome j: for $j = J$, $F_J := \{\varepsilon < |\hat{r}' - r|\}$ where $\hat{r}'$ is the large-norm estimate evaluated at the basis window projection $\pi'_{T_1,n}$; for $j < J$, $F_j := \{\varepsilon/2 < |\hat{r}_j - r|\}$ where $\hat{r}_j$ is the estimate of Algorithm 1 with $r_0 = t_j$ evaluated at $\pi_{T_1,s(t_j),K(t_j)}$. By Lemma **??**(ii)–(iii),

$$\{\varepsilon < |\hat{r} - r|\} \subseteq G^c \cup \bigcup_{1 \leq j \leq J} \left(G^{cc} \cap \{j_* = j\} \cap F_j\right) :$$

on $G \cap \{j_* = 0\}$, $r < 2\varepsilon$ by Theorem **??**(a) and $\hat{r} = \varepsilon$, so $|\hat{r} - r| \leq \varepsilon$; on $\{j_* = j\}$ with $j \geq 1$ the estimate is $\hat{r}_j$ or $\hat{r}'$. For each $j \geq 1$: if $G \cap \{j_* = j\} = \emptyset$ the term vanishes; otherwise Theorem **??**(b)–(c) gives the hypotheses of Theorem **??** ($\varepsilon \leq t_j$, $r/2 < t_j \leq 2r$) or of Theorem **??** ($r \geq \sqrt{d}$), and $\mathbb{P}_{\mathrm{seed}}(F_j) \leq 3\delta/8 \leq \delta/2$ (resp. $\leq \delta/2$) by Lemma **??**(i)–(ii) (the law of the window projection is the block, resp. noise, law). Since $\{j_* = j\}$ is a function of the prefix $(\omega_t)_{t<T_1}$ (Lemma **??**(iv)) and F_j is a function of a window projection with $a = T_1$, they are independent (Lemma **??**), so $\mathbb{P}_{\mathrm{seed}}(\{j_* = j\} \cap F_j) = \mathbb{P}_{\mathrm{seed}}(j_* = j)\, \mathbb{P}_{\mathrm{seed}}(F_j) \leq \mathbb{P}_{\mathrm{seed}}(j_* = j)\, \delta/2$. Summing over j (the events $\{j_* = j\}$ are disjoint, so their probabilities add up to at most 1), $\mathbb{P}_{\mathrm{seed}}(\varepsilon < |\hat{r} - r|) \leq 3\delta/8 + \delta/2 = 7\delta/8$.

Budget. Lemma **??** gives $N_{\mathrm{ms}} \leq 63736\, dL/\varepsilon^2$ with $L = \log(4/\delta)$. For $j < J$, $t_j \in [\varepsilon, 2\sqrt{d})$, so Theorem **??** (budget clause with $r_0 = t_j$) gives $s(t_j)K(t_j) \leq 32776\, dL/\varepsilon^2$, and Theorem **??** gives $dn \leq 49\, dL/\varepsilon^2$; hence $N_2 \leq 32776\, dL/\varepsilon^2$ and $N_{\mathrm{norm}} \leq (63736 + 32776)\, dL/\varepsilon^2 \leq 10^5\, dL/\varepsilon^2$. ☐

Remark 186 (Reading of the constants). The budget is $O(d\log(1/\delta)/\varepsilon^2)$ as in the paper: for $\delta \leq 1/4$, $\log(4/\delta) \leq 2\log(1/\delta)$. The constant $C_{\mathrm{norm}} = 10^5$ is dominated by the multi-scale phase (63736), itself driven by the test constant $c'_1 = 5120$; the constants are what the union bounds produce and were not optimized. In the paper the sample complexity of the multi-scale phase is written $\sum_j s_j t_j$, a typo for $\sum_j s_j K_j$. The paper's failure probabilities are $\delta/2$ for the multi-scale phase and $\delta/2$ for the second phase; the sharper $3\delta/8$ of Lemma **??** is the paper's own computation $\sum_j 3\delta_j/4$.

Chapter 9

A polynomial separation between adaptive and non-adaptive algorithms

This chapter formalizes Section 2.5 and Appendix I of the paper: an action set $\mathcal{X} \subset \mathbb{R}^{kd}$, a union of k unit balls living in k disjoint blocks of d coordinates, on which every non-adaptive algorithm needs $\Omega(kd^2/\varepsilon^2)$ samples while an adaptive algorithm succeeds with $O(kd \log(k/\delta)/\varepsilon^2 + d^2/\varepsilon^2)$ samples. The target is Theorem 7 of the paper (Theorem **??**). The adaptive algorithm first runs the norm estimation procedure of Chapter **??** on each block to locate the block containing a near-optimal arm, then identifies a good arm in that block with the basis design of Lemma **??**; as in Chapter **??**, the phases are composed by a deterministic round schedule of a seeded algorithm (Lemma **??**), and the analysis takes place on the seed space. The chapter is fully formalized. The non-adaptive lower bound deviates from the paper: instead of the cited adaptive lower bound of Chen et al. [**?**] we project a fixed design on the block receiving the fewest points (Lemma **??**, an instance of the transport Lemma **??**) and apply the ingredients of our own non-adaptive lower bound (the Bayesian step, Lemma **??**) with the width of the unit ball for a matrix (Proposition **??**, $w(\mathbb{B}_d; A) \geq \sqrt{2/\pi}\, d/\sqrt{\operatorname{tr} A}$). The class of non-adaptive algorithms is therefore that of Definition **??**: deterministic fixed designs with an arbitrary (possibly randomized) recommendation kernel, whereas the paper's Theorem 7 speaks of "possibly randomized non-adaptive" algorithms (Remark **??**). The block-ball set is spanning (Lemma **??**), which is the hypothesis needed wherever Theorem **??**, Theorem **??** or the Gaussian width (Definition **??**, Proposition **??** on the spanning set $\mathbb{B}_d$) is used below. Throughout, $k, d \geq 1$ are integers with $kd \geq 2$; the comparison of the two bounds uses $d \leq k \leq d^2$.

9.1 The block-ball action set

Definition 187 (Block-ball action set)**.** Maiti2026Power.blockBallSetdef:arm$_{s}etFor$i $\in \{1, \ldots, k\}$ let $B_i := \{(i-1)d+1, \ldots, id\}$ be the i-th block of coordinates of $\mathbb{R}^{kd}$, let $\iota_i : \mathbb{R}^d \to \mathbb{R}^{kd}$ be the isometric embedding which places a vector on the coordinates of B_i and 0 elsewhere, and let $\pi_i : \mathbb{R}^{kd} \to \mathbb{R}^d$ be the projection on the coordinates of B_i, so that $\pi_i \circ \iota_i = \mathrm{id}$, $\langle \iota_i u, v\rangle = \langle u, \pi_i v\rangle$ and $\|\iota_i u\| = \|u\|$. Define

$$\mathcal{X}_i := \{x \in \mathbb{R}^{kd} : \operatorname{supp}(x) \subseteq B_i,\ \|x\| \le 1\} = \iota_i(\mathbb{B}_d), \qquad \mathcal{X} := \bigcup_{i=1}^{k} \mathcal{X}_i.$$

For $\theta \in \mathbb{R}^{kd}$ we write $\theta^{(i)} := \pi_i\theta \in \mathbb{R}^d$ for its i-th block, so $\theta = \sum_i \iota_i\theta^{(i)}$ and $\|\theta\|^2 = \sum_i \|\theta^{(i)}\|^2$.

Lean remark. With $\mathbb{R}^{kd}$ encoded as `EuclideanSpace (Fin k × Fin d)` the blocks are $\{i\} \times \mathrm{Fin}\ d$ (0-indexed) and ι_i, π_i are the obvious linear maps; $\mathcal{X}_i \cap \mathcal{X}_j = \{0\}$ for $i \ne j$, so an element of $\mathcal{X}$ determines its block unless it is 0. Where a block index is needed for a point of $\mathcal{X}$ we use the smallest one.

Lemma 188 (Basic properties of the block-ball set)**.** *Maiti2026Power.blockEmb, Maiti2026Power.blockProj, Maiti2026Power.inner$_b$lockEmb$_l$eft, Maiti2026Power.norm$_b$lockEmb, Maiti2026* $\bigcup_i \iota_i(\mathbb{B}_d)$ *is a compact subset of* $\mathbb{B}_{kd}$ *with* $\operatorname{span}(\mathcal{X}) = \mathbb{R}^{kd}$*, hence an action set in the sense of Definition* **??***. Every* $x \in \mathcal{X}$ *satisfies* $\pi_j x = 0$ *for all* j *but one. For every* $\theta \in \mathbb{R}^{kd}$*,*

$$\max_{x\in\mathcal{X}} \langle x, \theta\rangle = \max_{1\le i\le k} \|\theta^{(i)}\|,$$

and for $\vartheta \in \mathbb{R}^d$ *and* $\hat{x} \in \mathbb{R}^{kd}$*,* $\max_{x\in\mathcal{X}} \langle x, \iota_i\vartheta\rangle = \|\vartheta\| = \max_{u\in\mathbb{B}_d} \langle u, \vartheta\rangle$ *and*

$$r_{\mathcal{X}}(\hat{x}, \iota_i\vartheta) = \|\vartheta\| - \langle \pi_i\hat{x}, \vartheta\rangle = r_{\mathbb{B}_d}(\pi_i\hat{x}, \vartheta).$$

Proof. def:block$_{b}all_{s}etEach\iota_i(\mathbb{B}_d)$ is the continuous image of a compact set, and a finite union of compact sets is compact; $x \in \mathcal{X}$ lies in $\iota_i(\mathbb{B}_d)$ for its block i, with $x = \iota_i\pi_i x$. $\mathcal{X}$ contains every basis vector e_m ($m \in B_i$ for some i), so it spans $\mathbb{R}^{kd}$; and $\|\iota_i u\| = \|u\| \le 1$ gives $\mathcal{X} \subseteq \mathbb{B}_{kd}$. For $x = \iota_i u$ with $u \in \mathbb{B}_d$, $\langle x, \theta\rangle = \langle u, \theta^{(i)}\rangle \le \|\theta^{(i)}\|$ by Cauchy–Schwarz, with equality for $u = \theta^{(i)}/\|\theta^{(i)}\|$ (any u if $\theta^{(i)} = 0$). Taking the maximum over i gives the second identity; for $\theta = \iota_i\vartheta$ the blocks are $\theta^{(i)} = \vartheta$ and $\theta^{(j)} = 0$ for $j \ne i$, and the same Cauchy–Schwarz argument on $\mathbb{B}_d$ gives $\max_{u\in\mathbb{B}_d}\langle u, \vartheta\rangle = \|\vartheta\|$. The regret formula follows from Definition **??** and $\langle \hat{x}, \iota_i\vartheta\rangle = \langle \pi_i\hat{x}, \vartheta\rangle$. $\square$

9.2 Lower bound for non-adaptive algorithms

Lemma 189 (Restriction of a fixed design to a block)**.** *Maiti2026Power.blockProjBall, Maiti2026Power.blockDesign, Maiti2026Power.blockRounds, Maiti2026Power.isPAC$_f$ixedDesignTransport$_b$lo* $\mathcal{X}$ *be a fixed design with recommendation kernel* ρ *(from* $\mathbb{R}^T$ *to* $\mathcal{X}$*) which is* (ε, δ)*-PAC on* $\mathcal{X}$*, and fix a block* i*. The projected design* $x'_t := \pi_i x_t \in \mathbb{B}_d$ *(*$t < T$*), with* *the recommendation kernel* $(\pi_i)_*\rho$ *(draw* $\hat{x} \sim \rho(y)$*, the history being relabelled with the actions* x_t*, and output* $\pi_i\hat{x} \in \mathbb{B}_d$*), is a fixed-design algorithm with budget* T *on* $\mathbb{B}_d$ *which is* (ε, δ)*-PAC on* $\mathbb{B}_d$*. Its design matrix* $\Sigma'_T := \sum_{t<T}(\pi_i x_t)(\pi_i x_t)^\top$ *satisfies* $\operatorname{tr}\Sigma'_T \le T_i$*, where* $T_i := |\{t < T : \pi_i x_t \neq 0\}|$ *is the number of rounds played in block* i*; the sets* $\{t < T : \pi_i x_t \neq 0\}$ *are pairwise disjoint, so* $\sum_i T_i \le T$ *and some block* i^* *has* $kT_{i^*} \le T$*.*

Proof. lem:fixed$_d esign_t ransport, lem : block_b all_b asic This is Lemma$ **??**$with$L $=\iota_i$ and $\pi = \pi_i$: by Lemma **??**, $\pi_i x_t \in \mathbb{B}_d$ (as $\|\pi_i x\| \le \|x\| \le 1$), $\langle \pi_i x_t, \vartheta\rangle = \langle x_t, \iota_i\vartheta\rangle$ (adjointness) and $r_{\mathbb{B}_d}(\pi_i\hat{x}, \vartheta) = r_{\mathcal{X}}(\hat{x}, \iota_i\vartheta)$ for every $\vartheta \in \mathbb{R}^d$. The trace bound is $\operatorname{tr}\Sigma'_T = \sum_{t<T}\|\pi_i x_t\|^2 \le T_i$ since each term is at most 1 and vanishes outside the rounds played in block i. The sets of rounds are disjoint because each x_t lies in a single block (Lemma **??**), so $\sum_i T_i \le T$, and the pigeonhole principle gives i^* with $kT_{i^*} \le T$. ☐

Lean remark. The paper (and an earlier version of this blueprint) restricts the design to the rounds t with x_t in block i; keeping all T rounds with the projected actions $\pi_i x_t$ (which are 0 outside the block and contribute pure noise) gives a fixed-design algorithm with the same budget T, which is exactly the situation of Lemma **??**, and the number T_i of rounds in the block enters only through the trace of the design matrix. The composition of ρ with the relabelling of the history and with π_i is where the randomness allowed in Definition **??** is used.

Theorem 190 (Non-adaptive lower bound for the block-ball set)**.** *Maiti2026Power.blockBallSet$_l e_b udget_o f_i sFix$*
$(0, 1]$*. Recall that* $\mathcal{X}$ *is a spanning action set in* $\mathbb{R}^{kd}$ *(Lemma* **??***).*

(i) If a fixed-design algorithm (Definition **??***: a deterministic design* $x_0, \ldots, x_{T-1} \in \mathcal{X}$ *with an arbitrary, possibly randomized, recommendation kernel; budget* $T \ge 0$*) is* (ε, δ)*-PAC on* $\mathcal{X}$ *with* $\delta \le 1/10$*, then*

$$T \ge \frac{kd^2}{210\,\varepsilon^2}.$$

(ii) If $kd \ge 2$ *and any identification algorithm (adaptive or not) with budget* T *is* (ε, δ)*-PAC on* $\mathcal{X}$ *with* $\delta < 1/16$*, then* $T \ge kd\log(1/\delta)/(20000\,\varepsilon^2)$ *(and* $T \ge c_{\mathrm{ad}}\,kd\log(1/\delta)/\varepsilon^2$ *with the constant* c_{ad} *of Theorem* **??***).*

In particular every fixed-design (ε, δ)-PAC algorithm with $\delta < 1/16$ has budget at least $\frac{1}{2}\big(c_{\mathrm{ad}}\,kd\log(1/\delta) + kd^2/210\big)/\varepsilon^2$.

Proof. lem:block$_r estriction, lem : lower_n onadaptive_p osdef, lem : lower_n onadaptive_b ayes_s tep, prop : width_b all, thm : lower_a daptive, lem : block_b all_b asic (i) If$k $= 0 or d = 0 there is nothing to prove, so assume$k, d ≥ 1. With the notation of Lemma **??**, pick a block i^* with $kT_{i^*} \le T$ and consider the projected design $x'_t = \pi_{i^*}x_t$ on $\mathbb{B}_d$, with design matrix Σ'_T, which is (ε, δ)-PAC on $\mathbb{B}_d$ with budget T.

Step 1: $\Sigma'_T \succ 0$. Since $0 \in \mathbb{B}_d$, $\mathbb{B}_d$ is spanning and $\delta \le 1/10 < 1/2$, Lemma **??** applies to the projected design (with any budget $T \ge 0$, including $T = 0$). In particular $T_{i^*} \ge 1$ (as $d \ge 1$ and $\operatorname{tr}\Sigma'_T \le T_{i^*}$, with $\operatorname{tr}\Sigma'_T > 0$).

Step 2: the Bayesian step. Lemma **??** on $\mathbb{B}_d$ gives $0.12\, w(\mathbb{B}_d; \Sigma'_T) \le \varepsilon$.

Step 3: the width of the ball. By Proposition **??**, $w(\mathbb{B}_d; \Sigma'_T) \ge \sqrt{2/\pi}\, d/\sqrt{\operatorname{tr} \Sigma'_T} \ge \sqrt{2/\pi}\, d/\sqrt{T_{i^*}}$.

Step 4: arithmetic. Combining, $0.12\sqrt{2/\pi}\, d \le \varepsilon\sqrt{T_{i^*}}$; squaring, $0.0144 \cdot (2/\pi)\, d^2 \le \varepsilon^2 T_{i^*}$, and since $\pi \le 4$, $2/\pi \ge 1/2$ and $0.0072\, d^2 \le \varepsilon^2 T_{i^*}$, i.e. $T_{i^*} \ge d^2/(139\, \varepsilon^2) \ge d^2/(210\, \varepsilon^2)$. Hence $T \ge kT_{i^*} \ge kd^2/(210\varepsilon^2)$.

(ii) $\mathcal{X}$ is a spanning action set in $\mathbb{R}^{kd}$ with $kd \ge 2$ (Lemma **??**), so Theorem **??** applies directly with dimension kd.

The last claim is $T \ge \max(a, b) \ge (a + b)/2$. □

9.3 The adaptive block algorithm

Lemma 191 (Transporting an algorithm on $\mathbb{B}_d$ to a block)**.** *Maiti2026Power.blockEmbBall, Maiti2026Power.blockProjBall$_b$lockEmbBall*, $Maiti2026Power.BlockParam.F_b lockEmbBall, Maiti2026Powe$ $\mathbb{B}_d \to \mathcal{X}$ *be the embedding of the unit ball on block i (so $\pi_i \circ \iota_i = \mathrm{id}$). For every $\theta \in \mathbb{R}^{kd}$ and $x \in \mathbb{B}_d$, the reward kernel of ν_θ at $\iota_i x$ is that of $\nu_{\theta^{(i)}}$ at x: $\langle \iota_i x, \theta \rangle + \eta = \langle x, \theta^{(i)} \rangle + \eta$. Consequently, the block algorithm of Definition* **??** simulates *the norm estimation meta-algorithm on each block: on the seed space, for $i < k$ and $t < N$, the step (action, observation) of round $iN + t$ of the block algorithm against ν_θ is $(\iota_i x_t, y_t)$ where (x_t, y_t) is the step of round t of* $\mathrm{NormEst}(\varepsilon/4, \delta/(2k))$ *against $\nu_{\theta^{(i)}}$ on the seed sequence shifted by iN, $(\omega_{iN+t})_t$. In particular the estimate $\hat{r}_i$ of block i is the estimate of the meta-algorithm against $\nu_{\theta^{(i)}}$ on the shifted seed sequence, and the shifted seed sequence has the law $\mathbb{P}_{\mathrm{seed}}$ of the seed sequence.*

Proof. def:env, lem:block$_b$all$_b$asic, $lem : seeded, lem : norm_m eta_p lan, lem : statistic_s tructureThekernelidentityisLemma$ **??**$(\langle \iota_i x, \theta \rangle = \langle x, \pi_i \theta \rangle)$. The simulation is a strong induction on $t < N$: the action of round $iN + t$ of the block algorithm is ι_i applied to the next action of the meta-algorithm computed from the past actions (projected by π_i) and observations of the rounds $iN + r$, $r \le t$, of the block, which by the induction hypothesis are the past actions and observations of the meta-algorithm on the shifted seed sequence; the observation follows from the kernel identity. The shift invariance of $\mathbb{P}_{\mathrm{seed}}$ is Lemma **??** (the coordinates $(\omega_{a+t})_t$ of an i.i.d. sequence form an i.i.d. sequence with the same law). □

Lean remark. The block algorithm is not obtained by transporting the meta-algorithm as an LML algorithm (which would require mapping its policy kernels by ι_i and its histories by π_i): it is a seeded algorithm whose next-action map, in the rounds of block i, applies ι_i to the next-action map of the meta-algorithm evaluated on the π_i-projected history of the block. The simulation lemma is then a statement about two recursively defined runs on the same seed space.

Proof. def:env, lem:block$_b$all$_b$$asicTherewardkernelof\nu_\theta$ at $\iota_i(x)$ is $\mathcal{N}(\langle \iota_i x, \theta \rangle, 1) = \mathcal{N}(\langle x, \theta^{(i)} \rangle, 1)$ (Lemma **??**), which is the reward kernel of $\nu_{\theta^{(i)}}$ at x. The policy of $\iota_i$alg is the image by ι_i of the policy of alg applied to the projected history, and $\pi_i \circ \iota_i = \mathrm{id}$, so the projected process satisfies the defining conditional-law conditions of an algorithm-environment sequence for $(\mathrm{alg}, \nu_{\theta^{(i)}})$ (LML `IsAlgEnvSeq`); uniqueness of the trajectory law gives the claim about estimates. □

Lean remark. This is the analogue, for the non-invertible isometric embedding ι_i, of the transformation lemma **??** used for the adaptive lower bound; it is a statement about mapping actions by a measurable map ϕ such that the environments correspond ($\kappa_\theta \circ \phi = \kappa_{\theta'}$), which holds for any stationary environments and could be stated at that level of generality. The map $\phi : \mathbb{B}_d \to \mathcal{X}$, $u \mapsto \iota_i u$, is continuous hence measurable between the subtypes, and the policy kernels of $\iota_i$alg are the images of those of alg by ϕ (`Kernel.map`), applied to the history mapped by π_i on its action coordinates; the action type of $\iota_i$alg is the subtype $\mathcal{X}$, as required to compose it with other algorithms on $\mathcal{X}$ (Lemma **??**).

Definition 192 (Block algorithm)**.** Maiti2026Power.BlockParam, Maiti2026Power.BlockParam.P,
Maiti2026Power.BlockParam.N, Maiti2026Power.BlockParam.n , Maiti2026Power.BlockParam.T ,
Maiti2026Power.BlockParam.T, Maiti2026Power.BlockParam.blockOf, Maiti2026Power.BlockParam.basisRou
Maiti2026Power.BlockParam.estN, Maiti2026Power.BlockParam.iHat, Maiti2026Power.BlockParam.thetaHat,
Maiti2026Power.normalizeBall, Maiti2026Power.BlockParam.recommend, Maiti2026Power.BlockParam.nextA
Maiti2026Power.BlockParam.alg, Maiti2026Power.BlockParam.outputdef:block$_ball_set, def : meta_algorithm, lem : ball_identification, lem : seededLet \varepsilon \in (0,1]$ and $\delta \in (0,1)$, let $N := N_{\mathrm{norm}}(\varepsilon/4, \delta/(2k))$ be the budget of the meta-algorithm $\mathrm{NormEst}(\varepsilon/4, \delta/(2k))$ (Definition **??**) and $n_2 := \lceil 4(8d + 48\log(2/\delta))/\varepsilon^2 \rceil$. The *block algorithm* on $\mathcal{X}$ is the seeded algorithm (Lemma **??**) with uniform sign-vector seeds and budget $N_{\mathrm{sep}}(\varepsilon, \delta) := kN + dn_2$ defined by the following round schedule.

(1) For $i = 0, \ldots, k-1$, the rounds $[iN, (i+1)N)$ run $\mathrm{NormEst}(\varepsilon/4, \delta/(2k))$ on block i: the action of round $iN + t$ is ι_i applied to the action of round t of the meta-algorithm computed from the (π_i-projected) history of the block and the seed of the round. The estimate $\hat{r}_i$ is the estimate of the meta-algorithm computed from the observations of the rounds of block i, and $\hat{i} := \arg\max_i \hat{r}_i$ (smallest index in case of ties).

(2) The rounds $kN + mn_2 + \ell$, $m < d$, $\ell < n_2$, play $\iota_{\hat{i}}(e_m)$ (the basis design of Lemma **??** on block $\hat{i}$); let $\hat{\vartheta} \in \mathbb{R}^d$ be the vector of the d empirical means of the observations at each e_m, and recommend $\hat{x} := \iota_{\hat{i}}\big(\hat{\vartheta}/\|\hat{\vartheta}\|\big)$ (with $\hat{\vartheta}/\|\hat{\vartheta}\| := 0$ if $\hat{\vartheta} = 0$).

Lean remark. The parameters are gathered in a structure `BlockParam` ($k \geq 1$, ε, δ), with the meta-algorithm parameters `BlockParam.P` $= (\varepsilon/4, \delta/(2k))$. The block of a round $t < kN$ is $\lfloor t/N \rfloor$ (`blockOf`), the basis vector of a round of phase (2) is given by `basisRound`. The estimate of block m (`estN`) is the estimate of Definition **??** applied to the observations y_{mN+r}; the selected block $\hat{i}$ (`iHat`, defined by `Nat.find`) and the empirical mean vector (`thetaHat`) are measurable functions of the observation sequence, hence so is the recommendation (`recommend`, `output`); the normalization $v \mapsto v/\|v\|$ (`normalizeBall`, with $0 \mapsto 0$) is measurable but not continuous. The seeded algorithm `alg` has next-action map `nextAction`. The recommendation 0 when $\hat{\vartheta} = 0$ replaces the paper's e_1; both have regret at most $2\|\hat{\vartheta} - \vartheta\|$.

Lean remark. Phase (1) is itself a k-fold composition of algorithms with action type $\mathcal{X}$ (each ι_iNormEst has action type $\mathcal{X}$ by Lemma **??**); the index $\hat{\imath}$ is a measurable function of the phase-(1) history h with values in $\{1, \ldots, k\}$, and phase (2) is the fixed design of Lemma **??** transported by $\iota_{\hat{\imath}}$, again with action type $\mathcal{X}$: a deterministic algorithm parametrized by $\hat{\imath}$, which is the "$\mathrm{alg}_2(h)$" of Lemma **??**. The map $h \mapsto \iota_{\hat{\imath}(h)}\mathrm{alg}$ is measurable in h because $\hat{\imath}$ takes finitely many values: on each of the measurable sets $\{\hat{\imath} = i\}$ it is the constant algorithm $\iota_i\mathrm{alg}$, so its policy kernels are finite case distinctions of kernels (`Kernel.piecewise`), and Lemma **??** applies.

Lemma 193 (Selection of a good block)**.** *$Maiti2026Power.BlockParam.estN_l e_i Hat, Maiti2026Power.norm_s ub$ $\mathbb{R}^{kd}$ and let y be an observation sequence such that $|\hat{r}_i - \|\theta^{(i)}\|| \leq \varepsilon/4$ for all $i < k$, where $\hat{r}_i$ is the estimate of block i computed from y. Then, with $\hat{\imath}$ the selected block,*

$$\max_{i<k} \|\theta^{(i)}\| \leq \|\theta^{(\hat{\imath})}\| + \frac{\varepsilon}{2};$$

and if moreover the empirical mean vector $\hat{\vartheta}$ computed from y satisfies $\|\hat{\vartheta} - \theta^{(\hat{\imath})}\| \leq \varepsilon/4$, then the recommendation $\hat{x} = \iota_{\hat{\imath}}(\hat{\vartheta}/\|\hat{\vartheta}\|)$ has simple regret $r(\hat{x}, \theta) \leq \varepsilon$ on $\mathcal{X}$.

Proof. def:block$_a lgorithm, lem : block_b all_b asicWith$i$_* \in \arg\max_i \|\theta^{(i)}\|$ and $\hat{r}_{i_*} \leq \hat{r}_{\hat{\imath}}$ (definition of $\hat{\imath}$), $\|\theta^{(\hat{\imath})}\| \geq \hat{r}_{\hat{\imath}} - \varepsilon/4 \geq \hat{r}_{i_*} - \varepsilon/4 \geq \|\theta^{(i_*)}\| - \varepsilon/2$. For the regret, by Lemma **??** the support function of $\mathcal{X}$ at θ is $\max_i \|\theta^{(i)}\|$ and $\langle \iota_{\hat{\imath}}\hat{x}', \theta\rangle = \langle \hat{x}', \theta^{(\hat{\imath})}\rangle$ with $\hat{x}' := \hat{\vartheta}/\|\hat{\vartheta}\|$; and for every $\vartheta, v \in \mathbb{R}^d$, $\|\vartheta\| - \langle v/\|v\|, \vartheta\rangle \leq 2\|v - \vartheta\|$ (if $v \neq 0$, $\langle v/\|v\|, \vartheta\rangle = \|v\| - \langle v/\|v\|, v - \vartheta\rangle \geq \|v\| - \|v - \vartheta\| \geq \|\vartheta\| - 2\|v - \vartheta\|$; if $v = 0$ the left side is $\|\vartheta\| = \|v - \vartheta\|$). Hence $r(\hat{x}, \theta) \leq \big(\|\theta^{(\hat{\imath})}\| + \varepsilon/2\big) - \big(\|\theta^{(\hat{\imath})}\| - 2 \cdot \varepsilon/4\big) = \varepsilon$. ☐

Proof. thm:norm$_e stimation, lem : separation_b lock_e mbedding, lem : compositionByLemma$ **??**$andTheorem$ **??** against $\nu_{\theta^{(i)}}$ on $\mathbb{B}_d$, so $\mathbb{P}(|\hat{r}_i - \|\theta^{(i)}\|| > \varepsilon/4) \leq \delta/(2k)$; by Lemma **??** this bound holds conditionally on the history of the earlier blocks, hence unconditionally. A union bound over the k blocks gives the first claim with probability at least $1 - \delta/2$. On this event, with $i_* \in \arg\max_i \|\theta^{(i)}\|$,

$$\|\theta^{(\hat{\imath})}\| \geq \hat{r}_{\hat{\imath}} - \frac{\varepsilon}{4} \geq \hat{r}_{i_*} - \frac{\varepsilon}{4} \geq \|\theta^{(i_*)}\| - \frac{\varepsilon}{2}.$$

☐

Theorem 194 (Polynomial separation; Theorem 7 of the paper)**.** *$Maiti2026Power.BlockParam.seedAction_o f_{Tl}$ $(0, 1]$ and $\delta \in (0, 1)$. The block algorithm is (ε, δ)-PAC on $\mathcal{X}$ (with failure probability at most $15\delta/16$), with budget*

$$N_{\mathrm{sep}}(\varepsilon, \delta) \leq \frac{1600192\, kd \log(8k/\delta) + 33\, d^2}{\varepsilon^2} \leq C_{\mathrm{sep}} \frac{kd \log(8k/\delta) + d^2}{\varepsilon^2}, \qquad C_{\mathrm{sep}} := 2 \cdot 10^6.$$

*If $d \le k$, this is at most $2C_{\mathrm{sep}}\, kd\big(\log(1/\delta) + \log(8k)\big)/\varepsilon^2$, whereas by Theorem **??** every non-adaptive (ε, δ)-PAC algorithm with $\delta < 1/16$ needs at least $\frac{1}{2}\big(c_{\mathrm{ad}}\, kd \log(1/\delta) + kd^2/210\big)/\varepsilon^2$ samples. Here "non-adaptive" means a deterministic fixed design with an arbitrary (possibly randomized) recommendation kernel, the class of Definition **??** and of Theorem **??**; see Remark **??**.*

Corollary 195 (Theorem 7 of the paper, existential form of the upper bound)**.** $Maiti2026Power.exists_isPAC_blockBallSetthm : separation, def : pacLet k \ge 1$, *$\varepsilon \in (0, 1]$ and $\delta \in (0, 1)$. There exist a budget $T \le C_{\mathrm{sep}}\,(kd \log(8k/\delta) + d^2)/\varepsilon^2$ and an identification algorithm with budget T that is (ε, δ)-PAC on the block-ball set $\mathcal{X} \subset \mathbb{R}^{kd}$.*

Proof. thm:separation, lem:seeded Take the block algorithm of Theorem **??**. (For $d = 0$ every reward vector is 0, every recommendation is optimal, and the budget 0 suffices.) ☐

Proof. lem:block$_selection, lem : ball_identification, lem : separation_block_embedding, lem : seeded, lem : block_ball_basic, thm : norm_estimation, thm : separation_lower_nonadaptive, lem : statistic_structureCorrectness.ByLemma$ **??**$itsufficestoprovetheboundontheseedspace(\Omega_{\mathrm{seed}}, \mathbb{P}_{\mathrm{seed}})$ of Lemma **??**, where the recommendation of the run is the recommendation computed from the observation sequence (it only reads the rounds $t < N_{\mathrm{sep}}$). Consider the two failure events

$$E_1 := \bigcup_{i<k} \Big\{ |\hat{r}_i - \|\theta^{(i)}\|| > \frac{\varepsilon}{4} \Big\}, \qquad E_2 := \Big\{ \|\Delta\| > \frac{\varepsilon}{4} \Big\}, \quad \Delta := \pi'_{kN,n_2}\text{-average of the noises,}$$

where $\Delta \in \mathbb{R}^d$ has coordinates $\Delta_m := \frac{1}{n_2} \sum_{\ell < n_2} \eta_{kN+mn_2+\ell}$. By Lemma **??**, $\hat{r}_i$ is the estimate of $\mathrm{NormEst}(\varepsilon/4, \delta/(2k))$ against $\nu_{\theta^{(i)}}$ on the shifted seed sequence, whose law is $\mathbb{P}_{\mathrm{seed}}$, so by Theorem **??** each event of the union has probability at most $\frac{7}{8} \cdot \delta/(2k)$ and $\mathbb{P}_{\mathrm{seed}}(E_1) \le 7\delta/16$. In phase (2) the actions are $\iota_{\hat{i}}(e_m)$, so the empirical mean vector is $\hat{\vartheta} = \theta^{(\hat{i})} + \Delta$, where Δ does not depend on $\hat{i}$; the law of the noises of the basis window is $\mathcal{N}(0, 1)^{\otimes dn_2}$ (Lemma **??**), so by the chi-square tail bound of Lemma **??** with $c = \varepsilon/4$ and the choice of n_2 ($n_2 \varepsilon^2/16 \ge 2d + 12 \log(2/\delta)$), $\mathbb{P}_{\mathrm{seed}}(E_2) \le \delta/2$. Outside $E_1 \cup E_2$, Lemma **??** gives $r(\hat{x}, \theta) \le \varepsilon$. Hence $\mathbb{P}_{\mathrm{seed}}(r(\hat{x}, \theta) \le \varepsilon) \ge 1 - 7\delta/16 - \delta/2 = 1 - 15\delta/16 \ge 1 - \delta$. No conditioning on the first phase is needed: the second failure event only involves the noises of the basis window, whatever the selected block.

Budget. By Theorem **??** with $(\varepsilon/4, \delta/(2k))$ (note $\log(4/(\delta/(2k))) = \log(8k/\delta)$), each run of phase (1) uses at most $10^5\, d \log(8k/\delta) \cdot 16/\varepsilon^2$ rounds, so phase (1) uses at most $1.6 \cdot 10^6\, kd \log(8k/\delta)/\varepsilon^2$ rounds. Phase (2) uses $dn_2 \le d\big((32d + 192 \log(2/\delta))/\varepsilon^2 + 1\big) \le (33d^2 + 192\, d \log(2/\delta))/\varepsilon^2$ (using $d \le d^2/\varepsilon^2$), and $192\, d \log(2/\delta) \le 192\, kd \log(8k/\delta)$. Summing gives the first bound, and $1600192 \le C_{\mathrm{sep}}$, $33 \le C_{\mathrm{sep}}$ give the second. If $d \le k$ then $d^2 \le kd \le kd \log(8k/\delta)$ (as $\log(8k/\delta) \ge \log 8 > 1$), whence the third bound with $\log(8k/\delta) = \log(1/\delta) + \log(8k)$. The lower bound is Theorem **??**. ☐

Remark 196 (The size of the gap). For $d \le k \le d^2$ and a fixed δ, the adaptive budget is $O(kd \log k/\varepsilon^2) = O(kd \log d/\varepsilon^2)$ (since $\log k \le 2 \log d$) while every non-adaptive algorithm needs $\Omega(kd^2/\varepsilon^2)$: adaptivity gains a factor $\Omega(d/\log d)$, polynomial in the ambient dimension $kd \le d^3$. The paper phrases the lower bound as $\Omega(kd\log(1/\delta)/\varepsilon^2 + kd^2/\varepsilon^2)$ using the adaptive lower bound of Chen et al. [**?**] on each block together with a padding argument for expected budgets; the route above through Theorem **??** gives the same kd^2 term with the explicit constant $1/210$ for deterministic budgets and $\delta \le 1/10$, without any citation.

Restriction of the comparison class. The paper's Theorem 7 states the lower bound for "possibly randomized non-adaptive" algorithms. The lower bound proved here (Theorem **??**(i)) covers only *deterministic* fixed designs, with an arbitrary randomized recommendation rule: this is the class of Definition **??** and of Theorem **??**, and it is the class in which Theorem **??** compares the block algorithm with non-adaptive ones. A non-adaptive algorithm whose design is drawn at random (independently of the observations) is not covered: for such an algorithm the pigeonhole step selects a block depending on the realized design, and the fixed-design lower bound would have to be applied conditionally on the design and integrated, which is not done here. The adaptive lower bound (ii) applies to every algorithm, randomized designs included, but only gives the $kd\log(1/\delta)$ term. This restriction is recorded in the list of deviations from the paper in the overview.

The constant C_{sep} inherits the (unoptimized) constant C_{norm} of Theorem **??** multiplied by 16 because the norm-estimation accuracy is $\varepsilon/4$.

Part II

Prerequisites to be added to Mathlib and LML

Overview

The proofs of Part **??** rely on several results of probability theory, information theory and bandit theory that are standard but not yet available in Mathlib or in LML. This part develops them, at the level of generality of a library: each chapter is a self-contained area that can be formalized (and contributed upstream) independently of the rest.

- Chapter **??**: Gaussian vectors, linear Gaussian processes, suprema over compact index sets, comparison by covariance domination, expectation of maxima and of norms.
- Chapter **??**: Gaussian concentration for Lipschitz functions (Maurey–Pisier) and the Borell–TIS inequality for linear Gaussian processes.
- Chapter **??**: sub-exponential random variables, squares of Gaussian and Rademacher linear forms, Bernstein-type bounds.
- Chapter **??**: Pinsker and Bretagnolle–Huber inequalities, convexity of the Kullback–Leibler divergence, and the divergence decomposition for algorithm-environment sequences.
- Chapter **??**: the Sudakov–Fernique inequality and the expectation of the maximum of independent Gaussians (lower bound).
- Chapter **??**: the Median Elimination algorithm for multi-armed bandits.
- Chapter **??**: likelihood ratios and Gaussian posteriors for adaptively collected linear-Gaussian data, and the adaptive simple-regret lower bound on the unit ball.

Chapter 10

Gaussian vectors and linear Gaussian processes

This chapter collects the facts about Gaussian vectors in $\mathbb{R}^n$ and about *linear Gaussian processes* $x \mapsto \langle x, G \rangle$ (G a centered Gaussian vector, x ranging over a compact index set $\mathcal{X} \subset \mathbb{R}^d$) that are used throughout Part **??**: closure of Gaussian vectors under affine maps, identification of the law by mean and covariance, rotation invariance of pairs of independent standard vectors, independence of uncorrelated jointly Gaussian vectors, moments of Gaussian norms, tails of empirical means, measurability and integrability of suprema over compact index sets, the expectation of a maximum of m Gaussians, and comparison by covariance domination (the special case of the Sudakov–Fernique inequality that suffices for the monotonicity of the Gaussian width, Lemma **??**). Everything here is classical [**?**, **?**]; the point is to fix the statements at the level of generality of a library and to name what exists.

What Mathlib provides. The standard Gaussian measure on a finite-dimensional inner product space, `ProbabilityTheory.stdGaussian`, and the multivariate Gaussian `multivariateGaussian   S` on `EuclideanSpace`, defined as the image of `stdGaussian` under $x \mapsto \mu + S^{1/2}x$ with $S^{1/2} =$ `CFC.sqrt S` (`Mathlib/Probability/Distributions/Gaussian/Multivariate.lean`), with `integral_id_multivariateGaussian`, `covarianceBilin_multivariateGaussian`, `charFun_multivariateGaussian`, `multivariateGaussian_zero_one`, `stdGaussian_map` (invariance under linear isometry equivalences), `map_pi_eq_stdGaussian` and `stdGaussian_eq_map_pi_orthonormalBasis` (coordinates are i.i.d. standard). The class `IsGaussian` of Gaussian measures (`Gaussian/Basic.lean`: every continuous linear functional has a real Gaussian law), closed under continuous linear maps (`isGaussian_map`), translations, negation and products, characterized by its characteristic function (`IsGaussian.charFun_eq`, `isGaussian_iff_charFun_eq`); the predicate `HasGaussianLaw X P` on random variables (`Gaussian/HasGaussianLaw/Basic.lean`: `HasGaussianLaw.map`, `.smul`, `.neg`, `.add`, `.eval`, `.integrable`, `.memLp`), and the independence results of `Gaussian/HasGaussianLaw/Independence.lean` (iIndepFun.hasGaussianLaw, IndepFun.hasGaussianLaw, HasGaussianLaw.indepFun_of_covariance_eval _inner, _eq_zero). Fernique's theorem IsGaussian.exists_integrable_exp_sq (Gaussian/Fernique.lean) and the rotation invariance IsGaussian.map_rotation_eq_self of $\mu \otimes \mu$ for centered μ. On the real line, gaussianReal, mgf_gaussianReal, integral_id_gaussianReal, variance_id_gaussianReal, gaussianReal_map_const_mul, gaussianReal_add_gaussianReal_of_indepFun (Gaussian/Real.lean). Sub-Gaussian tails: HasSubgaussianMGF with measure_ge_le (Mathlib/Probability/Moments/SubGaussian.lea and measure_le_le (LML, ForMathlib/Probability/Moments/SubGaussian.lean).

What is missing. The matrix form of the linear-image lemma (mean and covariance matrix of $b + LV$), the uniqueness of the law given mean and covariance matrix in the matrix language of Part **??**, the fourth moment of a Gaussian norm, the lower bound $\mathbb{E}\|Y\| \geq \sqrt{\operatorname{tr} S}/\sqrt{3}$, the expectation bound for a maximum of m Gaussians, the countable-dense reduction for suprema over compact index sets, and the covariance-domination comparison lemma. All are short.

Lean remark. Vectors live in EuclideanSpace (Fin d) and matrices in Matrix (Fin d) (Fin d) , acting through Matrix.toEuclideanLin / toEuclideanCLM. The positive semidefinite square root of $S \in \mathcal{S}_+^d$ is CFC.sqrt S, the one used by multivariateGaussian; for $S \in \mathcal{S}_{++}^d$ it agrees with the square root of Lemma **??**. The random vectors of this chapter are functions Ω → EuclideanSpace (Fin d) with HasLaw a given Gaussian measure, or with HasGaussianLaw together with prescribed mean and covariance.

10.1 Gaussian vectors

Definition 197 (Gaussian measures and Gaussian vectors)**.** Let $n \geq 1$, $\mu \in \mathbb{R}^n$ and $S \in \mathcal{S}_+^d$ (an $n \times n$ positive semidefinite matrix). The *Gaussian measure* $\mathcal{N}(\mu, S)$ on $\mathbb{R}^n$ is the image of the standard Gaussian measure $\mathcal{N}(0, I_n)$ (Mathlib stdGaussian) under $x \mapsto \mu + S^{1/2}x$, where $S^{1/2}$ is the positive semidefinite square root of S (Mathlib multivariateGaussian S). A random vector $V : \Omega \to \mathbb{R}^n$ satisfies $V \sim \mathcal{N}(\mu, S)$ if its law is $\mathcal{N}(\mu, S)$. More generally V is a *Gaussian vector* if $\langle t, V \rangle$ has a (possibly degenerate) real Gaussian law for every $t \in \mathbb{R}^n$ (Mathlib HasGaussianLaw V P); its *mean* is $\mathbb{E}V$ and its *covariance matrix* is $\operatorname{Cov}(V) := \mathbb{E}[(V - \mathbb{E}V)(V - \mathbb{E}V)^\top]$. For $n = 1$, $\mathcal{N}(\mu, \sigma^2)$ is Mathlib's gaussianReal ²; for $\sigma = 0$ it is the Dirac mass at μ.

We recall the facts that Mathlib attaches to this definition: $\mathcal{N}(\mu, S)$ is a probability measure with mean μ (integral_id_multivariateGaussian), covariance bilinear form $(s, t) \mapsto s^\top St$ (covarianceBilin_multivariateGaussian, for $S \in \mathcal{S}_+^d$), and characteristic function $t \mapsto \exp(i\langle t, \mu\rangle - \frac{1}{2}t^\top St)$ (charFun_multivariateGaussian); $\mathcal{N}(0, I_n) =$ stdGaussian (multivariateGaussian_zero_one); every Gaussian vector is integrable with all moments (HasGaussianLaw.memLp).

Lemma 198 (Standard Gaussian vectors have i.i.d. coordinates)**.** $ProbabilityTheory.hasLaw_pi_of_hasCondDistri$
≥ 1.

(i) If $\eta_0, \dots, \eta_{T-1}$ are independent real random variables, each with law $\mathcal{N}(0,1)$, then the random vector $\eta := (\eta_t)_{t<T} \in \mathbb{R}^T$ satisfies $\eta \sim \mathcal{N}(0, I_T)$.

(ii) Conversely, if $\eta \sim \mathcal{N}(0, I_T)$ then its coordinates $\eta_0, \dots, \eta_{T-1}$ are independent with law $\mathcal{N}(0,1)$.

Proof. def:pg$_g aussian_v ector(i) The joint law of independent random variables is the product of their laws (Mathlib$ seen in $\mathbb{R}^T =$ `EuclideanSpace (Fin T)`, is the image of $\bigotimes_{t<T} \mathcal{N}(0,1)$ under the identification `WithLp.toLp 2`, which is `stdGaussian` by `map_pi_eq_stdGaussian`; conclude with `multivariateGaussian_zero_one`. (ii) The same identity read backwards through the measurable equivalence `WithLp.toLp 2`: the law of $(\eta_t)_t$ as an element of $\mathrm{Fin}\, T \to \mathbb{R}$ is the product measure, which is the independence of the coordinates with the stated marginals (`iIndepFun_iff_map_fun_eq_pi_map` again, or `measurePreserving_eval_multivariateGaussian` for the marginals). □

Lean remark. Part (i) is Mathlib's `iIndepFun.hasLaw_pi` together with `map_pi_eq_stdGaussian`, and (ii) is `iIndepFun_iff_hasLaw_pi_pi`. The form in which (i) is used is sequential: if $\eta_0 \sim \nu$ and, for every n, the conditional law of η_{n+1} given $(\eta_0, \dots, \eta_n)$ (or given any variable of which this vector is a measurable function) is the constant ν, then $(\eta_0, \dots, \eta_{n-1})$ has the product law $\nu^{\otimes n}$ and the η_t are independent with law ν (`ProbabilityTheory.hasLaw_pi_of_hasCondDistrib_const`, `iIndepFun_of_hasCondDistrib_const`; proof by induction through the measurable equivalence `MeasurableEquiv.piFinSuccAbove`). This is what Lemma **??** provides for the noise of a linear Gaussian run.

Lemma 199 (The law of a Gaussian vector is determined by its mean and covariance)**.** *def:pg$_g aussian_v ector Let V, V' be Gaussian vectors in R^n$ (possibly on different probability spaces).*

(i) If $\mathbb{E}V = \mathbb{E}V'$ and $\mathrm{Cov}(V) = \mathrm{Cov}(V')$, then V and V' have the same law.

(ii) If $\mathbb{E}V = \mu$ and $\mathrm{Cov}(V) = S$, then $V \sim \mathcal{N}(\mu, S)$.

(iii) For $S \in \mathcal{S}_+^d$ and $g \sim \mathcal{N}(0, I_n)$, $S^{1/2} g \sim \mathcal{N}(0, S)$.

(iv) If $G \sim \mathcal{N}(0, S)$ then $-G \sim \mathcal{N}(0, S)$, i.e. $-G \overset{d}{=} G$.

Proof. def:pg$_g aussian_v ector(i) The characteristic function of a Gaussian vector is$ t $\mapsto \exp\left(i\,\mathbb{E}\langle t, V\rangle - \frac{1}{2}\mathrm{Var}\langle t, V\rangle\right)$ (Mathlib `IsGaussian.charFun_eq`, `HasGaussianLaw.charFun_map_eq`), and $\mathbb{E}\langle t, V\rangle = \langle t, \mathbb{E}V\rangle$, $\mathrm{Var}\langle t, V\rangle = t^\top \mathrm{Cov}(V)\, t$ (bilinearity of the covariance, Mathlib `covarianceBilin`). Hence V and V' have the same characteristic function, and a finite measure on a finite-dimensional inner product space is determined by its characteristic function (`Measure.ext_of_charFun`). (ii) The measure $\mathcal{N}(\mu, S)$ is Gaussian (`isGaussian_multivariateGaussian`) with mean μ and covariance S (the facts recalled after Definition **??**); apply (i) to V and to the identity on $(\mathbb{R}^n, \mathcal{N}(\mu, S))$. (iii) is the definition of $\mathcal{N}(0, S)$ (Mathlib `multivariateGaussian`, as the image of `stdGaussian`), using that the law of $S^{1/2} g$ is the image of the law of g. (iv) $-G$ is a Gaussian vector

(`HasGaussianLaw.neg`) with mean 0 and covariance $\mathbb{E}[(-G)(-G)^\top] = S$; apply (ii). ☐

Lemma 200 (Affine images of Gaussian vectors)**.** *ProbabilityTheory.multivariateGaussian$_m$ap$_t$oEuclideanCLM* $\sim \mathcal{N}(\mu, S)$ *in* $\mathbb{R}^n$, $L \in \mathbb{R}^{m\times n}$ *and* $b \in \mathbb{R}^m$. *Then* $b+LV \sim \mathcal{N}(b+L\mu,\ LSL^\top)$. *In particular, for* $g \sim \mathcal{N}(0, I_n)$, $M \in \mathbb{R}^{m\times n}$ *and* $\mu \in \mathbb{R}^m$*:* $\mu + Mg \sim \mathcal{N}(\mu, MM^\top)$*; and for* $a \in \mathbb{R}^n$, $\langle a, g\rangle \sim \mathcal{N}(0, \|a\|^2)$.

Proof. def:pg$_g$aussian$_v$ector, $lem : gauss_law_of_cov$b + LV$isaGaussianvector :$V$isone,$L$isacontinuouslinearma$ + Lμ by linearity of the integral, and for $s, t \in \mathbb{R}^m$, $\mathrm{Cov}(\langle s, b+LV\rangle, \langle t, b+LV\rangle) = \mathrm{Cov}(\langle L^\top s, V\rangle, \langle L^\top t, V\rangle) = (L^\top s)^\top S(L^\top t) = s^\top(LSL^\top)t$, i.e. the covariance matrix is $LSL^\top$. Conclude with Lemma **??**(ii). The special cases are $S = I_n$ and $L = a^\top$ (with $m = 1$, $\mathcal{N}(0, a^\top a)$ on $\mathbb{R}$ being `gaussianReal 0 ‖a‖²`). ☐

Lean remark. With V given by `HasLaw V (multivariateGaussian   S) P`, the conclusion is `HasLaw (fun    b + L (V  )) (multivariateGaussian (b + L  ) (L * S * L )) P`. The proof through characteristic functions is an alternative to the one through `HasGaussianLaw`: $\mathbb{E}e^{i\langle t, b+LV\rangle} = e^{i\langle t,b\rangle}\,\mathbb{E}e^{i\langle L^\top t, V\rangle}$ and `charFun_multivariateGaussian` on both sides.

Lemma 201 (Independent Gaussian vectors are jointly Gaussian)**.** *ProbabilityTheory.multivariateGaussian$_c$on* $\mathcal{N}(\mu_1, S_1)$ *in* $\mathbb{R}^{n_1}$ *and* $V_2 \sim \mathcal{N}(\mu_2, S_2)$ *in* $\mathbb{R}^{n_2}$ *be independent, on the same probability space. Then the pair* (V_1, V_2)*, seen in* $\mathbb{R}^{n_1+n_2}$*, satisfies* $(V_1, V_2) \sim \mathcal{N}\big((\mu_1, \mu_2), \mathrm{diag}(S_1, S_2)\big)$. *If* $n_1 = n_2$ *then* $V_1 + V_2 \sim \mathcal{N}(\mu_1 + \mu_2, S_1 + S_2)$.

Proof. def:pg$_g$aussian$_v$ector, $lem : gauss_law_of_cov, lem : gauss_linear_imageThepairisaGaussianvector(Mathl$ $thelawofthepairistheproductmeasure, andaproductofGaussianmeasuresisGaussian).Itsmeanis(\mu_1, \mu_2)$ and its covariance matrix is block diagonal: the cross covariances $\mathrm{Cov}(\langle s, V_1\rangle, \langle t, V_2\rangle)$ vanish because $\langle s, V_1\rangle$ and $\langle t, V_2\rangle$ are independent and square-integrable (the covariance of independent variables vanishes: Mathlib `IndepFun.covariance_eq_zero` in `Mathlib/Probability/Moments/Covariance.lean`, or the product formula `IndepFun.integral_mul_eq_mul_integral`). Conclude with Lemma **??**(ii). For the sum apply Lemma **??** with $L = [I_n\ I_n]$: $L\,\mathrm{diag}(S_1, S_2)L^\top = S_1 + S_2$. ☐

Lemma 202 (Orthogonal invariance and rotation of pairs)**.** *def:pg$_g$aussian$_v$ector*

(i) If $g \sim \mathcal{N}(0, I_d)$ *and* $U \in \mathbb{R}^{d\times d}$ *is orthogonal (*$U^\top U = I_d$*), then* $Ug \sim \mathcal{N}(0, I_d)$.

(ii) If g, g' *are independent, each with law* $\mathcal{N}(0, I_d)$*, then for every* $\theta \in \mathbb{R}$ *the pair*

$$(g_\theta, g'_\theta) := \big(g \sin\theta + g' \cos\theta,\ g\cos\theta - g' \sin\theta\big)$$

has the same law as (g, g')*, namely* $\mathcal{N}(0, I_{2d})$ *on* $\mathbb{R}^d \times \mathbb{R}^d$*; in particular* g_θ *and* g'_θ *are independent* $\mathcal{N}(0, I_d)$ *vectors.*

Proof. def:pg$_g$aussian$_v$ector, $lem : pg_indep_gaussian_pair, lem : gauss_pi, lem : gauss_uncorrelated_indep(i)Uisalinearisometryequivalenceof$R^d and the standard Gaussian measure is invariant under linear isometry equivalences (Mathlib `stdGaussian_map`); the law of Ug is the image of the law of g. (ii) By

Lemma **??** the pair (g, g') has law $\mathcal{N}(0, I_{2d})$. The map $R_\theta(u, v) := (u \sin\theta + v\cos\theta,\ u\cos\theta - v\sin\theta)$ is linear and

$$\|u\sin\theta+v\cos\theta\|^2+\|u\cos\theta-v\sin\theta\|^2 = \|u\|^2(\sin^2\theta+\cos^2\theta)+\|v\|^2(\cos^2\theta+\sin^2\theta)+2\langle u, v\rangle(\sin\theta\cos\theta-\cos\theta\sin\theta)$$

so R_θ is a linear isometry of $\mathbb{R}^{2d}$ (bijective since it is an isometry of a finite-dimensional space, or explicitly $R_\theta^{-1} = R_\theta^\top$). Apply (i) in dimension $2d$. (Mathlib's `IsGaussian.map_rotation_eq_self` is the same statement for the map `ContinuousLinearMap.rotation` of `Mathlib/Probability/Distributions/Fernique.lean`, namely $(u, v) \mapsto (u\cos\theta+v\sin\theta,\ -u\sin\theta+v\cos\theta)$, applied to $\mu\otimes\mu$ with μ a centered Gaussian measure. Since $\mathrm{rotation}(-\theta)(u, v) = (u\cos\theta - v\sin\theta,\ u\sin\theta + v\cos\theta)$, one has $R_\theta = \mathrm{swap}\circ\mathrm{rotation}(-\theta)$ with $\mathrm{swap}(u, v) := (v, u)$, and the swap also preserves $\mu\otimes\mu$ (`Measure.prod_swap`).) Independence of the components of a $\mathcal{N}(0, I_{2d})$ vector is Lemma **??**(ii) applied blockwise, or Lemma **??**. □

Lean remark. In the Lean development the pair of Gaussian vectors of Lemmas **??** and **??** is always of the form $(A_1g+A_2\eta,\ B_1g+B_2\eta)$ for a standard Gaussian pair (g, η) of $\mathbb{R}^d\times\mathbb{R}^T$ (`isGaussian_prod_stdGaussian`: the product of two standard Gaussian measures is a Gaussian measure on the product space) and continuous linear maps A_i, B_i; the cross-covariance is then $\mathrm{Cov}(\langle a, A_1g + A_2\eta\rangle, \langle b, B_1g+B_2\eta\rangle) = \langle a, (A_1B_1^*+A_2B_2^*)b\rangle$ (`covariance_inner_add_prod_stdGaussian`), and the independence criterion becomes $A_1B_1^*+A_2B_2^* = 0$ (`indepFun_add_prod_stdGaussian_of_comp_adjoi`

Lemma 203 (Joint law of two linear images)**.** *$ProbabilityTheory.isGaussian_prod_stdGaussian, ProbabilityTheo$ $dimensionalrealinnerproductspacesE,F$: $theirproductisthestandardGaussianmeasureofE$ $\times F$ with the inner product $\langle(a, b),(x, y)\rangle = \langle a, x\rangle+\langle b, y\rangle$. $def{:}pg_gaussian_vectorLetV$ $\sim \mathcal{N}(\mu, S)$ in $\mathbb{R}^n$ and $L_1 \in \mathbb{R}^{m_1\times n}$, $L_2 \in \mathbb{R}^{m_2\times n}$. Then (L_1V, L_2V) is a Gaussian vector in $\mathbb{R}^{m_1}\times\mathbb{R}^{m_2}$ with*

$$(L_1V, L_2V) \sim \mathcal{N}\Big((L_1\mu, L_2\mu), \begin{pmatrix} L_1SL_1^\top & L_1SL_2^\top \\ L_2SL_1^\top & L_2SL_2^\top \end{pmatrix}\Big),$$

and in particular the cross covariance is $\mathrm{Cov}(L_1V, L_2V) := \mathbb{E}[(L_1V - L_1\mu)(L_2V - L_2\mu)^\top] = L_1SL_2^\top$.

Proof. def:pg$_gaussian_vector, lem : gauss_linear_image(\mathrm{L_1}V, L_2V) = LV$ with $L := \binom{L_1}{L_2} \in \mathbb{R}^{(m_1+m_2)\times n}$; Lemma **??** gives $LV \sim \mathcal{N}(L\mu, LSL^\top)$ and $LSL^\top$ is the displayed block matrix. The cross covariance is its off-diagonal block. □

Lemma 204 (Uncorrelated jointly Gaussian vectors are independent)**.** *$ProbabilityTheory.indepFun_add_prod_stdG$ be a Gaussian vector in $\mathbb{R}^{m_1}\times\mathbb{R}^{m_2}$ (i.e. the pair is jointly Gaussian) such that* $\mathrm{Cov}(V_1, V_2) = 0$, *i.e.* $\mathrm{Cov}((V_1)_i, (V_2)_j) = 0$ *for all* $i \le m_1$, $j \le m_2$. *Then* V_1 *and* V_2 *are independent.*

Proof. def:pg$_gaussian_vectorMathlib$: `HasGaussianLaw.indepFun_of_covariance_eval`$(coordinates)or$`HasG`
0, which follows from the coordinatewise hypothesis by bilinearity). □

Lemma 205 (Expectation of an inner product with an independent centered vector)**.** *$ProbabilityTheory.IndepFun.integral_inner_comp_eq_zero, ProbabilityTheory.integrable_inner_compLetW$*

: $\Omega \to \mathbb{R}^d$ be integrable with $\mathbb{E}W = 0$, let $V : \Omega \to \mathcal{V}$ be a random variable with values in a measurable space, independent of W, and let $f : \mathcal{V} \to \mathbb{R}^d$ be bounded and measurable. Then $\langle f(V), W\rangle$ is integrable and $\mathbb{E}\langle f(V), W\rangle = 0$.

Proof. (The Lean proof is basis-free: the law of (V, W) is the product of the marginals, so by Fubini $\mathbb{E}\langle f(V), W\rangle = \int \langle f(v), \mathbb{E}W\rangle \, \mathbb{P}_V(dv) = 0$.) $\langle f(V), W\rangle = \sum_{i<d} f_i(V)\, W_i$. For each i, $f_i(V)$ is bounded and measurable and W_i is integrable, so their product is integrable; $f_i(V)$ and W_i are independent (compositions of independent variables with measurable maps, Mathlib `IndepFun.comp`), so $\mathbb{E}[f_i(V)W_i] = \mathbb{E}[f_i(V)]\,\mathbb{E}[W_i] = 0$ (product formula for independent integrable variables, Mathlib `IndepFun.integral_mul_eq_mul_integral`). Sum over i. $\square$

10.2 Moments and tails

Lemma 206 (Moments of a centered real Gaussian)**.** *ProbabilityTheory.integral$_a$bs$_g$aussianRealdef : pg$_g$aussi* *0 and $Z \sim \mathcal{N}(0, \sigma^2)$. Then Z has finite moments of all orders and*

$$\mathbb{E}Z = 0, \quad \mathbb{E}Z^2 = \sigma^2, \quad \mathbb{E}Z^4 = 3\sigma^4, \quad \mathbb{E}|Z| = \sigma\sqrt{2/\pi}, \quad \mathbb{E}e^{\lambda Z} = e^{\lambda^2\sigma^2/2}\ (\lambda \in \mathbb{R}).$$

Proof. def:pg$_g$*aussian$_v$ectorMathlib* : `memLp_id_gaussianReal`(*allmoments*), `integral_id_gaussianReal`, `v` $\mathbb{E}e^{\lambda Z} = \exp(0 \cdot \lambda + \sigma^2\lambda^2/2)$). For the fourth moment and the absolute moment, reduce to $\sigma = 1$ by $Z \overset{d}{=} \sigma\xi$, $\xi \sim \mathcal{N}(0,1)$ (`gaussianReal_map_const_mul`; for $\sigma = 0$ both sides vanish), and compute with the density $\varphi(x) = e^{-x^2/2}/\sqrt{2\pi}$: $\int x^4\varphi = 3\int x^2\varphi = 3$ by integration by parts ($x^3 \cdot x\varphi(x) = -x^3\varphi'(x)$, `integral_mul_deriv_eq_deriv_mul`), and $\int |x|\varphi = 2\int_0^\infty x\varphi(x)\,dx = 2\varphi(0) = \sqrt{2/\pi}$ since $x\varphi(x) = -\varphi'(x)$. Alternatively, $\mathbb{E}Z^4$ is the fourth derivative at 0 of the MGF $e^{\lambda^2\sigma^2/2}$ (Mathlib `iteratedDeriv_mgf_zero`).

(Lean: the absolute moment is `integral_abs_gaussianReal`, $\mathbb{E}|Z| = \sqrt{2\sigma^2/\pi}$, computed with the density and $\int_0^\infty xe^{-bx^2}\,dx = 1/(2b)$ (`integral_Ioi_mul_exp_neg_mul_sq`); the fourth moment is not formalized.) $\square$

Lemma 207 (Moments of the norm of a Gaussian vector)**.** *ProbabilityTheory.integral$_n$orm$_s$q$_m$ultivariateGauss* *$\in \mathcal{S}_+^d$ $(d \times d)$ and $Y \sim \mathcal{N}(0, S)$; equivalently $Y = S^{1/2}g$ with $g \sim \mathcal{N}(0, I_d)$. Then*

(i) $\mathbb{E}\|Y\|^2 = \operatorname{tr} S$;

(ii) $\mathbb{E}\|Y\|^4 = (\operatorname{tr} S)^2 + 2\operatorname{tr}(S^2) \le 3(\operatorname{tr} S)^2$;

(iii) $\mathbb{E}\|Y\| \le \sqrt{\operatorname{tr} S}$;

(iv) $\mathbb{E}\|Y\| \ge (\operatorname{tr} S)^{3/2}/\sqrt{(\operatorname{tr} S)^2 + 2\operatorname{tr}(S^2)} \ge \sqrt{\operatorname{tr} S}/\sqrt{3}$.

In the special case $S = I_d$: $\mathbb{E}\|g\|^2 = d$, $\mathbb{E}\|g\|^4 = d^2 + 2d$, $\mathbb{E}\|g\| \le \sqrt{d}$ and $\mathbb{E}\|g\| \ge d/\sqrt{d+2} \ge \sqrt{d/3}$.

Proof. def:pg$_g aussian_v ector, lem : gauss_l aw_o f_c ov, lem : gauss_r otation, lem : gauss_p i, lem : gauss_1 d_m omentsAllquantitiesdependonlyonthelawof$Y$, sobyLemma$ **??**$(iii)wemaytake$Y $=$ S$^{1/2}g$. Diagonalize $S = U \operatorname{diag}(s_1, \dots, s_d)U^\top$ with U orthogonal and $s_i \geq 0$ (spectral theorem for symmetric matrices, Mathlib `Matrix.IsHermitian.spectral_theorem`; nonnegativity of the eigenvalues from $S \in \mathcal{S}_+^d$). Then $\|Y\|^2 = g^\top S g = \sum_i s_i h_i^2$ with $h := U^\top g$, and $h \sim \mathcal{N}(0, I_d)$ by Lemma **??**(i), so the h_i are i.i.d. $\mathcal{N}(0,1)$ (Lemma **??**(ii)). (i) $\mathbb{E}\sum_i s_i h_i^2 = \sum_i s_i = \operatorname{tr} S$ (Lemma **??**). (ii) Expanding the square and using independence, $\mathbb{E}h_i^4 = 3$ and $\mathbb{E}h_i^2 = 1$,

$$\mathbb{E}\Big(\sum_i s_i h_i^2\Big)^2 = \sum_i s_i^2\, \mathbb{E}h_i^4 + \sum_{i\neq j} s_i s_j\, \mathbb{E}h_i^2\, \mathbb{E}h_j^2 = 3\sum_i s_i^2 + \Big(\big(\sum_i s_i\big)^2 - \sum_i s_i^2\Big) = (\operatorname{tr} S)^2 + 2\operatorname{tr}(S^2),$$

using $\operatorname{tr}(S^2) = \sum_i s_i^2$. Since $s_i \geq 0$, $\sum_i s_i^2 \leq (\sum_i s_i)^2$, whence the bound $3(\operatorname{tr} S)^2$. (iii) is Jensen (or Cauchy–Schwarz): $(\mathbb{E}\|Y\|)^2 \leq \mathbb{E}\|Y\|^2$. (iv) Hölder's inequality with exponents $3/2$ and 3 applied to $\|Y\|^2 = \|Y\|^{2/3} \cdot \|Y\|^{4/3}$ gives $\mathbb{E}\|Y\|^2 \leq (\mathbb{E}\|Y\|)^{2/3}(\mathbb{E}\|Y\|^4)^{1/3}$, i.e. $(\mathbb{E}\|Y\|)^{2/3} \geq \operatorname{tr} S \,/\, ((\operatorname{tr} S)^2 + 2\operatorname{tr}(S^2))^{1/3}$; raise to the power $3/2$ and use (ii) for the last inequality. For $S = I_d$, $\operatorname{tr} S = d$, $\operatorname{tr}(S^2) = d$ and $d^{3/2}/\sqrt{d^2 + 2d} = d/\sqrt{d+2}$, and $d/\sqrt{d+2} \geq \sqrt{d/3}$ since $d \geq 1$ gives $d \geq (d+2)/3$. ☐

Lemma 208 (Lower bound on the expected norm of a Gaussian vector)**.** *$ProbabilityTheory.integral_a bs_e val_m ultivariateGaussian, ProbabilityTheory.sum_m ul_s qrt_d iag_l e_i ntegral_n orm_m$ $\in \mathcal{S}_+^d$ and $Y \sim \mathcal{N}(0, S)$. For every $c \in \mathbb{R}^d$ with $c_i \geq 0$ and $\sum_i c_i^2 \leq 1$,*

$$\mathbb{E}\|Y\| \;\geq\; \sum_i c_i\, \mathbb{E}|Y_i| \;=\; \sqrt{2/\pi}\sum_i c_i \sqrt{S_{ii}}.$$

Proof. lem:gauss$_1 d_m oments, lem : gauss_l inear_i magePointwise, byCauchy--Schwarz,$$\sum_i c_i|y_i| \leq (\sum_i c_i^2)^{1/2}(\sum_i y_i^2)^{1/2} \leq \|y\|$; integrate. Each coordinate Y_i has law $\mathcal{N}(0, S_{ii})$ (Lemma **??**), so $\mathbb{E}|Y_i| = \sqrt{2S_{ii}/\pi}$ by Lemma **??**. ☐

Lemma 209 (Gaussian tail)**.** *$ProbabilityTheory.hasSubgaussianMGF_f un_i d_g aussianRealdef : pg_g aussian_v ecto$ $\mathbb{R}$, $\sigma > 0$ and $Z \sim \mathcal{N}(\mu, \sigma^2)$. Then $Z - \mu$ has a sub-Gaussian moment generating function with constant σ^2 (Mathlib `HasSubgaussianMGF (Z - ) ²:` $\mathbb{E}e^{\lambda(Z-\mu)} \leq e^{\sigma^2\lambda^2/2}$ for all λ, with equality), and for every $t \geq 0$,*

$$\mathbb{P}(Z-\mu \geq t) \leq e^{-t^2/(2\sigma^2)}, \qquad \mathbb{P}(Z-\mu \leq -t) \leq e^{-t^2/(2\sigma^2)}, \qquad \mathbb{P}(|Z-\mu| \geq t) \leq 2e^{-t^2/(2\sigma^2)}.$$

Proof. def:pg$_g aussian_v ector$Z - $\mu \sim \mathcal{N}(0, \sigma^2)$ (`gaussianReal_sub_const`) and `mgf_gaussianReal` gives $\mathbb{E}e^{\lambda(Z-\mu)} = e^{\sigma^2\lambda^2/2}$, with integrability from `integrable_exp_mul_gaussianReal`; this is the structure `HasSubgaussianMGF`. The tails are the Chernoff bounds `HasSubgaussianMGF.measure_ge_le` (Mathlib) and `HasSubgaussianMGF.measure_le_le` (LML); the two-sided bound is a union bound. ☐

Lemma 210 (Concentration of a Gaussian empirical mean)**.** *$def{:}pg_g aussian_v ector, lem : gauss_t ailLetn \geq 1$ and $\eta_1, \dots, \eta_n$ be i.i.d. $\mathcal{N}(0,1)$. Then $\bar\eta := \frac{1}{n}\sum_{i\leq n}\eta_i \sim \mathcal{N}(0, 1/n)$ and for all $t \geq 0$, $\mathbb{P}(|\bar\eta| \geq t) \leq 2e^{-nt^2/2}$. Consequently, for $\varepsilon > 0$ and $\delta \in (0,1)$, if $n \geq 8\log(2/\delta)/\varepsilon^2$ then $\mathbb{P}(|\bar\eta| \leq \varepsilon/2) \geq 1 - \delta$. The same holds for $\bar y - \mu$ when $y_i = \mu + \eta_i$.*

Proof. lem:gauss$_tail, lem : gauss_pi, lem : gauss_linear_image \sum_i \eta_i \sim \mathcal{N}(0, n)$ (sums of independent real Gaussians, Mathlib `gaussianReal_add_gaussianReal_of_indepFun` by induction; or Lemmas **??** and **??** with $a = (1, \dots, 1)$), and $\bar{\eta} \sim \mathcal{N}(0, 1/n)$ (`gaussianReal_map_const_mul`). Lemma **??** with $\sigma^2 = 1/n$ gives $\mathbb{P}(|\bar{\eta}| \geq t) \leq 2e^{-t^2/(2/n)} = 2e^{-nt^2/2}$. With $t = \varepsilon/2$: $2e^{-n\varepsilon^2/8} \leq 2e^{-\log(2/\delta)} = \delta$ when $n\varepsilon^2/8 \geq \log(2/\delta)$. ☐

10.3 Suprema of linear Gaussian processes

Lemma 211 (Suprema over a compact set reduce to a countable dense subset)**.** *supportFn, lipschitzWith$_s$upportFn, convexOn$_s$upportFn, continuous$_s$upportFn, abs$_s$upportFn$_l$e, exists$_s$uppo* $\mathbb{R}^d$ *be nonempty and compact and* $R := \max_{x \in \mathcal{X}} \|x\|$*.*

(i) There is a countable dense subset $D \subseteq \mathcal{X}$*; fix an enumeration* $D = \{q_n : n \in \mathbb{N}\}$ *(repetitions allowed).*

(ii) For every $v \in \mathbb{R}^d$ *the supremum* $m(v) := \sup_{x \in \mathcal{X}} \langle x, v \rangle$ *is attained (it is a maximum), and* $m(v) = \sup_{n \in \mathbb{N}} \langle q_n, v \rangle = \lim_{n \to \infty} \max_{i \leq n} \langle q_i, v \rangle$*, the sequence* $\max_{i \leq n} \langle q_i, v \rangle$ *being nondecreasing in* n*.*

(iii) $m : \mathbb{R}^d \to \mathbb{R}$ *is convex, R-Lipschitz (*$|m(v) - m(v')| \leq R\|v - v'\|$*), hence continuous and Borel measurable, and* $|m(v)| \leq R\|v\|$*.*

(iv) For every random vector $G : \Omega \to \mathbb{R}^d$*,* $\sup_{x \in \mathcal{X}} \langle x, G \rangle = m(G)$ *is a real random variable with* $|m(G)| \leq R\|G\|$*; it is integrable as soon as* $\|G\|$ *is.*

Proof. (i) $\mathcal{X}$ is a separable metric space (Mathlib `TopologicalSpace.exists_countable_dense` for the subtype, which is second countable as a subspace of $\mathbb{R}^d$). (ii) For fixed v the map $x \mapsto \langle x, v \rangle$ is continuous, so it attains its maximum on the compact $\mathcal{X}$ (`IsCompact.exists_isMaxOn`). Clearly $\sup_n \langle q_n, v \rangle \leq m(v)$. Conversely, if $x \in \mathcal{X}$ attains $m(v)$, density gives $q_{n_k} \to x$ and continuity gives $\langle q_{n_k}, v \rangle \to \langle x, v \rangle = m(v)$, so $\sup_n \langle q_n, v \rangle \geq m(v)$. The partial maxima are nondecreasing and converge to the supremum of the sequence (`tendsto_atTop_ciSup` for monotone bounded sequences). (iii) m is a supremum of linear functions, hence convex. For v, v' and $x \in \mathcal{X}$ attaining $m(v)$, $m(v) - m(v') \leq \langle x, v \rangle - \langle x, v' \rangle \leq \|x\|\|v - v'\| \leq R\|v - v'\|$; symmetrize. $m(0) = 0$ gives $|m(v)| \leq R\|v\|$. Measurability follows from continuity, or directly from (ii): a countable supremum of measurable functions is measurable (`Measurable.iSup`). (iv) is (ii)–(iii) composed with G. ☐

Lean remark. In Lean, $\sup_{x \in \mathcal{X}} \langle x, v \rangle$ is best written as `x : Xs, (x : E), v` (a `ciSup` over the compact subtype, bounded above), and item (ii) provides the equality with `n, q n, v` for a sequence `q : → Xs` with dense range (`TopologicalSpace.exists_dense_seq`). All measure-theoretic statements about $\sup_{x \in \mathcal{X}} \langle x, G \rangle$ in this blueprint are proved through this countable form, and Chapter **??** uses the partial maxima $\max_{i \leq n} \langle q_i, G \rangle$ as the finite approximations.

Lemma 212 (Expected maximum of m sub-Gaussian variables)**.** *ProbabilityTheory.integral$_i$Sup$_l$e$_o$f$_h$asSubgau* ≥ 1, $\sigma \geq 0$ *and* $Z_1, \ldots, Z_m$ *be real random variables on a common probability space, each with a sub-Gaussian moment generating function with constant* σ^2 *(Mathlib* `HasSubgaussianMGF (Z i)` ^{2}*:* $e^{\beta Z_i}$ *is integrable and* $\mathbb{E}e^{\beta Z_i} \leq e^{\beta^2\sigma^2/2}$ *for every* $\beta \in \mathbb{R}$*; no assumption on their joint law). Then* $\max_{i\leq m} Z_i$ *is integrable and*

$$\mathbb{E}\max_{i\leq m} Z_i \leq \sigma\sqrt{2\log m}.$$

In particular this applies to centered Gaussian variables $Z_i \sim \mathcal{N}(0, \sigma_i^2)$ *with* $\sigma_i^2 \leq \sigma^2$ *for every* i*, whatever their joint law (Lemma* **??***):* $\mathbb{E}\max_{i\leq m} Z_i \leq \sigma\sqrt{2\log m}$*.*

Proof. lem:gauss$_t$*ailIntegrability* $:|Z_i| \leq e^{Z_i} + e^{-Z_i}$ is integrable ($\beta = \pm 1$), and $|\max_i Z_i| \leq \sum_i |Z_i|$. Let $\beta > 0$. By Jensen's inequality for the convex function exp (Mathlib `ConvexOn.map_integral_le` with `convexOn_exp`; the integrand $e^{\beta\max_i Z_i} \leq \sum_i e^{\beta Z_i}$ is integrable by assumption),

$$\exp\left(\beta\,\mathbb{E}\max_i Z_i\right) \leq \mathbb{E}\exp\left(\beta\max_i Z_i\right) = \mathbb{E}\max_i e^{\beta Z_i} \leq \sum_{i\leq m}\mathbb{E}e^{\beta Z_i} \leq m\,e^{\beta^2\sigma^2/2},$$

where $\max_i e^{\beta Z_i} = e^{\beta\max_i Z_i}$ because exp is increasing. Taking logarithms, $\mathbb{E}\max_i Z_i \leq \frac{\log m}{\beta} + \frac{\beta\sigma^2}{2}$ for every $\beta > 0$. If $m \geq 2$ and $\sigma > 0$, choose $\beta := \sqrt{2\log m}/\sigma$: both terms equal $\sigma\sqrt{2\log m}/2$. If $m = 1$, let $\beta \downarrow 0$: $\mathbb{E}Z_1 \leq 0 = \sigma\sqrt{2\log 1}$. If $\sigma = 0$, let $\beta \to \infty$: $\mathbb{E}\max_i Z_i \leq 0$. The Gaussian case: by Lemma **??** (with $\mu = 0$), $Z_i \sim \mathcal{N}(0, \sigma_i^2)$ with $\sigma_i > 0$ has a sub-Gaussian MGF with constant $\sigma_i^2 \leq \sigma^2$, hence with constant σ^2 (the bound $e^{\beta^2\sigma_i^2/2} \leq e^{\beta^2\sigma^2/2}$ is monotone in the constant); if $\sigma_i = 0$ then $Z_i = 0$ a.s. and the constant-σ^2 bound is trivial. ☐

Lemma 213 (Comparison by covariance domination)**.** *ProbabilityTheory.gaussianWidth$_m$ultivariateGaussian* $\mathbb{R}^d$ *be nonempty and compact, let* $S_X, S_Y \in \mathcal{S}_+^d$ *with* $S_X \preceq S_Y$ *(i.e.* $S_Y - S_X \in \mathcal{S}_+^d$*), and let* $G_X \sim \mathcal{N}(0, S_X)$*,* $G_Y \sim \mathcal{N}(0, S_Y)$ *(on arbitrary probability spaces). Then* $\sup_{x\in\mathcal{X}}\langle x, G_X\rangle$ *and* $\sup_{x\in\mathcal{X}}\langle x, G_Y\rangle$ *are integrable and*

$$\mathbb{E}\sup_{x\in\mathcal{X}}\langle x, G_X\rangle \leq \mathbb{E}\sup_{x\in\mathcal{X}}\langle x, G_Y\rangle.$$

In particular (finite index set $\mathcal{X} = \{e_1, \ldots, e_m\}$*, the canonical basis of* $\mathbb{R}^m$*): if* $Z \sim \mathcal{N}(0, \Sigma_Z)$ *and* $W \sim \mathcal{N}(0, \Sigma_W)$ *in* $\mathbb{R}^m$ *with* $\Sigma_Z \preceq \Sigma_W$*, then* $\mathbb{E}\max_{i\leq m} Z_i \leq \mathbb{E}\max_{i\leq m} W_i$*.*

Proof. lem:sup$_c$*ountable$_d$ense, lem : gauss$_n$orm$_m$oments, lem : pg$_i$ndep$_g$aussian$_p$air, lem : gauss$_l$aw$_o$f$_c$ovWrite*m(v) $:= \sup_{x\in\mathcal{X}}\langle x, v\rangle$, which is continuous with $|m(v)| \leq R\|v\|$, $R := \max_{x\in\mathcal{X}}\|x\|$ (Lemma **??**); integrability of $m(G_X)$ and $m(G_Y)$ follows from $\mathbb{E}\|G\| < \infty$ for Gaussian vectors (Lemma **??**(iii)). Both expectations depend only on the laws, so we may work on the product space $(\mathbb{R}^d \times \mathbb{R}^d, \mathcal{N}(0, S_X) \otimes \mathcal{N}(0, S_Y - S_X))$ with coordinate projections G_X and G_Z, which are independent with $G_Z \sim \mathcal{N}(0, S_Y - S_X)$ ($S_Y - S_X \in \mathcal{S}_+^d$ by assumption).

By Lemma **??**, $G_X + G_Z \sim \mathcal{N}(0, S_X + (S_Y - S_X)) = \mathcal{N}(0, S_Y)$, so $\mathbb{E}m(G_Y) = \mathbb{E}m(G_X + G_Z)$ (Lemma **??** is not even needed: this is equality of laws). By Fubini on the product measure (Mathlib `MeasureTheory.integral_prod`; the integrand $(u,z) \mapsto m(u+z)$ is continuous and dominated by $R(\|u\| + \|z\|)$, which is integrable on the product),

$$\mathbb{E}m(G_X + G_Z) = \int \Big(\int m(u+z)\,\mathcal{N}(0, S_Y - S_X)(dz) \Big) \mathcal{N}(0, S_X)(du).$$

Fix u. For every $x \in \mathcal{X}$ and every z, $\langle x, u\rangle + \langle x, z\rangle \le m(u+z)$; integrating in z and using $\int \langle x, z\rangle\,\mathcal{N}(0, S_Y - S_X)(dz) = \langle x, 0\rangle = 0$ (centered), monotonicity of the integral gives $\langle x, u\rangle \le \int m(u+z)\,\mathcal{N}(0, S_Y - S_X)(dz)$. Taking the supremum over $x \in \mathcal{X}$: $m(u) \le \int m(u+z)\,\mathcal{N}(0, S_Y - S_X)(dz)$. Integrating in u yields $\mathbb{E}m(G_X) \le \mathbb{E}m(G_X + G_Z) = \mathbb{E}m(G_Y)$. The finite-index statement is the case $\mathcal{X} = \{e_1, \dots, e_m\}$, since $\langle e_i, G\rangle = G_i$. $\square$

Remark 214 (Relation with Sudakov–Fernique). The Sudakov–Fernique inequality (Chapter **??**) only requires $\mathbb{E}(Z_i - Z_j)^2 \le \mathbb{E}(W_i - W_j)^2$ for all pairs, which is weaker than $\Sigma_Z \preceq \Sigma_W$. The Loewner-order version above is all that the monotonicity of the Gaussian width (Lemma **??**) needs, and its proof is elementary; this is one of the deviations from the paper listed in the introduction.

Chapter 11

Gaussian concentration and the Borell–TIS inequality

This chapter proves that the supremum $M = \sup_{x\in\mathcal{X}}\langle x, G\rangle$ of a linear Gaussian process over a compact index set concentrates around its mean at the scale $\sigma := \sup_{x\in\mathcal{X}}\sqrt{x^\top S x}$ of the largest individual standard deviation (the Borell–TIS inequality, Theorem **??**), in the form used by the fixed-design upper bound (Corollaries **??** and **??**). The route is the Maurey–Pisier argument [**?**] (see also [**?**, Section 5.4] and [**?**, Chapter 5]): a C^1 function f of a standard Gaussian vector with $\|\nabla f\| \le L$ has a sub-Gaussian moment generating function with constant $c_{\mathrm{G}}L^2$, where

$$c_{\mathrm{G}} := \pi^2/4,$$

and the supremum of a finite family of linear forms is approximated by the smooth "log-sum-exp" function. The optimal constant is 1 (Gaussian isoperimetry, or the Gaussian log-Sobolev inequality, see [**?**]); the factor c_{G} is the price of an argument that uses only Jensen's inequality, the fundamental theorem of calculus, Fubini and the rotation invariance of Gaussian pairs, and it propagates as an explicit absolute constant into the upper bounds of Part **??**.

What Mathlib provides. The structure `ProbabilityTheory.HasSubgaussianMGF X c` (`Mathlib/Probability/Moments/SubGaussian.lean`): integrability of e^{tX} for all t together with $\mathbb{E}e^{tX} \le e^{ct^2/2}$; its Chernoff bound `HasSubgaussianMGF.measure_ge_le`, and `measure_le_le` in LML (`ForMathlib/Probability/Moments/SubGaussian.lean`). Jensen's inequality `ConvexOn.map_integral_le` and `ConvexOn.map_average_le` (`Mathlib/Analysis/Convex/Integral.lean`) with `convexOn_exp`. The fundamental theorem of calculus `intervalIntegral.integral_eq_sub_of_hasDerivAt`, the chain rule `HasFDerivAt.comp_hasDerivAt`, the mean value inequality `lipschitzWith_of_nnnorm_fderiv_`, Fubini `MeasureTheory.integral_prod` and `integral_integral_swap`, dominated convergence `tendsto_integral_of_dominated_convergence`, Fernique's theorem `IsGaussian.exists_integrable_exp_sq`, the rotation invariance of Gaussian pairs (Lemma **??**, Mathlib `stdGaussian_map` and `IsGaussian.map_rotation_eq_self`), and the Gaussian MGF `mgf_gaussianReal`.

What is missing. Everything specific to concentration: the Maurey–Pisier inequality (Lemma **??**), the log-sum-exp smoothing (Lemma **??**), the passage from finite to compact index sets (Lemma **??**) and the Borell–TIS inequality itself. Mathlib has no Gaussian concentration inequality for Lipschitz functions at the time of writing.

Lean remark. Sub-Gaussianity of $M - \mathbb{E}M$ is stated with Mathlib's `HasSubgaussianMGF (fun    M   - P[M]) (cG *  ²) P`, whose constant is an `0`; all constants below are therefore nonnegative reals coerced to `0`. Gradients are `gradient f x` in `EuclideanSpace    (Fin d)`, with $\langle \nabla f(x), v\rangle = f'(x)(v)$ (`HasGradientAt / HasFDerivAt`), and "$\|\nabla f\| \le L$ everywhere" is `x, ‖gradient f x‖   L` together with differentiability; the class C^1 is `ContDiff    1 f`.

11.1 Preliminaries

Definition 215 (The Gaussian concentration constant)**.** ProbabilityTheory.gaussianConcentrationConst$c_{\mathrm{G}} :=$ $\pi^2/4$ (≈ 2.4674).

Lemma 216 (MGF of a linear form of a standard Gaussian vector)**.** *ProbabilityTheory.stdGaussian$_m$ap$_t$oDual,* $\sim \mathcal{N}(0, I_d)$ *and* $a \in \mathbb{R}^d$*. Then* $e^{\langle a,g'\rangle}$ *is integrable and* $\mathbb{E}e^{\langle a,g'\rangle} = e^{\|a\|^2/2}$*. More generally, for* $s \in \mathbb{R}$*,* $\mathbb{E}e^{s\langle a,g'\rangle} = e^{s^2\|a\|^2/2}$*.*

Proof. lem:gauss$_l$*inear$_i$mage, lem : gauss$_1$d$_m$oments*$\langle a, g'\rangle \sim \mathcal{N}(0, \|a\|^2)$ by Lemma **??**, and Lemma **??** (Mathlib `mgf_gaussianReal` at the point s, with `integrable_exp_mul_gaussianReal`) gives the value. Alternatively, by Lemma **??**, $\langle a, g'\rangle = \sum_i a_i g_i'$ with independent coordinates, and $\mathbb{E}\prod_i e^{s a_i g_i'} = \prod_i e^{s^2 a_i^2/2}$ (`iIndepFun.mgf_sum`). ☐

Lemma 217 (Exponential moments of Gaussian norms)**.** *ProbabilityTheory.IsGaussian.integrable$_e$xp$_m$ul$_n$orm,* $\sim \mathcal{N}(\mu, S)$ *in* $\mathbb{R}^n$ *and* $c \ge 0$*. Then* $e^{c\|G\|}$ *is integrable. Consequently, if* $F : \mathbb{R}^n \to \mathbb{R}$ *is measurable with* $|F(x)| \le A + B\|x\|$ *for constants* $A, B \ge 0$*, then* $F(G)$ *is integrable and* $e^{\lambda F(G)}$ *is integrable for every* $\lambda \in \mathbb{R}$*.*

Proof. lem:gauss$_1$*d$_m$oments, lem : gauss$_p$iByFernique′stheorem(Mathlib*`IsGaussian.exists_integrable_`> 0*with*$e^{C\|x\|^2}$ integrable under the law of G; since $c\|x\| \le C\|x\|^2 + c^2/(4C)$ (AM–GM), $e^{c\|G\|} \le e^{c^2/(4C)} e^{C\|G\|^2}$ is integrable. (Elementary alternative for $G = \mu + S^{1/2}g$: $\|G\| \le \|\mu\| + \|S^{1/2}\|_{\mathrm{op}} \sum_i |g_i|$, $e^{c'|g_i|} \le e^{c' g_i} + e^{-c' g_i}$, and by independence of the coordinates (Lemma **??**) and Lemma **??** the expectation of the product is a finite product of finite numbers.) For F: $|F(G)| \le A + B\|G\|$ is integrable, and $e^{\lambda F(G)} \le e^{|\lambda| A} e^{|\lambda| B\|G\|}$ is integrable by the first part with $c = |\lambda| B$. ☐

Lemma 218 (Bounded gradient implies Lipschitz and linear growth)**.** *lipschitzWith$_o$f$_n$orm$_g$radient$_l$e, abs$_l$e$_o$f$_n$* $: R^d \to \mathbb{R}$ *be differentiable with* $\|\nabla f(x)\| \le L$ *for every* x*, where* $L \ge 0$*. Then* $|f(x) - f(y)| \le L\|x - y\|$ *for all* x, y*,* $|f(x)| \le |f(0)| + L\|x\|$*, and* $|\langle \nabla f(x), v\rangle| \le L\|v\|$ *for all* x, v*. In particular, for* $g \sim \mathcal{N}(0, I_d)$*,* $f(g)$ *is integrable and* $e^{\lambda f(g)}$ *is integrable for every* $\lambda \in \mathbb{R}$*.*

Proof. lem:pc$_{e}xp_{n}orm_{i}ntegrableTheLipschitzboundisthemeanvalueinequality(Mathlib$`lipschitzWith_of_n`
in an inner product space); $y = 0$ gives the growth bound; the last inequality is Cauchy–Schwarz. Integrability is Lemma **??** with $A = |f(0)|$, $B = L$. ☐

Lemma 219 (The log-sum-exp smoothing of a finite maximum)**.** *innerPi, softmax, logSumExp, logSumExp$_{e}q_{c}omp, sum_{e}xp_{i}nner_{p}os, softmax_{n}onneg, sum_{s}oftmax, contDiff_{l}ogSumExp, h$*
≥ 1, $x_1, \dots, x_m \in \mathbb{R}^d$, $\beta > 0$ and

$$f_\beta(v) := \frac{1}{\beta} \log \sum_{i \leq m} e^{\beta \langle x_i, v \rangle}, \qquad v \in \mathbb{R}^d.$$

(i) f_β is C^∞ on $\mathbb{R}^d$, with gradient $\nabla f_\beta(v) = \sum_{i \leq m} p_i(v)\, x_i$ where $p_i(v) := e^{\beta\langle x_i, v\rangle} / \sum_{j \leq m} e^{\beta \langle x_j, v \rangle}$ satisfies $p_i(v) \geq 0$, $\sum_i p_i(v) = 1$; hence $\|\nabla f_\beta(v)\| \leq \max_{i \leq m} \|x_i\|$ for all v.

(ii) For every v, $\max_{i \leq m} \langle x_i, v \rangle \leq f_\beta(v) \leq \max_{i \leq m} \langle x_i, v \rangle + \frac{\log m}{\beta}$.

Proof. (i) $v \mapsto \sum_i e^{\beta \langle x_i, v \rangle}$ is a finite sum of compositions of exp with linear forms, hence C^∞ and strictly positive; log is C^∞ on $(0, \infty)$ (Mathlib `ContDiff.log`, `contDiff_exp`, `ContDiff.inner`). The chain rule gives $\nabla f_\beta(v) = \frac{1}{\beta} \cdot \frac{\sum_i \beta e^{\beta \langle x_i, v\rangle} x_i}{\sum_j e^{\beta \langle x_j, v \rangle}} = \sum_i p_i(v) x_i$. The $p_i(v)$ are positive with sum 1, so by the triangle inequality $\|\nabla f_\beta(v)\| \leq \sum_i p_i(v) \|x_i\| \leq \max_i \|x_i\|$. (ii) Let $M(v) := \max_i \langle x_i, v \rangle$. Since exp and log are increasing, $e^{\beta M(v)} \leq \sum_i e^{\beta \langle x_i, v \rangle} \leq m\, e^{\beta M(v)}$; take logarithms and divide by β. ☐

11.2 The Maurey–Pisier argument

Throughout this section $f : \mathbb{R}^d \to \mathbb{R}$ is of class C^1 with $\|\nabla f(x)\| \leq L$ for all x, and g, g' are independent $\mathcal{N}(0, I_d)$ vectors on $(\Omega, \mathbb{P})$. For $\theta \in \mathbb{R}$ we write, as in Lemma **??**,

$$g_\theta := g \sin\theta + g' \cos\theta, \qquad g'_\theta := g \cos\theta - g' \sin\theta,$$

so that $g_0 = g'$, $g_{\pi/2} = g$ and $\frac{d}{d\theta} g_\theta = g'_\theta$.

Lemma 220 (Interpolation along a quarter circle)**.** *$hasDerivAt_{c}omp_{q}uarterCircle, sub_{e}q_{i}ntegral_{q}uarterCircle$*
$: R^d \to \mathbb{R}$ be of class C^1 and $u, v \in \mathbb{R}^d$. Set $\gamma(\theta) := u \sin\theta + v \cos\theta$ and $\gamma'(\theta) := u \cos\theta - v \sin\theta$. Then $\theta \mapsto f(\gamma(\theta))$ is differentiable on $\mathbb{R}$ with derivative $\theta \mapsto \langle \nabla f(\gamma(\theta)), \gamma'(\theta) \rangle$, this derivative is continuous, and

$$f(u) - f(v) = \int_0^{\pi/2} \langle \nabla f(\gamma(\theta)), \gamma'(\theta) \rangle \, d\theta.$$

Proof. γ is differentiable with $\gamma'(\theta) = u \cos\theta - v \sin\theta$ (`HasDerivAt.smul_const` for sin and cos, `HasDerivAt.add`). The chain rule (`HasFDerivAt.comp_hasDerivAt`) gives that $f \circ \gamma$ has derivative $f'(\gamma(\theta))(\gamma'(\theta)) = \langle \nabla f(\gamma(\theta)), \gamma'(\theta) \rangle$ at θ. This derivative is continuous in θ as a composition of continuous maps (∇f is continuous since f is C^1), hence interval integrable on $[0, \pi/2]$ (`ContinuousOn.intervalIntegrable`). The fundamental theorem of calculus (`intervalIntegral.integral_eq_sub_of_hasDerivAt`) gives $\int_0^{\pi/2}(f \circ \gamma)' = f(\gamma(\pi/2)) - f(\gamma(0)) = f(u) - f(v)$. $\square$

Lemma 221 (Jensen's inequality for the uniform measure on $[0, \pi/2]$)**.** $exp_{m}ul_{i}ntegral_{l}e_{i}ntegral_{e}xpLeth$ *: [0,π/2]* $\to \mathbb{R}$ *be continuous and* $s \in \mathbb{R}$*. Then*

$$\exp\Big(s\int_0^{\pi/2} h(\theta)\, d\theta\Big) \le \frac{2}{\pi}\int_0^{\pi/2} \exp\Big(\frac{\pi s}{2}\, h(\theta)\Big) d\theta.$$

Proof. Let $\nu := \frac{2}{\pi}\,\mathrm{Leb}|_{[0,\pi/2]}$, a probability measure, and $k(\theta) := \frac{\pi s}{2} h(\theta)$, which is continuous, hence bounded on the compact interval; so k and e^k are ν-integrable. Jensen's inequality for the convex continuous function exp on the closed convex set $\mathbb{R}$ (Mathlib `ConvexOn.map_integral_le` with `convexOn_exp`, `continuous_exp`, `isClosed_univ`) gives $\exp(\int k\, d\nu) \le \int e^k\, d\nu$. Finally $\int k\, d\nu = \frac{2}{\pi} \cdot \frac{\pi s}{2} \int_0^{\pi/2} h = s\int_0^{\pi/2} h$ and $\int e^k\, d\nu = \frac{2}{\pi}\int_0^{\pi/2} e^{k(\theta)} d\theta$. $\square$

Lemma 222 (MGF of the rotated gradient term)**.** $ProbabilityTheory.integral_{c}omp_{r}otation_{s}wap_{p}rod, Probabilit$ *with* $\|\nabla f\| \le L$ *everywhere,* g, g' *independent* $\mathcal{N}(0, I_d)$ *vectors,* $s \in \mathbb{R}$ *and* $\theta \in \mathbb{R}$*. Then* $\omega \mapsto \exp(s\langle \nabla f(g_\theta), g'_\theta\rangle)$ *is integrable and*

$$\mathbb{E}\exp\big(s\langle \nabla f(g_\theta), g'_\theta\rangle\big) = \mathbb{E}\exp\big(s\langle \nabla f(g), g'\rangle\big) = \mathbb{E}\exp\Big(\frac{s^2\|\nabla f(g)\|^2}{2}\Big) \le \exp\Big(\frac{s^2L^2}{2}\Big).$$

Proof. lem:gauss$_{r}otation, lem : gauss_{m}gf_{i}nner, lem : pc_{l}ipschitz_{o}f_{g}rad, lem : pc_{e}xp_{n}orm_{i}ntegrableLet\Phi(u, v) := \exp(s\langle \nabla f(u), v\rangle)$, a continuous function on $\mathbb{R}^d \times \mathbb{R}^d$ with $0 < \Phi(u, v) \le \exp(|s|L\|v\|)$ (Lemma **??**). By Lemma **??**(ii), (g_θ, g'_θ) has the same law as (g, g'), so $\mathbb{E}\Phi(g_\theta, g'_\theta) = \mathbb{E}\Phi(g, g')$ (equality of laws; both sides are finite by the domination and Lemma **??**). The law of (g, g') is the product $\mathcal{N}(0, I_d) \otimes \mathcal{N}(0, I_d)$ (independence), and Φ is integrable for it (dominated by $\exp(|s|L\|v\|)$, which is integrable in v and constant in u), so Fubini (`MeasureTheory.integral_prod`) gives

$$\mathbb{E}\Phi(g, g') = \int\Big(\int \exp\big(\langle s\nabla f(u), v\rangle\big)\,\mathcal{N}(0, I_d)(dv)\Big)\mathcal{N}(0, I_d)(du) = \int \exp\Big(\frac{s^2\|\nabla f(u)\|^2}{2}\Big)\mathcal{N}(0, I_d)(du),$$

by Lemma **??** with $a = s\nabla f(u)$. The last integrand is at most $\exp(s^2L^2/2)$ pointwise, and $\mathcal{N}(0, I_d)$ is a probability measure. $\square$

Lemma 223 (Maurey–Pisier Gaussian concentration)**.** $ProbabilityTheory.hasSubgaussianMGF_{s}ub_{i}ntegral_{s}tdG$ ≥ 0 *and let* $f : \mathbb{R}^d \to \mathbb{R}$ *be of class* C^1 *with* $\|\nabla f(x)\| \le L$ *for every* $x \in \mathbb{R}^d$*. Let* $g \sim \mathcal{N}(0, I_d)$*. Then* $f(g)$ *is integrable, and for every* $\lambda \in \mathbb{R}$ *the variable* $e^{\lambda(f(g) - \mathbb{E}f(g))}$ *is integrable with*

$$\mathbb{E}\exp\big(\lambda(f(g) - \mathbb{E}f(g))\big) \le \exp\Big(\frac{\pi^2\lambda^2L^2}{8}\Big) = \exp\Big(\frac{c_{\mathrm{G}}L^2\,\lambda^2}{2}\Big).$$

In Mathlib's terms, $f(g) - \mathbb{E}f(g)$ *satisfies* `HasSubgaussianMGF` *with constant* $c_{\mathrm{G}}L^2$*.*

Proof. lem:$\mathrm{pc}_l ipschitz_o f_g rad, lem : pc_e xp_n orm_i ntegrable, lem : pc_i nterpolation_f tc, lem :$
$pc_j ensen_e xp_a verage, lem : pc_r otated_m g f_b ound, lem : gauss_r otation$ *Step 0 (integrability).ByLemma* **??**,$f$is L$-L$
$|f(0)| + L\|x\|$, $f(g)$ is integrable, and $e^{\lambda f(g)}$ is integrable for every λ; hence so is $e^{\lambda(f(g)-\mathbb{E}f(g))}$. Fix $\lambda \in \mathbb{R}$ (for $\lambda = 0$ there is nothing to prove). Enlarging the probability space if necessary (all quantities depend only on the law of g), let g' be an independent copy of g; the law of (g, g') is $\gamma \otimes \gamma$ with $\gamma := \mathcal{N}(0, I_d)$.

Step 1 (symmetrization by Jensen in g'). For fixed $u \in \mathbb{R}^d$, the variable $\lambda(f(u) - f(g'))$ is integrable, and $e^{\lambda(f(u)-f(g'))} \le e^{|\lambda|(|f(u)|+|f(0)|+L\|g'\|)}$ is integrable (Lemma **??**); Jensen's inequality for exp (`ConvexOn.map_integral_le`, `convexOn_exp`) gives

$$\exp\big(\lambda(f(u) - \mathbb{E}f(g'))\big) \le \mathbb{E}\exp\big(\lambda(f(u) - f(g'))\big).$$

Since $\mathbb{E}f(g') = \mathbb{E}f(g)$, evaluating at $u = g$ and integrating in g (Fubini for $\gamma \otimes \gamma$, `MeasureTheory.integral_prod`; the function $(u, v) \mapsto e^{\lambda(f(u)-f(v))}$ is dominated by $e^{2|\lambda||f(0)|}e^{|\lambda|L\|u\|}e^{|\lambda|L\|v\|}$, integrable on the product by Lemma **??**),

$$\mathbb{E}\exp\big(\lambda(f(g) - \mathbb{E}f(g))\big) \le \mathbb{E}\exp\big(\lambda(f(g) - f(g'))\big).$$

Step 2 (interpolation). Pointwise in ω, Lemma **??** with $u = g(\omega)$, $v = g'(\omega)$ gives

$$f(g) - f(g') = \int_0^{\pi/2} h_\omega(\theta)\, d\theta, \qquad h_\omega(\theta) := \langle \nabla f(g_\theta), g'_\theta \rangle,$$

where $\theta \mapsto h_\omega(\theta)$ is continuous.

Step 3 (Jensen for the uniform measure on $[0, \pi/2]$). By Lemma **??** with $s = \lambda$, pointwise in ω,

$$\exp\big(\lambda(f(g) - f(g'))\big) \le \frac{2}{\pi}\int_0^{\pi/2} \exp\Big(\frac{\pi\lambda}{2}\, h_\omega(\theta)\Big) d\theta.$$

Step 4 (Fubini and rotation invariance). The function $(\omega, \theta) \mapsto \exp(\frac{\pi\lambda}{2} h_\omega(\theta))$ is measurable (a continuous function of $(g(\omega), g'(\omega), \theta)$) and satisfies, by Lemma **??** and $\|g'_\theta\| \le \|g\| + \|g'\|$,

$$0 < \exp\Big(\frac{\pi\lambda}{2} h_\omega(\theta)\Big) \le \exp\Big(\frac{\pi|\lambda|L}{2}\big(\|g(\omega)\| + \|g'(\omega)\|\big)\Big) =: D(\omega),$$

where D does not depend on θ and is integrable on Ω (product of two independent integrable variables, Lemma **??**); hence the function is integrable on $\Omega \times [0, \pi/2]$ and Fubini (`integral_integral_swap`) applies:

$$\mathbb{E}\exp\big(\lambda(f(g) - f(g'))\big) \le \frac{2}{\pi}\int_0^{\pi/2} \mathbb{E}\exp\Big(\frac{\pi\lambda}{2}\langle \nabla f(g_\theta), g'_\theta \rangle\Big) d\theta.$$

For each fixed θ, Lemma **??** with $s = \pi\lambda/2$ gives

$$\mathbb{E}\exp\Big(\frac{\pi\lambda}{2}\langle \nabla f(g_\theta), g'_\theta \rangle\Big) \le \exp\Big(\frac{(\pi\lambda/2)^2 L^2}{2}\Big) = \exp\Big(\frac{\pi^2\lambda^2 L^2}{8}\Big).$$

Step 5 (conclusion). Integrating the constant bound over $\theta \in [0, \pi/2]$ and multiplying by $2/\pi$,

$$\mathbb{E} \exp\big(\lambda(f(g) - \mathbb{E}f(g))\big) \le \frac{2}{\pi} \cdot \frac{\pi}{2} \cdot \exp\Big(\frac{\pi^2\lambda^2 L^2}{8}\Big) = \exp\Big(\frac{\pi^2\lambda^2 L^2}{8}\Big).$$

Finally $\frac{\pi^2 L^2}{8}\lambda^2 = \frac{(\pi^2/4)L^2\,\lambda^2}{2} = \frac{c_{\mathrm{G}} L^2 \lambda^2}{2}$, which together with Step 0 is the statement `HasSubgaussianMGF` with constant $c_{\mathrm{G}} L^2$. ☐

Remark 224 (On the constant). The factor $\pi^2/8$ in the exponent comes from Step 3: Jensen's inequality replaces $\lambda \int_0^{\pi/2} h$ by an average of $\exp(\frac{\pi}{2}\lambda h(\theta))$, and the length $\pi/2$ of the interval is squared in the Gaussian MGF of Step 4. With the standard Gaussian interpolation $g_t = \sqrt{t}\, g + \sqrt{1-t}\, g'$ one would need the derivative $\frac{d}{dt} g_t$, which is singular at the endpoints; the quarter-circle parametrization avoids this at the cost of the constant. Sharper routes (Gaussian log-Sobolev + Herbst, or the Gaussian isoperimetric inequality) give $c_{\mathrm{G}} = 1$ [**?**, **?**], and the Borell–TIS inequality with $c_{\mathrm{G}} = 1$ is the form usually quoted; none of the results of Part **??** is sensitive to the value of c_{G} beyond the absolute constants.

11.3 Suprema of linear Gaussian processes

Lemma 225 (Concentration of a finite maximum of Gaussian linear forms)**.** $abs_c iSup_i nner_l e, ProbabilityTheory.measurable_c iSup_i nner, ProbabilityTheory.hasSubgaussianMGF_i Sup_i n$ ≥ 1, *$x_1, \dots, x_m \in \mathbb{R}^d$, $S \in \mathcal{S}_+^d$ and $G \sim \mathcal{N}(0, S)$. Let $M := \max_{i \le m}\langle x_i, G\rangle$ and $\sigma^2 := \max_{i \le m} x_i^\top S x_i$. Then M is integrable and $M - \mathbb{E}M$ has a sub-Gaussian MGF with constant $c_{\mathrm{G}}\sigma^2$: for every $\lambda \in \mathbb{R}$, $e^{\lambda(M - \mathbb{E}M)}$ is integrable and $\mathbb{E}e^{\lambda(M-\mathbb{E}M)} \le \exp(c_{\mathrm{G}}\sigma^2\lambda^2/2)$.*

Proof. lem:gauss$_l aw_o f_c ov, lem : maurey_p isier, lem : logsumexp_p rops, lem :$ $pc_e xp_n orm_i ntegrable The statement only involves the law of \mathrm{M}, a measurable function of \mathrm{G}, so by Lemma$ **??**$(iii) we$ $= \mathrm{S}^{1/2} g$ with $g \sim \mathcal{N}(0, I_d)$ (Mathlib `HasSubgaussianMGF.congr_identDistrib` / `of_map` transport the conclusion). Since $S^{1/2}$ is symmetric, $\langle x_i, S^{1/2} g\rangle = \langle y_i, g\rangle$ with $y_i := S^{1/2} x_i$, and $\|y_i\|^2 = x_i^\top S x_i \le \sigma^2$. Thus $M = F(g)$ with $F(v) := \max_{i \le m}\langle y_i, v\rangle$, and $|F(v)| \le \sigma\|v\|$, so M and $e^{\lambda M}$ are integrable (Lemma **??**).

Fix $\lambda \in \mathbb{R}$ and $\beta > 0$, and let f_β be the log-sum-exp function of Lemma **??** for the vectors $y_1, \dots, y_m$. It is C^∞ with $\|\nabla f_\beta\| \le \max_i \|y_i\| \le \sigma$, so Lemma **??** with $L = \sigma$ gives $\mathbb{E}e^{\lambda(f_\beta(g) - \mathbb{E}f_\beta(g))} \le \exp(c_{\mathrm{G}}\sigma^2\lambda^2/2)$. By Lemma **??**(ii), $0 \le f_\beta(g) - M \le (\log m)/\beta$ pointwise, hence also $0 \le \mathbb{E}f_\beta(g) - \mathbb{E}M \le (\log m)/\beta$, and therefore

$$|(M - \mathbb{E}M) - (f_\beta(g) - \mathbb{E}f_\beta(g))| \le \frac{2\log m}{\beta} \quad \text{pointwise, so} \quad \mathbb{E}e^{\lambda(M-\mathbb{E}M)} \le e^{2|\lambda|(\log m)/\beta}\, \mathbb{E}e^{\lambda(f_\beta(g) - \mathbb{E}f_\beta(g))} \le e^{2|\lambda|(\log$$

The left-hand side does not depend on β; letting $\beta \to \infty$ (or taking the infimum over $\beta > 0$: a real number bounded by ae^ε for every $\varepsilon > 0$ is bounded by a)

gives $\mathbb{E}e^{\lambda(M-\mathbb{E}M)} \le \exp(c_{\mathrm{G}}\sigma^2\lambda^2/2)$. No limit theorem for integrals is needed here. ☐

Lemma 226 (Concentration of the supremum over a compact index set)**.** *$ProbabilityTheory.hasSubgaussianMGF_supportFn_stdGaussiandef : pg_gaussian_vector, def : pc_gaussian_consta$* $\mathbb{R}^d$ *be nonempty and compact,* $S \in \mathcal{S}_+^d$ *and* $G \sim \mathcal{N}(0, S)$*. Let* $M := \sup_{x\in\mathcal{X}}\langle x, G\rangle$ *(a random variable by Lemma* **??***) and* $\sigma^2 := \sup_{x\in\mathcal{X}} x^\top Sx$ *(finite, attained). Then* M *is integrable and* $M - \mathbb{E}M$ *has a sub-Gaussian MGF with constant* $c_{\mathrm{G}}\sigma^2$*: for every* $\lambda \in \mathbb{R}$*,* $e^{\lambda(M-\mathbb{E}M)}$ *is integrable and* $\mathbb{E}e^{\lambda(M-\mathbb{E}M)} \le \exp(c_{\mathrm{G}}\sigma^2\lambda^2/2)$*.*

Proof. lem:sup$_c ountable_d ense, lem : finite_s up_s ubgaussian, lem : pc_e xp_n orm_i ntegrableLet$R $:= \max_{x\in\mathcal{X}}\|x\|$ and let $(q_n)_{n\in\mathbb{N}}$ enumerate a countable dense subset of $\mathcal{X}$ (Lemma **??**(i)). σ^2 is the maximum of the continuous function $x \mapsto x^\top Sx = \|S^{1/2}x\|^2$ on the compact $\mathcal{X}$. Define the partial maxima $M_n := \max_{i\le n}\langle q_i, G\rangle$, $n \in \mathbb{N}$. By Lemma **??**(ii) applied pointwise to $v = G(\omega)$, $M_n \uparrow M$ everywhere, and $|M_n| \le R\|G\|$, $|M| \le R\|G\|$.

Bound for each n. Lemma **??** applied to $q_0, \dots, q_n$ gives, with $\sigma_n^2 := \max_{i\le n} q_i^\top Sq_i \le \sigma^2$, that $\mathbb{E}e^{\lambda(M_n-\mathbb{E}M_n)} \le \exp(c_{\mathrm{G}}\sigma_n^2\lambda^2/2) \le \exp(c_{\mathrm{G}}\sigma^2\lambda^2/2)$ for every λ.

Passage to the limit. Fix $\lambda \in \mathbb{R}$. The dominating function $e^{|\lambda|R\|G\|}$ is integrable (Lemma **??**) and $e^{\lambda M_n} \le e^{|\lambda|R\|G\|}$ for all n, while $e^{\lambda M_n} \to e^{\lambda M}$ pointwise (continuity of exp). Dominated convergence (`tendsto_integral_of_dominated_convergence`) gives $\mathbb{E}e^{\lambda M_n} \to \mathbb{E}e^{\lambda M}$, and $e^{\lambda M}$ is integrable. Likewise $|M_n| \le R\|G\|$, $M_n \to M$, and $R\|G\|$ is integrable, so $\mathbb{E}M_n \to \mathbb{E}M$ (dominated convergence again, or monotone convergence `integral_tendsto_of_tendsto_of_monotone` since M_n is nondecreasing); in particular M is integrable. Therefore

$$\mathbb{E}e^{\lambda(M-\mathbb{E}M)} = e^{-\lambda\mathbb{E}M}\,\mathbb{E}e^{\lambda M} = \lim_{n\to\infty} e^{-\lambda\mathbb{E}M_n}\,\mathbb{E}e^{\lambda M_n} = \lim_{n\to\infty}\mathbb{E}e^{\lambda(M_n-\mathbb{E}M_n)} \le \exp\left(\frac{c_{\mathrm{G}}\sigma^2\lambda^2}{2}\right),$$

as a limit of numbers bounded by the right-hand side (`le_of_tendsto`). Integrability of $e^{\lambda(M-\mathbb{E}M)} = e^{-\lambda\mathbb{E}M}e^{\lambda M}$ was shown above. ☐

Lean remark. The formalization states Lemmas **??**, **??** and Theorem **??** for the standard Gaussian measure $\mathcal{N}(0, I_d)$ (Mathlib `stdGaussian E`, on any finite-dimensional inner product space E) and a bound σ on the norms of the index vectors, which is the case $S = I_d$; the case of a general covariance $S \in \mathcal{S}_+^d$ is the image of this one under $S^{1/2}$ (Lemma **??**(iii) and `supportFn_image`), see `ProbabilityTheory.hasSubgaussianMGF_supportFn_multivariateGaussian`. The supremum over $\mathcal{X}$ is `supportFn K` and the partial maxima are `partialSupInner q n ( i : Fin (n+1), q i, v )` for the dense sequence of Lemma **??**; the two limits are `Filter.Tendsto` statements along `atTop` proved by dominated convergence. Lemma **??** is stated for a family `y : → E` indexed by a nonempty finite type, and the limit $\beta \to \infty$ is `ge_of_tendsto`.

Theorem 227 (Borell–TIS inequality for linear Gaussian processes)**.** *$ProbabilityTheory.measureReal_supportFn$* $\mathbb{R}^d$ *be nonempty and compact,* $S \in \mathcal{S}_+^d$*,* $G \sim \mathcal{N}(0, S)$*,* $M := \sup_{x\in\mathcal{X}}\langle x, G\rangle$ *and*

$\sigma^2 := \sup_{x\in\mathcal{X}} x^\top Sx$, assumed positive. Then M is integrable and for every $u \geq 0$,

$$\mathbb{P}\big(M - \mathbb{E}M \geq u\big) \leq \exp\Big(-\frac{u^2}{2c_{\mathrm{G}}\sigma^2}\Big), \qquad \mathbb{P}\big(M - \mathbb{E}M \leq -u\big) \leq \exp\Big(-\frac{u^2}{2c_{\mathrm{G}}\sigma^2}\Big).$$

(If $\sigma = 0$ then $\langle x, G\rangle = 0$ a.s. for every $x \in \mathcal{X}$, $M = 0$ a.s., and both probabilities vanish for $u > 0$.)

Proof. lem:compact$_s up_s ubgaussianByLemma$ **??**,M - EM$satisfies$`HasSubgaussianMGF`$withconstant$c := c$_{\mathrm{G}}\sigma^2$. The upper tail is the Chernoff bound `HasSubgaussianMGF.measure_ge_le`: $\mathbb{P}(X \geq u) \leq \exp(-u^2/(2c))$ for $u \geq 0$; the lower tail is `HasSubgaussianMGF.measure_le_le` (LML; it is `measure_ge_le` applied to $-X$, which is sub-Gaussian with the same constant by `HasSubgaussianMGF.neg`). The degenerate case: $\mathrm{Var}\langle x, G\rangle = x^\top Sx = 0$ forces $\langle x, G\rangle = 0$ a.s. for each x, hence for all x in the countable dense set simultaneously, hence $M = 0$ a.s. by Lemma **??**(ii). ☐

Remark 228 (The classical statement). The Borell–TIS inequality [**?**, Theorem 2.1.1] holds for every almost surely bounded centered Gaussian process with c_{G} replaced by 1, and its proof goes through the Gaussian isoperimetric inequality. Only the linear processes $x \mapsto \langle x, G\rangle$ over compact $\mathcal{X} \subset \mathbb{R}^d$ are needed in this blueprint, and only up to the absolute constant c_{G}; the general statement is not pursued.

Corollary 229 (High-probability upper bound on the supremum)**.** *thm:borell$_t isInthesettingofTheorem$* **??**, *$fc$
$(0, 1)$,*

$$\mathbb{P}\Big(M \leq \mathbb{E}M + \sigma\sqrt{2c_{\mathrm{G}}\log(1/\delta)}\Big) \geq 1-\delta, \qquad \mathbb{P}\Big(M \geq \mathbb{E}M - \sigma\sqrt{2c_{\mathrm{G}}\log(1/\delta)}\Big) \geq 1-\delta.$$

Proof. thm:borell$_t isLet$u := $\sigma\sqrt{2c_{\mathrm{G}}\log(1/\delta)} \geq 0$; then $u^2/(2c_{\mathrm{G}}\sigma^2) = \log(1/\delta)$ and Theorem **??** gives $\mathbb{P}(M - \mathbb{E}M \geq u) \leq e^{-\log(1/\delta)} = \delta$. The event $\{M \leq \mathbb{E}M + u\}$ contains the complement of $\{M - \mathbb{E}M \geq u\}$ (in fact equals the complement of $\{M - \mathbb{E}M > u\}$), so its probability is at least $1 - \delta$. The second bound is the lower tail. If $\sigma = 0$ both events have probability one. ☐

Corollary 230 (Symmetric index set)**.** *$Maiti2026Power.integral_d iffSup_m ultivariateGaussian, Maiti2026Po$
$\mathbb{R}^d$ be nonempty and compact, $S \in \mathcal{S}^d_+$, $G \sim \mathcal{N}(0, S)$, $\sigma^2 := \sup_{x\in\mathcal{X}} x^\top Sx$, and consider the difference set $\mathcal{X} - \mathcal{X} := \{x - x' : x, x' \in \mathcal{X}\}$, which is nonempty and compact. Let $D := \sup_{x,x'\in\mathcal{X}}\langle x - x', G\rangle = \sup_{z\in\mathcal{X}-\mathcal{X}}\langle z, G\rangle$. Then*

(i) $\mathbb{E}D = 2\,\mathbb{E}\sup_{x\in\mathcal{X}}\langle x, G\rangle$;

(ii) $\sigma^2_{\mathcal{X}-\mathcal{X}} := \sup_{z\in\mathcal{X}-\mathcal{X}} z^\top Sz \leq 4\sigma^2$;

(iii) for every $\delta \in (0, 1)$, $\mathbb{P}\big(D \leq \mathbb{E}D + 2\sigma\sqrt{2c_{\mathrm{G}}\log(1/\delta)}\big) \geq 1 - \delta$.

Proof. cor:sup$_b ound_w hp, lem : gauss_l aw_o f_c ov, lem : sup_c ountable_d ense$X- X$istheimageofthecompact$X$\times\mathcal{X}$ under the continuous map $(x, x') \mapsto x - x'$, hence compact. (i) Pointwise, $\sup_{x,x'}(\langle x, G\rangle - \langle x', G\rangle) = \sup_x\langle x, G\rangle + \sup_{x'}\langle x', -G\rangle$ (the two suprema are over independent variables x, x'), and $-G \sim \mathcal{N}(0, S)$ (Lemma **??**(iv)), so $\mathbb{E}\sup_{x'}\langle x', -G\rangle = \mathbb{E}\sup_x\langle x, G\rangle$; both terms are integrable by Lemma **??**. (ii) For $z = x - x'$, $z^\top Sz = \|S^{1/2}x - S^{1/2}x'\|^2 \le (\|S^{1/2}x\| + \|S^{1/2}x'\|)^2 \le (2\sigma)^2$. (iii) Apply Corollary **??** to the index set $\mathcal{X} - \mathcal{X}$: with probability at least $1-\delta$, $D \le \mathbb{E}D + \sigma_{\mathcal{X}-\mathcal{X}}\sqrt{2c_{\mathrm{G}}\log(1/\delta)} \le \mathbb{E}D + 2\sigma\sqrt{2c_{\mathrm{G}}\log(1/\delta)}$ by (ii). (If $\sigma_{\mathcal{X}-\mathcal{X}} = 0$ the bound holds with probability one.) $\square$

Lean remark. In the applications (Lemmas **??** and **??**) the index set is $\mathcal{X} - \mathcal{X}$ for the action set or for one region of a partition, $S = \Sigma_T^{-1}$ for a fixed design, and $\sigma^2 = \max_x\|x\|^2_{\Sigma_T^{-1}}$; the compact set $\mathcal{X} - \mathcal{X}$ is `Set.image2 (· - ·) Xs Xs`, which is the image of the product set $\mathcal{X} \times \mathcal{X}$ (`Set.prod`) under `fun p   p.1 - p.2` (`Set.image_prod`); with `hK : IsCompact Xs` it is compact by `(hK.prod hK).image continuous_sub` (Mathlib `IsCompact.image`), or equivalently `(hK.prod hK).image_of_continuousOn continuous_sub.continuousOn`.

Chapter 12

Sub-exponential random variables

This chapter develops the Bernstein-type concentration used by the ℓ_2-norm estimation of Chapter **??** and by the unit-ball identification algorithm (Lemma **??**): a notion of sub-exponential random variable with two explicit parameters (V, b) (Appendix D.1 of the paper), its stability under scaling and independent sums, the corresponding tail bound with the explicit constant $c = 1/2$, and the sub-exponential parameters of the centered squares of Gaussian variables (the paper's Lemma 1 in Appendix A, with its complete-the-square proof), of chi-square variables and of Rademacher linear forms $\langle \varepsilon, \theta \rangle$. For the last one the paper invokes the Hanson–Wright inequality with unspecified constants; we replace it by a direct argument valid for every sub-Gaussian variable Z with $\mathbb{E}Z^2$ equal to its sub-Gaussian constant (a Gaussian trick for positive λ, a fourth-moment bound for negative λ, the fourth moment being itself controlled by the Gaussian trick), which gives explicit parameters $(16\|\theta\|^4, 4\|\theta\|^2)$. All constants are explicit and tracked. References: [**?**, Section 2.7], [**?**, Section 2.4].

What Mathlib provides. The sub-Gaussian structure `ProbabilityTheory.HasSubgaussianMGF X c` (`Mathlib/Probability/Moments/SubGaussian.lean`) with `measure_ge_le`, `neg`, `add_of_indepFun`, `sum_of_iIndepFun`, `of_map`, `congr_identDistrib`, Hoeffding's lemma `hasSubgaussianMGF_of_mem_Icc_of_integral_eq_zero`, and LML's `measure_le_le`; its versions with respect to a kernel and a measure (`Kernel.HasSubgaussianMGF`) and conditionally on a sub-σ-algebra (`HasCondSubgaussianMGF`, through `condExpKernel`), with the Azuma–Hoeffding inequality `measure_sum_ge_le_of_hasCondSubgaussian` Moment generating functions: `ProbabilityTheory.mgf`, `iIndepFun.mgf_sum`, `iIndepFun.integrable_exp_mul_sum`, the Chernoff bound `measure_ge_le_exp_mul_mgf` (`Mathlib/Probability/Moments/Basic.lean`), the derivative of the MGF `hasDerivAt_mgf` (`MGFAnalytic.lean`), integrability of all moments from the integrability of $e^{\lambda X}$ near $\lambda = 0$ (`integrable_pow_of_mem_interior_integrableExpSet`, `IntegrableExpMul.lean`), `mgf_gaussianReal`. The Gaussian integral `integral_gaussian` ($\int e^{-bx^2} dx = \sqrt{\pi/b}$, `Mathlib/Analysis/SpecialFunctions/Gaussian/GaussianIntegral.lean`);

the real logarithm series bound `Real.abs_log_sub_add_sum_range_le` (`Mathlib/Analysis/SpecialFunction`
`Real.add_one_le_exp` and `Real.quadratic_le_exp_of_nonneg`; independence of the coordinates of a product measure `ProbabilityTheory.iIndepFun_pi`; the standard Gaussian vector `ProbabilityTheory.stdGaussian` as the image of a product of `gaussianReal 0 1` (`map_pi_eq_stdGaussian`).

What is missing. The sub-exponential structure itself and all its properties (Section **??**), its kernel and conditional versions with the Azuma–Bernstein inequality (Section **??**), the MGF of the square of a real Gaussian (Lemma **??**), and the sub-exponential bounds of Sections **??** and **??**. The sub-exponential structure and its API now live in LML (`LeanMachineLearning/ForMathlib/Probability/Moments/SubExpone`
only the Gaussian and Rademacher applications of Sections **??** and **??** are formalized here.

Lean remark. The definition below mirrors `HasSubgaussianMGF`: a structure `HasSubexponentialMGF (X : Ω → ) (V b : ) ( : Measure Ω) : Prop` with fields `integrable_exp_mul : t : , b * |t| 1 → Integrable (fun exp (t * X ))` and `mgf_le : t : , b * |t| 1 → mgf X t exp (V * t^2 / 2)` (`LeanMachineLearning/ForMathlib/Probability/Moments/SubExponential.lean`). The range of t is written as $b|t| \leq 1$ rather than $|t| \leq 1/b$ so that `b = 0` imposes no restriction on t (the hypothesis $0{\cdot}|t| \leq 1$ is always true): `HasSubexponentialMGF X V 0` is then equivalent to `HasSubgaussianMGF X V` (same two fields, quantified over all `t`), and Mathlib's sub-Gaussian API is the $b = 0$ case of the statements below (Lemma **??**). Unlike Mathlib's sub-Gaussian constant, the parameters are real numbers (not `0`), which avoids coercions in the arithmetic of Chapter **??**; the lemmas that need $V > 0$ or $b > 0$ assume it explicitly. As Mathlib does for sub-Gaussian variables, the file also provides the version with respect to a kernel and a measure (`Kernel.HasSubexponentialMGF`), the conditional version (`HasCondSubexponentialMGF`) and the Azuma–Bernstein inequality for martingale differences (Section **??**); the measure version is derived from the kernel version wherever Mathlib does so, and the tail bounds below therefore hold for any measure, not only finite ones. None of Section **??** is used in Part I.

12.1 Definition and basic properties

Definition 231 (Sub-exponential moment generating function)**.** ProbabilityTheory.HasSubexponentialMGFLet $X : \Omega \to \mathbb{R}$ be a random variable on $(\Omega, \mathbb{P})$ and $V, b \in \mathbb{R}$. We say that X *has a sub-exponential MGF with parameters* (V, b), written HasSubexponentialMGF(X, V, b), if for every $\lambda \in \mathbb{R}$ with $b|\lambda| \leq 1$ (i.e. $|\lambda| \leq 1/b$ when $b > 0$, and every $\lambda \in \mathbb{R}$ when $b \leq 0$), the variable $e^{\lambda X}$ is integrable and

$$\mathbb{E}e^{\lambda X} \leq \exp\left(\frac{\lambda^2 V}{2}\right).$$

The case $b = 0$ is exactly Mathlib's `HasSubgaussianMGF X V` (Lemma **??**). No centering is assumed; it is a consequence (Lemma **??**). In practice $V \geq 0$ and $b \geq 0$.

Lemma 232 (Sub-exponential variables are integrable and centered)**.** *ProbabilityTheory.HasSubexponentialMG*
= 0.

Proof. def:subexp The set $\{\lambda : b|\lambda| \leq 1\}$ is a neighbourhood of 0 (for every b, since $\lambda \mapsto b|\lambda|$ is continuous and vanishes at 0), so 0 lies in the interior of the set of λ such that $e^{\lambda X}$ is integrable. Mathlib then provides the integrability of $|X|^p$ for every p (`integrable_pow_of_mem_interior_integrableExpSet`) and the differentiability of $\lambda \mapsto \mathbb{E}e^{\lambda X}$ at 0 with derivative $\mathbb{E}X$ (`hasDerivAt_mgf`). Centering: the function $\varphi(\lambda) := \mathbb{E}e^{\lambda X} - e^{\lambda^2 V/2}$ is nonpositive on a neighbourhood of 0 and vanishes at 0 (as $\mathbb{E}e^0 = 1$ for a probability measure), so it has a local maximum at 0; it is differentiable there with derivative $\mathbb{E}X - 0$, which must vanish (Fermat's rule, Mathlib `IsLocalMax.hasDerivAt_eq_zero`). Hence $\mathbb{E}X = 0$. ☐

Lemma 233 (Monotonicity in the parameters)**.** *ProbabilityTheory.HasSubexponentialMGF.monodef:subexp If* HasSubexponentialMGF(X, V, b)*,* $V \leq V'$ *and* $b \leq b'$*, then* HasSubexponentialMGF(X, V', b').

Proof. def:subexp $b'|\lambda| \leq 1$ implies $b|\lambda| \leq b'|\lambda| \leq 1$, and $e^{\lambda^2 V/2} \leq e^{\lambda^2 V'/2}$. ☐

Lemma 234 (Sub-Gaussian is sub-exponential with $b = 0$)**.** *ProbabilityTheory.HasSubgaussianMGF.hasSubexponentialMGF, ProbabilityTheory.HasSubexponentialMGF.hasSubgaussianM lem:ps*$_s$*ubexp*$_m$*onoXhasasub−GaussianMGFwithconstant* $c \geq 0$ *(Mathlib* `HasSubgaussianMGF X c`*:* $e^{\lambda X}$ *integrable and* $\mathbb{E}e^{\lambda X} \leq e^{c\lambda^2/2}$ *for all* $\lambda \in \mathbb{R}$*) if and only if* HasSubexponentialMGF$(X, c, 0)$*. Consequently, if* X *has a sub-Gaussian MGF with constant* c*, then* HasSubexponentialMGF(X, c, b) *for every* $b \in \mathbb{R}$*.*

Proof. def:subexp, lem:ps$_s$*ubexp*$_m$*onoFor*b $= 0$*thecondition*b$|\lambda| \leq 1$ holds for every $\lambda \in \mathbb{R}$, so the two requirements of Definition **??** are literally the two fields of `HasSubgaussianMGF`. The second claim is the same argument: the condition $b|\lambda| \leq 1$ only restricts the set of λ for which the two fields are required. ☐

Lemma 235 (Scaling)**.** *ProbabilityTheory.HasSubexponentialMGF.const*$_m$*ul, ProbabilityTheory.HasSubexpon*
$\in \mathbb{R}$*, then* HasSubexponentialMGF$(aX,\ a^2V,\ |a|\,b)$*. In particular (*$a = -1$*),* HasSubexponentialMGF$(-X, V, b)$*.*

Proof. def:subexp Let $(|a|b)|\lambda| \leq 1$, i.e. $b|\lambda a| \leq 1$; then $e^{\lambda(aX)} = e^{(\lambda a)X}$ is integrable and $\mathbb{E}e^{(\lambda a)X} \leq e^{(\lambda a)^2 V/2} = e^{\lambda^2(a^2V)/2}$ (for $a = 0$ this reads $\mathbb{E}e^0 = 1 \leq 1$). ☐

Lemma 236 (Transport)**.** *ProbabilityTheory.HasSubexponentialMGF.congr, ProbabilityTheory.HasSubexponen*
if $X = Y$*almostsurely, orif XandYhavethesamelaw(onpossiblydifferentprobabilityspaces), thenXhasparame*
: $\Omega' \to \Omega$ *is measurable and* $X : \Omega \to \mathbb{R}$ *has parameters* (V, b) *under the image measure* $\mathbb{P}' \circ Y^{-1}$*, then* $X \circ Y$ *has parameters* (V, b) *under* $\mathbb{P}'$*.*

Proof. def:subexp Both fields of Definition **??** are expressed through integrals of functions of X, which are unchanged by these operations (change of variables `integrable_map_measure`, `mgf_map`). This is the same proof as for Mathlib's `HasSubgaussianMGF.of_map` and `congr_identDistrib`. ☐

Lemma 237 (Bernstein-type tail bound)**.** *ProbabilityTheory.HasSubexponentialMGF.measure*$_g e_l e_e xp_n eg_s q, Pr$
≥ 0. *Then*

$$\mathbb{P}(X \geq t) \leq \begin{cases} \exp\left(-\frac{t^2}{2V}\right) & \text{if } tb \leq V \text{ and } V > 0, \\ \exp\left(-\frac{t}{2b}\right) & \text{if } tb \geq V \text{ and } b > 0, \end{cases}$$

and consequently, when $V > 0$ *and* $b > 0$*,*

$$\mathbb{P}(X \geq t) \leq \exp\left(-\min\left(\frac{t^2}{2V}, \frac{t}{2b}\right)\right), \quad \mathbb{P}(X \leq -t) \leq \exp\left(-\min\left(\frac{t^2}{2V}, \frac{t}{2b}\right)\right), \quad \mathbb{P}(|X| \geq t) \leq 2\exp\left(-\min\left(\frac{t^2}{2V},\right.\right.$$

If $b = 0$ *(the sub-Gaussian case) and* $V > 0$*, the first bound applies for every* $t \geq 0$ *(the condition* $tb \leq V$ *is automatic): this is Mathlib* ***`HasSubgaussianMGF.measure_ge_le`*** *(and LML* ***`measure_le_le`*** *for the lower tail), i.e. the* min *form with the convention* $t/(2b) = +\infty$*.*

Proof. def:subexp, lem:subexp$_c onst_m ulFor \lambda \geq 0$ with $b\lambda \leq 1$, Markov's inequality applied to the nonnegative integrable variable $e^{\lambda X}$ gives (Chernoff, Mathlib `measure_ge_le_exp_mul_mgf`)

$$\mathbb{P}(X \geq t) = \mathbb{P}(e^{\lambda X} \geq e^{\lambda t}) \leq e^{-\lambda t}\, \mathbb{E}e^{\lambda X} \leq \exp\Big(-\lambda t + \frac{\lambda^2 V}{2}\Big).$$

Case $tb \leq V$, $V > 0$*:* take $\lambda := t/V \geq 0$, which satisfies $b\lambda \leq 1$; the exponent is $-t^2/V + t^2/(2V) = -t^2/(2V)$. *Case* $tb \geq V$, $b > 0$*:* take $\lambda := 1/b$; the exponent is $-t/b + V/(2b^2) \leq -t/b + t/(2b) = -t/(2b)$ since $V \leq tb$. For the min form it is not necessary to identify which term realizes the minimum: in each case the bound obtained is $\exp(-u)$ for one of the two terms u of the minimum, and $\exp(-u) \leq \exp(-\min(\cdot,\cdot))$. The lower tail is the upper tail of $-X$, which has the same parameters (Lemma **??** with $a = -1$); the two-sided bound is a union bound over $\{X \geq t\} \cup \{X \leq -t\}$. □

Remark 238 (The constant c of the paper)*.* The paper's Lemma 2 (Appendix A) states this bound with an unspecified absolute constant c in the exponent; the computation above, which is the one given there, yields $c = 1/2$. This value is used in Chapter **??** to make all sample-complexity constants explicit.

Lemma 239 (Independent sums)**.** *ProbabilityTheory.HasSubexponentialMGF.add*$_o f_i ndepFun, ProbabilityThe$
be independent with HasSubexponentialMGF(X_i, V_i, b) *for every* $i \leq K$ *(the same* b*). Then* HasSubexponentialMGF$\big(\sum_{i\leq K} X_i,\ \sum_{i\leq K} V_i,\ b\big)$*.*

Proof. def:subexp Let $b|\lambda| \leq 1$. Each $e^{\lambda X_i}$ is integrable and they are independent, so $e^{\lambda \sum_i X_i} = \prod_i e^{\lambda X_i}$ is integrable (Mathlib `iIndepFun.integrable_exp_mul_sum`) and $\mathbb{E}e^{\lambda \sum_i X_i} = \prod_i \mathbb{E}e^{\lambda X_i}$ (`iIndepFun.mgf_sum`) $\leq \prod_i e^{\lambda^2 V_i/2} = e^{\lambda^2 \sum_i V_i/2}$. □

Corollary 240 (Tail of an average)**.** *ProbabilityTheory.HasSubexponentialMGF.average*$_o f_i IndepFun, Probabil$
≥ 1 *and* $X_1, \ldots, X_K$ *be independent with* HasSubexponentialMGF(X_i, V_i, b),

and let $\bar{V} := \frac{1}{K}\sum_{i \le K} V_i$. *Then* $\mathrm{HasSubexponentialMGF}\big(\frac{1}{K}\sum_{i\le K} X_i,\ \bar{V}/K,\ b/K\big)$ *and, if* $\bar{V} > 0$ *and* $b > 0$*, for every* $t \ge 0$*,*

$$\mathbb{P}\Big(\Big|\frac{1}{K}\sum_{i\le K} X_i\Big| \ge t\Big) \le 2\exp\Big(-K\min\Big(\frac{t^2}{2\bar{V}}, \frac{t}{2b}\Big)\Big).$$

When all the V_i *are equal to* V*,* $\bar{V} = V$*.*

Proof. lem:subexp$_s um, lem : subexp_c onst_m ul, lem : subexp_t ail Lemma$ **??**$gives parameters(\sum_i V_i, b) =$ $(K\bar{V}, b)$ for the sum, and Lemma **??** with $a = 1/K$ gives $(K\bar{V}/K^2, b/K) =$ $(\bar{V}/K, b/K)$ for the average. Lemma **??** (with $b/K > 0$ and $\bar{V}/K > 0$) then gives the two-sided bound $2\exp(-\min(t^2/(2\bar{V}/K), t/(2b/K))) = 2\exp(-K\min(t^2/(2\bar{V}), t/(2b)))$. (For $b = 0$ the same computation with the $b = 0$ clause of Lemma **??** gives $2\exp(-Kt^2/(2\bar{V}))$, Mathlib's `measure_sum_ge_le_of_iIndepFun`.) □

12.2 Kernel and conditional versions

Mathlib defines sub-Gaussian variables first with respect to a kernel and a measure (`Kernel.HasSubgaussianMGF`), then derives the conditional notion (`HasCondSubgaussianMGF`, the kernel being the conditional expectation kernel of a sub-σ-algebra) and the notion with respect to a measure (a constant kernel); the payoff is the Azuma–Hoeffding inequality for martingale differences. This section follows the same layering for the sub-exponential property, so that the sub-exponential file has the same interface as Mathlib's sub-Gaussian file. It also records Bernstein's condition (Lemma **??**), the sub-exponential counterpart of Hoeffding's lemma.

Definition 241 (Sub-exponential MGF with respect to a kernel)**.** ProbabilityTheory.Kernel.HasSubexponentialMGF, ProbabilityTheory.hasSubexponentialMGF$_i ff_k erneldef : subexpLe$ be a measure on Ω', κ a kernel from Ω' to Ω, $X : \Omega \to \mathbb{R}$ and $V, b \in \mathbb{R}$. We say that X has a sub-exponential MGF with parameters (V, b) with respect to κ and ν if for every λ with $b|\lambda| \le 1$, $e^{\lambda X}$ is integrable with respect to the composition $\kappa \circ \nu = \int \kappa(\omega', \cdot)\, d\nu(\omega')$, and for ν-almost every ω' and every such λ, $\int e^{\lambda X}\, d\kappa(\omega', \cdot) \le e^{\lambda^2 V/2}$. The property of Definition **??** with respect to a measure $\mathbb{P}$ is the case of the constant kernel $\mathbb{P}$ on the one-point space.

Lemma 242 (Basic properties of the kernel version)**.** *ProbabilityTheory.Kernel.HasSubexponentialMGF.aestron* *ProbabilityTheory.Kernel.HasSubexponentialMGF.ae*$_i ntegrable_e xp_m ul, ProbabilityTheory.Kernel.HasSubexp$ *exponentialMGFwithparameters(V,b)withrespectto*κ *and* ν*. Then* X *is almost everywhere strongly measurable for* $\kappa \circ \nu$*, and for* ν*-almost every* ω'*:* $\kappa(\omega', \cdot)$ *is a finite measure of mass at most* 1*;* $e^{\lambda X}$ *is* $\kappa(\omega', \cdot)$*-integrable for every admissible* λ *simultaneously, and* $e^{\lambda X} \in L^p(\kappa(\omega', \cdot))$ *whenever* $p\lambda$ *is admissible;* X *has a sub-exponential MGF with parameters* (V, b) *with respect to the measure* $\kappa(\omega', \cdot)$ *(Definition* **??***). In particular, if* κ *is a Markov kernel,* $\int X\, d\kappa(\omega', \cdot) = 0$ *for* ν*-almost every* ω'*, and if moreover* $V = 0$ *then* $X = 0$ $\kappa(\omega', \cdot)$*-a.s. for* ν*-almost every* ω'*. If* $V \ge 0$*, the cumulant generating function is at most* $\lambda^2 V/2$ *on the admissible set. Conversely, to establish the property it suffices to check the MGF bound for each admissible* λ *separately,* ν*-almost everywhere.*

Proof. def:subexp$_kernel, lem : ps_subexp_integral_zeroThemeasurability, finitenessandintegrabilityclaimsaret$
$b|\lambda| \leq 1\}$ in place of $\mathbb{R}$: for $b \leq 0$ it is $\mathbb{R}$ and integrability for all λ follows from integrability at the integers; for $b > 0$ it is the interval $[-1/b, 1/b]$ and integrability at the two endpoints suffices, the set of λ such that $e^{\lambda X}$ is integrable being convex. The converse statement ("for all λ, a.e." implies "a.e., for all λ") uses the countability of the rationals: almost everywhere, the bound holds at every admissible rational and at the (at most two) endpoints $\pm 1/b$, and $e^{\lambda X}$ is integrable on the admissible set; on the open set $\{b|\lambda| < 1\}$ the MGF is continuous (Mathlib `continuousOn_mgf` on the interior of the integrability set), so the bound extends from the rationals by density. The case $V = 0$: the MGF is at most 1 on a neighbourhood of 0 and, by Jensen's inequality and the centering of Lemma **??**, at least $e^{\lambda \mathbb{E} X} = 1$, so it is constant there; its second derivative at 0, which is $\mathbb{E}X^2$ (Mathlib `iteratedDeriv_mgf_zero`), vanishes. ☐

Lemma 243 (Stability of the kernel version)**.** *ProbabilityTheory.Kernel.HasSubexponentialMGF.fun$_z$ero, Proba*
monotonicityin(V,b), thesub$-$Gaussiancase$b = 0$, scaling$aX \mapsto (a^2V, |a|\, b)$ and negation, almost-sure modification of X (for $\kappa \circ \nu$), and transport along a measurable map Y (if X has the property for the kernel κ pushed forward by Y, then $X \circ Y$ has it for κ). The zero variable has parameters $(0, b)$ for every b when κ is zero or Markov and ν is finite, and every X has every parameters for the zero kernel or the zero measure.

Proof. def:subexp$_kernelSameproofsasforthemeasureversion, insidethealmost-$
$everywherequantifier.$ ☐

Lemma 244 (Tail bounds for the kernel version)**.** *ProbabilityTheory.Kernel.HasSubexponentialMGF.measure$_g$e*
in place of $\mathbb{P}$, for ν-almost every ω'.

Proof. def:subexp$_kernel, lem : subexp_tailChernoff'sbound\kappa(\omega', \{X \geq t\}) \leq$
$e^{-st} \int e^{sX} d\kappa(\omega', \cdot) \leq \exp(-st + s^2V/2)$ for every admissible $s \geq 0$, almost surely, then the choices of s of Lemma **??**. ☐

Lemma 245 (Sum of two dependent sub-exponential variables)**.** *ProbabilityTheory.Kernel.HasSubexponentialM*
and (V_Y, b_Y), with respect to a kernel and a measure or with respect to a measure, and let $p, q > 1$ with $1/p + 1/q = 1$. Then $X + Y$ has parameters $(pV_X + qV_Y,\ \max(pb_X, qb_Y))$. In particular ($p = q = 2$), $X + Y$ has parameters $(2(V_X + V_Y),\ 2\max(b_X, b_Y))$.

Proof. def:subexp$_kernel, lem : subexp_kernel_basicHlder'sinequality : for \max(pb_X, qb_Y)|\lambda| \leq$
1, $p\lambda$ is admissible for X and $q\lambda$ for Y, so $e^{\lambda X} \in L^p$ and $e^{\lambda Y} \in L^q$ (Lemma **??**) and $\mathbb{E}e^{\lambda(X+Y)} \leq (\mathbb{E}e^{p\lambda X})^{1/p}(\mathbb{E}e^{q\lambda Y})^{1/q} \leq \exp(pV_X\lambda^2/2 + qV_Y\lambda^2/2)$. The integrability of $e^{\lambda(X+Y)}$ for $\kappa \circ \nu$ is that of the product of an L^p and an L^q function. ☐

Lemma 246 (Sum along a composition of kernels)**.** *ProbabilityTheory.Kernel.HasSubexponentialMGF.prodMkL*
be an s-finite measure on Ω', κ a kernel from Ω' to Ω and η a zero-or-Markov kernel from $\Omega' \times \Omega$ to Ω''. If $X : \Omega \to \mathbb{R}$ has parameters (V_X, b) with respect to κ and ν, and $Y : \Omega'' \to \mathbb{R}$ has parameters (V_Y, b) with respect to η and $\nu \otimes \kappa$,

then $(x,y) \mapsto X(x) + Y(y)$ has parameters $(V_X + V_Y, b)$ with respect to $\kappa \otimes \eta$ and ν. The same holds for a kernel η from Ω to Ω'' and Y with parameters (V_Y, b) with respect to η and $\kappa \circ \nu$.

Proof. def:subexp$_k$$ernel, lem : subexp_kernel_basic This is Mathlib's$`Kernel.HasSubgaussianMGF.add_compProd`
The bound on the MGF, for a fixed admissible λ and ν-almost every ω', is Fubini for $(\kappa \otimes \eta)(\omega', \cdot)$: $\int e^{\lambda(X(x)+Y(y))} = \int e^{\lambda X(x)} \big(\int e^{\lambda Y(y)}\, d\eta((\omega', x), y) \big) d\kappa(\omega', x) \le e^{\lambda^2 V_Y/2} \int e^{\lambda X} d\kappa(\omega', \cdot) \le e^{\lambda^2 (V_X+V_Y)/2}$; the "for all λ" form then follows from Lemma **??**. The integrability of $e^{\lambda(X+Y)}$ for $(\kappa \otimes \eta) \circ \nu$ cannot use the Cauchy–Schwarz argument of the sub-Gaussian case, which needs 2λ to be admissible: it is proved on the Lebesgue integral of the nonnegative integrand, which by the same Fubini computation is at most $e^{\lambda^2 V_Y/2} \int e^{\lambda X} d(\kappa \circ \nu) < \infty$. ☐

Definition 247 (Conditionally sub-exponential variable)**.** ProbabilityTheory.HasCondSubexponentialMGF, ProbabilityTheory.HasCondSubexponentialMGF.mgf$_l$$e, ProbabilityTheory.HasCondSubexponentialMGF.cg$
be a standard Borel space, $\mathbb{P}$ a finite measure and $\mathcal{G}$ a sub-σ-algebra. A random variable X is conditionally sub-exponential with parameters (V, b) given $\mathcal{G}$ if it has the property of Definition **??** with respect to the conditional expectation kernel of $\mathcal{G}$ (Mathlib `condExpKernel`) and the restriction of $\mathbb{P}$ to $\mathcal{G}$. Concretely, $e^{\lambda X}$ is integrable for every admissible λ and $\mathbb{E}[e^{\lambda X} \mid \mathcal{G}] \le e^{\lambda^2 V/2}$ almost surely; the case $b = 0$ is Mathlib's `HasCondSubgaussianMGF`.

Lemma 248 (Bernstein's condition)**.** *$ProbabilityTheory.exp_le_one_add_add_of_abs_le_one, ProbabilityTheory.hasSub$*
M almost surely, $\mathbb{E}X = 0$ and $\mathbb{E}X^2 \le v$. Then HasSubexponentialMGF$(X, 3v/2, M)$*.*
Consequently, if $|X| \le M$ almost surely then $X - \mathbb{E}X$ has parameters $(3\mathrm{Var}(X)/2,\ 2M)$*.*

Proof. def:subexp For $|x| \le 1$, $e^x \le 1 + x + \frac{3}{4}x^2$ (Mathlib `Real.exp_bound` with two terms of the series). For $M|\lambda| \le 1$, $|\lambda X| \le 1$ almost surely, hence $\mathbb{E}e^{\lambda X} \le 1 + \lambda \mathbb{E}X + \frac{3}{4}\lambda^2 \mathbb{E}X^2 \le 1 + \frac{3}{4}\lambda^2 v \le e^{\frac{3}{4}\lambda^2 v} = e^{\lambda^2 (3v/2)/2}$. For the second claim, $|X - \mathbb{E}X| \le |X| + |\mathbb{E}X| \le 2M$ and $\mathbb{E}(X - \mathbb{E}X)^2 = \mathrm{Var}(X)$. ☐

Lemma 249 (Further properties of the measure version)**.** *$ProbabilityTheory.HasSubexponentialMGF.aestrongly$*
$ProbabilityTheory.HasSubexponentialMGF.memLp_exp_mul, ProbabilityTheory.HasSubexponentialMGF.cgf_le$
are independent with parameters (V_i, b), $\sum_i V_i > 0$ and $b > 0$, then for $t \ge 0$, $\mathbb{P}(\sum_i X_i \ge t)$ and $\mathbb{P}(\sum_i X_i \le -t)$ are at most $\exp(-\min(t^2/(2\sum_i V_i),\ t/(2b)))$; for K variables with the same parameters (V, b) the bound is $\exp(-\min(t^2/(2KV), t/(2b)))$.
(ii) If $X - \mathbb{E}X$ and $Y - \mathbb{E}Y$ have parameters (V_X, b) and (V_Y, b), X and Y are independent and $\mathbb{E}Y \le \mathbb{E}X$, then $\mathbb{P}(X \le Y) \le \exp(-\min((\mathbb{E}X - \mathbb{E}Y)^2/(2(V_X + V_Y)),\ (\mathbb{E}X - \mathbb{E}Y)/(2b)))$. (iii) The property is preserved by restricting $\mathbb{P}$ to a sub-σ-algebra for which X is measurable; a difference of independent sub-exponential variables has parameters $(V_X + V_Y, b)$; the zero variable has parameters $(0, b)$, and conversely a variable with parameters $(0, b)$ on a probability space vanishes almost surely; $e^{\lambda X} \in L^p$ whenever $p\lambda$ is admissible; and the cumulant generating function is at most $\lambda^2 V/2$ on the admissible set when $V \ge 0$.

Proof. def:subexp, lem:subexp$_s$$um, lem : subexp_tail, lem : subexp_kernel_basic (i) is Lemma$ **??** $followedbyLemm$
- EY) - (X - EX), $which has parameters$ $(V_X + V_Y, b)$, at $t = \mathbb{E}X - \mathbb{E}Y \ge 0$; (iii)

are the corresponding statements of Lemma **??** for the constant kernel, or direct. ☐

Lemma 250 (Sums of conditionally sub-exponential increments)**.** *ProbabilityTheory.HasSubexponentialMGF.a be standard Borel and $\mathbb{P}$ finite. If X has parameters (V_X, b) with respect to the restriction of $\mathbb{P}$ to a sub-σ-algebra $\mathcal{G}$, and Y is conditionally sub-exponential with parameters (V_Y, b) given $\mathcal{G}$, then $X+Y$ has parameters (V_X+V_Y, b) with respect to $\mathbb{P}$. Consequently, if $\mathbb{P}$ is a probability measure, $(Y_i)_{i\in\mathbb{N}}$ is a process adapted to a filtration $(\mathcal{F}_i)_{i\in\mathbb{N}}$, Y_0 has parameters (V_0, b) and, for $i < n-1$, Y_{i+1} is conditionally sub-exponential with parameters (V_{i+1}, b) given $\mathcal{F}_i$, then $\sum_{i<n} Y_i$ has parameters $(\sum_{i<n} V_i,\ b)$.*

Proof. def:subexp$_cond, lem: subexp_add_compProd, lem: subexp_measure_more The first claim is Lemma$ **??** $for th$ and the conditional expectation kernel, whose composition-product is the law of $\omega \mapsto (\omega, \omega)$ under $\mathbb{P}$ (Mathlib `compProd_trim_condExpKernel`); $X+Y$ is the image of $(x, y) \mapsto X(x)+Y(y)$ under this diagonal map. The second is an induction on n, the partial sum $\sum_{i\le n} Y_i$ being $\mathcal{F}_n$-measurable, hence sub-exponential for $\mathbb{P}|_{\mathcal{F}_n}$ by Lemma **??**(iii). ☐

Lemma 251 (Azuma–Bernstein inequality)**.** *ProbabilityTheory.measure$_sum_ge_le_of_hasCondSubexponentialMC$ 0 and $b > 0$, then for every $t \geq 0$,*

$$\mathbb{P}\Big(\sum_{i<n} Y_i \geq t\Big) \leq \exp\Big(-\min\Big(\frac{t^2}{2\sum_{i<n} V_i}, \frac{t}{2b}\Big)\Big).$$

Proof. lem:subexp$_martingale, lem: subexp_tail Lemmas$ **??** *and* **??**. ☐

12.3 Squares of Gaussian variables

Lemma 252 (MGF of the square of a real Gaussian)**.** *integral$_exp_quadratic, integrable_exp_quadratic, Probability$ < 0and$c,\ d \in \mathbb{R}$,*

$$\int_{\mathbb{R}} \exp(bx^2 + cx + d)\, dx = \sqrt{\frac{\pi}{-b}}\, \exp\Big(d - \frac{c^2}{4b}\Big),$$

the integrand being integrable. Consequently, let $\mu \in \mathbb{R}$, $\sigma \geq 0$, $X \sim \mathcal{N}(\mu, \sigma^2)$ and $\lambda \in \mathbb{R}$ with $2\sigma^2\lambda < 1$. Then $e^{\lambda X^2}$ is integrable and

$$\mathbb{E}e^{\lambda X^2} = (1-2\sigma^2\lambda)^{-1/2} \exp\Big(\frac{\mu^2\lambda}{1-2\sigma^2\lambda}\Big).$$

Proof. def:pg$_gaussian_vector Complete the square :bx^2+cx+d = b\,(x+\frac{c}{2b})^2 + d - \frac{c^2}{4b}$, so by the translation invariance of Lebesgue measure (`integral_add_right_eq_self`) and Mathlib `integral_gaussian` ($\int e^{-\beta y^2} dy = \sqrt{\pi/\beta}$ with $\beta = -b > 0$), the first identity follows; integrability is `integrable_exp_neg_mul_sq` after the same translation. If $\sigma = 0$ then $X = \mu$ a.s. and both sides of the second identity equal $e^{\lambda\mu^2}$. Let $\sigma > 0$. With the density $\frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)}$ (Mathlib `gaussianReal`, `gaussianPDFReal`, `integral_gaussianReal_eq_integral_smul`),

$$\mathbb{E}e^{\lambda X^2} = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{\mathbb{R}} \exp\Big(\lambda x^2 - \frac{(x-\mu)^2}{2\sigma^2}\Big) dx,$$

and $\lambda x^2 - \frac{(x-\mu)^2}{2\sigma^2} = bx^2 + cx + d$ with $b = \lambda - \frac{1}{2\sigma^2} = -\frac{1-2\sigma^2\lambda}{2\sigma^2} < 0$, $c = \frac{\mu}{\sigma^2}$, $d = -\frac{\mu^2}{2\sigma^2}$. The first identity gives $\frac{1}{\sqrt{2\pi\sigma^2}}\sqrt{\frac{2\pi\sigma^2}{1-2\sigma^2\lambda}} = (1-2\sigma^2\lambda)^{-1/2}$ and the exponent $d - \frac{c^2}{4b} = -\frac{\mu^2}{2\sigma^2} + \frac{\mu^2}{2\sigma^2(1-2\sigma^2\lambda)} = \frac{\mu^2\lambda}{1-2\sigma^2\lambda}$. $\square$

Lemma 253 (Two elementary inequalities)**.** $ProbabilityTheory.neg_{l}og_{o}ne_{s}ub_{l}e_{a}dd_{s}q, ProbabilityTheory.one_{d}i$ $\mathbb{R}$ *with* $|\nu| \le 1/2$*:*

(i) $-\log(1-\nu) \le \nu + \nu^2$*, equivalently* $-\frac{1}{2}\log(1-\nu) - \frac{\nu}{2} \le \frac{\nu^2}{2}$*;*

(ii) $\frac{1}{1-\nu} \le 2$*.*

Proof. (i) It is enough to show $(1-\nu)e^{\nu+\nu^2} \ge 1$ and take logarithms. For $\nu \ge 0$ use $e^y \ge 1+y+y^2/2$ (Mathlib `Real.quadratic_le_exp_of_nonneg`) with $y = \nu+\nu^2$: $(1-\nu)\big(1+(\nu+\nu^2)+(\nu+\nu^2)^2/2\big) = 1+\frac{\nu^2}{2}(1-\nu-\nu^2-\nu^3) \ge 1$, since $\nu+\nu^2+\nu^3 \le \frac{1}{2}+\frac{1}{4}+\frac{1}{8} < 1$. For $\nu \le 0$ use $e^y \ge 1+y$: $(1-\nu)(1+\nu+\nu^2) = 1-\nu^3 \ge 1$. The bound is the truncation of $-\log(1-\nu)-\nu = \sum_{k\ge 2}\nu^k/k \le \frac{\nu^2}{2(1-\nu)} \le \nu^2$. (ii) $1-\nu \ge 1-|\nu| \ge \frac{1}{2}$. $\square$

Lemma 254 (Centered square of a Gaussian is sub-exponential)**.** $ProbabilityTheory.hasSubexponentialMGF_{s}q_{s}$ $\mathbb{R}$*,* $\sigma \ge 0$ *and* $X \sim \mathcal{N}(\mu,\sigma^2)$*. Then*

$$\mathrm{HasSubexponentialMGF}\big(X^2 - (\mu^2+\sigma^2),\ 4\sigma^4 + 8\mu^2\sigma^2,\ 4\sigma^2\big),$$

i.e. for all $|\lambda| \le \frac{1}{4\sigma^2}$*,* $\log \mathbb{E}e^{\lambda(X^2-(\mu^2+\sigma^2))} \le (2\sigma^4+4\mu^2\sigma^2)\lambda^2$ *(and* $\mathbb{E}X^2 = \mu^2+\sigma^2$ *by Lemma* **??***).*

Proof. lem:gaussian$_{s}quare_{m}gf, lem : log_{i}neq_{n}uLet 4\sigma^2|\lambda| \le 1$ and $\nu := 2\sigma^2\lambda$, so that $|\nu| \le 1/2$ and in particular $2\sigma^2\lambda < 1$: Lemma **??** applies, $e^{\lambda X^2}$ is integrable (hence so is $e^{\lambda(X^2-\mu^2-\sigma^2)}$), and

$$\log \mathbb{E}e^{\lambda(X^2-\mu^2-\sigma^2)} = -\frac{1}{2}\log(1-\nu) + \frac{\mu^2\lambda}{1-\nu} - \lambda(\mu^2+\sigma^2) = \Big(-\frac{1}{2}\log(1-\nu) - \frac{\nu}{2}\Big) + \mu^2\lambda\Big(\frac{1}{1-\nu} - 1\Big),$$

using $\lambda\sigma^2 = \nu/2$. By Lemma **??**(i), the first bracket is at most $\frac{\nu^2}{2} = 2\sigma^4\lambda^2$. The second term equals $\mu^2\lambda\nu/(1-\nu) = 2\mu^2\sigma^2\lambda^2 \cdot \frac{1}{1-\nu} \le 4\mu^2\sigma^2\lambda^2$ by Lemma **??**(ii), since $\lambda\nu = 2\sigma^2\lambda^2 \ge 0$. Hence $\log \mathbb{E}e^{\lambda(X^2-\mu^2-\sigma^2)} \le (2\sigma^4+4\mu^2\sigma^2)\lambda^2 = \frac{\lambda^2}{2}(4\sigma^4 + 8\mu^2\sigma^2)$, which is the claim with $V = 4\sigma^4 + 8\mu^2\sigma^2$ and $b = 4\sigma^2$. Both constants are essentially optimal at $b = 4\sigma^2$, where the exact requirement is $V \ge (16\log 2 - 8)\sigma^4 + 8\mu^2\sigma^2 \approx 3.09\,\sigma^4 + 8\mu^2\sigma^2$; and no finite V works for $b \le 2\sigma^2$, where $\mathbb{E}e^{\lambda X^2} = \infty$ at $\lambda = 1/b$. $\square$

Definition 255 (Chi-squared distribution)**.** ProbabilityTheory.chiSqPDFReal, ProbabilityTheory.chiSqPDF, ProbabilityTheory.chiSqMeasure, ProbabilityTheory.chiSqPDFReal$_o f_n onneg, P$ is the gamma distribution with shape $d/2$ and rate $1/2$, that is, the measure on $\mathbb{R}$ with density

$$x \mapsto \frac{x^{d/2-1}e^{-x/2}}{2^{d/2}\,\Gamma(d/2)} \quad (x \geq 0), \qquad 0 \quad (x < 0)$$

with respect to the Lebesgue measure. It is a probability measure for $d \geq 1$.

Lemma 256 (Moment generating function of the gamma and chi-squared distributions)**.** *ProbabilityTheory.integral$_e xp_m ul_g ammaMeasure$, ProbabilityTheory.mgf$_c hiSqMeasuredef : chi_s q$ with $a, r > 0$ and every $t < r$, $\int e^{tx}\, d\gamma(a,r)(x) = \left(\frac{r}{r-t}\right)^a$. In particular, for $d \geq 1$ and $t < 1/2$, the moment generating function of χ^2_d is $(1-2t)^{-d/2}$.*

Proof. The density of $\gamma(a,r)$ vanishes on $(-\infty, 0)$, so $\int e^{tx}\, d\gamma(a,r) = \frac{r^a}{\Gamma(a)} \int_0^\infty x^{a-1}e^{-(r-t)x}\, dx = \frac{r^a}{\Gamma(a)} \cdot \frac{\Gamma(a)}{(r-t)^a}$, using Mathlib's `integral_rpow_mul_exp_neg_mul_Ioi` with the rate $r - t > 0$. The chi-squared case is $a = d/2$, $r = 1/2$. Mathlib has the gamma distribution but not its moment generating function. □

Lemma 257 (The squared norm of a Gaussian vector is chi-squared)**.** *ProbabilityTheory.integral$_e xp_m ul_n orm_s q_s$ $\sim \mathcal{N}(0, I_d)$ with $d \geq 1$. For every $t < 1/2$, $\mathbb{E}e^{t\|g\|^2} = (1-2t)^{-d/2}$, which is the moment generating function of χ^2_d (Lemma ??). This is what justifies calling $\|g\|^2 - d$ a centered chi-square variable in Lemmas ?? and ??.*

Proof. Writing $\mathcal{N}(0, I_d)$ as the image of $\mathcal{N}(0,1)^{\otimes d}$ (Lemma ??) and $e^{t\|g\|^2} = \prod_{i<d} e^{tg_i^2}$, the integral factorizes over the product measure, and each factor is $(1-2t)^{-1/2}$ by Lemma ?? with $\mu = 0$, $\sigma = 1$. □

Theorem 258 (Moment generating functions determine distributions)**.** *MeasureTheory.Measure.ext$_o f_e qOn_m g$ Yberealrandomvariablesunderaprobabilitymeasure P andafinitemeasure P'.Assumethat $\lambda \mapsto \mathbb{E}e^{\lambda X}$ is finite in a neighbourhood of 0, that the same holds for Y, and that $\mathbb{E}e^{\lambda X} = \mathbb{E}e^{\lambda Y}$ for λ in a neighbourhood of 0. Then X and Y have the same law: $\mathbb{P} \circ X^{-1} = \mathbb{P}' \circ Y^{-1}$.*

Proof. Mathlib's `eqOn_complexMGF_of_mgf` extends the equality of the moment generating functions to the complex moment generating functions $z \mapsto \mathbb{E}e^{zX}$, which are holomorphic on the vertical strips over the interiors of their domains of definition; here we run the same analytic continuation on the common strip, whose base is the (convex, open) intersection of the two interiors and contains 0. Equality of two functions analytic on a preconnected open set, knowing they agree along a real sequence tending to 0, is Mathlib's `AnalyticOnNhd.eqOn_of_preconnected_of_frequently_eq`. Since 0 is in the base, the strip contains the whole imaginary axis, on which the complex moment generating function is the characteristic function ($\mathbb{E}e^{itX} = \mathbb{E}e^{zX}$ at $z = it$), so the characteristic functions coincide and `Measure.ext_of_charFun` concludes. No analytic continuation beyond the strip is needed, which is why agreeing near 0 suffices. □

Corollary 259 (The squared norm of a Gaussian vector is chi-squared, as laws)**.** *ProbabilityTheory.map$_n$orm$_s$q$_s$tdGaussian$_e$q$_c$hiSqMeasurethm : ext$_o$f$_m$gf, lem : norm$_s$q$_c$hi$_s$quared, def : ch*
$\sim \mathcal{N}(0, I_d)$ *with* $d \geq 1$*, the law of* $\|g\|^2$ *is* χ^2_d*.*

Proof. lem:mgf$_c$*hi$_s$quaredByLemma* **??***thetwomomentgeneratingfunctionsagreeon*(-$\infty, 1/2)$, and both are infinite for $t \geq 1/2$, where e^{tx} is not integrable: for χ^2_d because the density times e^{tx} dominates $c\,x^{d/2-1}$ on $(1, \infty)$, whose integral diverges, and for $\|g\|^2$ because $e^{t\|g\|^2} \geq e^{tg_j^2}$ and the one-dimensional integrand $e^{tx^2}\varphi(x) \geq (2\pi)^{-1/2}$ is bounded below by a positive constant. Both moment generating functions are finite on $(-\infty, 1/2)$, a neighbourhood of 0, so Theorem **??** applies. □

Lean generality. The Lean statements of this section are for a standard Gaussian vector of an arbitrary finite-dimensional real inner product space 'E' (with $d = \operatorname{finrank}_{\mathbb{R}} E$ in place of the number of coordinates), not just for $\mathbb{R}^d$: the proofs pick an orthonormal basis of E, which is what `stdGaussian` is defined from. They therefore apply verbatim to a subspace or to a product of such spaces.

Lemma 260 (Chi-square variables are sub-exponential)**.** *ProbabilityTheory.hasSubexponentialMGF$_s$um$_s$q$_s$ub$_o$r*
$\sim \mathcal{N}(0, I_d)$*. Then* $\operatorname{HasSubexponentialMGF}\big(\|g\|^2 - d,\ 4d,\ 4\big)$*.*

Proof. lem:gauss$_p$*i, lem : gaussian$_s$quare$_s$ubexp, lem : subexp$_s$um, lem : ps$_s$ubexp$_m$ap*$\|g\|^2 - d = \sum_{i<d}(g_i^2 - 1)$ with $g_0, \dots, g_{d-1}$ i.i.d. $\mathcal{N}(0,1)$ (Lemma **??**; in Lean, $\mathcal{N}(0, I_d)$ is the image of the product measure $\mathcal{N}(0,1)^{\otimes d}$ on $\mathbb{R}^d$, `map_pi_eq_stdGaussian`, and the statement is transported by Lemma **??**). Each $g_i^2 - 1$ has parameters $(4, 4)$ by Lemma **??** with $\mu = 0$, $\sigma = 1$; the summands are independent (`iIndepFun_pi`), so Lemma **??** gives $(4d, 4)$. These are the parameters behind the Laurent–Massart form of the chi-square tail: the optimal V at $b = 4$ is $(16 \log 2 - 8)d \approx 3.09\,d$, and $V \geq 2d = \operatorname{Var}(\|g\|^2)$ for every b. □

Lemma 261 (Chi-square tail)**.** *ProbabilityTheory.measureReal$_n$orm$_s$q$_s$ub$_g$e$_l$e$_s$tdGaussian, ProbabilityTheory.*
$\sim \mathcal{N}(0, I_d)$ *with* $d \geq 1$*. For every* $t \geq 0$*,*

$$\mathbb{P}\big(\|g\|^2 - d \geq t\big) \leq \exp\Big(-\min\Big(\frac{t^2}{8d}, \frac{t}{8}\Big)\Big),$$

and for every $\delta \in (0, 1)$*,* $\mathbb{P}\big(\|g\|^2 \geq 2d + 12\log(1/\delta)\big) \leq \delta$*.*

Proof. lem:chi$_s$*quare$_s$ubexp, lem : subexp$_t$ailThefirstboundisLemma* **??***with*(V,b) = (4d,4)*fromLemma* **??** :t$^2/(2V) = t^2/(8d)$ and $t/(2b) = t/8$. For the second, let $L := \log(1/\delta) > 0$ and $t := \max(4\sqrt{dL}, 8L)$. Then $t^2/(8d) \geq 16dL/(8d) = 2L \geq L$ and $t/8 \geq L$, so $\mathbb{P}(\|g\|^2 - d \geq t) \leq e^{-L} = \delta$. Moreover, by AM–GM, $4\sqrt{dL} = 2\sqrt{d \cdot 4L} \leq d + 4L$, so $t \leq \max(d + 4L, 8L) \leq d + 8L \leq d + 12L$. Hence $\{\|g\|^2 \geq 2d + 12L\} = \{\|g\|^2 - d \geq d + 12L\} \subseteq \{\|g\|^2 - d \geq t\}$ has probability at most δ. □

Lemma 262 (Gaussian quadratic forms are sub-exponential)**.** *def:subexp, def:pg$_g$aussian$_v$ectorLetS* $\in \mathcal{S}_+^d$ *($d \times d$, $S \neq 0$) and $g \sim \mathcal{N}(0, I_d)$. Then $\mathbb{E}[g^\top Sg] = \operatorname{tr} S$ and*

$$\operatorname{HasSubexponentialMGF}\big(g^\top Sg - \operatorname{tr} S,\ 8\operatorname{tr}(S^2),\ 4\|S\|_{\mathrm{op}}\big),$$

where $\|S\|_{\mathrm{op}}$ is the largest eigenvalue of S.

Proof. lem:gauss$_r$*otation, lem : gauss$_p$i, lem : gaussian$_s$quare$_s$ubexp, lem : subexp$_c$onst$_m$ul, lem : ps$_s$ubexp$_m$ono, lem : subexp$_s$um, lem : gauss$_n$orm$_m$omentsDiagonalize*S $= \mathrm{U}\mathrm{diag}(s_1, \ldots, s_d)U^\top$ with U orthogonal and $s_i \geq 0$ (Mathlib `Matrix.IsHermitian.spectral_theorem`), so that $\operatorname{tr} S = \sum_i s_i$, $\operatorname{tr}(S^2) = \sum_i s_i^2$ and $\|S\|_{\mathrm{op}} = s_{\max} := \max_i s_i > 0$. Let $h := U^\top g$; then $h \sim \mathcal{N}(0, I_d)$ (Lemma **??**(i)), its coordinates are i.i.d. $\mathcal{N}(0,1)$ (Lemma **??**), and $g^\top Sg - \operatorname{tr} S = \sum_i s_i(h_i^2 - 1)$. The expectation is Lemma **??**(i). For each i with $s_i > 0$, Lemma **??** gives $(8, 4)$ for $h_i^2 - 1$ and Lemma **??** gives $(8s_i^2, 4s_i)$ for $s_i(h_i^2 - 1)$, hence $(8s_i^2, 4s_{\max})$ by Lemma **??**; for $s_i = 0$ the summand is 0, which has parameters $(0, 4s_{\max})$. The summands are independent, so Lemma **??** gives $(8\sum_i s_i^2, 4s_{\max}) = (8\operatorname{tr}(S^2), 4\|S\|_{\mathrm{op}})$. ☐

Remark 263 (Not used)*.* Lemma **??** is recorded for completeness (it is the general form of Lemma **??**, which is the case $S = I_d$) but none of the main results uses it: the large-norm regime of the norm estimation uses a deterministic basis design, for which only the chi-square case is needed. It is not formalized.

12.4 Squares of sub-Gaussian variables and of Rademacher linear forms

Lemma 264 (MGF of the square of a sub-Gaussian variable, positive parameter)**.** *ProbabilityTheory.HasSubgaussianMGF.lintegral$_e$xp$_m$ul$_s$q$_l$e, ProbabilityTheory.HasSubgaussianMGF.in GaussianMGFwithconstant$c \geq 0$ (Mathlib* ***`HasSubgaussianMGF Z c`****), and let $0 \leq \lambda$ with $2c\lambda < 1$ (any $\lambda \geq 0$ if $c = 0$). Then $e^{\lambda Z^2}$ is integrable and*

$$\mathbb{E}e^{\lambda Z^2} \leq (1 - 2c\lambda)^{-1/2}.$$

Proof. lem:gaussian$_s$*quare$_m$gf, lem : gauss$_1$d$_m$omentsGaussian trick:foreveryrealz, bytheMGFofg*' $\sim \mathcal{N}(0,1)$ at $s = \sqrt{2\lambda}\, z$ (Mathlib `mgf_gaussianReal`),

$$e^{\lambda z^2} = e^{(\sqrt{2\lambda}\, z)^2/2} = \mathbb{E}_{g'} \exp\big(\sqrt{2\lambda}\, z\, g'\big).$$

Hence, by Tonelli's theorem for the nonnegative measurable function $(\omega, v) \mapsto \exp(\sqrt{2\lambda}\, Z(\omega)\, v)$ on the product of $(\Omega, \mathbb{P})$ with $(\mathbb{R}, \mathcal{N}(0,1))$ (`MeasureTheory.lintegral_lintegral_swap`),

$$\mathbb{E}e^{\lambda Z^2} = \int \Big(\int \exp\big(\sqrt{2\lambda}\, Z(\omega)\, v\big)\, \mathcal{N}(0,1)(dv) \Big) \mathbb{P}(d\omega) = \int \Big(\int \exp\big((\sqrt{2\lambda}\, v)\, Z(\omega)\big) \mathbb{P}(d\omega) \Big) \mathcal{N}(0,1)(dv).$$

For fixed v, the inner integral is the MGF of Z at $s := \sqrt{2\lambda}\, v$, which is at most $e^{cs^2/2} = e^{c\lambda v^2}$ by sub-Gaussianity. Therefore

$$\mathbb{E}e^{\lambda Z^2} \leq \int e^{c\lambda v^2}\, \mathcal{N}(0,1)(dv) = (1 - 2c\lambda)^{-1/2}$$

by Lemma **??** with $\mu = 0$, $\sigma = 1$ and the parameter $c\lambda < 1/2$. Since the right-hand side is finite, $e^{\lambda Z^2}$ is integrable: all integrals above are Lebesgue integrals in $[0, \infty]$ of nonnegative functions, and `Integrable` is the conjunction of `AEStronglyMeasurable` (clear) and `HasFiniteIntegral`, which for a nonnegative function f is $\int^{-}$ ENNReal.ofReal$(f)\,d\mathbb{P} < \infty$ (Mathlib `hasFiniteIntegral_iff_ofReal`). $\square$

Lemma 265 (Fourth moment of a sub-Gaussian variable)**.** *ProbabilityTheory.HasSubgaussianMGF.integral$_{p}$ow* *GaussianMGFwithconstant* $c \geq 0$ *under a probability measure. Then* $\mathbb{E}Z^4 \leq 14c^2$*.*

Proof. lem:subgaussian$_{s}quare_{m}gf_{p}osIf$c$=0then$Z$=0a.s.$ $(Mathlib$`ae_eq_zero_of_hasSubgaussianMGF_zer` $> 0 and \lambda := 1/(4c)$, so that $2c\lambda = 1/2$ and Lemma **??** gives $\mathbb{E}e^{\lambda Z^2} \leq \sqrt{2}$. For $x \geq 0$, $1+x+x^2/2 \leq e^x$ (Mathlib `Real.quadratic_le_exp_of_nonneg`), so with $x = \lambda Z^2 \geq 0$, pointwise $\frac{\lambda^2}{2} Z^4 \leq e^{\lambda Z^2} - 1$. Taking expectations, $\frac{\lambda^2}{2}\mathbb{E}Z^4 \leq \sqrt{2} - 1$, i.e. $\mathbb{E}Z^4 \leq 32c^2(\sqrt{2} - 1) \leq 14c^2$ since $\sqrt{2} \leq 23/16$. $\square$

Lemma 266 (A quadratic upper bound for the exponential on the negative half-line)**.** *Real.exp$_{l}e_{o}ne_{a}dd_{a}dd_{s}q_{d}iv_{t}wo_{o}f_{n}onposForevery$* $x \leq 0$*,* $e^x \leq 1 + x + \frac{x^2}{2}$*.*

Proof. Let $y := -x \geq 0$. By Mathlib `Real.quadratic_le_exp_of_nonneg`, $e^y \geq 1+y+y^2/2 > 0$, so $e^x = 1/e^y \leq 1/(1+y+y^2/2)$. Finally $1/(1+y+y^2/2) \leq 1-y+y^2/2$, because $(1+y+y^2/2)(1-y+y^2/2) = (1+y^2/2)^2 - y^2 = 1+y^4/4 \geq 1$ and $1-y+y^2/2 = ((1-y)^2+1)/2 > 0$. Since $1-y+y^2/2 = 1+x+x^2/2$, this is the claim. $\square$

Lemma 267 (Fourth moment when the variance saturates)**.** *ProbabilityTheory.HasSubgaussianMGF.integral$_{p}o$* *GaussianMGFwithconstant* $c \geq 0$ *under a probability measure and assume* $\mathbb{E}Z^2 = c$*. Then* $\mathbb{E}Z^4 \leq 4c^2$*.*

Proof. lem:subgaussian$_{s}quare_{m}gf_{p}osIf$c$=0then$E$Z^2 = 0$ forces $Z = 0$ a.s. Let $c > 0$ and $\lambda := 1/(8c)$, so $2c\lambda = 1/4$. Keeping all three terms of $1+x+x^2/2 \leq e^x$ at $x = \lambda Z^2 \geq 0$ and taking expectations,

$$1 + \lambda c + \frac{\lambda^2}{2}\mathbb{E}Z^4 \leq \mathbb{E}e^{\lambda Z^2} \leq (1-2c\lambda)^{-1/2} = \tfrac{2}{\sqrt{3}} \leq \tfrac{500}{433},$$

by Lemma **??**. With $\lambda c = 1/8$ and $\lambda^2 = 1/(64c^2)$ this gives $\mathbb{E}Z^4 \leq 128c^2(\frac{500}{433} - \frac{9}{8}) \leq 4c^2$. The sharp constant is $3c^2$, the Gaussian value: letting $\lambda \to 0$ above gives $\mathbb{E}Z^4 \leq 3c^2$, which is also what matching the λ^4 coefficients of $\mathbb{E}e^{\lambda Z} \leq e^{c\lambda^2/2}$ at $\lambda = 0$ gives, so $\mathbb{E}Z^2 = c$ forces a kurtosis of at most 3. Discarding the linear term instead, as Lemma **??** does, loses a factor of more than three. $\square$

Lemma 268 (Centered square of a sub-Gaussian variable is sub-exponential)**.** *ProbabilityTheory.HasSubgaussianMGF.hasSubexponentialMGF$_{s}q_{s}ubdef$: $subexpLetZhaveasub-$* *GaussianMGFwithconstant* $c \geq 0$ *under a probability measure, and assume* $\mathbb{E}Z^2 = c$*. Then*

$$\mathrm{HasSubexponentialMGF}\big(Z^2 - c,\ 4c^2,\ 4c\big).$$

Proof. lem:subgaussian$_{s}quare_{m}gf_{p}os, lem : subgaussian_{f}ourth_{m}oment_{e}q, lem :$
$log_{i}neq_{n}u, lem : ps_{e}xp_{l}e_{q}uadraticZhasallmoments(Mathlib$ `integrable_pow_of_mem_interior_integrableE`
$1/(4c)$; we show that $e^{\lambda(Z^2-c)}$ is integrable and $\mathbb{E}e^{\lambda(Z^2-c)} \le e^{2c^2\lambda^2} = e^{\lambda^2 \cdot 4c^2/2}$.

Case $0 \le \lambda \le 1/(4c)$. Let $\nu := 2c\lambda \in [0, 1/2]$, so that $2c\lambda < 1$ and Lemma **??** gives the integrability of $e^{\lambda Z^2}$ (hence of $e^{\lambda(Z^2-c)} = e^{-\lambda c} e^{\lambda Z^2}$) and $\mathbb{E}e^{\lambda Z^2} \le (1-\nu)^{-1/2}$. Hence, using $\lambda c = \nu/2$ and Lemma **??**(i),

$$\log \mathbb{E}e^{\lambda(Z^2-c)} \le -\frac{1}{2}\log(1-\nu) - \frac{\nu}{2} \le \frac{\nu^2}{2} = 2c^2\lambda^2.$$

Case $-1/(4c) \le \lambda < 0$. Pointwise $\lambda Z^2 \le 0$, so $e^{\lambda(Z^2-c)} \le e^{-\lambda c}$ is bounded, hence integrable, and Lemma **??** gives $e^{\lambda Z^2} \le 1 + \lambda Z^2 + \frac{\lambda^2 Z^4}{2}$. Taking expectations and using $\mathbb{E}Z^2 = c$ and Lemma **??** ($\mathbb{E}Z^4 \le 4c^2$),

$$\mathbb{E}e^{\lambda Z^2} \le 1 + \lambda c + 2c^2\lambda^2 \le \exp\left(\lambda c + 2c^2\lambda^2\right),$$

by $1 + y \le e^y$. Multiplying by $e^{-\lambda c}$: $\mathbb{E}e^{\lambda(Z^2-c)} \le e^{2c^2\lambda^2}$.

In both cases $\mathbb{E}e^{\lambda(Z^2-c)} \le \exp(\lambda^2 \cdot 4c^2/2)$, which is the claim with $V = 4c^2$ and $b = 4c$. The parameters are close to optimal: $Z \sim \mathcal{N}(0, c)$ saturates Lemma **??**, and for it the smallest V that works on $|\lambda| \le 1/(4c)$ is $(16\log 2 - 8)c^2 \approx 3.09\, c^2$; moreover $V \ge 2c^2 = \mathrm{Var}(Z^2)$ for every b, and no V works for $b \le 2c$. □

Definition 269 (Rademacher vector)**.** ProbabilityTheory.signMeasure, ProbabilityTheory.rademacherMeasure, ProbabilityTheory.integral$_{s}ignMeasure, ProbabilityTheory.iIndepFun_{e}val$
is a random vector $\varepsilon = (\varepsilon_1, \dots, \varepsilon_d)$ whose coordinates are independent with $\mathbb{P}(\varepsilon_i = 1) = \mathbb{P}(\varepsilon_i = -1) = 1/2$; its law is the product $\left(\frac{1}{2}\delta_1 + \frac{1}{2}\delta_{-1}\right)^{\otimes d}$ (Lean: `signMeasure := 2 ¹ • (dirac 1 + dirac (-1))` and `rademacherMeasure := Measure.pi (fun _   signMeasure)` on → ; the coordinates are independent by `iIndepFun_pi`, each with law `signMeasure`, $\mathbb{E}\varepsilon_i = 0$, $\mathbb{E}\varepsilon_i^2 = 1$ and $|\varepsilon_i| \le 1$ a.s.). For $\theta \in \mathbb{R}^d$ we consider the linear form $Z := \langle \varepsilon, \theta \rangle = \sum_i \varepsilon_i \theta_i$, which satisfies $|Z| \le \sum_i |\theta_i|$.

Lemma 270 (Rademacher linear forms are sub-Gaussian)**.** *ProbabilityTheory.hasSubgaussianMGF*$_{s}um_{m}ul_{r}ade$
be a Rademacher vector in $\mathbb{R}^d$ *and* $\theta \in \mathbb{R}^d$. *Then* $\langle \varepsilon, \theta \rangle$ *has a sub-Gaussian MGF with constant* $\|\theta\|^2$ *(Mathlib* `HasSubgaussianMGF   ,     ‖ ‖ ²`*):* $\mathbb{E}e^{\lambda\langle \varepsilon, \theta \rangle} \le e^{\|\theta\|^2\lambda^2/2}$ *for all* $\lambda \in \mathbb{R}$.

Proof. def:ps$_{r}ademacherEach\theta_i\varepsilon_i$ takes values in $[-|\theta_i|, |\theta_i|]$ and has mean zero, so by Hoeffding's lemma (Mathlib `hasSubgaussianMGF_of_mem_Icc_of_integral_eq_zero`) it has a sub-Gaussian MGF with constant $((2|\theta_i|)/2)^2 = \theta_i^2$. The $\theta_i \varepsilon_i$ are independent, so `HasSubgaussianMGF.sum_of_iIndepFun` gives the constant $\sum_i \theta_i^2 = \|\theta\|^2$ for the sum. □

Lemma 271 (First and second moments of a Rademacher linear form)**.** *ProbabilityTheory.integral*$_{e}val_{m}ul_{e}val_{r}$
be a Rademacher vector in $\mathbb{R}^d$, $\theta \in \mathbb{R}^d$ *and* $Z := \langle \varepsilon, \theta \rangle$. *Then* $\mathbb{E}[\varepsilon_i \varepsilon_j] = \mathbf{1}\{i = j\}$, $\mathbb{E}Z = 0$ *and* $\mathbb{E}Z^2 = \|\theta\|^2$.

Proof. def:$\mathrm{ps}_r ademacher Allthevariablesinvolvedarebounded, henceintegrable.$ $\mathrm{E}[\varepsilon_i^2] = 1$ and, for $i \neq j$, $\mathbb{E}[\varepsilon_i\varepsilon_j] = \mathbb{E}\varepsilon_i\,\mathbb{E}\varepsilon_j = 0$ by independence. Then $\mathbb{E}Z = \sum_i \theta_i \mathbb{E}\varepsilon_i = 0$ and, expanding the square, $\mathbb{E}Z^2 = \sum_{i,j}\theta_i\theta_j\mathbb{E}[\varepsilon_i\varepsilon_j] = \sum_i \theta_i^2$. ☐

Lemma 272 (Centered square of a Rademacher linear form is sub-exponential)**.** *$ProbabilityTheory.hasSubexponentialMGF_sq_sum_mul_sub_rademacherMeasuredef : subexp, def : ps_rademacher$ be a Rademacher vector in $\mathbb{R}^d$, $\theta \in \mathbb{R}^d$ with $r := \|\theta\|$, and $Z := \langle \varepsilon, \theta\rangle$. Then*

$$\mathrm{HasSubexponentialMGF}\big(Z^2 - r^2,\ 4r^4,\ 4r^2\big).$$

Proof. lem:$\mathrm{rademacher}_l inear_s ubgaussian, lem : rademacher_f ourth_m oment, lem : subgaussian_s quare_s ubexpByLemma$ **??**,$Zissub-Gaussianwithconstant$c = r^2, and $\mathbb{E}Z^2 = r^2 = c$ by Lemma **??**. This is exactly the hypothesis of Lemma **??**, which gives the parameters $(16c^2, 4c) = (16r^4, 4r^2)$. ☐

Remark 273 (Constants, and the case $r = 0$)*.* The proof of Lemma **??** shows the stronger parameters $(8c^2, 4c)$ (the bound $e^{7c^2\lambda^2}$ for negative λ could be improved by using the exact fourth moment $\mathbb{E}Z^4 = 3r^4 - 2\sum_i \theta_i^4$ of a Rademacher form instead of the general bound $14c^2$, but this is not needed); the value $V = 16r^4$ is kept because it is the one used by Chapter **??** (Lemma **??**). The paper obtains parameters (Cr^4, Cr^2) with an unspecified C from the Hanson–Wright inequality (its Lemma *lem:hanson-wright*, Appendix A) and a tail-to-MGF conversion; the argument above is self-contained and explicit. If $r = 0$ then $Z = 0$ a.s. and $Z^2 - r^2 = 0$, which has parameters $(0, 0)$, consistent with the statement; the norm-estimation chapter treats this case separately anyway.

Lean remark. The Gaussian trick of Lemma **??** needs an auxiliary standard Gaussian independent of Z; it is encoded by Tonelli's theorem on the product measure `.prod (gaussianReal 0 1)`, without introducing a new random variable. The Rademacher results are stated for the product measure `rademacherMeasure` on `→` and the coordinate sum `i, i * i`; the identification with the inner product on `EuclideanSpace` is `inner_toLp_eq_sum`.

Chapter 13

Information-theoretic tools and the divergence decomposition

This chapter collects the information-theoretic facts used by the lower bounds of Chapters **??** and **??**: the Kullback–Leibler divergence of two Gaussians, the Pinsker and Bretagnolle–Huber inequalities, convexity of the divergence in its second argument, and, most importantly, the *divergence decomposition* for algorithm-environment sequences (the “chain rule” of the paper, Appendix A): the divergence between the laws of the histories produced by one algorithm against two stationary environments is the expected sum of the one-step divergences along the trajectory. The decomposition is stated for the LML framework, for arbitrary reward kernels, and the linear Gaussian case of the paper is a corollary.

What Mathlib/LML provides. Mathlib defines the Kullback–Leibler divergence `InformationTheory.klDiv` as an `0∞`-valued `irreducible_def` (`Mathlib/InformationTheory/KullbackLeibler/Basic.lean`): it equals $\int \mathrm{llr}(\mu,\nu)\,d\mu + \nu(\Omega) - \mu(\Omega)$ when $\mu \ll \nu$ and the log-likelihood ratio $\mathrm{llr}(\mu,\nu) = \log \frac{d\mu}{d\nu}$ is μ-integrable, and ∞ otherwise (`klDiv_of_ac_of_integrable`, `klDiv_of_not_ac`, `klDiv_of_not_integrable`). It is also the f-divergence $\int \mathrm{klFun}(\frac{d\mu}{d\nu})\,d\nu$ with $\mathrm{klFun}(x) = x\log x + 1 - x$ (`klDiv_eq_lintegral_klFun_of_ac`). The chain rule is available in the form `klDiv_compProd_eq_add`: $\mathrm{KL}(\mu\otimes\kappa \,\|\, \nu\otimes\eta) = \mathrm{KL}(\mu\|\nu) + \mathrm{KL}(\mu\otimes\kappa \,\|\, \mu\otimes\eta)$, together with `klDiv_compProd_left` ($\mathrm{KL}(\mu\otimes\kappa \,\|\, \nu\otimes\kappa) = \mathrm{KL}(\mu\|\nu)$), and the data-processing inequalities `klDiv_map_le`, `klDiv_trim_le`, `klDiv_comp_right_le`. The lower bound `mul_log_le_klDiv` and the converse Gibbs inequality `klDiv_eq_zero_iff` are also there. Radon–Nikodym derivatives of composition-products are in `Mathlib/Probability/Kernel/Composition/RadonNikodym.lean` (`rnDeriv_measure_compProd`) and kernel derivatives `Kernel.rnDeriv` are measurable (`Mathlib/Probability/Kernel/RadonNikodym.lean`, under the hypothesis `MeasurableSpace.CountableOrCountablyGenerated`, which is implied by

a countably generated target space); the same file proves that $\{x : \kappa_x \ll \eta_x\}$ is measurable (`Kernel.measurableSet_absolutelyContinuous`, via `Kernel.singularPart_eq_zero_iff_abso`
Real Gaussians are `ProbabilityTheory.gaussianReal` `v` with density `gaussianPDFReal` (`Mathlib/Probability/Distributions/Gaussian/Real.lean`), with `gaussianReal_absolutelyContinuous` `rnDeriv_gaussianReal`, `integral_id_gaussianReal`. LML provides the trajectory measure `Learning.trajMeasure alg env`, the step decomposition of histories (`history_succ`, `hasCondDistrib_step`, `hasCondDistrib_action`), stationary environments `stationaryEnv` (feedback kernel `feedback_stationaryEnv`: $(h, a) \mapsto \nu(a)$), and uniqueness of the law of the history (`eq_trajMeasure_map_frestrictLe_of_isAlgEnvSeqU`

What is missing. Mathlib has neither the divergence of two Gaussians, nor Pinsker's inequality, nor the Bretagnolle–Huber inequality, nor convexity of KL in its second argument, nor the "integrated" form of the chain rule $\mathrm{KL}(\mu \otimes \kappa \,\|\, \mu \otimes \eta) = \int \mathrm{KL}(\kappa_x \| \eta_x)\, \mu(dx)$ (the Mathlib file explicitly avoids it because $x \mapsto \mathrm{KL}(\kappa_x \| \eta_x)$ need not be measurable in general; it is measurable when the target space is countably generated, which is all we need). The integrated chain rule and the invariance of KL under measurable embeddings now live in LML (`LeanMachineLearning/ForMathlib/InformationTheory/KullbackLeibler/ChainRule.lean`); the policy/reward specialization of Lemma **??** and the divergence decomposition for algorithm-environment sequences are new to LML. Mathlib also lacks the behaviour of KL under restrictions (Lemma **??**), the invariance under sufficient statistics (Lemma **??**) and the continuity of KL along a generating sequence of σ-algebras (Lemma **??**), which the stopping-time and trajectory versions of the decomposition (Section **??**) rely on.

13.1 Basic inequalities

Lemma 274 (Invariance under measurable embeddings)**.** *InformationTheory.klDiv$_m$ap$_m$easurableEmbedding,* *be finite measures on Ω and $f : \Omega \to \Omega'$ a measurable embedding (for example a measurable equivalence). Then* $\mathrm{KL}(f_*\mu \,\|\, f_*\nu) = \mathrm{KL}(\mu \| \nu)$.

Proof. Two applications of the data-processing inequality (Mathlib `klDiv_map_le`): with f we get $\leq$, and with a measurable left inverse g of f applied to $f_*\mu, f_*\nu$ we get $\geq$, since $g_* f_* \mu = (g \circ f)_* \mu = \mu$. If Ω is empty then $\mu = \nu = 0$ and both sides vanish; otherwise such a g is given by `MeasurableEmbedding.invFun`. ☐

Lemma 275 (Divergence of two Gaussians with the same variance)**.** *ProbabilityTheory.gaussianReal$_a$bsolutelyC* *$\mathbb{R}$ and $v > 0$. Then $\mathcal{N}(m_1, v) \ll \mathcal{N}(m_2, v)$, the log-likelihood ratio is* $\mathrm{llr}(y) = \frac{(m_1 - m_2)}{v}\big(y - \frac{m_1 + m_2}{2}\big)$ *for Lebesgue-a.e. y, it is integrable, and*

$$\mathrm{KL}\big(\mathcal{N}(m_1, v) \,\big\|\, \mathcal{N}(m_2, v)\big) = \frac{(m_1 - m_2)^2}{2v} < \infty.$$

In particular $\mathrm{KL}(\mathcal{N}(m_1, 1) \| \mathcal{N}(m_2, 1)) = (m_1 - m_2)^2/2$.

Proof. lem:gauss$_1d_m$*omentsBothmeasuresare*Leb*withthedensities*$\mathrm{p}_i(y) = (2\pi v)^{-1/2} \exp(-(y - m_i)^2/(2v))$ (Mathlib `gaussianReal_of_var_ne_zero`), which are positive everywhere, so each is absolutely continuous with respect to the other and $\frac{d\mathcal{N}(m_1,v)}{d\mathcal{N}(m_2,v)} =$

p_1/p_2 a.e. (Mathlib `Measure.rnDeriv_withDensity` and the chain rule `Measure.rnDeriv_mul_rnDeriv` for Leb $\to \mathcal{N}(m_2, v) \to \mathcal{N}(m_1, v)$, or directly the uniqueness of the Radon–Nikodym derivative, `Measure.eq_rnDeriv`). Taking logarithms,

$$\log \frac{p_1(y)}{p_2(y)} = \frac{(y-m_2)^2-(y-m_1)^2}{2v} = \frac{(m_1-m_2)\,(2y-m_1-m_2)}{2v},$$

an affine function of y, hence integrable under $\mathcal{N}(m_1, v)$ (Gaussians have a first moment, Lemma **??**; Mathlib `integral_id_gaussianReal`). Its integral under $\mathcal{N}(m_1, v)$ is obtained by replacing y by its mean m_1: $\frac{(m_1-m_2)(2m_1-m_1-m_2)}{2v} = \frac{(m_1-m_2)^2}{2v}$. Conclude with Mathlib `klDiv_of_ac_of_integrable` (both measures are probability measures so the correction term $\nu(\Omega) - \mu(\Omega)$ vanishes). □

Lemma 276 (Divergence of two Bernoulli measures)**.** *InformationTheory.klBin,* $InformationTheory.klDiv_{b}ernoulliMeasure, InformationTheory.ofReal_{l}e_{k}lDiv_{b}ernoulliMeasure, Probabilit$
$\in [0,1]$ *write* $\mathrm{B}(p) := p\,\delta_1 + (1-p)\,\delta_0$ *for the Bernoulli measure on* $\{0,1\}$*. Let* $p \in [0,1]$ *and* $q \in (0,1)$*. Then*

$$\mathrm{KL}(\mathrm{B}(p) \,\|\, \mathrm{B}(q)) = p \log \frac{p}{q} + (1-p)\log\frac{1-p}{1-q} =: \mathrm{kl}(p,q),$$

with the convention $0 \log 0 = 0$*. If* $q \in \{0,1\}$ *and* $p \neq q$ *then* $\mathrm{KL}(\mathrm{B}(p)\|\mathrm{B}(q)) = \infty$*.*

Proof. For $q \in (0,1)$, $\mathrm{B}(q)$ charges both points so $\mathrm{B}(p) \ll \mathrm{B}(q)$, with Radon–Nikodym derivative p/q at 1 and $(1-p)/(1-q)$ at 0; every function on a finite set is integrable. By Mathlib `klDiv_eq_lintegral_klFun_of_ac`, the divergence is $q\,\mathrm{klFun}(p/q) + (1-q)\,\mathrm{klFun}((1-p)/(1-q))$ with $\mathrm{klFun}(x) = x \log x + 1 - x$, which expands to $p \log(p/q) + (1-p)\log((1-p)/(1-q)) + (q-p) + ((1-q)-(1-p)) = \mathrm{kl}(p,q)$; the convention $0 \log 0 = 0$ is exactly $\mathrm{klFun}(0) = 1$ (Mathlib `klFun_zero`). If $q = 0$ and $p > 0$ then $\mathrm{B}(p)(\{1\}) > 0 = \mathrm{B}(0)(\{1\})$ so $\mathrm{B}(p) \not\ll \mathrm{B}(q)$ and the divergence is ∞ (`klDiv_of_not_ac`); the case $q = 1$ is symmetric. □

Lemma 277 (Pinsker's inequality for Bernoulli measures)**.** $InformationTheory.klBin_{p}inskerlem : pi_{k}l_{b}ernoulli$
$\in [0,1]$ *and* $q \in (0,1)$*,* $\mathrm{kl}(p,q) \geq 2(p-q)^2$*.*

Proof. lem:pi$_k l_b ernoulliFix$p $\in (0,1)$ first and let $g(q) := \mathrm{kl}(p,q) - 2(p-q)^2$ on $(0,1)$. Then $g(p) = 0$ and

$$g'(q) = -\frac{p}{q} + \frac{1-p}{1-q} + 4(p-q) = \frac{q-p}{q(1-q)} - 4(q-p) = (q-p)\Big(\frac{1}{q(1-q)} - 4\Big).$$

Since $q(1-q) \leq 1/4$, the last factor is nonnegative, so $g' \leq 0$ on $(0,p]$ and $g' \geq 0$ on $[p,1)$: g is nonincreasing then nondecreasing (Mathlib `antitoneOn_of_deriv_nonpos`, `monotoneOn_of_deriv_nonneg`) and its minimum is $g(p) = 0$. For $p = 1$ the claim is $-\log q \geq 2(1-q)^2$: the function $h(q) := -\log q - 2(1-q)^2$ satisfies $h(1) = 0$ and $h'(q) = -1/q + 4(1-q) = -(2q-1)^2/q \leq 0$, so $h \geq 0$ on $(0,1]$. The case $p = 0$ is the case $p = 1$ with q replaced by $1-q$. □

Lemma 278 (Pinsker's inequality)**.** $InformationTheory.pinsker_measureReal, InformationTheory.abs_sub_le_sq$ *be probability measures on a measurable space and* A *a measurable set. Then*

$$2\left(\mu(A)-\nu(A)\right)^2 \le \mathrm{KL}(\mu\|\nu)$$

(as elements of $[0,\infty]$*). In particular, if* $\mathrm{KL}(\mu\|\nu)<\infty$ *then* $|\mu(A)-\nu(A)| \le \sqrt{\mathrm{KL}(\mu\|\nu)/2}$*, and if* $\mathrm{KL}(\mu\|\nu)=\infty$ *the statement is empty.*

Proof. lem:pi$_k l_b ernoulli, lem : kl_b ernoulli_p insker Let g := 1_A : \Omega \to \{0,1\}$, measurable. Then $g_*\mu = \mathrm{B}(p)$ and $g_*\nu = \mathrm{B}(q)$ with $p := \mu(A)$, $q := \nu(A)$, and the data-processing inequality (Mathlib `klDiv_map_le`) gives $\mathrm{KL}(\mathrm{B}(p)\|\mathrm{B}(q)) \le \mathrm{KL}(\mu\|\nu)$. If $p = q$ there is nothing to prove. If $q \in \{0,1\}$ and $p \ne q$, then $\mathrm{KL}(\mathrm{B}(p)\|\mathrm{B}(q)) = \infty$ by Lemma **??**, so $\mathrm{KL}(\mu\|\nu)=\infty$ and the inequality holds trivially. Otherwise $q \in (0,1)$ and Lemmas **??** and **??** give $2(p-q)^2 \le \mathrm{kl}(p,q) = \mathrm{KL}(\mathrm{B}(p)\|\mathrm{B}(q)) \le \mathrm{KL}(\mu\|\nu)$. ☐

Lemma 279 (Bretagnolle–Huber inequality)**.** $InformationTheory.bretagnolle_huberLet\mu,\nu$ *be probability measures on a measurable space and* A *a measurable set. Then*

$$\mu(A)+\nu(A^c) \ge \tfrac{1}{2}\exp\big(-\mathrm{KL}(\mu\|\nu)\big),$$

where the right-hand side is 0 *if* $\mathrm{KL}(\mu\|\nu)=\infty$*.*

Proof. If $\mathrm{KL}(\mu\|\nu)=\infty$ there is nothing to prove; assume $\mu \ll \nu$ and that $\mathrm{llr}(\mu,\nu)$ is μ-integrable, so that $\mathrm{KL}(\mu\|\nu) = \int \mathrm{llr}\,d\mu$ (Mathlib `klDiv_of_ac_of_integrable`). Let $\lambda := \mu+\nu$ and $p := d\mu/d\lambda$, $q := d\nu/d\lambda$, which exist (Mathlib `Measure.rnDeriv`, `withDensity_rnDeriv_eq` since $\mu,\nu \ll \lambda$) and satisfy $p+q = 1$ λ-a.e. (`Measure.rnDeriv_add`, `rnDeriv_self`), in particular $p, q \le 1$. The proof has three steps.

(i) $\mu(A)+\nu(A^c) = \int_A p\,d\lambda + \int_{A^c} q\,d\lambda \ge \int \min(p,q)\,d\lambda$.

(ii) By the Cauchy–Schwarz inequality in $L^2(\lambda)$ applied to $\sqrt{\min(p,q)}$ and $\sqrt{\max(p,q)}$, whose product is $\sqrt{pq}$,

$$\Big(\int \sqrt{pq}\,d\lambda\Big)^2 \le \int \min(p,q)\,d\lambda \cdot \int \max(p,q)\,d\lambda \le 2\int \min(p,q)\,d\lambda,$$

since $\max(p,q) \le p+q$ and $\int (p+q)\,d\lambda = \mu(\Omega)+\nu(\Omega) = 2$.

(iii) Since $\mu \ll \nu$, we have $p = 0$ λ-a.e. on $\{q=0\}$, and $\frac{d\mu}{d\nu} = p/q$ μ-a.e. (Mathlib `Measure.rnDeriv_mul_rnDeriv`: $\frac{d\mu}{d\nu}\cdot\frac{d\nu}{d\lambda} = \frac{d\mu}{d\lambda}$ λ-a.e.). Hence

$$\int \sqrt{pq}\,d\lambda = \int_{\{p>0\}} \sqrt{q/p}\,p\,d\lambda = \int \sqrt{q/p}\,d\mu = \int \exp\left(-\tfrac{1}{2}\,\mathrm{llr}(\mu,\nu)\right)d\mu \ge \exp\Big(-\tfrac{1}{2}\int \mathrm{llr}(\mu,\nu)\,d\mu\Big) = e^{-\mathrm{K}}$$

by Jensen's inequality for the convex function exp and the probability measure μ (Mathlib `ConvexOn.map_integral_le`, with llr integrable).

Combining, $\mu(A)+\nu(A^c) \ge \int \min(p,q)\,d\lambda \ge \frac{1}{2}(\int \sqrt{pq}\,d\lambda)^2 \ge \frac{1}{2}e^{-\mathrm{KL}(\mu\|\nu)}$. This is Theorem 14.2 of [**?**]; see also [**?**]. ☐

Lemma 280 (Convexity of the divergence in its second argument)**.** *$MeasureTheory.isProbabilityMeasure_sum_sm$ be a probability measure, $\nu_1, \dots, \nu_J$ probability measures on the same space, $c_1, \dots, c_J \geq 0$ with $\sum_j c_j = 1$, and $\nu := \sum_j c_j \nu_j$. Then*

$$\mathrm{KL}(\mu \| \nu) \leq \sum_{j=1}^{J} c_j \, \mathrm{KL}(\mu \| \nu_j) \quad \textit{in } [0, \infty].$$

Proof. If some j with $c_j > 0$ has $\mathrm{KL}(\mu \| \nu_j) = \infty$ the right-hand side is ∞. Otherwise, for every j with $c_j > 0$ we have $\mu \ll \nu_j$ and $\mathrm{llr}(\mu, \nu_j)$ is μ-integrable; drop the indices with $c_j = 0$. Then $\mu \ll \nu$ (a ν-null set is ν_j-null for all j). Let $g_j := d\nu_j / d\mu$ and $g := d\nu / d\mu$ (Radon–Nikodym derivatives with respect to μ; they exist for σ-finite measures without absolute continuity, as the density of the absolutely continuous part). By linearity of the Lebesgue decomposition, $g = \sum_j c_j g_j$ μ-a.e. (Mathlib `Measure.rnDeriv_add`, `Measure.rnDeriv_smul_left`). Since $\mu \ll \nu_j$, Mathlib `Measure.inv_rnDeriv` gives $g_j = (d\mu/d\nu_j)^{-1}$ μ-a.e., and $d\mu/d\nu_j$ is finite μ-a.e. (`Measure.rnDeriv_lt_top`) and positive μ-a.e. (`Measure.rnDeriv_pos`), so $0 < g_j < \infty$ μ-a.e. and $\mathrm{llr}(\mu, \nu_j) = -\log g_j$ μ-a.e.; likewise $\mathrm{llr}(\mu, \nu) = -\log g$. By concavity of log on $(0, \infty)$ (Mathlib `strictConcaveOn_log_Ioi`, `ConcaveOn.le_map_sum`),

$$\mathrm{llr}(\mu, \nu) = -\log \Big(\sum_j c_j g_j \Big) \leq \sum_j c_j (-\log g_j) = \sum_j c_j \, \mathrm{llr}(\mu, \nu_j) \qquad \mu\text{-a.e.}$$

Integrability of $\mathrm{llr}(\mu, \nu)$: its positive part is dominated by the integrable $\sum_j c_j \, \mathrm{llr}(\mu, \nu_j)^+$, and its negative part is always integrable for probability measures, because $\log^-(d\mu/d\nu) = \log^+(g) \leq g$ and $\int g \, d\mu \leq \nu(\Omega) = 1$ (Mathlib `Measure.lintegral_rnDeriv_le`). Integrating the inequality gives the claim, using `klDiv_of_ac_of_integrable` on both sides (all measures are probability measures). $\square$

Lean remark. In Mathlib, KL takes values in `0∞`; the statements above are therefore stated in $[0, \infty]$ with `ENNReal.ofReal` on the real side (e.g. `ENNReal.ofReal (2 * ( .real A -  .real A)^2)   klDiv    `), and their real versions are obtained through `klDiv_ne_top_iff` and `ENNReal.toReal`. The mixture in Lemma **??** is the measure ` j, (c j :   0) •   j`.

13.2 The integrated chain rule

Lemma 281 (Chain rule with an integrated conditional divergence)**.** *$InformationTheory.measurable_klDiv_kerne$ be a finite measure on $\mathcal{H}$ and κ, η be Markov kernels from $\mathcal{H}$ to a countably generated measurable space $\mathcal{Y}$. Then $x \mapsto \mathrm{KL}(\kappa_x \| \eta_x)$ is measurable and*

$$\mathrm{KL}(\mu \otimes \kappa \,\|\, \mu \otimes \eta) = \int^{-} \mathrm{KL}(\kappa_x \| \eta_x) \, \mu(dx),$$

where $\int^{-}$ denotes the Lebesgue integral of a $[0, \infty]$-valued function. Consequently, for any finite measure ν on $\mathcal{H}$, $\mathrm{KL}(\mu \otimes \kappa \,\|\, \nu \otimes \eta) = \mathrm{KL}(\mu \| \nu) + \int^{-} \mathrm{KL}(\kappa_x \| \eta_x) \, \mu(dx)$.

Proof. Mathlib `Measure.absolutelyContinuous_compProd_right_iff` gives $\mu \otimes \kappa \ll \mu \otimes \eta$ iff $\kappa_x \ll \eta_x$ for μ-a.e. x.

Measurability. Mathlib's kernel Radon–Nikodym theory (`Mathlib/Probability/Kernel/RadonNikodym.le` is developed under the hypothesis `MeasurableSpace.CountableOrCountablyGenerated` , which is implied by `CountablyGenerated` (an instance). Under it, the kernel Radon–Nikodym derivative $\frac{d\kappa}{d\eta} : \mathcal{H} \times \mathcal{Y} \to [0, \infty]$ is jointly measurable (`Kernel.rnDeriv`, `Kernel.measurable_rnDeriv`) and $\frac{d\kappa}{d\eta}(x, \cdot) = \frac{d\kappa_x}{d\eta_x}$ η_x-a.e. for every x (`Kernel.rnDeriv_eq_rnDeriv_measure`). The set $\{x : \kappa_x \ll \eta_x\}$ is measurable: this is exactly Mathlib `Kernel.measurableSet_absolutelyContinuous` (for finite kernels), whose proof rewrites the condition as $\kappa.\mathrm{singularPart}(\eta)(x) = 0$ by `Kernel.singularPart_eq_zero_iff_absolutelyContinuous` and uses the measurability of $x \mapsto \kappa.\mathrm{singularPart}(\eta)(x)(\mathcal{Y})$. On this set, $\mathrm{KL}(\kappa_x \| \eta_x) = \int^{-} \mathrm{klFun}\big(\frac{d\kappa}{d\eta}(x, y)\big)\, \eta_x(dy)$ (`klDiv_eq_lintegral_klFun_of_ac`), which is measurable in x by `Measurable.lintegral_kernel_prod_right`; off this set it is the constant ∞.

Equality. If $\mu \otimes \kappa \not\ll \mu \otimes \eta$, then $\kappa_x \not\ll \eta_x$ on a set of positive μ-measure, where $\mathrm{KL}(\kappa_x \| \eta_x) = \infty$: both sides are ∞. Otherwise, by `klDiv_eq_lintegral_klFun_of_ac` and the chain rule for derivatives of composition-products (`rnDeriv_measure_compProd`, again under `CountableOrCountablyGenerated`: $\frac{d(\mu \otimes \kappa)}{d(\mu \otimes \eta)}(x, y) = \frac{d\kappa}{d\eta}(x, y)$ for $(\mu \otimes \eta)$-a.e. (x, y)),

$$\mathrm{KL}(\mu \otimes \kappa \,\|\, \mu \otimes \eta) = \int^{-} \mathrm{klFun}\Big(\frac{d\kappa}{d\eta}(x, y)\Big)\, (\mu \otimes \eta)(d(x, y)) = \int^{-}\!\!\int^{-} \mathrm{klFun}\Big(\frac{d\kappa}{d\eta}(x, y)\Big)\, \eta_x(dy)\, \mu(dx)$$

by Tonelli for composition-products (`Measure.lintegral_compProd`), and the inner integral is $\mathrm{KL}(\kappa_x \| \eta_x)$ for μ-a.e. x (those with $\kappa_x \ll \eta_x$). The last claim is Mathlib `klDiv_compProd_eq_add` followed by the equality just proved. $\square$

Lemma 282 (Sufficient statistics and the divergence)**.** $InformationTheory.klDiv_map_of_leftInverse, Informatio$ *be finite measures on* Ω.

- *(i) If* $g : \Omega \to \Omega'$ *is measurable with a measurable left inverse, then* $\mathrm{KL}(g_*\mu \| g_*\nu) = \mathrm{KL}(\mu \| \nu)$.
- *(ii) If* $\mu = (f \circ g) \cdot \nu$ *for a measurable* $g : \Omega \to \Omega'$ *and a measurable* $f : \Omega' \to [0, \infty]$ *(the density factors through* g*, i.e.* g *is a sufficient statistic), then* $\mathrm{KL}(g_*\mu \| g_*\nu) = \mathrm{KL}(\mu \| \nu)$.
- *(iii) Let* κ, η *be finite kernels from* $\mathcal{B}$ *to* $\mathcal{C}$ *and* $f : \Omega \to \mathcal{B}$ *measurable. Then* $\mathrm{KL}(\mu \otimes (\kappa \circ f) \,\|\, \mu \otimes (\eta \circ f)) = \mathrm{KL}(f_*\mu \otimes \kappa \,\|\, f_*\mu \otimes \eta)$*: the conditional divergence of two kernels which depend on the conditioning variable only through* f *is the conditional divergence given* f*.*

No countable generation hypothesis is needed.

Proof. lem:pi$_k l_m easurable_e quiv (i) is two applications of the data-processing inequality (as for measurable embec$ $f \cdot g_*\nu$ (change of variables), so both pairs are absolutely continuous with densities $f \circ g$ and f, and the klFun integrals agree by the change of variables g.

(iii): if $f_*\mu \otimes \kappa \not\ll f_*\mu \otimes \eta$ then, transporting along $(\omega, c) \mapsto (f(\omega), c)$, neither is the left-hand pair, and both divergences are infinite. Otherwise let D be the density of $f_*\mu \otimes \kappa$ with respect to $f_*\mu \otimes \eta$; the sections of D integrate to $\kappa(b, t)$ for $f_*\mu$-almost every b, for each fixed measurable t, so $\mu \otimes (\kappa \circ f)$ and $(D \circ (f \times \mathrm{id})) \cdot (\mu \otimes (\eta \circ f))$ agree on rectangles, hence everywhere, and (ii) applies with $g = f \times \mathrm{id}$. ☐

Lemma 283 (One step of an algorithm-environment pair)**.** *ProbabilityTheory.Kernel.compProd$_p$rodMkLeft$_a$pp a Markov kernel from $\mathcal{H}$ to a countably generated space $\mathcal{A}$ ("policy"), and κ, κ' Markov kernels from $\mathcal{A}$ to a countably generated space $\mathcal{Y}$ ("reward kernels"). Write $\tilde\kappa$ for the kernel $(h, a) \mapsto \kappa(a)$ from $\mathcal{H} \times \mathcal{A}$ to $\mathcal{Y}$ (LML/Mathlib `Kernel.prodMkLeft`), and similarly $\tilde\kappa'$. Then*

$$\mathrm{KL}\big(P\otimes(\pi\otimes\tilde\kappa) \,\big\|\, P\otimes(\pi\otimes\tilde\kappa')\big) = \int^- \mathrm{KL}(\kappa(a)\|\kappa'(a))\,(\pi\circ P)(da) = \int^-\!\!\int^- \mathrm{KL}(\kappa(a)\|\kappa'(a))\,\pi(h, da)\,P(dh),$$

where $\pi \circ P := \int \pi(h, \cdot)\, P(dh)$ is the law of the next action. In composition-product form, without any countable generation hypothesis, $\mathrm{KL}\big(P\otimes(\pi\otimes\tilde\kappa) \,\big\|\, P\otimes (\pi \otimes \tilde\kappa')\big) = \mathrm{KL}\big((\pi \circ P) \otimes \kappa \,\big\|\, (\pi \circ P) \otimes \kappa'\big)$*.*

Proof. lem:kl$_c$*ompProd$_k$ernelApplyLemma* **??***to*P*andthekernels*$\pi \otimes \tilde\kappa$, $\pi \otimes \tilde\kappa'$, Markov kernels from $\mathcal{H}$ to the countably generated space $\mathcal{A} \times \mathcal{Y}$ (Mathlib `MeasurableSpace.CountablyGenerated` is stable under products). For each h, the measure $(\pi \otimes \tilde\kappa)(h)$ is the composition-product $\pi(h) \otimes \kappa$ of the measure $\pi(h)$ on $\mathcal{A}$ with the kernel κ (Mathlib `Kernel.compProd_apply` and `Kernel.prodMkLeft_apply`). By `klDiv_compProd_eq_add`, `klDiv_self` and Lemma **??** again,

$$\mathrm{KL}\big(\pi(h)\otimes\kappa \,\big\|\, \pi(h)\otimes\kappa'\big) = \mathrm{KL}(\pi(h)\|\pi(h)) + \int^- \mathrm{KL}(\kappa(a)\|\kappa'(a))\,\pi(h, da) = \int^- \mathrm{KL}(\kappa(a)\|\kappa'(a))\,\pi(h, da).$$

Integrating in h and using $\int^-\!\int^- f(a)\,\pi(h, da)\,P(dh) = \int^- f\, d(\pi \circ P)$ (Mathlib `Measure.lintegral_bind` / `Kernel.lintegral_comp`) gives the claim. ☐

13.3 The divergence decomposition

Definition 284 (A pair of stationary environments and the laws of histories)**.** Let $\mathcal{A}$ (actions) and $\mathcal{Y}$ (observations) be countably generated measurable spaces (for instance Borel subsets of $\mathbb{R}^d$, $\mathbb{R}$, or finite types; in particular $\mathcal{A}$ may be any action set $\mathcal{X}$ in the sense of Definition **??**, i.e. a nonempty compact subset of $\mathbb{R}^d$, spanning or not). Let κ, κ' be Markov kernels from $\mathcal{A}$ to $\mathcal{Y}$ and let $\nu := \mathrm{stationaryEnv}(\kappa)$, $\nu' := \mathrm{stationaryEnv}(\kappa')$ be the corresponding stationary environments (LML `Learning.stationaryEnv`: the observation at every round has law $\kappa(x_t)$ given the past and the action x_t). Let alg be an LML algorithm with actions in $\mathcal{A}$ and observations in $\mathcal{Y}$, with policy kernels π_n (from histories $(x_t, y_t)_{t\le n}$ to $\mathcal{A}$) and initial law p_0. For $n \ge 0$ let

$$\mathbb{P}^{(n)} := \mathrm{law}\big((x_t, y_t)_{t\le n}\big) = (\mathrm{trajMeasure}(\mathrm{alg}, \nu))_*\mathrm{hist}_n, \qquad \mathbb{P}'^{(n)} := (\mathrm{trajMeasure}(\mathrm{alg}, \nu'))_*\mathrm{hist}_n,$$

the laws of the history up to time n (a history of *length* $T = n + 1$) under the two environments. We write $\mathbb{E}^{(n)}$ for the expectation under $\mathbb{P}^{(n)}$ and, for $T \geq 1$, $\mathbb{P}^T := \mathbb{P}^{(T-1)}$ for the law of a history of length T. By LML `eq_trajMeasure_map_frestrictLe_of_isAlgEnvSeqUntil`, $\mathbb{P}^{(n)}$ is the law of $(x_t, y_t)_{t \leq n}$ for *any* algorithm-environment sequence of (alg, ν) on any probability space.

Theorem 285 (Divergence decomposition)**.** $Learning.IsAlgEnvSeq.map_h istory_s ucc, Learning.IsAlgEnvSeq.k$ ≥ 0,

$$\mathrm{KL}\big(\mathbb{P}^{(n)} \,\big\|\, \mathbb{P}'^{(n)}\big) = \sum_{t=0}^{n} \int^{-} \mathrm{KL}\big(\kappa(x_t) \,\big\|\, \kappa'(x_t)\big)\, d\mathbb{P}^{(n)} \qquad \textit{in } [0, \infty],$$

where x_t denotes the t-th action of the history. Equivalently, for a history of length $T \geq 1$, $\mathrm{KL}(\mathbb{P}^T \| \mathbb{P}'^T) = \sum_{t<T} \mathbb{E}^{(T-1)}[\mathrm{KL}(\kappa(x_t) \| \kappa'(x_t))]$. If moreover $\mathrm{KL}(\kappa(a) \| \kappa'(a)) \leq C < \infty$ for all $a \in \mathcal{A}$, then $\mathrm{KL}(\mathbb{P}^T \| \mathbb{P}'^T) \leq TC < \infty$ and the identity holds in $\mathbb{R}$ with ordinary expectations.

Proof. lem:pi$_k l_m easurable_e quiv, lem : kl_c ompProd_k ernel, lem : pi_k l_s tep, lem :$ $stopped_h istory_c hain_r uleIntheformalizationthetheoremisthespecialcase \mathrm{S} = \emptyset$ (no stopping) of the chain rule for stopped histories, Lemma **??**: the stopping time is $+\infty$, the history stopped at $\min(\tau, T)$ is the history of the first T rounds, and the one-step terms are those of Lemma **??**. We nevertheless record the direct induction, which is the argument of the paper.

Write $\mathcal{H}_n := (\{0, \ldots, n\} \to \mathcal{A} \times \mathcal{Y})$ for the space of histories up to time n. By definition of the stationary environment, the feedback kernel at every time is $\tilde{\kappa} : (h, a) \mapsto \kappa(a)$ (LML `feedback_stationaryEnv`, `0_stationaryEnv`). We prove the identity by induction on n.

Base case $n = 0$. The pair (x_0, y_0) has law $p_0 \otimes \kappa$ under ν and $p_0 \otimes \kappa'$ under ν' (LML `hasLaw_step_zero`), and hist_0 is the image of (x_0, y_0) by the measurable equivalence $\mathcal{A} \times \mathcal{Y} \simeq \mathcal{H}_0$. By Lemma **??**, Mathlib `klDiv_compProd_eq_add`, `klDiv_self` and Lemma **??**,

$$\mathrm{KL}(\mathbb{P}^{(0)} \| \mathbb{P}'^{(0)}) = \mathrm{KL}(p_0 \otimes \kappa \,\|\, p_0 \otimes \kappa') = 0 + \int^{-} \mathrm{KL}(\kappa(a) \| \kappa'(a))\, p_0(da) = \int^{-} \mathrm{KL}(\kappa(x_0) \| \kappa'(x_0))\, d\mathbb{P}^{(0)},$$

the last step because $x_0 \sim p_0$ under $\mathbb{P}^{(0)}$ (`hasLaw_action_zero`) and the integrand only depends on x_0 (change of variables for the Lebesgue integral, `lintegral_map`).

Induction step. Assume the identity for n. By LML `history_succ` and `hasCondDistrib_step`, $\mathrm{hist}_{n+1} = e^{-1}(\mathrm{hist}_n, (x_{n+1}, y_{n+1}))$ for the measurable equivalence $e : \mathcal{H}_{n+1} \simeq \mathcal{H}_n \times (\mathcal{A} \times \mathcal{Y})$ (`MeasurableEquiv.IicSuccProd`), and the law of $(\mathrm{hist}_n, (x_{n+1}, y_{n+1}))$ is the composition-product $\mathbb{P}^{(n)} \otimes (\pi_n \otimes \tilde{\kappa})$ (the step kernel `stepKernel alg env n = alg.policy n    env.feedback n`); similarly with primes, with the *same* policy kernel π_n. Hence, by Lemma **??**, `klDiv_compProd_eq_add` and Lemma **??**,

$$
\begin{aligned}
\mathrm{KL}(\mathbb{P}^{(n+1)} \| \mathbb{P}'^{(n+1)}) &= \mathrm{KL}\big(\mathbb{P}^{(n)} \otimes (\pi_n \otimes \tilde{\kappa}) \,\big\|\, \mathbb{P}'^{(n)} \otimes (\pi_n \otimes \tilde{\kappa}')\big) \\
&= \mathrm{KL}(\mathbb{P}^{(n)} \| \mathbb{P}'^{(n)}) + \mathrm{KL}\big(\mathbb{P}^{(n)} \otimes (\pi_n \otimes \tilde{\kappa}) \,\big\|\, \mathbb{P}^{(n)} \otimes (\pi_n \otimes \tilde{\kappa}')\big) \\
&= \sum_{t=0}^{n} \int^{-} \mathrm{KL}(\kappa(x_t) \| \kappa'(x_t))\, d\mathbb{P}^{(n)} + \int^{-} \mathrm{KL}(\kappa(a) \| \kappa'(a))\, (\pi_n \circ \mathbb{P}^{(n)})(da).
\end{aligned}
$$

Finally $\pi_n \circ \mathbb{P}^{(n)}$ is the law of x_{n+1} under $\mathbb{P}^{(n+1)}$ (LML `hasCondDistrib_action`: the conditional law of x_{n+1} given hist_n is $\pi_n(\mathrm{hist}_n)$), and $\mathbb{P}^{(n)}$ is the image of $\mathbb{P}^{(n+1)}$ under the restriction $\mathcal{H}_{n+1} \to \mathcal{H}_n$ (consistency of the laws of histories, `hist_eq_frestrictLe`), so every term is an integral under $\mathbb{P}^{(n+1)}$ of a function of the history, and the sum runs over $t \le n+1$.

Finite case. If the one-step divergences are bounded by C, the right-hand side is at most $(n+1)C$, so all quantities are finite and `ENNReal.toReal` turns the identity into a real identity, the integrals becoming Bochner integrals of bounded measurable functions (`integral_eq_lintegral_of_nonneg_ae`). □

Lean remark. The formalization (`Learning.IsAlgEnvSeq.klDiv_map_history`) states the theorem for two algorithm-environment sequences `IsAlgEnvSeq X Y alg (stationaryEnv  ) P` and `IsAlgEnvSeq X' Y' alg (stationaryEnv  ') P'` on arbitrary probability spaces, as `klDiv (P.map (history X Y n)) (P'.map (history X' Y' n)) =   t   range (n + 1),    , klDiv (  (X t  )) ( ' (X t  ))  P`; this covers the trajectory measures (which are such sequences) and avoids the uniqueness-of-law step. The finite form is only recorded for the Gaussian case (Corollary **??**). Every conditional divergence is recorded in two forms: the *integral form* above, which needs the hypothesis `MeasurableSpace.CountablyGenerated` for the measurability of the integrands (Lemma **??**; it holds for $\mathcal{Y} = \mathbb{R}$), and the *composition-product form* `  t   range (n + 1), klDiv (P.map (X t) ) (P.map (X t)    ')` (`klDiv_map_history_compProd`), valid on arbitrary measurable spaces. Versions indexed by the number of rounds played (`finHistory X Y M`, the history of the first M rounds) are `klDiv_map_finHistory` and `klDiv_map_finHistory_compProd`, and the chain rule for two arbitrary environments, with the conditional divergences of the step kernels, is `klDiv_map_finHistory_stepKernelFin`. Only the environment changes: the policy kernels are shared, which is what makes `klDiv_compProd_left` (inside Lemma **??**) applicable. This is the environment analogue of LML's `Algorithm.density`, and of Lemma **??** below, which give the Radon–Nikodym derivative rather than its expected logarithm.

Lemma 286 (Appending the recommendation does not change the divergence)**.** *Learning.IdentAlg.IsFixedBudget.stoppingTime*$_e$*q*, *Learning.IdentAlg.IsFixedBudget.stoppedHist*$_e$*q*, *Learnin* ≥ 1 *and let* ρ *be a Markov kernel from histories of length* T *to a measurable space* $\mathcal{X}$ *(a recommendation rule). Then*

$$\mathrm{KL}\big(\mathbb{P}^T \otimes \rho \,\big\|\, \mathbb{P}'^T \otimes \rho\big) = \mathrm{KL}(\mathbb{P}^T \| \mathbb{P}'^T),$$

and for every measurable set E *of pairs (history, recommendation),*

$$(\mathbb{P}^T \otimes \rho)(E) - (\mathbb{P}'^T \otimes \rho)(E) \le \sqrt{\mathrm{KL}(\mathbb{P}^T \| \mathbb{P}'^T)/2}, \qquad (\mathbb{P}^T \otimes \rho)(E) + (\mathbb{P}'^T \otimes \rho)(E^c) \ge \tfrac{1}{2} \exp\big(-\mathrm{KL}(\mathbb{P}^T \| \mathbb{P}'^T)\big).$$

*When the environments are those of Definition **??** on an arbitrary action set $\mathcal{X}$ (nonempty compact, not assumed spanning) and* (alg, ρ) *is an identification algorithm (Definition **??**),* $\mathbb{P}^T \otimes \rho$ *is the law* $\mathbb{P}_\theta$ *of* $(H_T, \hat{x})$*.*

Proof. lem:pinsker, lem:bretagnolle$_h uber The equality is Mathlib$`klDiv_compProd_left`$(same kernel on both side$
ρ, $\mathbb{P}'^T \otimes \rho$ (trivial if the divergence is infinite). The last sentence is the definition of $\mathbb{P}_\theta$ in Definition **??** together with the uniqueness of the law of the history. □

Lean remark. In the formalization the recommendation is the output of a run (Definition **??**, `IsRun`): for a fixed-budget algorithm the stopping time is T and the history at the stopping time is the history of the T rounds (`IsFixedBudget.stoppingTime_eq`, `IsFixedBudget.stoppedHist_eq`); the law of (history, output) is the composition-product with the output kernel, which is a probability kernel on the stopping set, so `klDiv_compProd_left` applies in the form `klDiv_compProd_left_of_ae`. Only the inequality $\mathrm{KL}(\mathrm{law}(\hat{x}) \| \mathrm{law}'(\hat{x})) \le \mathrm{KL}(\mathbb{P}^T \| \mathbb{P}'^T)$ is recorded (`IsRun.klDiv_map_out_le`, by data processing), which is what the lower bounds use, and the Bretagnolle–Huber consequence for linear Gaussian environments, $\mathbb{P}_\theta(\hat{x} \in B) + \mathbb{P}_{\theta'}(\hat{x} \notin B) \ge \frac{1}{2}\exp(-TR^2\|\theta - \theta'\|^2/2)$, is `LinearBandit.IsRun.exp_neg_le_measureReal_add`.

13.3.1 Stopped histories and infinite trajectories

The paper only needs the divergence decomposition for a fixed number of rounds. For the library we also prove it for histories stopped at a stopping time (the change-of-measure lemma of fixed-confidence lower bounds, Kaufmann, Cappé and Garivier 2016) and for whole trajectories. The results below are formalized in `Maiti2026Power/LeanMachineLearning/{FinHistory, StoppedHistory, DivergenceDecomposition, RunDivergence}.lean` and `Maiti2026Power/Mathlib/Infor`
`KLRestrict, KLFiltration}.lean`.

Definition 287 (Histories of the first n rounds, step kernels)**.** Learning.finHistory, Learning.toFinHistory, Learning.ofFinHistory, Learning.finHistoryEquiv, MeasurableEquiv.finSnoc, Learning.Algorithm.policyFin, Learning.Environment.feedbackFin, Learning.stepKernelFin, Learning.IsAlgEnvSeq.hasCondDistrib$_a ction_f in History, Learning.IsAlgEnvSeq.has$
$Y on \Omega$, the *history of the first n rounds* is $\mathrm{finHist}_n := (X_t, Y_t)_{t<n} : \Omega \to (\{0, \dots, n-1\} \to \mathcal{A} \times \mathcal{Y})$ ($n = 0$ allowed); it is a measurable bijection of the LML history up to time $n-1$ when $n \ge 1$, and $\mathrm{finHist}_{n+1} = (\mathrm{finHist}_n, (X_n, Y_n))$ (`Fin.snoc`). For an algorithm alg and an environment ν, the *policy at round n* π_n^{fin} is the kernel from histories of n rounds to actions equal to p_0 for $n = 0$ and to the LML policy π_{n-1} otherwise; the *feedback kernel at round n* $\tilde{\nu}_n$ is ν_0 for $n = 0$ and the LML feedback kernel otherwise; the *step kernel at round n* is $\sigma_n := \pi_n^{\mathrm{fin}} \otimes \tilde{\nu}_n$, a Markov kernel from histories of n rounds to $\mathcal{A} \times \mathcal{Y}$. For an algorithm-environment sequence, X_n, Y_n and (X_n, Y_n) have conditional laws π_n^{fin}, $\tilde{\nu}_n$ and σ_n given $\mathrm{finHist}_n$ (and X_n), and the law of $\mathrm{finHist}_{n+1}$ is $\mathrm{law}(\mathrm{finHist}_n) \otimes \sigma_n$. For a stationary environment with reward kernel κ, $\tilde{\nu}_n = \tilde{\kappa} : (h, a) \mapsto \kappa(a)$ at every round.

Definition 288 (Stopping rules, stopping times, stopped histories)**.** Learning.stoppingTime, Learning.stoppedHist, Learning.stoppingTime$_{le}$$_i$$ff$, $Learning.lt_stoppingTime_iff, Learning$
$\subseteq \bigsqcup_n(\{0,\dots,n-1\} \to \mathcal{A} \times \mathcal{Y})$: the interaction stops after n rounds if the history of these n rounds belongs to S. Its *stopping time* $\tau_S := \inf\{n : \mathrm{finHist}_n \in S\} \in \mathbb{N} \cup \{\infty\}$ (Mathlib `hittingAfter`) is the number of rounds played; it is a stopping time of the history filtration of any algorithm-environment sequence, $\{n < \tau_S\}$ is determined by $\mathrm{finHist}_n$, and $\tau_S = \infty$ for $S = \emptyset$. For a random time τ, the *stopped history* $\mathrm{finHist}_\tau := \langle \tau, \mathrm{finHist}_\tau \rangle$ is the history of the first τ rounds as a history of variable length (of length 0 if $\tau = \infty$); it belongs to S when $\tau = \tau_S < \infty$. The *truncation* trunc_M keeps the first M rounds of a history of variable length: $\mathrm{trunc}_M(\mathrm{finHist}_\tau) = \mathrm{finHist}_{\min(\tau,M)}$ when $\tau < \infty$, so that the law of $\mathrm{finHist}_{\min(\tau,M)}$ is the image of the law of $\mathrm{finHist}_\tau$ by trunc_M when τ is almost surely finite, and both laws agree on histories of length $< M$. The stopping time and stopped history of an identification algorithm (Definition **??**) are those of its stopping rule.

Lemma 289 (Restrictions and the divergence)**.** $MeasureTheory.Measure.rnDeriv_restrict_restrict, Information$
be finite measures on Ω *and* s *a measurable set. Then* $\mathrm{KL}(\mu\|\nu) = \mathrm{KL}(\mu|_s \,\|\, \nu|_s) + \mathrm{KL}(\mu|_{s^c} \,\|\, \nu|_{s^c})$*; in particular* $\mathrm{KL}(\mu|_s\|\nu|_s) \le \mathrm{KL}(\mu\|\nu)$*, and the divergence of two sums of measures carried by* s *and* s^c *respectively is the sum of the divergences of the two parts. If* $s_n \uparrow \Omega$ *is an increasing sequence of measurable sets,* $\mathrm{KL}(\mu\|\nu) = \sup_n \mathrm{KL}(\mu|_{s_n}\|\nu|_{s_n})$.

Proof. The Radon–Nikodym derivative of $\mu|_s$ with respect to $\nu|_s$ is that of μ with respect to ν (both are $\nu|_s$-densities of $\mu|_s = (\nu|_s) \cdot \frac{d\mu}{d\nu}$), so when $\mu \ll \nu$ the identities are those of the integral $\int \mathrm{klFun}(d\mu/d\nu)\, d\nu$ over s, s^c and s_n (monotone convergence for the last one, $\mathrm{klFun} \ge 0$). When $\mu \not\ll \nu$, a set A with $\nu(A) = 0 < \mu(A)$ meets s or s^c, and some s_n, with positive μ-measure, so the corresponding restricted divergences are infinite too. □

Lemma 290 (Chain rule for stopped histories)**.** $Learning.map_stoppedHist_min_eq_add, Learning.map_stoppedHis$
two environments, (X, Y) *and* (X', Y') *algorithm-environment sequences of* (alg, ν) *and* (alg, ν') *on probability spaces* $(\Omega, \mathbb{P})$ *and* $(\Omega', \mathbb{P}')$*,* S *a stopping rule with stopping times* τ*,* τ' *on the two spaces, and* $M \ge 0$*. Then*

$$\mathrm{KL}\big(\mathrm{law}(\mathrm{finHist}_{\min(\tau,M)}) \,\big\|\, \mathrm{law}(\mathrm{finHist}'_{\min(\tau',M)})\big) = \sum_{t<M} \mathrm{KL}\big(\mathbb{P}|_{\{t<\tau\}} \circ \mathrm{finHist}_t^{-1} \otimes \sigma_t \,\big\|\, \mathbb{P}|_{\{t<\tau\}} \circ \mathrm{finHist}_t^{-1} \otimes \sigma'_t\big),$$

the sum over the rounds of the conditional divergences of the step kernels σ_t, σ'_t *of the two environments given the first* t *rounds, on the event* $\{t < \tau\}$ *(the round is played). For two stationary environments with reward kernels* κ, κ' *the* t*-th term is* $\mathrm{KL}(\mathbb{P}|_{\{t<\tau\}} \circ X_t^{-1} \otimes \kappa \,\|\, \mathbb{P}|_{\{t<\tau\}} \circ X_t^{-1} \otimes \kappa')$ *(composition-product form) and, when* $\mathcal{Y}$ *is countably generated, the right-hand side is* $\mathbb{E}\big[\sum_{t<\min(\tau,M)} \mathrm{KL}(\kappa(X_t)\|\kappa'(X_t))\big]$ *(integral form).*

Proof. lem:kl$_restrict, lem : pi_kl_measurable_equiv, lem : pi_kl_step, lem : kl_sufficient_statistic, lem :$
$kl_compProd_kernelInductiononM.ForM = 0bothlawsaretheDiracmassattheemptyhistory.Forthestep M\to$
$M + 1$, write $Z_M := \mathrm{finHist}_{\min(\tau,M)}$. The law of Z_{M+1} splits according to

whether $\tau \le M$: on $\{\tau \le M\}$, $Z_{M+1} = Z_M$, and on $\{M < \tau\}$, Z_{M+1} is the history of the first $M+1$ rounds; the two parts are carried by the disjoint sets of histories of length $\le M$ and of length $M+1$. Likewise the law of Z_M splits into its restriction to $\{\tau \le M\}$ (carried by S, as $Z_M = \mathrm{finHist}_\tau \in S$ there) and the law of $\mathrm{finHist}_M$ on $\{M < \tau\}$ (carried by S^c). By the additivity of Lemma **??** over disjoint supports, the invariance under the embeddings $h \mapsto \langle n, h\rangle$ (Lemma **??**) and the chain rule `klDiv_compProd_eq_add`, since on $\{M < \tau\}$ (an event determined by $\mathrm{finHist}_M$) the law of $\mathrm{finHist}_{M+1}$ is $\mathrm{law}(\mathrm{finHist}_M) \otimes \sigma_M$,

$$\mathrm{KL}(Z_{M+1}) = \mathrm{KL}(Z_M|_{\tau\le M}) + \mathrm{KL}(\rho_M) + \mathrm{KL}(\rho_M \otimes \sigma_M \,\|\, \rho_M \otimes \sigma'_M) = \mathrm{KL}(Z_M) + \mathrm{KL}(\rho_M \otimes \sigma_M \,\|\, \rho_M \otimes \sigma'_M),$$

where ρ_M is the law of $\mathrm{finHist}_M$ on $\{M < \tau\}$ (primes omitted on the second arguments). The stationary case is Lemma **??** in composition-product form, the law of X_t on $\{t < \tau\}$ being the policy applied to the law of $\mathrm{finHist}_t$ on that event; the integral form is Lemma **??** and the exchange of the finite sum with the integral. ☐

Theorem 291 (Divergence decomposition at a stopping time)**.** *Learning.IsAlgEnvSeq.klDiv*$_m$*ap*$_s$*toppedHist*$_s$*te* $\sum_{t=0}^{\infty} \mathrm{KL}\big(\mathbb{P}|_{\{t<\tau\}} \circ \mathrm{finHist}_t^{-1} \otimes \sigma_t \,\big\|\, \mathbb{P}|_{\{t<\tau\}} \circ \mathrm{finHist}_t^{-1} \otimes \sigma'_t\big)$, and for two stationary environments with reward kernels κ, κ',

$$\mathrm{KL}\big(\mathrm{law}(\mathrm{finHist}_\tau) \,\big\|\, \mathrm{law}(\mathrm{finHist}'_{\tau'})\big) = \sum_{t=0}^{\infty} \mathrm{KL}\big(\mathbb{P}|_{\{t<\tau\}} \circ X_t^{-1} \otimes \kappa \,\big\|\, \mathbb{P}|_{\{t<\tau\}} \circ X_t^{-1} \otimes \kappa'\big) = \mathbb{E}\Big[\sum_{t<\tau} \mathrm{KL}(\kappa(X_t)\|\kappa'(X_t))\Big]$$

the last form when $\mathcal{Y}$ is countably generated.

Proof. lem:stopped$_h$*istory*$_c$*hain*$_r$*ule*, *lem* : *kl*$_r$*estrictTheseriesisthesupremumover*M*ofthepartialsums*, *which* by Lemma **??**. Since the stopping times are almost surely finite, these laws are the images of the laws μ, μ' of the stopped histories by the truncation trunc_M, so their divergences are at most $\mathrm{KL}(\mu\|\mu')$ (data processing). Conversely, the images agree with μ, μ' on histories of length $< M$, and these sets increase to the whole space, so by Lemma **??** (monotone convergence, then monotonicity under restriction) $\mathrm{KL}(\mu\|\mu') = \sup_M \mathrm{KL}(\mu|_{\mathrm{length}<M} \| \mu'|_{\mathrm{length}<M}) \le \sup_M \mathrm{KL}(\mathrm{trunc}_{M*}\mu \| \mathrm{trunc}_{M*}\mu')$. The integral form follows from the composition-product form as in Lemma **??**, the series and the integral being exchanged by monotone convergence, and $\sum_t \mathbf{1}_{t<\tau}(\cdot) = \sum_{t<\tau}(\cdot)$ on $\{\tau < \infty\}$. ☐

Lemma 292 (Change of measure at a stopping time for runs)**.** *Learning.IdentAlg.IsRun.ae*$_s$*toppedHist*$_m$*em*$_s$*top* *and* $(\Omega', \mathbb{P}')$*, whose stopping times are almost surely finite. Then the divergence between the laws of (stopped history, output) is the divergence between the laws of the stopped histories, and the divergence between the laws of the outputs is at most the latter. For two stationary environments with reward kernels* κ, κ'*,*

$$\mathrm{KL}\big(\mathrm{law}(\hat{x}) \,\big\|\, \mathrm{law}'(\hat{x}')\big) \le \mathbb{E}\Big[\sum_{t<\tau} \mathrm{KL}(\kappa(X_t)\|\kappa'(X_t))\Big]$$

(when $\mathcal{Y}$ *is countably generated; the composition-product form holds in general).*

Proof. thm:divergence$_{d}ecomposition_{s}topping, lem : kl_{d}ata_{p}rocessing_{r}ecommendationAsinLemma$ **??** :
$thestoppedhistorybelongstothestoppingsetalmostsurely, onwhichtheoutputruleisaprobabilitykernel, so$`klDiv_`

Lemma 293 (The divergence along a generating sequence of maps)**.** *InformationTheory.klDiv*$_{m}ap_{e}q_{l}integral_{k}l$ *be finite measures on* Ω *and* $g_n : \Omega \to \Omega_n$ *be measurable maps whose* σ*-algebras* $\sigma(g_n)$ *increase to the* σ*-algebra of* Ω*. Then* $\mathrm{KL}(\mu\|\nu) = \sup_n \mathrm{KL}(g_{n*}\mu \,\|\, g_{n*}\nu)$*. The projections of a sequence space* $\mathbb{N} \to E$ *on the coordinates* $\leq n$ *satisfy the hypothesis.*

Proof. lem:kl$_{r}estrictTheinequality$$\geq$ is the data-processing inequality. For $\leq$, if $\mu \ll \nu$ with density f, then $\mathrm{KL}(g_{n*}\mu\|g_{n*}\nu) = \int \mathrm{klFun}(\mathbb{E}_\nu[f \mid \sigma(g_n)])\, d\nu$ (Mathlib `toReal_rnDeriv_map`), the conditional expectations converge ν-almost everywhere to f by Lévy's upward theorem (Mathlib `Integrable.tendsto_ae_condExp`), and Fatou's lemma gives $\int \mathrm{klFun}(f)\, d\nu \leq \liminf_n \int \mathrm{klFun}(\mathbb{E}_\nu[f \mid \sigma(g_n)])\, d\nu$. If $\mu \not\ll \nu$, pick A with $\nu(A) = 0 < \mu(A)$; by Lévy's theorem under $\mu + \nu$ applied to $\mathbf{1}_A$, the $\sigma(g_n)$-measurable sets $B_n := \{\mathbb{E}_{\mu+\nu}[\mathbf{1}_A \mid \sigma(g_n)] > 1/2\}$ satisfy $\mu(B_n) \to \mu(A)$ and $\nu(B_n) \to 0$, and the lower bound $\mu(B)\log(\mu(B)/\nu(B)) + \nu(B) - \mu(B) \leq \mathrm{KL}(\mu|_B\|\nu|_B) \leq \mathrm{KL}(\mu\|\nu)$ (Mathlib `mul_log_le_klDiv` and Lemma **??**), applied to the images and the sets $B_n = g_n^{-1}(C_n)$, shows that the divergences of the images tend to infinity. $\square$

Theorem 294 (Divergence decomposition for trajectories)**.** *Learning.klDiv*$_{m}ap_{t}rajectory_{e}q_{i}Sup, Learning.IsA$ *environmentsequenceson*$(\Omega, \mathbb{P})$*,* $(\Omega', \mathbb{P}')$*, the divergence between the laws of the whole trajectories* $(X_t, Y_t)_{t\in\mathbb{N}}$ *is*

$$\sum_{t=0}^{\infty} \mathrm{KL}\big(\mathbb{P} \circ X_t^{-1} \otimes \kappa \,\big\|\, \mathbb{P} \circ X_t^{-1} \otimes \kappa'\big) = \sum_{t=0}^{\infty} \mathbb{E}\big[\mathrm{KL}(\kappa(X_t)\|\kappa'(X_t))\big]$$

(the last form when $\mathcal{Y}$ *is countably generated), and for two arbitrary environments it is the series of the conditional divergences of the step kernels given the past.*

Proof. thm:divergence$_{d}ecomposition, lem : kl_{i}Sup_{m}apByLemma$ **??**$appliedtotheprojectionsonthefirst$n+1$co$

Corollary 295 (Divergence decomposition for linear Gaussian environments)**.**
Learning.LinearBandit.klDiv$_{l}inearGaussianKernel, Learning.LinearBandit.klDiv_{m}ap_{h}istory, Learning.Lin$
nonemptycompact, not*assumedspanning*; *Part* **??***appliesthiscorollarytonon* $-$ *spanningsets)and*$\theta, \theta' \in \mathbb{R}^d$*. For every algorithm with actions in* $\mathcal{X}$ *and every* $T \geq 1$*, writing* $\mathbb{P}_\theta^T$ *for the law of the history of length* T *under* ν_θ *(Definition* **??***; this is* $\mathbb{P}^T$ *of Definition* **??** *with* $\mathcal{A} = \mathcal{X}$*,* $\mathcal{Y} = \mathbb{R}$ *and* $\kappa(x) = \mathcal{N}(\langle x, \theta\rangle, 1)$*),*

$$\mathrm{KL}\big(\mathbb{P}_\theta^T \,\big\|\, \mathbb{P}_{\theta'}^T\big) = \frac{1}{2}\sum_{t<T} \mathbb{E}_\theta\big[\langle x_t, \theta - \theta'\rangle^2\big] \leq \frac{T}{2}\Big(\sup_{x\in\mathcal{X}}\|x\|\Big)^2 \|\theta - \theta'\|^2 < \infty.$$

Proof. thm:divergence$_{d}ecomposition, lem : kl_{g}aussianThesubtype$X$of$R^d is countably generated, so Definition **??** applies with $\mathcal{A} = \mathcal{X}$ (compactness and spanning play no role there). The reward kernels are $\kappa(x) = \mathcal{N}(\langle x, \theta\rangle, 1)$ and $\kappa'(x) = \mathcal{N}(\langle x, \theta'\rangle, 1)$, so by Lemma **??** $\mathrm{KL}(\kappa(x)\|\kappa'(x)) = \frac{1}{2}\langle x, \theta - \theta'\rangle^2 \le \frac{1}{2}R^2\|\theta - \theta'\|^2$ with $R := \sup_{x\in\mathcal{X}}\|x\| < \infty$ (compactness). Apply Theorem **??** in its finite form. $\square$

Lean remark. The action type is the subtype $\mathcal{X}$ of `EuclideanSpace (Fin d)` and the kernel is `fun x  gaussianReal  x,   1`, measurable by `measurable_gaussianReal` composed with the continuous inner product. The expectation $\mathbb{E}_\theta[\langle x_t, \theta - \theta'\rangle^2]$ is an integral of a bounded measurable function of the history, so no integrability side condition arises.

Chapter 14

The Sudakov–Fernique inequality

This chapter proves the Sudakov–Fernique comparison inequality for finite families of centered Gaussian random variables, by the Gaussian interpolation method ([**?**, Theorem 7.2.11]), and a lower bound on the expected maximum of independent standard Gaussians. Both are only used for the secondary lower bound $w(\mathcal{X}) \geq c\sqrt{d \log d}$ (Proposition **??**); the covariance-domination special case needed everywhere else has a direct proof (Lemma **??**). The chapter is written so that each step is a small declaration: Gaussian integration by parts (Stein's identity), the derivative of an interpolated expectation, the Hessian of the soft-max, and elementary tail estimates.

What Mathlib/LML provides. Gaussian vectors: `ProbabilityTheory.stdGaussian`, `multivariateGaussian`, `IsGaussian`, and their linear images (see Chapter **??**). Differentiation under the integral sign: `hasDerivAt_integral_of_dominated_loc_of_deriv_le` (`Mathlib/Analysis/Calculus/ParametricIntegral.lean`). Integration by parts on $\mathbb{R}$ with limits at $\pm\infty$: `MeasureTheory.integral_mul_deriv_eq_deriv_mul` and `integral_deriv_mul_eq_sub` in `MeasureTheory/Integral/IntegralEqImproper.lean`. Monotonicity from the sign of the derivative: `antitoneOn_of_deriv_nonpos`. The Gaussian density `gaussianPDFReal` and `gaussianReal_apply_eq_integral`. Numerical bounds on π, e and $\log 2$: `Real.pi_lt_d2`, `Real.pi_gt_d2`, `Real.exp_one_lt_d9`, `Real.log_two_lt_d9`, `Real.log_two_gt_d9`.

What is missing. Stein's identity for Gaussian vectors, the interpolation formula, the Sudakov–Fernique inequality itself, the lower Mills-ratio bound and the lower bound on the expected maximum are all absent from Mathlib; they are formalized in this project in `Maiti2026Power/Mathlib/Probability/{SteinReal, SteinIdentity, GaussianInterpolation, SudakovFernique, GaussianMaxLower}.lean` and `Maiti2026Power/Mathlib/Analysis/{SpecialFunctions, InnerProductSpace}/LogSumExp.lean`.

Lean remark (organisation). The Lean development is coordinate-free where possible. Stein's identity is proved for the standard Gaussian measure of a finite-dimensional inner product space E in the form $\mathbb{E}[\langle a, g\rangle F(g)] = \mathbb{E}[DF(g)\, a]$

(`integral_inner_mul_stdGaussian`, from the one-dimensional identity by Fubini along the coordinates of an orthonormal basis), and for $X \sim \mathcal{N}(0, S)$ in the form $\mathbb{E}[\langle a, X\rangle F(X)] = \mathbb{E}[DF(X)(Sa)]$ (`integral_inner_mul_multivariateGaussian`); the version used by the interpolation is `integral_fderiv_apply_stdGaussian`: $\mathbb{E}[DF(Lg)(L'g)] = \mathbb{E}[\sum_k D^2F(Lg)(Le_k, L'e_k)]$ for linear maps $L, L' : E' \to E$ and an orthonormal basis (e_k) of E'. The interpolation $Z_t = \sqrt{t}\,X + \sqrt{1-t}\,Y$ with X, Y independent is realized as $Z_t = \sqrt{t}\,L_1 g + \sqrt{1-t}\,L_0 g$ for a single standard Gaussian vector g of E' (`gaussianInterp`); for the Sudakov–Fernique inequality, $E' = \mathbb{R}^{\iota \sqcup \iota}$, $L_1 g = S_X^{1/2}(g \circ \mathrm{inl})$ and $L_0 g = S_Y^{1/2}(g \circ \mathrm{inr})$ (`blockInl`, `blockInr`), so that the derivative of $\psi(t) = \mathbb{E}F(Z_t)$ is $\mathbb{E}[\sum_k D^2F(Z_t)(Z_t e_k, Z_t' e_k)]$ with the cross terms vanishing block by block (`hasDerivAt_integral_comp_gaussianInterp`, `sum_apply_gaussianInterp_block`). The soft-max is `logSumExp   (EuclideanSpace.basisFun`
`)`: the coordinate soft-max `Real.logSumExp` of `Maiti2026Power/Mathlib/Analysis/SpecialFunctions/Lo`
(defined on $\mathbb{R}^m$, with its Hessian `Real.fderiv_fderiv_logSumExp_single`) composed with the coordinates, as an instance of the general log-sum-exp smoothing of a maximum of linear forms of `Maiti2026Power/Mathlib/Analysis/InnerProductSpace/LogSumExp.lea`

14.1 Gaussian integration by parts

Throughout, "$F : \mathbb{R}^m \to \mathbb{R}$ is C^1 with $\|\nabla F\| \le L$" means that F is continuously differentiable and $\|\nabla F(x)\| \le L$ for every x; such an F is L-Lipschitz with $|F(x)| \le |F(0)| + L\|x\|$ (Lemma **??**), so that for a Gaussian vector X the variables $F(X)$, $X_i F(X)$ and $\partial_j F(X)$ are integrable (Lemma **??**, or Lemma **??** for the second moments needed for $X_i F(X)$). Similarly, "F is C^2 with $\|\mathrm{Hess}\, F\|_{\mathrm{op}} \le L$" means that the Hessian matrix of F has operator norm at most L at every point; for a C^2 function this is equivalent to ∇F being L-Lipschitz (mean value inequality, Mathlib `lipschitzWith_of_nnnorm_fderiv_le`).

Lemma 296 (One-dimensional Stein identity)**.** $ProbabilityTheory.integral_sub_mul_gaussianReal, ProbabilityThe$
$\mathcal{N}(0,1)$, $L \ge 0$, and $f : \mathbb{R} \to \mathbb{R}$ be C^1 with $|f'(x)| \le L$ for every x. Then $\xi f(\xi)$ and $f'(\xi)$ are integrable and $\mathbb{E}[\xi f(\xi)] = \mathbb{E}[f'(\xi)]$. (In Lean the identity is proved for every $\mathcal{N}(\mu, v)$, $v \ge 0$, in the form $\mathbb{E}[(\xi - \mu) f(\xi)] = v\,\mathbb{E}[f'(\xi)]$, for differentiable f with bounded derivative.)

Proof. lem:gauss$_1d_moments, lem : pc_lipschitz_of_gradLet\varphi(x) := (2\pi)^{-1/2} e^{-x^2/2}$, so that $\varphi'(x) = -x\varphi(x)$ and $\mathbb{E}[\xi f(\xi)] = \int x f(x)\varphi(x)\,dx = -\int f(x)\varphi'(x)\,dx$. By Lemma **??** (in dimension one), $|f(x)| \le |f(0)| + L|x|$. Integration by parts on $\mathbb{R}$ (Mathlib `integral_mul_deriv_eq_deriv_mul`) requires: $f\varphi'$ and $f'\varphi$ integrable, which holds since $|f(x)\varphi'(x)| \le (|f(0)| + L|x|)|x|\varphi(x)$ and $|f'\varphi| \le L\varphi$, and ξ has a second moment (Lemma **??**); and $f(x)\varphi(x) \to 0$ as $x \to \pm\infty$, which holds since $|f(x)\varphi(x)| \le (|f(0)| + L|x|)\varphi(x) \to 0$. It gives $-\int f\varphi' = \int f'\varphi = \mathbb{E}[f'(\xi)]$. ∎

Lemma 297 (Multivariate Stein identity)**.** $ProbabilityTheory.integral_inner_mul_stdGaussian, ProbabilityTheory$
$\in \mathcal{S}_+^d$ (of size m), $X \sim \mathcal{N}(0, S)$ on $\mathbb{R}^m$, $L \ge 0$, and $F : \mathbb{R}^m \to \mathbb{R}$ be C^1 with

$\|\nabla F(x)\| \le L$ *for every* x*. Then* $X_i F(X)$ *and* $\partial_j F(X)$ *are integrable and, for every* $i \le m$*,*

$$\mathbb{E}\big[X_i F(X)\big] = \sum_{j=1}^{m} S_{ij}\, \mathbb{E}\big[\partial_j F(X)\big].$$

Proof. lem:gauss$_p i, lem : gauss_l inear_i mage, lem : gauss_l aw_o f_c ov, lem : psf_i bp_1 d, lem : pc_l ipschitz_o f_g rad, lem : pc_e xp_n orm_i ntegrable, lem : gauss_n orm_m omentsIntegrability.ByLemma$ **??**,$|F(x)| \le |F(0)| + L\|x\|$, so $|X_i F(X)| \le \|X\|(|F(0)| + L\|X\|)$ is integrable by Lemma **??** (first and second moments; or Lemma **??**), and $|\partial_j F(X)| \le L$ is bounded.

Standard case $S = I_m$*.* By Lemma **??** the coordinates of $X = g$ are i.i.d. standard Gaussians, so the law of g is the product measure. Fix i and write $g = (g_i, g_{-i})$. By Fubini (justified by the integrability just shown), $\mathbb{E}[g_i F(g)] = \mathbb{E}_{g_{-i}}\big[\mathbb{E}_{g_i}[g_i F(g_i, g_{-i})]\big]$, and for fixed g_{-i} the function $x \mapsto F(x, g_{-i})$ is C^1 with derivative $\partial_i F(x, g_{-i})$ bounded by L in absolute value (as $|\partial_i F| \le \|\nabla F\| \le L$). Lemma **??** gives $\mathbb{E}_{g_i}[g_i F(g_i, g_{-i})] = \mathbb{E}_{g_i}[\partial_i F(g_i, g_{-i})]$; integrate in g_{-i}.

General case. Let $M := S^{1/2}$, so that $MM^\top = S$ and $Mg \sim \mathcal{N}(0, S)$ has the law of X (Lemmas **??** and **??**); expectations of measurable functions only depend on the law, so we may take $X = Mg$. Then $X_i = \sum_k M_{ik} g_k$ and $G := F \circ M$ is C^1 with $\nabla G(g) = M^\top \nabla F(Mg)$, i.e. $\partial_k G(g) = \sum_j M_{jk} (\partial_j F)(Mg)$ (chain rule), and $\|\nabla G\| \le \|M\|_{\mathrm{op}} L$: G has a bounded gradient. By the standard case,

$$\mathbb{E}[X_i F(X)] = \sum_k M_{ik} \mathbb{E}[g_k G(g)] = \sum_k M_{ik} \mathbb{E}[\partial_k G(g)] = \sum_k \sum_j M_{ik} M_{jk} \mathbb{E}[\partial_j F(Mg)] = \sum_j S_{ij} \mathbb{E}[\partial_j F(X)].$$

□

14.2 Gaussian interpolation

Lemma 298 (Derivative of an interpolated expectation)**.** *ProbabilityTheory.gaussianInterp, ProbabilityTheory.hasDerivAt$_i$ntegral$_c$omp$_g$aussianInterp, ProbabilityTheory.continuousOn$_i$ntegral$_c$omp$_g$a* $\mathcal{S}_+^d$ *(size* m*),* $X \sim \mathcal{N}(0, S_X)$ *and* $Y \sim \mathcal{N}(0, S_Y)$ *be independent,* $L \ge 0$*, and* $F : \mathbb{R}^m \to \mathbb{R}$ *be* C^2 *with* $\|\operatorname{Hess} F(z)\|_{\mathrm{op}} \le L$ *for every* z *(so that* ∇F *is* L*-Lipschitz). For* $t \in [0,1]$ *set* $Z_t := \sqrt{t}\, X + \sqrt{1-t}\, Y$ *and* $\psi(t) := \mathbb{E}[F(Z_t)]$*. Then* ψ *is continuous on* $[0,1]$*, differentiable on* $(0,1)$*, and for* $t \in (0,1)$*,*

$$\psi'(t) = \frac{1}{2} \sum_{i,j=1}^{m} (S_X - S_Y)_{ij}\, \mathbb{E}\big[\partial_{ij} F(Z_t)\big].$$

Proof. lem:gauss$_p i, lem : gauss_l inear_i mage, lem : gauss_l aw_o f_c ov, lem : gauss_i bp, lem : gauss_n orm_m omentsSince$ $\|\operatorname{Hess} F\|_{\mathrm{op}} \le L$ everywhere, ∇F is L-Lipschitz (mean value inequality applied to the C^1 map ∇F, whose derivative is the Hessian; Mathlib `lipschitzWith_of_nnnorm_fderiv_le`), so $\|\nabla F(z)\| \le \|\nabla F(0)\| + L\|z\|$ and $|F(z)| \le |F(0)| + \|\nabla F(0)\|\|z\| + L\|z\|^2/2$. Hence $F(Z_t)$, $\partial_i F(Z_t)$ and $X_i \partial_i F(Z_t)$ are integrable for every t (Lemma **??**: X and Y have second moments), and each $\partial_i F$ is C^1 with $\|\nabla \partial_i F(z)\| = \|\operatorname{Hess} F(z)\, e_i\| \le L$.

Continuity. $|F(Z_t)-F(Z_s)| \le (\|\nabla F(0)\|+L(\|X\|+\|Y\|))\,(|\sqrt{t}-\sqrt{s}|\|X\|+|\sqrt{1-t}-\sqrt{1-s}|\|Y\|)$, whose expectation tends to 0 as $s \to t$ (second moments, Lemma **??**).

Differentiability. For $t \in (0,1)$, $\frac{d}{dt}F(Z_t) = \left\langle \nabla F(Z_t),\ \frac{X}{2\sqrt{t}} - \frac{Y}{2\sqrt{1-t}} \right\rangle$, which on any interval $[a,b] \subset (0,1)$ is dominated by $(\|\nabla F(0)\|+L(\|X\|+\|Y\|))(\|X\|+\|Y\|)/(2\min(\sqrt{a},\sqrt{1-b}))$, an integrable function. Differentiation under the integral sign (Mathlib `hasDerivAt_integral_of_dominated_loc_of_deriv_le`) gives

$$\psi'(t) = \frac{1}{2\sqrt{t}}\sum_i \mathbb{E}\big[X_i\,\partial_i F(Z_t)\big] - \frac{1}{2\sqrt{1-t}}\sum_i \mathbb{E}\big[Y_i\,\partial_i F(Z_t)\big].$$

Stein's identity. Write $X = S_X^{1/2}g$, $Y = S_Y^{1/2}g'$ with (g,g') a standard Gaussian vector of $\mathbb{R}^{2m}$ (Lemmas **??**, **??**; independence of X and Y is the product structure of the law of (g,g')). Then $W := (X,Y)$ is the linear image of (g,g') by the block-diagonal matrix $\operatorname{diag}(S_X^{1/2}, S_Y^{1/2})$, hence $W \sim \mathcal{N}(0,\operatorname{diag}(S_X,S_Y))$ (Lemma **??**). Fix i and apply Lemma **??** to W and the C^1 function $G(x,y) := \partial_i F(\sqrt{t}\,x+\sqrt{1-t}\,y)$, whose gradient is $\nabla G(x,y) = (\sqrt{t}\,\nabla\partial_i F(Z), \sqrt{1-t}\,\nabla\partial_i F(Z))$ with $Z = \sqrt{t}x+\sqrt{1-t}y$, so that $\|\nabla G(x,y)\|^2 = (t+(1-t))\|\operatorname{Hess}F(Z)\,e_i\|^2 \le L^2$: G has a gradient bounded by L, as Lemma **??** requires. The covariance of W is block-diagonal, so

$$\mathbb{E}[X_iG(W)] = \sum_j (S_X)_{ij}\mathbb{E}[\partial_{x_j}G(W)] = \sqrt{t}\sum_j (S_X)_{ij}\mathbb{E}[\partial_{ij}F(Z_t)], \qquad \mathbb{E}[Y_iG(W)] = \sqrt{1-t}\sum_j (S_Y)_{ij}\mathbb{E}[\partial_{ij}F(Z$$

Substituting in the formula for $\psi'(t)$, the factors $\sqrt{t}$ and $\sqrt{1-t}$ cancel and we obtain $\psi'(t) = \frac{1}{2}\sum_{ij}(S_X)_{ij}\mathbb{E}[\partial_{ij}F(Z_t)] - \frac{1}{2}\sum_{ij}(S_Y)_{ij}\mathbb{E}[\partial_{ij}F(Z_t)]$. $\square$

Lemma 299 (The soft-max and its Hessian)**.** *Real.logSumExp, Real.softmax, $Real.softmax_nonneg, Real.sum_softmax, Real.ciSup_le_logSumExp, Real.logSumExp_le_ciSup_add, Real.contDif$ 0 and $z \in \mathbb{R}^m$ let $F_\beta(z) := \beta^{-1}\log\sum_{i=1}^m e^{\beta z_i}$ and $p_i(z) := e^{\beta z_i}/\sum_k e^{\beta z_k}$, so that $p(z)$ is a probability vector. Then:*

(i) $\max_i z_i \le F_\beta(z) \le \max_i z_i + (\log m)/\beta$;

(ii) F_β *is* C^∞, $\partial_i F_\beta = p_i$, *hence* $\|\nabla F_\beta(z)\| \le 1$ *and* $|F_\beta(z)-F_\beta(z')| \le \|z-z'\|$;

(iii) $\partial_{ij}F_\beta = \beta(\delta_{ij}p_i - p_ip_j)$, *i.e.* $\operatorname{Hess}F_\beta(z) = \beta(\operatorname{diag}(p)-pp^\top)$, *a positive semidefinite matrix with* $\|\operatorname{Hess}F_\beta(z)\|_{\mathrm{op}} \le \beta$ *(and all entries bounded by* β *in absolute value);*

(iv) for every symmetric matrix $\Gamma \in \mathbb{R}^{m\times m}$ *and every* z,

$$\sum_{i,j}\Gamma_{ij}\,\partial_{ij}F_\beta(z) = \frac{\beta}{2}\sum_{i,j}p_i(z)p_j(z)\left(\Gamma_{ii}+\Gamma_{jj}-2\Gamma_{ij}\right).$$

Proof. (i): $e^{\beta\max_i z_i} \le \sum_i e^{\beta z_i} \le m\,e^{\beta\max_i z_i}$; take logarithms and divide by β. (ii): $\partial_i \log\sum_k e^{\beta z_k} = \beta e^{\beta z_i}/\sum_k e^{\beta z_k}$; $\|p\|_2 \le \|p\|_1 = 1$; the Lipschitz bound is the mean value inequality (Mathlib `Convex.norm_image_sub_le_of_norm_fderiv_le`). (iii): differentiate $p_i = e^{\beta z_i}/\sum_k e^{\beta z_k}$: $\partial_j p_i = \beta\delta_{ij}p_i - \beta p_i p_j$, and $|\delta_{ij}p_i - p_i p_j| \leq 1$. For the operator norm, for every $v \in \mathbb{R}^m$, with $\bar{v} := \sum_i p_i v_i$, $v^\top(\operatorname{diag}(p) - pp^\top)v = \sum_i p_i v_i^2 - \bar{v}^2 = \sum_i p_i(v_i - \bar{v})^2$, the variance of v under the probability vector p, which lies in $[0, \sum_i p_i v_i^2] \subseteq [0, \|v\|^2]$; so the symmetric matrix $\operatorname{Hess} F_\beta(z)$ has its eigenvalues in $[0, \beta]$. (iv): by (iii), $\sum_{ij} \Gamma_{ij}\partial_{ij}F_\beta = \beta\big(\sum_i \Gamma_{ii}p_i - \sum_{ij}\Gamma_{ij}p_i p_j\big)$. On the other hand, using $\sum_j p_j = 1$,

$$\sum_{i,j} p_i p_j(\Gamma_{ii}+\Gamma_{jj}-2\Gamma_{ij}) = \sum_i p_i\Gamma_{ii}+\sum_j p_j\Gamma_{jj}-2\sum_{ij} p_i p_j \Gamma_{ij} = 2\Big(\sum_i \Gamma_{ii}p_i - \sum_{ij}\Gamma_{ij}p_i p_j\Big),$$

and multiplying by $\beta/2$ gives the claim. $\square$

Theorem 300 (Sudakov–Fernique inequality)**.** *ProbabilityTheory.sudakov$_f$erniquedef : pg$_g$aussian$_v$ectorLetX with laws $\mathcal{N}(0, S_X)$ and $\mathcal{N}(0, S_Y)$, such that*

$$\mathbb{E}\big[(X_i - X_j)^2\big] \leq \mathbb{E}\big[(Y_i - Y_j)^2\big] \qquad \textit{for all } i, j \leq m.$$

Then $\mathbb{E}\big[\max_{i\leq m} X_i\big] \leq \mathbb{E}\big[\max_{i\leq m} Y_i\big]$. (Lean: `sudakov_fernique`, for the measures $\mathcal{N}(0, S_X)$, $\mathcal{N}(0, S_Y)$ on `EuclideanSpace` with S_X, S_Y positive semidefinite and the increment condition written as $(S_X)_{ii} + (S_X)_{jj} - 2(S_X)_{ij} \leq (S_Y)_{ii} + (S_Y)_{jj} - 2(S_Y)_{ij}$.)

Proof. lem:gauss$_l$aw$_o$f$_c$ov, *lem : interpolation$_d$erivative, lem : logsumexp$_h$essianBothsidesonlydependonthel* 0, let F_β be the soft-max of Lemma **??**, and $Z_t := \sqrt{t}\,X + \sqrt{1-t}\,Y$, $\psi(t) := \mathbb{E}[F_\beta(Z_t)]$, so that $Z_0 = Y$ and $Z_1 = X$. By Lemma **??**(iii) F_β is C^2 with $\|\operatorname{Hess} F_\beta\|_{\mathrm{op}} \leq \beta$, so Lemma **??** applies with $L = \beta$: ψ is continuous on $[0, 1]$, differentiable on $(0, 1)$, and with $\Gamma := S_X - S_Y$ (symmetric) and Lemma **??**(iv),

$$\psi'(t) = \frac{1}{2}\,\mathbb{E}\Big[\sum_{i,j}\Gamma_{ij}\partial_{ij}F_\beta(Z_t)\Big] = \frac{\beta}{4}\,\mathbb{E}\Big[\sum_{i,j} p_i(Z_t)p_j(Z_t)\big(\Gamma_{ii} + \Gamma_{jj} - 2\Gamma_{ij}\big)\Big].$$

Now $\Gamma_{ii}+\Gamma_{jj}-2\Gamma_{ij} = \big((S_X)_{ii}+(S_X)_{jj}-2(S_X)_{ij}\big)-\big((S_Y)_{ii}+(S_Y)_{jj}-2(S_Y)_{ij}\big) = \mathbb{E}[(X_i - X_j)^2] - \mathbb{E}[(Y_i - Y_j)^2] \leq 0$ by hypothesis, and $p_i p_j \geq 0$, so $\psi'(t) \leq 0$ on $(0, 1)$. Hence ψ is nonincreasing on $[0, 1]$ (Mathlib `antitoneOn_of_deriv_nonpos` with continuity on the closed interval), and $\mathbb{E}[F_\beta(X)] = \psi(1) \leq \psi(0) = \mathbb{E}[F_\beta(Y)]$. Finally, by Lemma **??**(i),

$$\mathbb{E}\big[\max_i X_i\big] \leq \mathbb{E}[F_\beta(X)] \leq \mathbb{E}[F_\beta(Y)] \leq \mathbb{E}\big[\max_i Y_i\big] + \frac{\log m}{\beta},$$

and letting $\beta \to \infty$ concludes (all expectations are finite since $|\max_i X_i| \leq \|X\|$). $\square$

Remark 301 (On the sign of the interpolation derivative)*.* The outline of this chapter hesitated about the sign of ψ'. The computation above is the correct one: with $Z_t = \sqrt{t}X + \sqrt{1-t}Y$ the derivative is $\frac{1}{2}\sum_{ij}(S_X - S_Y)_{ij}\mathbb{E}[\partial_{ij}F_\beta(Z_t)]$ (no minus sign), and the Hessian identity of Lemma **??**(iv) carries the factor

$+\beta/2$ (the sign error in the outline came from writing $-\beta/2$ there). Smaller increments for X therefore make ψ *nonincreasing* from $\psi(0) = \mathbb{E}F_\beta(Y)$ to $\psi(1) = \mathbb{E}F_\beta(X)$, which is exactly $\mathbb{E}\max X \le \mathbb{E}\max Y$, the standard statement ([**?**, Theorem 7.2.11]). The compact-index-set version (suprema over $\mathcal{X}$) follows by density, Lemma **??**, but is not needed.

14.3 Expected maximum of independent Gaussians

Lemma 302 (Lower Mills-ratio bound)**.** $ProbabilityTheory.mul_{g}aussianPDFReal_{l}e_{m}easureReal_{I}ciLet\xi \sim$ *$\mathcal{N}(0,1)$ and $\varphi(t) := (2\pi)^{-1/2}e^{-t^2/2}$. For every $t \ge 0$,*

$$\mathbb{P}(\xi \ge t) \ge \frac{t}{1+t^2}\,\varphi(t) = \frac{t}{1+t^2} \cdot \frac{e^{-t^2/2}}{\sqrt{2\pi}}.$$

(In Lean for $t > 0$; the case $t = 0$ is trivial.)

Proof. (The Lean proof integrates $\frac{d}{dx}\big[-\varphi(x)/x\big] = \varphi(x)(1+x^{-2})$ over $[t,\infty)$: $\varphi(t)/t = \int_t^\infty \varphi(x)(1+x^{-2})\,dx \le (1+t^{-2})\mathbb{P}(\xi \ge t)$, which is the claim. The argument below is the one of the outline.) Let $h(t) := \mathbb{P}(\xi \ge t) - \frac{t}{1+t^2}\varphi(t)$ for $t \in \mathbb{R}$. The tail $t \mapsto \mathbb{P}(\xi \ge t) = \int_t^\infty \varphi$ is differentiable with derivative $-\varphi(t)$ (fundamental theorem of calculus, Mathlib `intervalIntegral.integral_hasDerivAt_right` after writing the tail as $1 - \int_{-\infty}^t \varphi$; the tail is `gaussianReal_apply_eq_integral`). Using $\varphi'(t) = -t\varphi(t)$,

$$\frac{d}{dt}\Big[\frac{t}{1+t^2}\varphi(t)\Big] = \varphi(t)\Big[\frac{1-t^2}{(1+t^2)^2} - \frac{t^2}{1+t^2}\Big], \qquad h'(t) = -\varphi(t)\Big[1 + \frac{1-t^2}{(1+t^2)^2} - \frac{t^2}{1+t^2}\Big] = -\frac{2\,\varphi(t)}{(1+t^2)^2} < 0,$$

since $1 - \frac{t^2}{1+t^2} = \frac{1}{1+t^2}$ and $\frac{1+t^2}{(1+t^2)^2} + \frac{1-t^2}{(1+t^2)^2} = \frac{2}{(1+t^2)^2}$. Thus h is decreasing (Mathlib `antitone_of_deriv_nonpos`) and $h(t) \to 0$ as $t \to \infty$ (both terms tend to 0: the tail by dominated convergence, the second term since $\varphi(t) \to 0$). Hence $h(t) \ge \lim_{s\to\infty} h(s) = 0$ for every t. ☐

Lemma 303 (Expected maximum of two standard Gaussians)**.** $ProbabilityTheory.integral_{m}ax_{p}i_{g}aussianReal,$ *are i.i.d. $\mathcal{N}(0,1)$ then $\mathbb{E}[\max(\xi_1,\xi_2)] = 1/\sqrt{\pi} \ge 0.56$. (Lean: for two distinct coordinates of a product of standard Gaussians, via $\mathbb{E}[\xi^+] = (2\pi)^{-1/2}$, `integral_posPart_gaussianReal_zero_one`.)*

Proof. lem:gauss$_{p}i, lem : gauss_{l}inear_{i}mage, lem : gauss_{1}d_{m}oments$$\max(a,b) = \frac{a+b}{2} + \frac{|a-b|}{2}$. The first term has expectation 0. By Lemmas **??** and **??**, $\xi_1 - \xi_2 \sim \mathcal{N}(0,2)$, so $\mathbb{E}|\xi_1 - \xi_2| = \sqrt{2}\cdot\sqrt{2/\pi} = 2/\sqrt{\pi}$ (Lemma **??**), and $\mathbb{E}\max(\xi_1,\xi_2) = 1/\sqrt{\pi}$; finally $\pi < 3.15$ gives $1/\sqrt{\pi} > 0.563$. ☐

Lemma 304 (Monotonicity in the number of variables)**.** $le_{c}iSupLet$ *$(\xi_i)_{i\ge1}$ be real random variables with $\mathbb{E}|\xi_i| < \infty$. For $1 \le m \le m'$, $\mathbb{E}[\max_{i\le m}\xi_i] \le \mathbb{E}[\max_{i\le m'}\xi_i]$. (In Lean this is used inline: $\max(\xi_i,\xi_j) \le \sup_k \xi_k$ pointwise, `le_ciSup`, and `integral_mono`.)*

Proof. Pointwise, $\max_{i \le m} \xi_i \le \max_{i \le m'} \xi_i$; both are integrable since $|\max_{i \le m} \xi_i| \le \sum_{i \le m} |\xi_i|$. $\square$

Lemma 305 (Lower bound on the expected maximum, large m)**.** *$ProbabilityTheory.sqrt_{l}og_{d}iv_{f}our_{l}e_{i}ntegral_{i}S$ be i.i.d. $\mathcal{N}(0,1)$ with $m \ge 128$. Then $\mathbb{E}[\max_{i \le m} \xi_i] \ge \frac{1}{4}\sqrt{\log m}$.*

Proof. lem:gauss$_t ail_l ower, lem\ :\ gauss_1 d_m omentsLet$t $:= \sqrt{\log m}$ and $p_t := \mathbb{P}(\xi_1 \ge t)$. We proceed in steps.

(i) Since $m \ge 128 = 2^7$, $t^2 = \log m \ge 7 \log 2 > 4.85$ (Mathlib `Real.log_two_gt_d9`), so $t > 2.2 > 1$.

(ii) By Lemma **??** and $e^{-t^2/2} = m^{-1/2}$, $p_t \ge \frac{t}{1+t^2} \cdot \frac{m^{-1/2}}{\sqrt{2\pi}} \ge \frac{m^{-1/2}}{2t\sqrt{2\pi}}$, using $\frac{t}{1+t^2} \ge \frac{1}{2t}$ for $t \ge 1$ (equivalent to $t^2 \ge 1$).

(iii) Hence $mp_t \ge \frac{\sqrt{m}}{2\sqrt{2\pi}\sqrt{\log m}} \ge 1$: the inequality $\sqrt{m} \ge 2\sqrt{2\pi}\sqrt{\log m}$ is equivalent to $m \ge 8\pi \log m$; at $m = 128$ it reads $128 \ge 8\pi \cdot 7 \log 2$, true since $8\pi \cdot 7 \log 2 < 8 \cdot 3.15 \cdot 7 \cdot 0.6932 < 122.3$; and $m \mapsto m/\log m$ is nondecreasing on $[e, \infty)$ (its derivative is $(\log m - 1)/(\log m)^2 \ge 0$), so $m \ge 8\pi \log m$ for all $m \ge 128$.

(iv) By independence, $\mathbb{P}(\max_i \xi_i < t) = (1-p_t)^m \le e^{-mp_t} \le e^{-1}$, so $\mathbb{P}(\max_i \xi_i \ge t) \ge 1-e^{-1} \ge \frac{1}{2}$ (Mathlib `Real.add_one_le_exp`, `Real.exp_one_gt_d9`).

(v) Write $M := \max_i \xi_i = M^+ - M^-$. Since $t > 0$, $M^+ \ge t\,\mathbf{1}\{M \ge t\}$, so $\mathbb{E}[M^+] \ge t\,\mathbb{P}(M \ge t) \ge t/2$. Since $M \ge \xi_1$, $M^- \le \xi_1^-$ and $\mathbb{E}[M^-] \le \mathbb{E}[\xi_1^-] = \frac{1}{2}\mathbb{E}|\xi_1| = \frac{1}{\sqrt{2\pi}} < 0.4$ (Lemma **??** and symmetry of ξ_1).

(vi) Therefore $\mathbb{E}[M] \ge \frac{t}{2} - 0.4 \ge \frac{t}{4}$, because $\frac{t}{4} \ge 0.4$ is implied by $t > 2.2$. That is, $\mathbb{E}[\max_i \xi_i] \ge \frac{1}{4}\sqrt{\log m}$.

$\square$

Lemma 306 (Lower bound on the expected maximum of i.i.d. Gaussians)**.** *$ProbabilityTheory.sqrt_{l}og_{d}iv_{f}our_{l}e_{i}ntegral_{i}Sup_{p}i_{g}aussianReallem : psf_{m}ax_{t}wo, lem : psf_{m}ax_{m}ono, lem : psf$ be i.i.d. $\mathcal{N}(0,1)$ with $m \ge 2$. Then*

$$\mathbb{E}\Big[\max_{i \le m} \xi_i\Big] \ge c_{\max}\sqrt{\log m}, \qquad c_{\max} := \frac{1}{4}.$$

Proof. lem:psf$_m ax_t wo, lem\ :\ psf_m ax_m ono, lem\ :\ psf_m ax_l ower_l argeIfm\ge$ 128 this is Lemma **??**. If $2 \le m < 128$, then by Lemmas **??** and **??**, $\mathbb{E}[\max_{i \le m} \xi_i] \ge \mathbb{E}[\max(\xi_1, \xi_2)] = 1/\sqrt{\pi}$, and $\frac{1}{4}\sqrt{\log m} \le \frac{1}{4}\sqrt{\log 128} = \frac{1}{4}\sqrt{7 \log 2}$. It remains to check $1/\sqrt{\pi} \ge \frac{1}{4}\sqrt{7 \log 2}$, i.e. $16/\pi \ge 7 \log 2$: indeed $16/\pi > 16/3.15 > 5.07$ and $7 \log 2 < 7 \cdot 0.6932 < 4.86$ (Mathlib `Real.pi_lt_d2`, `Real.log_two_lt_d9`). $\square$

Remark 307 (Constants). The constant $c_{\max} = 1/4$ holds for every $m \ge 2$; the argument is not sharp ($\mathbb{E}\max_i \xi_i \sim \sqrt{2 \log m}$), but only an absolute constant is needed for Proposition **??**. The splitting at $m = 128$ is chosen so that both branches only require the crude numerical bounds $\pi < 3.15$, $0.693 < \log 2 < 0.6932$ and $e^{-1} < 1/2$, which are available in Mathlib. The outline suggested the threshold $t = \sqrt{\log m}$ and a Paley–Zygmund argument on the number of exceedances; the direct bound $\mathbb{P}(\max \geq t) \geq 1 - e^{-mp_t}$ used above is simpler and gives the same order.

Lean remark. The i.i.d. family is `Fin m →` with law `Measure.pi (fun _ gaussianReal 0 1)`; the maximum is `Finset.sup'` or `i, i`. In Theorem **??** the vectors are `EuclideanSpace (Fin m)`-valued with laws `multivariateGaussian 0 S`, and the soft-max F_β is `fun z ¹ * Real.log ( i, Real.exp ( * z i))`, whose derivatives are best handled through `ContDiff` and `fderiv` of the composition $\log \circ \sum \circ \exp$.

Chapter 15

Median Elimination

This chapter formalizes the Median Elimination algorithm of Even-Dar, Mannor and Mansour [**?**, **?**] for ε-best arm identification in a K-armed bandit with sub-Gaussian rewards, with a fixed (deterministic) budget of $O(K \log(1/\delta)/\varepsilon^2)$ samples. It is the second phase of the algorithm of Chapter **??** (paper, Appendix C), where the arms are d candidate vectors of $\mathcal{X}$ selected from a first, non-adaptive phase. The algorithm is written as an LML `Algorithm` with a deterministic policy, and its guarantee is proved through the sequential composition of Lemma **??**, one elimination round at a time.

What Mathlib/LML provides. Sub-Gaussian random variables `ProbabilityTheory.HasSubgaussianM` `X c` with the tail bounds `measure_ge_le` (Mathlib) and `measure_le_le` (LML `ForMathlib/Probability/Moments/SubGaussian.lean`), closure under sums of independent variables `HasSubgaussianMGF.sum_of_iIndepFun` and scaling `HasSubgaussianMGF.const_mul`, and Hoeffding's inequality `measure_sum_ge_le_of_iIndepFun`. LML provides stationary environments `stationaryEnv` for a kernel `: Kernel (Fin K)`, deterministic algorithms `detAlgorithm nextA` and the class `IsDeterministicAlg`, the round-robin algorithm `roundRobinAlgorithm` (`SequentialLearning/Algorithms/RoundRobin.lean`), pull counts `pullCount`, cumulative sums and empirical means `sumRewards`, `empMean` (`SequentialLearning/SumRewards.lean`), the conditional law of the feedback given the history and the current action (the structure field `IsAlgEnvSeq.hasCondDistrib_feedback n : HasCondDistrib (Y (n+1)) (history A Y n, A (n+1)) (env.feedback n) P`, and `IsObliviousEnv.hasCondDistrib_feedback_history_action` in `SequentialLearning/Stationar` the lemma `hasCondDistrib_feedback_stationaryEnv` conditions on the current action only), and the sequential composition of Lemma **??**.

What is missing. The Median Elimination algorithm, its per-round analysis and its sample complexity are not in LML. The empirical means used by the algorithm are per-round averages (not the cumulative `empMean`), so a small variant of `empMean` restricted to a time window is needed.

Formalization note. The chapter is formalized for Gaussian rewards: Median Elimination runs on K candidate arms $x^{(1)}, \ldots, x^{(K)}$ of a linear Gaussian bandit (arm a has the reward law $\mathcal{N}(\langle x^{(a)}, \theta\rangle, 1)$), which is the only case used in Chapter **??**; the sub-Gaussian generality of Definition **??** is not formalized. The algorithm is a *phased algorithm* in the sense of Lemma **??** (LML `PhasedAlg`), whose round-by-round analysis replaces the sequential composition of Lemma **??**; the additive Gaussian noise of a phase is independent of the past (Lemma **??** and `IsAlgEnvSeq.hasCondDistrib_noise_window`), which is what makes the per-round analysis possible without conditioning on the history.

15.1 Sub-Gaussian bandits and fixed designs

Definition 308 (Sub-Gaussian K-armed bandit)**.** Let $K \geq 1$. A 1-*sub-Gaussian K-armed bandit* is a Markov kernel $\nu : \mathrm{Fin}\, K \to \mathbb{R}$ (LML `Kernel (Fin K)  `) such that, for every arm a, the mean $\mu_a := \int y\, \nu(a, dy)$ exists and the centered reward $y - \mu_a$ under $\nu(a)$ has sub-Gaussian moment generating function with constant 1: $\int e^{\lambda(y-\mu_a)} \nu(a, dy) \leq e^{\lambda^2/2}$ for all $\lambda \in \mathbb{R}$ (Mathlib `HasSubgaussianMGF (fun y   y -   a) 1 (  a)`). The associated environment is stationaryEnv(ν). The Gaussian bandit $\nu(a) = \mathcal{N}(\mu_a, 1)$ is 1-sub-Gaussian (Mathlib `mgf_gaussianReal`).

Lemma 309 (Rewards of a history-independent design are independent)**.** *Learning.IsAlgEnvSeq.hasCondDistri be a Markov kernel from $\mathcal{A}$ to $\mathcal{Y}$, $a_0, a_1, \dots \in \mathcal{A}$ a deterministic sequence, and* alg *the deterministic algorithm playing a_t at round t regardless of the history (LML `detAlgorithm` with a history-independent next action). Under any algorithm-environment sequence for* (alg, stationaryEnv(ν))*, the observations $(y_t)_{t\geq 0}$ are independent and $y_t \sim \nu(a_t)$ for every t.*

Proof. By induction on t. The structure field `IsAlgEnvSeq.hasCondDistrib_feedback n` of LML states that the conditional law of y_{n+1} given the *pair* (history up to time n, x_{n+1}) is the feedback kernel `env.feedback n` evaluated at that pair; for a stationary environment this kernel is $(h, a) \mapsto \nu(a)$ (`feedback_stationaryEnv`; equivalently `IsObliviousEnv.hasCondDistrib_feedback_history_action` in `SequentialLearning/StationaryEnv.lean`, whose kernel is `(feedbackCondAction env (n+1)).prodMkLeft _`). Since the algorithm is deterministic and history-independent, $x_{n+1} = a_{n+1}$ almost surely (`action_ae_eq`), so the conditional law of y_{n+1} given $(\mathrm{hist}_n, x_{n+1})$ is the *constant* kernel $\nu(a_{n+1})$: it depends neither on the past observations nor on the action. A conditional distribution given $(\mathrm{hist}_n, x_{n+1})$ that is a constant kernel means that y_{n+1} is independent of $(\mathrm{hist}_n, x_{n+1})$, hence of $(y_s)_{s\leq n}$, with law $\nu(a_{n+1})$ (Mathlib `HasCondDistrib` with a constant kernel gives `IndepFun` and the marginal law). Note that `hasCondDistrib_feedback_stationar` conditions on the current action only and would *not* give independence from the past. The case $t = 0$ is `hasCondDistrib_feedback_zero` (law $\nu(x_0) = \nu(a_0)$). Independence of y_{n+1} from $(y_s)_{s\leq n}$ for every n gives joint independence of the whole sequence (`iIndepFun`, by induction on finite subfamilies). This is the argument of Lemma **??** for a general kernel.

Lean. Formalized in the Gaussian case in the stronger form used by the phased analysis: for any algorithm, the noises $y_t - \langle x_t, \theta\rangle$ of the rounds $t \geq m$ are i.i.d. $\mathcal{N}(0, 1)$ conditionally on the history of the first m rounds (`hasCondDistrib_noise_window`, from the abstract statement `hasCondDistrib_pi_of_hasCondDistrib_const`:

a sequence whose conditional law given an extra variable Z and its past is a constant ν is i.i.d. ν and independent of Z); hence the noise of a phase is independent of the state at the start of the phase (`PhasedAlg.hasCondDistrib_phaseNoise`), and the observations of a design determined by that state are the means plus this noise. □

Lemma 310 (Tail of an empirical mean)**.** *Learning.MedianElim.empMean, Learning.MedianElim.hasSubgaussianMGF$_e$mpMean, Learning.MedianElim.measureReal$_l$e$_e$mpMean$_l$e, Lea be independent real random variables such that each $Y_i - \mu$ has sub-Gaussian moment generating function with constant* 1*, and let $\hat\mu := \frac{1}{n}\sum_i Y_i$. Then $\hat\mu - \mu$ has sub-Gaussian moment generating function with constant $1/n$ and, for every $u \geq 0$,*

$$\mathbb{P}(\hat\mu - \mu \geq u) \leq e^{-nu^2/2}, \qquad \mathbb{P}(\hat\mu - \mu \leq -u) \leq e^{-nu^2/2}.$$

In particular, if $n \geq 8\log(3/\delta)/\varepsilon^2$ then each of $\mathbb{P}(\hat\mu - \mu \geq \varepsilon/2)$ and $\mathbb{P}(\hat\mu - \mu \leq -\varepsilon/2)$ is at most $\delta/3$.

Proof. Mathlib `HasSubgaussianMGF.sum_of_iIndepFun` (independent summands, constants add up to n) and `HasSubgaussianMGF.const_mul` with the factor $1/n$ (constant $n \cdot (1/n)^2 = 1/n$), then `measure_ge_le` / `measure_le_le`: $\mathbb{P}(X \geq u) \leq \exp(-u^2/(2c))$ with $c = 1/n$. With $u = \varepsilon/2$ the bound is $\exp(-n\varepsilon^2/8) \leq \exp(-\log(3/\delta)) = \delta/3$. □

15.2 The algorithm

Definition 311 (Schedule of Median Elimination)**.** Learning.MedianElim.epsSched, Learning.MedianElim.deltaSched, Learning.MedianElim.numPullsFor $\varepsilon > 0$, $\delta \in (0,1)$ and $\ell \in \mathbb{N}$ (rounds are 0-indexed) let

$$\varepsilon_\ell := \frac{\varepsilon}{4}\Big(\frac{3}{4}\Big)^\ell, \qquad \delta_\ell := \frac{\delta}{2^{\ell+1}}, \qquad n_\ell := \Big\lceil \frac{8\log(3/\delta_\ell)}{\varepsilon_\ell^2} \Big\rceil,$$

so that $\varepsilon_0 = \varepsilon/4$, $\delta_0 = \delta/2$, $\varepsilon_{\ell+1} = 3\varepsilon_\ell/4$ and $\delta_{\ell+1} = \delta_\ell/2$.

Lemma 312 (Sums of the schedule)**.** *Learning.MedianElim.sum$_e$psSched$_l$e, Learning.MedianElim.sum$_d$eltaS ε and $\sum_{\ell\geq 0}\delta_\ell = \delta$; in particular every finite partial sum is at most ε, resp. δ. Moreover $n_\ell \geq 8\log(3/\delta_\ell)/\varepsilon_\ell^2$ and $\log(3/\delta_\ell) = \log(3/\delta) + (\ell+1)\log 2$.*

Proof. def:me$_s$*cheduleGeometricseries* :$\varepsilon_{4\sum_\ell(3/4)^\ell=\frac{\varepsilon}{4}\cdot 4}$ and $\frac{\delta}{2}\sum_\ell 2^{-\ell} = \delta$ (Mathlib `tsum_geometric_of_lt_one`); the rest is the definition and $\log(2^{\ell+1}) = (\ell+1)\log 2$. □

Definition 313 (Halving sizes and number of rounds)**.** Learning.MedianElim.survivors, Learning.MedianElim.numRounds, Learning.MedianElim.survivors$_s$ucc, Learning.MedianElim.two$_l$e$_s$urvivor 1 and $\ell \in \mathbb{N}$ let $s_\ell := \lceil K/2^\ell\rceil$ and $L := \min\{\ell : s_\ell = 1\} = \lceil \log_2 K\rceil$. Then $s_0 = K$, $s_{\ell+1} = \lceil s_\ell/2\rceil$, $s_\ell \geq 2$ for $\ell < L$, and $s_\ell \leq 2K/2^\ell$ for $\ell < L$ (since $\lceil x\rceil \leq x + 1 \leq 2x$ when $x \geq 1$, and $K/2^\ell > 1$ when $s_\ell \geq 2$).

Definition 314 (Top-half selection)**.** Finset.topBy, Finset.topBy$_subset, Finset.card_topBy, Finset.apply_le_of_n_o$
Fin K be finite with $|S| = s \geq 2$ and $v : S \to \mathbb{R}$. Order S by the strict total order $a \succ b$ iff $v_a > v_b$, or $v_a = v_b$ and $a < b$ (ties broken by index). $\mathrm{Top}(S, v)$ is the set of the $\lceil s/2 \rceil$ largest elements of S for $\succ$. It has the property that if $a \in S \setminus \mathrm{Top}(S, v)$ then $v_b \geq v_a$ for every $b \in \mathrm{Top}(S, v)$.

Definition 315 (Median Elimination)**.** Learning.MedianElim.roundArm, Learning.MedianElim.roundScore, Learning.MedianElim.roundNext, Learning.MedianElim.medianElim, Learning.MedianElim.meOut, Learning.MedianElim.budget, Learning.MedianElim.start$_medianElim_numRou$
1, $\varepsilon > 0$, $\delta \in (0, 1)$. Median Elimination $\mathrm{ME}(K, \varepsilon, \delta)$ is the deterministic algorithm with actions in $\mathrm{Fin}\, K$ and observations in $\mathbb{R}$ defined round by round. Set $S_0 := \mathrm{Fin}\, K$ and $\tau_0 := 0$. For $\ell < L$, given the surviving set S_ℓ (with $|S_\ell| = s_\ell$) and the starting time τ_ℓ:

(i) *Round* ℓ occupies the times $\tau_\ell \leq t < \tau_{\ell+1} := \tau_\ell + s_\ell n_\ell$; the action at time $\tau_\ell + j$ is the (j mod s_ℓ)-th element of S_ℓ in increasing order (round-robin over S_ℓ), so each arm of S_ℓ is pulled exactly n_ℓ times during the round, at times that do not depend on the observations.

(ii) For $a \in S_\ell$, $\hat{\mu}_a^{(\ell)}$ is the average of the n_ℓ observations received when pulling a during round ℓ (a measurable function of the history of length $\tau_{\ell+1}$).

(iii) $S_{\ell+1} := \mathrm{Top}(S_\ell, \hat{\mu}^{(\ell)})$, of size $s_{\ell+1}$.

The *budget* of the algorithm is the deterministic integer $N_{\mathrm{ME}}(K, \varepsilon, \delta) := \tau_L = \sum_{\ell<L} s_\ell n_\ell$, and its *recommendation* is the unique element $\hat{a}$ of S_L, a measurable function of the history of length N_{ME}. After time τ_L the algorithm plays an arbitrary fixed arm (its actions are irrelevant). Formally, $\mathrm{ME}(K, \varepsilon, \delta)$ *is* the LML deterministic algorithm `detAlgorithm nextA` whose next-action function $\mathrm{nextA}(n, h)$ recomputes $S_0, \ldots, S_\ell$ from the history h of length n by (ii)–(iii), where ℓ is the round with $\tau_\ell \leq n < \tau_{\ell+1}$, and returns the action prescribed by (i) (see the Lean remark below). Its relation to the sequential composition of Lemma **??** is the content of Lemma **??**; it is not part of the definition.

Lean remark. The algorithm is the phased algorithm (Lemma **??**, `MedianElim.medianElim`) with state (c, S), where $c : \mathrm{Fin}\, K \to \mathcal{X}$ maps arms to actions (fixed along the run) and S is the surviving set: phase ℓ has length $s_\ell n_\ell$, plays the round-robin design over S (`roundArm`: offset j decodes to $(i, r) \in \mathrm{Fin}\, s_\ell \times \mathrm{Fin}\, n_\ell$ and plays the i-th element of S in increasing order, `Finset.orderEmbOfFin`), and updates S to the $s_{\ell+1}$ arms with the largest per-round empirical means (`roundNext`, through the top selection `Finset.topBy` of Definition **??**). Measurability of the update holds because the surviving set is a measurable function of the scores (`Finset.measurable_topBy`) and the state space of surviving sets is finite. The recommendation is the surviving arm (`meOut`) and the budget is τ_L (`budget`, `start_medianElim_numRounds`). The lengths of the phases are deterministic while the size of S is only guaranteed by the analysis: the good states of Lemma **??** record $|S_\ell| = s_\ell$.

Lemma 316 (Median Elimination as a sequential composition, round by round)**.** *Learning.PhasedAlg.ae$_a$ction$_s$tart$_a$dd$_e$q, Learning.PhasedAlg.ae$_p$haseObs$_e$q, Learning.PhasedAlg.ae$_s$tatePr* *L and $T_1 := \tau_\ell$. Let* $\mathrm{alg}_1 := \mathrm{ME}(K, \varepsilon, \delta)$ *and, for a history h of length T_1, let* $\mathrm{alg}_2(h)$ *be the history-independent deterministic algorithm playing the round-robin design of Definition* **??***(i) over $S_\ell(h)$ (the set computed from h by (ii)–(iii)) for $s_\ell n_\ell$ rounds, then an arbitrary fixed arm. Then $h \mapsto \mathrm{alg}_2(h)$ is measurable in the sense of Lemma* **??***, and the policy of* $\mathrm{ME}(K, \varepsilon, \delta)$ *at every time $t < \tau_{\ell+1}$ coincides with the policy of the sequential composition of* alg_1 *and* alg_2 *(Lemma* **??***) with $T_1 = \tau_\ell$. Consequently, against every environment ν, the law of the history of length $\tau_{\ell+1}$ under* $\mathrm{ME}(K, \varepsilon, \delta)$ *is its law under the composition; in particular, conditionally on the history h of length τ_ℓ, the actions and observations of round ℓ form an algorithm-environment sequence for* $(\mathrm{alg}_2(h), \nu)$*, and the probability bound of Lemma* **??** *applies to events of the history of length $\tau_{\ell+1}$.*

Proof. def:median$_e$*limination, lem : composition, lem : phased***Lean.***Thisisthegeneralrunstructureofphased* *theactionsofphaseℓ* are the design of the state at the start of the phase, the observations are the means plus the noise of the phase, which is i.i.d. $\mathcal{N}(0, 1)$ and independent of that state, and the next state is the update applied to them. The composition argument below is the informal version of that lemma.

For $t < \tau_\ell$ both algorithms follow the policy of $\mathrm{alg}_1 = \mathrm{ME}$. For $\tau_\ell \le t < \tau_{\ell+1}$ and a history h' of length t with prefix h of length τ_ℓ, $\mathrm{nextA}(t, h')$ decodes the round ℓ and the offset $j = t - \tau_\ell$, recomputes $S_0, \dots, S_\ell$ from h', and plays the ($j \bmod s_\ell$)-th element of S_ℓ. The sets S_k, $k \le \ell$, only depend on the observations received before time τ_ℓ (Definition **??**(ii)–(iii)), i.e. on h, so this action is the action of $\mathrm{alg}_2(h)$ at its time j: the two policy kernels at time t are the same deterministic kernel of the same measurable function of h'. Measurability of $h \mapsto \mathrm{alg}_2(h)$: $\mathrm{alg}_2(h)$ is the fixed round-robin algorithm reparametrized by $S_\ell(h)$, a function with finitely many values whose fibers are finite Boolean combinations of the measurable sets $\{\hat\mu_a^{(k)} \ge \hat\mu_b^{(k)}\}$, $k < \ell$, so the policy of $\mathrm{alg}_2(h)$ at time j, as a function of $(h, \cdot)$, is a deterministic kernel of a measurable function. Since the trajectory measure is determined by the initial law and the policy kernels (`Learning.trajMeasure`) and the two algorithms agree at all times $t < \tau_{\ell+1}$, the laws of the histories of length $\tau_{\ell+1}$ coincide (the policies at later times do not affect this law, by the consistency of `trajMeasure` under `frestrictLe`). The conditional statement and the probability bound are then those of Lemma **??**. ☐

15.3 Analysis

Lemma 317 (The median counting argument)**.** *Finset.exists$_m$em$_t$opBy$_o$f$_c$ard$_f$ilter$_l$tdef : pm$_t$opLetSbefinite* *$s \ge 2$, $\mu, v : S \to \mathbb{R}$, $\varepsilon > 0$, and $a_* \in S$ with $\mu_{a_*} = \max_{a \in S} \mu_a$. Define the set of* bad *arms*

$$B := \{a \in S : \mu_a < \mu_{a_*} - \varepsilon \text{ and } v_a \ge v_{a_*}\}.$$

If $|B| < s/2$ then $\max_{a \in \mathrm{Top}(S, v)} \mu_a \ge \mu_{a_*} - \varepsilon$.

Proof. def:pm$_t opLet$S' := Top(S,v)$, of size \lceil s/2\rceil \geq s/2 > |B|$. If $a_* \in S'$ the conclusion holds since $\mu_{a_*} \geq \mu_{a_*} - \varepsilon$. Otherwise $a_* \in S \setminus S'$, so by the property of Definition **??** every $b \in S'$ satisfies $v_b \geq v_{a_*}$. Since $|S'| > |B|$, there is $b \in S' \setminus B$; as $v_b \geq v_{a_*}$ and $b \notin B$, the first condition defining B fails: $\mu_b \geq \mu_{a_*} - \varepsilon$. Hence $\max_{a \in S'} \mu_a \geq \mu_b \geq \mu_{a_*} - \varepsilon$. □

Lemma 318 (One round of Median Elimination)**.** *$Learning.MedianElim.one_sub_le_measureReal_round, Learning$* $s \geq 2$, $\varepsilon > 0$, $\delta \in (0,1)$, *and let* $(\hat{\mu}_a)_{a \in S}$ *be real random variables (not necessarily independent) and* $(\mu_a)_{a \in S}$ *real numbers such that for every* $a \in S$,

$$\mathbb{P}(\hat{\mu}_a - \mu_a \geq \varepsilon/2) \leq \delta/3 \qquad \textit{and} \qquad \mathbb{P}(\hat{\mu}_a - \mu_a \leq -\varepsilon/2) \leq \delta/3.$$

Then

$$\mathbb{P}\Big(\max_{a \in \mathrm{Top}(S,\hat{\mu})} \mu_a \geq \max_{a \in S} \mu_a - \varepsilon \Big) \geq 1 - \delta.$$

In particular this applies, with $\delta = \delta_\ell$ *and* $\varepsilon = \varepsilon_\ell$, *to the empirical means* $\hat{\mu}^{(\ell)}$ *of round* ℓ *of Median Elimination conditionally on the history before the round, by Lemmas* **??** *and* **??**.

Proof. lem:pm$_s election, lem: pm_e mpmean_t ail, lem: pm_f ixed_d esign_r ewards, def: median_e limination, lem: me_s chedule_s ums, lem: pm_p olicy_c ompositionFix$a$_* \in$ S with $\mu_{a_*} = \max_S \mu$ and let B be the (random) set of bad arms of Lemma **??** with $v = \hat{\mu}$. Consider the events

$$E_1 := \{\hat{\mu}_{a_*} < \mu_{a_*} - \varepsilon/2\}, \qquad N := \sum_{a \in S} \mathbf{1}\{\hat{\mu}_a - \mu_a \geq \varepsilon/2\}, \qquad E_2 := \{N \geq s/2\}.$$

By hypothesis $\mathbb{P}(E_1) \leq \delta/3$ and $\mathbb{E}[N] \leq s\delta/3$, so by Markov's inequality $\mathbb{P}(E_2) \leq \frac{s\delta/3}{s/2} = 2\delta/3$; hence $\mathbb{P}(E_1 \cup E_2) \leq \delta$. On E_1^c, every bad arm $a \in B$ satisfies $\hat{\mu}_a \geq \hat{\mu}_{a_*} \geq \mu_{a_*} - \varepsilon/2 > \mu_a + \varepsilon/2$, so $\mathbf{1}_B(a) \leq \mathbf{1}\{\hat{\mu}_a - \mu_a \geq \varepsilon/2\}$ and $|B| \leq N$. Therefore on $E_1^c \cap E_2^c$ we have $|B| \leq N < s/2$, and Lemma **??** gives $\max_{\mathrm{Top}(S,\hat{\mu})} \mu \geq \mu_{a_*} - \varepsilon$. The last sentence: conditionally on the history before round ℓ, the set S_ℓ is fixed and the actions of the round form a history-independent design (Definition **??**(i)), i.e. round ℓ is an algorithm-environment sequence for the fixed design $\mathrm{alg}_2(h)$ of Lemma **??**; so by Lemma **??** the n_ℓ observations of each arm $a \in S_\ell$ are independent with law $\nu(a)$, and Lemma **??** with $n_\ell \geq 8 \log(3/\delta_\ell)/\varepsilon_\ell^2$ (Lemma **??**) gives the two tail bounds with $\delta_\ell/3$. □

Theorem 319 (Median Elimination is (ε, δ)-PAC)**.** *$Learning.MedianElim.isPAC_medianElimdef: pm_subgauss$* *be a* 1*-sub-Gaussian* K*-armed bandit with means* $(\mu_a)_a$, $\varepsilon > 0$ *and* $\delta \in (0,1)$. *Run against* stationaryEnv(ν), *Median Elimination* $\mathrm{ME}(K, \varepsilon, \delta)$ *uses the deterministic budget* $N_{\mathrm{ME}}(K, \varepsilon, \delta)$ *and its recommendation* $\hat{a}$ *satisfies*

$$\mathbb{P}\Big(\mu_{\hat{a}} \geq \max_a \mu_a - \varepsilon\Big) \geq 1 - \delta.$$

Lean. *Formalized for the Gaussian bandit of candidate arms* $x^{(1)}, \ldots, x^{(K)} \in \mathcal{X}$ *of a linear Gaussian bandit (Corollary* **??** *with fixed candidates), i.e.* $\nu(a) =$

$\mathcal{N}(\langle x^{(a)}, \theta\rangle, 1)$, for every θ: the identification algorithm with budget N_{ME} built from the phased algorithm is δ-PAC for the goodness predicate $\langle x^{(\hat{a})}, \theta\rangle \geq$ $\max_a \langle x^{(a)}, \theta\rangle - \varepsilon$ *(`isPAC_medianElim`).*

Proof. lem:me$_r ound, lem : me_s chedule_s ums, lem : composition, lem : pm_f ixed_d esign_r ewards, lem : pm_p olicy_c omposition, lem : phasedIf \mathrm{K} = 1 then \mathrm{L} = 0, istheuniquearmandthereisnothingtoprove. Let \mathrm{K} \geq$ 2. For $\ell < L$ let $G_\ell := \{\max_{a \in S_\ell} \mu_a \leq \max_{a \in S_{\ell+1}} \mu_a + \varepsilon_\ell\}$, an event of the history of length $\tau_{\ell+1}$. By Lemma **??**, the history of length $\tau_{\ell+1}$ under ME has the law of the composition (Lemma **??**) of ME up to time τ_ℓ with the fixed design of round ℓ parametrized by S_ℓ, so by Lemma **??** the conditional probability of G_ℓ given the history of length τ_ℓ is at least $1 - \delta_\ell$ for every value of that history. Iterating the last clause of Lemma **??** over $\ell = 0, \ldots, L-1$ (with $E_h := G_\ell$ and $\delta' := \sum_{k<\ell} \delta_k$ at step ℓ), $\mathbb{P}(\bigcap_{\ell<L} G_\ell) \geq 1 - \sum_{\ell<L} \delta_\ell \geq 1 - \delta$ by Lemma **??**. On $\bigcap_{\ell<L} G_\ell$, telescoping,

$$\max_a \mu_a = \max_{a \in S_0} \mu_a \leq \max_{a \in S_L} \mu_a + \sum_{\ell<L} \varepsilon_\ell = \mu_{\hat{a}} + \sum_{\ell<L} \varepsilon_\ell \leq \mu_{\hat{a}} + \varepsilon,$$

again by Lemma **??**, since $S_L = \{\hat{a}\}$. □

Lemma 320 (Budget of Median Elimination)**.** *$Learning.MedianElim.budget_l e, Learning.MedianElim.budget_l$ 1, $\varepsilon > 0$, $\delta \in (0,1)$,*

$$N_{\mathrm{ME}}(K, \varepsilon, \delta) \leq \frac{256K}{\varepsilon^2}\Big(9 \log \frac{3}{\delta} + 81 \log 2\Big) + 4K.$$

If moreover $\varepsilon \leq 1$, then $N_{\mathrm{ME}}(K, \varepsilon, \delta) \leq C_{\mathrm{ME}}\, K \log(3/\delta)/\varepsilon^2$ with $C_{\mathrm{ME}} := 16000$.

Proof. lem:me$_s chedule_s umsByDefinition$ **??**,s$_\ell \leq 2K/2^\ell$ for $\ell < L$, and $n_\ell \leq 8 \log(3/\delta_\ell)/\varepsilon_\ell^2 + 1$ with $1/\varepsilon_\ell^2 = 16(16/9)^\ell/\varepsilon^2$ and $\log(3/\delta_\ell) = \log(3/\delta) + (\ell + 1) \log 2$ (Lemma **??**). Hence

$$s_\ell n_\ell \leq \frac{2K}{2^\ell} \cdot \frac{128\,(16/9)^\ell}{\varepsilon^2}\Big(\log \frac{3}{\delta} + (\ell+1) \log 2\Big) + \frac{2K}{2^\ell} = \frac{256K}{\varepsilon^2}\Big(\frac{8}{9}\Big)^\ell \Big(\log \frac{3}{\delta} + (\ell+1) \log 2\Big) + \frac{2K}{2^\ell}.$$

Summing over $\ell < L$ and extending the sums to all $\ell \geq 0$ (nonnegative terms), with $\sum_{\ell \geq 0} (8/9)^\ell = 9$, $\sum_{\ell \geq 0} (\ell + 1)(8/9)^\ell = 1/(1 - 8/9)^2 = 81$ (Mathlib `tsum_geometric_of_lt_one`, `tsum_coe_mul_geometric_of_norm_lt_one`) and $\sum_{\ell \geq 0} 2^{-\ell} = 2$,

$$N_{\mathrm{ME}} = \sum_{\ell<L} s_\ell n_\ell \leq \frac{256K}{\varepsilon^2}\Big(9 \log \frac{3}{\delta} + 81 \log 2\Big) + 4K.$$

For the second form, use $\log(3/\delta) \geq \log 3 > 1.09$ (as $\delta < 1$) and $\varepsilon \leq 1$: $81 \log 2 = \frac{81 \log 2}{\log 3} \log 3 \leq 51.2 \log(3/\delta)$ (since $81 \cdot 0.6932/1.0986 < 51.2$), so $9 \log(3/\delta) + 81 \log 2 \leq 60.2 \log(3/\delta)$ and $256 \cdot 60.2 < 15412$; and $4K \leq 4K \log(3/\delta)/(\varepsilon^2 \log 3) \leq 3.7K \log(3/\delta)/\varepsilon^2$. Altogether $N_{\mathrm{ME}} \leq (15412 + 3.7) K \log(3/\delta)/\varepsilon^2 \leq 16000\, K \log(3/\delta)/\varepsilon^2$. □

Remark 321 (Constants of the budget). The outline announced the bound $(128K/\varepsilon^2)(9\log(3/\delta)+81\log 2)+2K$; the factor 128 there implicitly uses $s_\ell \leq K/2^\ell$, which fails because of the ceilings ($s_\ell = \lceil K/2^\ell \rceil$). The correct bookkeeping uses $s_\ell \leq 2K/2^\ell$ (valid while $s_\ell \geq 2$), which doubles the constants: 256 and $4K$. The resulting $C_{\mathrm{ME}} = 16000$ is far from optimal (the schedule of Definition **??** is the one of [**?**] and was not tuned), but only its existence matters for Theorem **??**.

Corollary 322 (Median Elimination on candidate arms of a linear bandit)**.** *$Learning.MedianElim.isPAC_medianElim, Learning.MedianElim.one_sub_le_measureReal_roundNext_mem, Lea$* $\mathbb{R}^d$*,* $K \geq 1$*, and* $x^{(1)}, \ldots, x^{(K)} \in \mathcal{X}$*. Consider the algorithm with actions in* $\mathcal{X}$ *that runs* $\mathrm{ME}(K,\varepsilon,\delta)$ *on the arms* $\mathrm{Fin}\,K$ *and plays* $x^{(a)}$ *whenever* ME *pulls* a *(its next action is* $x^{(\mathrm{nextA}(n,\cdot))}$ *applied to the sequence of past observations). Against* ν_θ *(Definition* **??***) it uses the budget* $N_{\mathrm{ME}}(K,\varepsilon,\delta) \leq C_{\mathrm{ME}} K \log(3/\delta)/\varepsilon^2$ *(for* $\varepsilon \leq 1$*) and its recommendation* $\hat{a}$ *satisfies*

$$\mathbb{P}_\theta\Big(\langle x^{(\hat{a})}, \theta\rangle \geq \max_{a \leq K} \langle x^{(a)}, \theta\rangle - \varepsilon\Big) \geq 1-\delta.$$

The same holds when the candidates $x^{(a)} = x^{(a)}(h)$ *are measurable functions of a history* h *of a previous phase, conditionally on* h*, so that by Lemma* **??** *the composed algorithm satisfies the conclusion with* δ *replaced by* $\delta + \delta'$ *if the previous phase fails with probability at most* δ'*.*

Proof. thm:median$_elimination, lem : me_budget, lem : composition, lem : gauss_tail, lem :$
$phased$**Lean.**$The fixed-candidates statement is$ `isPAC_medianElim` $(this is the form in which Theorem$ **??** $is form$
$round guarantee$ `one_sub_le_measureReal_roundNext_mem` $applies to the candidates of the current state; the unio$
δ' directly, without a composition lemma; see the region algorithm of Definition **??**.

Let $\nu'(a) := \mathcal{N}(\langle x^{(a)}, \theta\rangle, 1)$, a 1-sub-Gaussian K-armed bandit with means $\mu_a = \langle x^{(a)}, \theta\rangle$ (Definition **??**; the Gaussian moment generating function is `mgf_gaussianReal`, see also Lemma **??**). The sequence of arms pulled and observations received by the $\mathcal{X}$-valued algorithm against ν_θ is an algorithm-environment sequence for $(\mathrm{ME}, \mathrm{stationaryEnv}(\nu'))$: the next arm is the same deterministic function of past observations, and the conditional law of the observation given the past and the arm a is $\mathcal{N}(\langle x^{(a)}, \theta\rangle, 1) = \nu'(a)$ (the structure field `IsAlgEnvSeq.hasCondDistrib_feedback` for ν_θ, whose kernel is $(h,x) \mapsto \mathcal{N}(\langle x, \theta\rangle, 1)$, composed with the map $a \mapsto x^{(a)}$). Hence Theorem **??** applies and gives the probability bound, and Lemma **??** the budget. The conditional version is Lemma **??** with $\mathrm{alg}_2(h)$ the algorithm just described for the candidates $x^{(a)}(h)$; measurability in h holds because $h \mapsto (x^{(a)}(h))_a$ is measurable and the policy of $\mathrm{alg}_2(h)$ is the deterministic kernel of a measurable function of $(h, \mathrm{observations})$. □

Lean remark. The $\mathcal{X}$-valued algorithm is `detAlgorithm` with next action `fun n h   x (nextA n (fun t   (h t).2))`, where `nextA` only reads observations; this avoids having to recover the arm index from the played vector $x^{(a)}$ (the map $a \mapsto x^{(a)}$ need not be injective). The identification of the two algorithm-environment sequences is an instance of the general fact that mapping the actions of a deterministic algorithm through a measurable map f turns an algorithm-environment sequence for the environment $\nu \circ f$ into one for ν, to be contributed to LML.

Chapter 16

Gaussian posteriors for adaptively collected data and the unit-ball lower bound

This chapter replaces the citation of [**?**] in Appendix H of the paper by a self-contained argument: for every (adaptive, possibly randomized) identification algorithm with budget T on the unit ball $\mathbb{B}_d$, there is a reward vector θ of norm $O(d/\sqrt{T})$ under which the expected simple regret is $\Omega(d/\sqrt{T})$; consequently an (ε, δ)-PAC algorithm on $\mathbb{B}_d$ needs $T = \Omega(d^2/\varepsilon^2)$ samples (Corollary **??**, used in Chapter **??**). The proof is Bayesian: under a Gaussian prior $\theta \sim \mathcal{N}(0, \sigma^2 I_d)$, the posterior mean of θ given the history is $m_T = (\sigma^{-2} I + S_T)^{-1} b_T$, and the recommendation cannot correlate with θ better than with m_T. We never use abstract conditional expectations: the key identity $\mathbb{E}\langle \varphi, \theta - m_T \rangle = 0$ is proved by Fubini, the likelihood ratio of the history with respect to the noise-only environment, and an explicit completion of the square in $\mathbb{R}^d$. The likelihood ratio is first established for general stationary environments whose kernels have densities, as the environment analogue of LML's `Algorithm.density`.

What Mathlib/LML provides. Measures with densities `Measure.withDensity`, `withDensity_mul`, `Measure.compProd_withDensity` (`Mathlib/Probability/Kernel/CompProdEqIff.lean`: $\mu \otimes (\kappa.\text{withDensity } f) = (\mu \otimes \kappa).\text{withDensity}(f)$), and in LML `ForMathlib/Probability/WithDensity.lean`: `compProd_withDensity_left`, `compProd_withDensity_withDensity`, `map_equiv_withDensity`, `Kernel.compProd_withDensity_left` (a density on the *left* factor of a kernel composition-product). The algorithm density `Algorithm.density` and `IsAlgEnvSeq.hasLaw_history_withDe` (`SequentialLearning/AlgorithmDensity.lean`) are the model for Lemma **??**. Bayesian sequences: `IsBayesAlgEnvSeq`, `bayesStationaryEnv`, `IT.bayesTrajMeasure`, `IsBayesAlgEnvSeq.ae_IsAlgEnvSeq` (`SequentialLearning/BayesStationaryEnv.lean`) give the joint law of a parameter drawn from a prior and the trajectory. Gaussians: `gaussianReal_of_var_ne_zero`, `gaussianPDFReal`, `integral_gaussian`, `multivariateGaussian`, `stdGaussian`, the linear change of variables for the Lebesgue measure `MeasureTheory.Measure.map_linearMap_addHaar_eq_smul_addHaar` and `MeasureTheory.integral_map_equiv`.

Formalization note. The chapter is formalized along the plan above, with two deviations. The posterior identities of Section **??** are obtained from Stein's identity (Gaussian integration by parts, Lemma **??**, extended to functions of exponential growth in Lemma **??**) applied to the tilt $\theta \mapsto \exp(\langle b, \theta\rangle - \frac{1}{2}\theta^\top S\theta)$, instead of the Lebesgue density of the prior and a change of variables; no density with respect to the Lebesgue measure on $\mathbb{R}^d$ is ever used. And the lower bound $\mathbb{E}\|\theta\| \geq \sigma d/\sqrt{d+2}$ of the outline is replaced by the sharper $\mathbb{E}\|\theta\| \geq \sqrt{2/\pi}\,\sigma\sqrt{d}$ of Lemma **??**, which improves the constants (we keep the outline's headline constants).

16.1 Likelihood ratio of the history with respect to a reference environment

Definition 323 (Environment density along a history)**.** Learning.envDensity, Learning.measurable$_e$$nvDensity, Learning.envDensity_z ero, Learning.envDensity_s nocLetA,Ybemeasurables$ $\mathcal{A} \times \mathcal{Y} \to [0, \infty]$ be measurable. For $n \geq 0$ and a history $h = (x_t, y_t)_{t \leq n}$ define $D^\rho_n(h) := \prod_{t=0}^n \rho(x_t, y_t)$, a measurable function of h. For a history of length $T \geq 1$ we write $D^\rho_T := D^\rho_{T-1}$.

Lemma 324 (Composition-product of a kernel with a kernel having a density)**.** *$ProbabilityTheory.Kernel.compProd_w ithDensity_r ight, ProbabilityTheory.Kernel.prodMkLeft_w ithDensityLe$ be an s-finite kernel from $\mathcal{H}$ to $\mathcal{A}$, η an s-finite kernel from $\mathcal{H} \times \mathcal{A}$ to $\mathcal{Y}$, and $f : (\mathcal{H} \times \mathcal{A}) \times \mathcal{Y} \to [0, \infty]$ measurable such that the kernel $\eta.\mathrm{withDensity}(f)$ is s-finite (in all applications below it is a Markov kernel). Then, as kernels from $\mathcal{H}$ to $\mathcal{A} \times \mathcal{Y}$,*

$$\pi \otimes \big(\eta.\mathrm{withDensity}(f)\big) = (\pi \otimes \eta).\mathrm{withDensity}\big((h, (a, y)) \mapsto f((h, a), y)\big).$$

In particular, if $f((h, a), y) = \rho(a, y)$ does not depend on h, the density of the composition-product is $(h, (a, y)) \mapsto \rho(a, y)$, i.e. $\rho \circ \mathrm{snd}$.

Proof. (The Lean proof is not fiberwise: both sides are evaluated on a measurable set through `Kernel.compProd_apply`, `Kernel.withDensity_apply'` and `Kernel.lintegral_compProd`, exactly as Mathlib's measure-level `Measure.compProd_withDensity`.) Fiberwise. Fix $h \in \mathcal{H}$. For every kernel η' from $\mathcal{H} \times \mathcal{A}$ to $\mathcal{Y}$, $(\pi \otimes \eta')(h) = \pi(h) \otimes \eta'_h$, where $\eta'_h := \eta'((h, \cdot))$ is the section kernel from $\mathcal{A}$ to $\mathcal{Y}$ (Mathlib `Kernel.compProd_apply_eq_compProd_sectR`). The section of a kernel with a density is the section with the sectioned density: $(\eta.\mathrm{withDensity}(f))_h = \eta_h.\mathrm{withDensity}\big((a, y) \mapsto f((h, a), y)\big)$, both sides being the measure $\eta((h, a)).\mathrm{withDensity}(f((h, a), \cdot))$ at each a (`Kernel.withDensity_apply`). The measure-level identity `Measure.compProd_withDensity` (`Mathlib/Probability/Kernel/CompProdEqIff.lean`) then gives

$$\pi(h) \otimes \big(\eta_h.\mathrm{withDensity}(f((h, \cdot), \cdot))\big) = \big(\pi(h) \otimes \eta_h\big).\mathrm{withDensity}\big((a, y) \mapsto f((h, a), y)\big),$$

and the right-hand side is $\big((\pi \otimes \eta).\mathrm{withDensity}(\tilde{f})\big)(h)$ with $\tilde{f}(h,(a,y)) := f((h,a),y)$, by `Kernel.withDensity_apply` again ($\tilde{f}$ is measurable, being f composed with a measurable rearrangement of the variables) and `Kernel.compProd_apply_eq_compProd_sectl`
Two kernels that agree at every point are equal (`Kernel.ext`). ☐

Lemma 325 (Environment likelihood ratio)**.** *Learning.IsAlgEnvSeq.map$_h$istory$_e$q$_w$ithDensity$_e$nv, Learning.I be Markov kernels from $\mathcal{A}$ to $\mathcal{Y}$ and $\rho : \mathcal{A} \times \mathcal{Y} \to [0,\infty]$ measurable with $\kappa(a) = \kappa_0(a).\mathrm{withDensity}(\rho(a,\cdot))$ for every $a \in \mathcal{A}$. Let* alg *be any algorithm and, for $n \geq 0$, let $\mathbb{P}^{(n)}$ and $\mathbb{P}_0^{(n)}$ be the laws of the history $(x_t, y_t)_{t \leq n}$ under* stationaryEnv(κ)*, resp.* stationaryEnv(κ_0) *(Definition* **??***). Then*

$$\mathbb{P}^{(n)} = \mathbb{P}_0^{(n)}.\mathrm{withDensity}\big(D_n^\rho\big),$$

so $\mathbb{P}^{(n)} \ll \mathbb{P}_0^{(n)}$ and, for every measurable $f \geq 0$ (or f integrable under $\mathbb{P}^{(n)}$), $\int f \, d\mathbb{P}^{(n)} = \int f \, D_n^\rho \, d\mathbb{P}_0^{(n)}$.

Proof. def:pb$_e$nv$_d$ensity, lem : pb$_k$ernel$_c$ompProd$_w$ithDensity$_r$ightInductiononn, exactlyasLML′s`hasLaw_hi`
for the kernels $(h,a) \mapsto \kappa(a)$, $(h,a) \mapsto \kappa_0(a)$ (the feedback kernels of the stationary environments, LML `feedback_stationaryEnv`), so that $\tilde{\kappa} = \tilde{\kappa}_0.\mathrm{withDensity}((h,a),y) \mapsto \rho(a,y))$.

Base case. (x_0, y_0) has law $p_0 \otimes \kappa = p_0 \otimes (\kappa_0.\mathrm{withDensity}\, \rho) = (p_0 \otimes \kappa_0).\mathrm{withDensity}((a,y) \mapsto \rho(a,y))$ (Mathlib `Measure.compProd_withDensity`), and hist_0 is its image by the measurable equivalence $\mathcal{A} \times \mathcal{Y} \simeq \mathcal{H}_0$, which commutes with `withDensity` (LML `map_equiv_withDensity`); $D_0^\rho(h) = \rho(x_0, y_0)$.

Induction step. By LML `history_succ` and `hasCondDistrib_step`, $\mathbb{P}^{(n+1)}$ is the image under $e^{-1} : \mathcal{H}_n \times (\mathcal{A} \times \mathcal{Y}) \simeq \mathcal{H}_{n+1}$ of $\mathbb{P}^{(n)} \otimes (\pi_n \otimes \tilde{\kappa})$, where π_n is the policy kernel. Now $\pi_n \otimes \tilde{\kappa} = (\pi_n \otimes \tilde{\kappa}_0).\mathrm{withDensity}((h,(a,y)) \mapsto \rho(a,y))$ (Lemma **??** with $f((h,a),y) := \rho(a,y)$; here $\tilde{\kappa} = \tilde{\kappa}_0.\mathrm{withDensity}(f)$ is a Markov kernel), and by the induction hypothesis $\mathbb{P}^{(n)} = \mathbb{P}_0^{(n)}.\mathrm{withDensity}(D_n^\rho)$. A composition-product of a measure with a density and a kernel with a density is the composition-product with the product density (LML `compProd_withDensity_withDensity`):

$$\mathbb{P}^{(n)} \otimes (\pi_n \otimes \tilde{\kappa}) = \big(\mathbb{P}_0^{(n)} \otimes (\pi_n \otimes \tilde{\kappa}_0)\big).\mathrm{withDensity}\big((h,(a,y)) \mapsto D_n^\rho(h)\rho(a,y)\big).$$

Pushing forward by e^{-1} (`map_equiv_withDensity`) gives $\mathbb{P}^{(n+1)} = \mathbb{P}_0^{(n+1)}.\mathrm{withDensity}(D_{n+1}^\rho)$ since $D_{n+1}^\rho(e^{-1}(h,(a,y))) = D_n^\rho(h)\rho(a,y)$. The integral formulas are `lintegral_withDensity_eq_lintegral_`
and `integral_withDensity_eq_integral_smul`. ☐

Lean remark. `Learning.envDensity   n : (Fin n →   ×  ) →  0∞` is the product $\prod_{t<n} \rho(x_t, y_t)$ over `Fin n` (so that it applies to the histories `finHistory` used throughout), with the recursion `envDensity_snoc`; the law identity is proved first for LML's `history X Y n : Iic n →   ×` by the induction of `hasLaw_history_withDensity` (`map_history_eq_withDensity_env`) and transported to `finHistory` through `toFinHistory` (`map_finHistory_eq_withDensity_env`). The kernel-level identity $\pi \otimes (\tilde{\kappa}_0.\mathrm{withDensity}\, \rho) = (\pi \otimes \tilde{\kappa}_0).\mathrm{withDensity}(\rho \circ \mathrm{snd})$ is Lemma **??**, the right-factor analogue of LML's `Kernel.compProd_withDensity_left`; it is not in Mathlib or LML and is proved fiberwise from `Measure.compProd_withDensity`.

Lemma 326 (Gaussian density ratio)**.** *ProbabilityTheory.gaussianReal$_e$q$_w$ithDensity$_e$xp, ProbabilityTheory.g* $\mathbb{R}$, $\mathcal{N}(m,1) = \mathcal{N}(0,1).\mathrm{withDensity}\big(y \mapsto e^{ym-m^2/2}\big)$.

Proof. Both are Lebesgue measure with densities (Mathlib `gaussianReal_of_var_ne_zero`): $\mathcal{N}(m,1) = \mathrm{Leb.withDensity}(\varphi(\cdot - m))$ with $\varphi(y) = (2\pi)^{-1/2}e^{-y^2/2}$, and $\varphi(y - m) = \varphi(y)\, e^{ym-m^2/2}$ since $-(y-m)^2/2 = -y^2/2 + ym - m^2/2$. Conclude with `withDensity_mul`: $(\mathrm{Leb.withDensity}\,\varphi).\mathrm{withDensity}(e^{ym-m^2/2}) = \mathrm{Leb.withDensity}(\varphi \cdot e^{ym-m^2/2})$. □

Corollary 327 (Likelihood ratio of a linear Gaussian history)**.** *Learning.LinearBandit.stepLogLR, Learning.LinearBandit.stepLR, Learning.LinearBandit.measurable$_u$ncurry$_s$tepLR, Learning.LinearBandit.lin* $\mathbb{R}^d$*, $T \geq 1$, and for a history $h = (x_t, y_t)_{t<T}$ define*

$$S_T(h) := \sum_{t<T} x_t x_t^\top \in \mathcal{S}_+^d, \qquad b_T(h) := \sum_{t<T} y_t x_t \in \mathbb{R}^d, \qquad \ell_\theta(h) := \exp\Big(\langle b_T(h), \theta\rangle - \tfrac{1}{2}\theta^\top S_T(h)\theta\Big).$$

*For every algorithm with actions in $\mathcal{X}$, the law $\mathbb{P}_\theta^T$ of the history of length T under ν_θ (Definition **??**) satisfies $\mathbb{P}_\theta^T = \mathbb{P}_0^T.\mathrm{withDensity}(\ell_\theta)$, where $\mathbb{P}_0^T$ is the law under ν_0 (pure noise: $y_t \sim \mathcal{N}(0,1)$). Moreover $(\theta, h) \mapsto \ell_\theta(h)$ is jointly measurable and positive. (In Lean, `withDensity` takes an 0∞-valued density, so the density is `fun h   ENNReal.ofReal (    h)`, and likewise $\mathrm{ENNReal.ofReal} \circ \rho_\theta$ in the proof; joint measurability of $(\theta, h) \mapsto \mathrm{ENNReal.ofReal}(\ell_\theta(h))$ follows from that of ℓ and `ENNReal.measurable_ofReal`.) In Lean, `likelihood   h` is the sum form $\exp(\sum_{t<T}(y_t\langle x_t, \theta\rangle - \frac{1}{2}\langle x_t, \theta\rangle^2))$, valid on any action set; the identification with $\langle b_T, \theta\rangle - \frac{1}{2}\theta^\top S_T\theta$ is `Maiti2026Power.likelihood_eq_gaussTilt` (stated on the unit ball, with `histB`, `histS` for b_T, S_T).*

Proof. lem:env$_l$*ikelihood*$_r$*atio, lem* : *pb*$_g$*aussian*$_d$*ensity*$_r$*atioLemma* **??***with*m $= \langle x, \theta\rangle$ gives $\mathcal{N}(\langle x, \theta\rangle, 1) = \mathcal{N}(0,1).\mathrm{withDensity}(\rho_\theta(x, \cdot))$ with $\rho_\theta(x,y) := \exp(y\langle x, \theta\rangle - \frac{1}{2}\langle x, \theta\rangle^2)$, jointly continuous in (θ, x, y). Lemma **??** then gives the density $D_T^{\rho_\theta}(h) = \prod_{t<T}\exp(y_t\langle x_t, \theta\rangle - \frac{1}{2}\langle x_t, \theta\rangle^2) = \ell_\theta(h)$, since $\sum_t y_t\langle x_t, \theta\rangle = \langle b_T, \theta\rangle$ and $\sum_t \langle x_t, \theta\rangle^2 = \theta^\top S_T\theta$. Since $\ell_\theta > 0$ is real-valued, the 0∞ density $\mathrm{ENNReal.ofReal}(\ell_\theta)$ is finite everywhere and $\mathrm{ENNReal.ofReal}(\prod_t \rho_\theta(x_t, y_t)) = \prod_t \mathrm{ENNReal.ofReal}(\rho_\theta(x_t, y_t))$ (`ENNReal.ofReal_prod_of_nonneg`). □

Lemma 328 (The law of the run as a Markov kernel in θ)**.** *Learning.LinearBandit.noiseHistLaw, Learning.LinearBandit.histKernel, Learning.LinearBandit.histKernel$_a$pply, Learning.IsAlgEnvSeq.map$_f$inHi = (alg,ρ) an identification algorithm with budget $T \geq 1$ on $\mathcal{X}$ (Definition **??**). Define, for $\theta \in \mathbb{R}^d$,*

$$K(\theta) := \mathbb{P}_0^T.\mathrm{withDensity}\big(\mathrm{ENNReal.ofReal} \circ \ell_\theta\big), \qquad \kappa(\theta) := K(\theta) \otimes \rho.$$

*Then K is a Markov kernel from $\mathbb{R}^d$ to histories of length T with $K(\theta) = \mathbb{P}_\theta^T$ for every θ, and κ is a Markov kernel from $\mathbb{R}^d$ to pairs (history, recommendation) with $\kappa(\theta) = \mathbb{P}_\theta^T \otimes \rho = \mathbb{P}_\theta$, the law of $(H_T, \hat{x})$ of Definition **??**, for* every θ*. In particular κ satisfies the hypothesis of Lemma **??** for every prior π, and for every bounded measurable g of $(\theta, H_T, \hat{x})$ the map $\theta \mapsto \mathbb{E}_\theta[g(\theta, H_T, \hat{x})] = \int g(\theta, \cdot)\, d\kappa(\theta)$ is measurable.*

Proof. cor:pb$_g aussian_l ikelihood$KisKernel.withDensity$of the constant kernel \theta \mapsto$ $\mathbb{P}_0^T$ with the density $(\theta, h) \mapsto$ ENNReal.ofReal$(\ell_\theta(h))$, which is jointly measurable by Corollary **??**; this joint measurability is exactly what `Kernel.withDensity` requires to produce a kernel, and $K(\theta) = \mathbb{P}_\theta^T$ by the same corollary (`Kernel.withDensity_apply`). Each $K(\theta)$ is a probability measure, so K is a Markov kernel. Then $\kappa := K \otimes \rho'$ with $\rho' := \rho$.prodMkLeft (the kernel $(\theta, h) \mapsto \rho(h)$) is a Markov kernel (`Kernel.compProd` of Markov kernels) with $\kappa(\theta) = K(\theta) \otimes \rho = \mathbb{P}_\theta^T \otimes \rho$ (`Kernel.compProd_apply_eq_compProd_sectR`, `Kernel.prodMkLeft_apply`), which is $\mathbb{P}_\theta$ by Definition **??** and the uniqueness of the law of the history. The measurability of $\theta \mapsto \int g \, d\kappa(\theta)$ is the measurability of integrals against a kernel (`Kernel.measurable_lintegral`, `Measurable.lintegral_kernel_prod_right`, and `StronglyMeasurable.integral_kernel_prod_right` for real-valued bounded g). □

Lean remark. This is the construction announced in the Lean remark of Lemma **??**: the measurability of $\theta \mapsto$ trajMeasure(alg, ν_θ) is never needed, because the kernel is built from the single reference law $\mathbb{P}_0^T$ and the explicit density ℓ_θ.

16.2 The joint law of the parameter, the history and the recommendation

Definition 329 (Bayesian joint law and posterior quantities)**.** Maiti2026Power.noisePairLaw, Maiti2026Power.pairKernel$_e q_w ithDensity, Maiti2026Power.bayesJointLaw, Maiti2026Power.bayesJointLa$
1, $\mathcal{X}$ an action set, $\mathsf{Alg} = (\mathrm{alg}, \rho)$ an identification algorithm with budget $T \geq 1$ on $\mathcal{X}$ (Definition **??**) and $\sigma > 0$. Let $\pi := \mathcal{N}(0, \sigma^2 I_d)$ (the prior). The *joint law* of $(\theta, H_T, \hat{x})$ is the probability measure on $\mathbb{R}^d \times \mathcal{H}_T \times \mathcal{X}$

$$Q := \pi \otimes \kappa, \qquad \kappa(\theta) = \mathbb{P}_\theta^T \otimes \rho = \mathbb{P}_\theta,$$

where κ is the Markov kernel of Lemma **??** (built from $\mathbb{P}_\theta^T = \mathbb{P}_0^T$.withDensity$(\ell_\theta)$, Corollary **??**). Equivalently, with $Q_0 := \mathbb{P}_0^T \otimes \rho$ the law of (history, recommendation) under pure noise, $\kappa(\theta) = Q_0$.withDensity$((h, x) \mapsto \ell_\theta(h))$ and

$$Q = (\pi \times Q_0).\mathrm{withDensity}\big((\theta, h, x) \mapsto \ell_\theta(h)\big).$$

We write $\mathbb{E}$ for the expectation under Q and, for a history h of length T,

$$V_T(h) := \big(\sigma^{-2} I_d + S_T(h)\big)^{-1} \in \mathcal{S}_{++}^d, \qquad m_T(h) := V_T(h)\, b_T(h) \in \mathbb{R}^d,$$

so that $(\sigma^{-2} I + S_T) m_T = b_T$ and $b_T - S_T m_T = \sigma^{-2} m_T$.

Lean remark. The pure-noise laws are `noiseHistLaw` and `noisePairLaw`, the kernel is `pairKernel`, the joint law is `bayesJointLaw A T` and its product-with-density form is `bayesJointLaw_eq_withDensity`. The posterior quantities are defined for a general tilt (b, S): `postPrec S` $= \sigma^{-2} I + S$, `postCov S` its inverse and `postMean b S` $= V b$; they are measurable functions of the history through b_T, S_T, but this measurability is never needed (see the Lean remark after Lemma **??**).

Lemma 330 (Fubini for the joint law)**.** $Maiti2026Power.integral_b ayesJointLaw, Maiti2026Power.integrable_b$
: $R^d \times \mathcal{H}_T \times \mathcal{X} \to [0, \infty]$,

$$\mathbb{E}[f(\theta, H_T, \hat{x})] = \iiint f(\theta, h, x)\, \rho(h, dx)\, \mathbb{P}_\theta^T(dh)\, \pi(d\theta) = \iint \ell_\theta(h)\, f(\theta, h, x)\, Q_0(dh, dx)\, \pi(d\theta) = \iint \ell_\theta(h)\, f(\theta,$$

and the same holds for real-valued f that are Q-integrable, i.e. such that $\ell \cdot f$ is $\pi \times Q_0$-integrable. In particular $\int \ell_\theta\, dQ_0 = 1$ for every θ, the marginal law of θ is π, the conditional law of $(H_T, \hat{x})$ given θ is $\mathbb{P}_\theta$, and for a function g of θ alone, $\int \ell_\theta(h) g(\theta)\, d(\pi \times Q_0) = \int g\, d\pi$.

Proof. cor:pb$_g aussian_l ikelihood The first equality is Tonelli/Fubini for composition-$
$products(Mathlib$`Measure.lintegral_compProd`, `Measure.integral_compProd`$), the second is the density of Co$
Q_0 (`integral_prod`, `integral_prod_symm`). Since $\kappa(\theta)$ is a probability measure, $\int \ell_\theta\, dQ_0 = \kappa(\theta)(\text{all}) = 1$. ☐

16.3 Posterior mean and posterior second moment

Lemma 331 (Stein's identity for functions of exponential growth)**.** $ProbabilityTheory.integral_s ub_m ul_g aussianR$
: $R^d \to \mathbb{R}$ *be* C^1 *with* $|F(x)| \le Ae^{B\|x\|}$ *and* $\|DF(x)\| \le Ae^{B\|x\|}$ *for all x (some constants A, B). Then for $g \sim \mathcal{N}(0, I_d)$ and every $a \in \mathbb{R}^d$, $\mathbb{E}[\langle a, g\rangle F(g)] = \mathbb{E}[DF(g)\, a]$, and for $X \sim \mathcal{N}(0, S)$ with $S \in \mathcal{S}_+^d$, $\mathbb{E}[\langle a, X\rangle F(X)] = \mathbb{E}[DF(X)\,(Sa)]$.*

Proof. lem:gauss$_i bp, lem : psf_i bp_1 d The one-dimensional identity \mathbb{E}[\xi f(\xi)] = \mathbb{E}[f'(\xi)]$ (Lemma **??**) only needs the integrability of f, $\xi f(\xi)$ and f', which holds for functions of exponential growth since $e^{B|t|}$ has all Gaussian moments. The multivariate identity follows by the coordinate-wise Fubini argument of Lemma **??**: the section $t \mapsto F(x + te_i)$ has growth $Ae^{B\|x\|}e^{B|t|}$, and the integrability of $\langle a, g\rangle F(g)$ and $DF(g)a$ under a Gaussian measure is Fernique's theorem. ☐

Lemma 332 (Gaussian tilt: derivative and growth)**.** *ProbabilityTheory.gaussTilt,*
$ProbabilityTheory.gaussTilt_p os, ProbabilityTheory.gaussTilt_l e, ProbabilityTheory.hasFDerivAt_g aussTilt, P$
$\mathcal{S}_+^d$*,* $b \in \mathbb{R}^d$ *and* $\ell(\theta) := \exp(\langle b, \theta\rangle - \frac{1}{2}\theta^\top S\theta)$*. Then $\ell > 0$, $\ell(\theta) \le e^{\|b\|\|\theta\|}$, ℓ is C^1 with $D\ell(\theta) = \ell(\theta)\,\langle b - S\theta, \cdot\rangle$, and for $c \in \mathbb{R}^d$, $r \in \mathbb{R}$ the function $\theta \mapsto (\langle c, \theta\rangle + r)\ell(\theta)$ and its derivative have exponential growth: they are bounded by $Ae^{(\|b\|+2)\|\theta\|}$ with $A = (\|c\| + |r|)(\|b\| + \|S\| + 1) + \|c\|$.*

Proof. $\theta^\top S\theta \ge 0$ gives the bound; the derivative is the chain rule, and the growth bounds use $\|\theta\| \le e^{\|\theta\|}$. ☐

Lemma 333 (Stein's identity for the affine tilt)**.** *ProbabilityTheory.integral$_i$nner$_m$ul$_a$ffine$_m$ul$_g$aussTilt, Prob* 0, $\pi = \mathcal{N}(0, \sigma^2 I_d)$, $S \in \mathcal{S}_+^d$, $b \in \mathbb{R}^d$ *and* ℓ *as above. For all* $a, c \in \mathbb{R}^d$ *and* $r \in \mathbb{R}$,

$$\int \langle a, \theta\rangle \,(\langle c, \theta\rangle + r)\,\ell(\theta)\,\pi(d\theta) = \sigma^2 \int \big(\langle c, a\rangle + (\langle c, \theta\rangle + r)\langle b - S\theta, a\rangle\big)\ell(\theta)\,\pi(d\theta).$$

All the functions $\theta \mapsto (\langle a, \theta\rangle + r)(\langle c, \theta\rangle + s)\ell(\theta)$ *are* π*-integrable,* $Z := \int \ell\, d\pi \in (0, \infty)$*, and* $\int \langle a, \theta\rangle \ell(\theta)\,\pi(d\theta) = \langle a, \int \ell(\theta)\theta\,\pi(d\theta)\rangle$.

Proof. lem:gauss$_i$bp$_e$xp, lem : pb$_c$omplete$_s$quare$_i$dentityLemma **??** for N$(0,\sigma^2 I)$ applied to $F = (\langle c, \cdot\rangle + r)\ell$, whose derivative is computed in Lemma **??**; $Sa = \sigma^2 a$ here. ☐

Lemma 334 (Gaussian posterior identities)**.** *ProbabilityTheory.integral$_g$aussTilt$_s$mul$_e$q, ProbabilityTheory.in* 0, $S \in \mathcal{S}_+^d$, $b \in \mathbb{R}^d$, $V := (\sigma^{-2} I_d + S)^{-1}$, $m := Vb$, $\pi := \mathcal{N}(0, \sigma^2 I_d)$, $\ell(\theta) := \exp(\langle b, \theta\rangle - \frac{1}{2}\theta^\top S\theta)$ *and* $Z := \int \ell\, d\pi > 0$. *Then*

$$\int \ell(\theta)\,\theta\,\pi(d\theta) = Z\,m, \qquad \int \langle c, \theta - m\rangle\,\ell(\theta)\,\pi(d\theta) = 0 \quad (c \in \mathbb{R}^d),$$

$$\int \|\theta - m\|^2 \ell(\theta)\,\pi(d\theta) = \mathrm{tr}(V)\,Z, \qquad \int \|\theta\|^2 \ell(\theta)\,\pi(d\theta) = (\mathrm{tr}\,V + \|m\|^2)\,Z.$$

(Equivalently, $\ell\, d\pi / Z$ *is the Gaussian* $\mathcal{N}(m, V)$*; we only need these moments.)*

Proof. lem:pb$_g$aussian$_i$ntegrals, def : bayes$_j$ointMean.Lemma **??** with c = 0, r = 1: writing u := $\int \ell(\theta)\theta\, d\pi$, for every a, $\langle a, u\rangle = \sigma^2(\langle b, a\rangle Z - \langle a, Su\rangle)$ (using $\langle S\theta, a\rangle = \langle \theta, Sa\rangle$). Hence $u + \sigma^2 Su = \sigma^2 Zb$, i.e. $(\sigma^{-2} I + S)u = Zb$ and $u = Zm$. The second identity is $\langle c, u\rangle - \langle c, m\rangle Z = 0$.

Second moment. Let $\Phi(a, c) := \int \langle a, \theta - m\rangle \langle c, \theta - m\rangle \ell\, d\pi$. Lemma **??** with $r = -\langle c, m\rangle$ (so that $\langle c, \theta\rangle + r = \langle c, \theta - m\rangle$), together with $b - S\theta = \sigma^{-2} m - S(\theta - m)$ (Definition **??**) and the mean identity, gives $\Phi(a, c) + \sigma^2 \Phi(Sa, c) = \sigma^2 \langle c, a\rangle Z$ for all a, c. With $a := \sigma^{-2} V e_i$ one has $a + \sigma^2 Sa = \sigma^2(\sigma^{-2} I + S)a = e_i$, so $\Phi(e_i, c) = \langle c, V e_i\rangle Z$; taking $c = e_i$ and summing over i gives $\int \|\theta - m\|^2 \ell\, d\pi = \mathrm{tr}(V)Z$. Finally $\|\theta\|^2 = \|\theta - m\|^2 + 2\langle m, \theta - m\rangle + \|m\|^2$ and the cross term integrates to 0. ☐

Lean remark. `PosteriorGaussian.lean` contains this section for the general tilt (b, S) (`gaussTilt b S`, `postCov  S`, `postMean  b S`); it is applied to $b = b_T(h)$, $S = S_T(h)$ for each fixed history h through `Maiti2026Power.likelihood_eq_gaussTilt`.

Lemma 335 (Posterior mean identity)**.** *Maiti2026Power.postU, Maiti2026Power.stronglyMeasurable$_p$ostU, Ma* $\mathbb{B}_d$ *and define, for a history* h, $u_T(h) := \int \ell_\theta(h)\,\theta\,\pi(d\theta) = Z_T(h)\,m_T(h)$ *with* $Z_T(h) := \int \ell_\theta(h)\,\pi(d\theta)$*;* u_T *is a measurable function of* h *with* $\int \|u_T\|\, dQ_0 < \infty$. *Then*

$$\mathbb{E}\langle \hat{x}, \theta\rangle = \int \langle x, u_T(h)\rangle\, Q_0(dh, dx) \le \int \|u_T(h)\|\, Q_0(dh, dx).$$

Proof. lem:pb$_j oint_f ubini, lem : gaussian_c omplete_s quareByLemma$ **??**$withthe\theta$-integral innermost (the integrand $\ell_\theta(h)\langle x, \theta\rangle$ is $\pi \times Q_0$-integrable, being dominated by $\ell_\theta(h)\|\theta\|$ whose integral is $\mathbb{E}_\pi\|\theta\|$), $\mathbb{E}\langle \hat{x}, \theta\rangle = \int\int \ell_\theta(h)\langle x, \theta\rangle\, \pi(d\theta)\, Q_0(dh, dx) = \int \langle x, u_T(h)\rangle\, Q_0(dh, dx)$ (Lemma **??**), and $\langle x, u_T(h)\rangle \le \|u_T(h)\|$ since $\|x\| \le 1$. Measurability of u_T is that of a parametric integral (Mathlib `StronglyMeasurable.integral_prod_right'`). ☐

Lemma 336 (Posterior second moment)**.** *Maiti2026Power.postZ, Maiti2026Power.postN, Maiti2026Power.stronglyMeasurable*$_p ostZ, Maiti2026Power.stronglyMeasurable_p ostN, Maiti2026Power.pos$
$\mathbb{B}_d$ *and* $N_T(h) := \int \|\theta\|^2 \ell_\theta(h)\, \pi(d\theta)$*. For every history* h*, with* $c := d^2/(d\sigma^{-2} + T)$*,*

$$\|u_T(h)\|^2 = Z_T(h)^2\|m_T(h)\|^2 \le Z_T(h)\big(N_T(h) - c\, Z_T(h)\big),$$

and $\int Z_T\, dQ_0 = 1$*,* $\int N_T\, dQ_0 = \mathbb{E}_\pi\|\theta\|^2 = \sigma^2 d$*. Consequently*

$$\int \|u_T(h)\|\, Q_0(dh, dx) \;\le\; \sqrt{\sigma^2 d - \frac{d^2}{d\sigma^{-2} + T}}.$$

Proof. lem:pb$_j oint_f ubini, lem : gaussian_c omplete_s quare, lem : trace_i nv_l owerByLemma$ **??**$forthefixedhistor$
$Z_T m_T$ and $N_T = (\operatorname{tr} V_T + \|m_T\|^2) Z_T$, so $Z_T\|m_T\|^2 = N_T - \operatorname{tr}(V_T) Z_T \le N_T - cZ_T$ by Lemma **??**; multiply by Z_T. The integrals of Z_T and N_T are Lemma **??** (Fubini, the θ-integral outermost). Finally, by the Cauchy–Schwarz inequality in $L^2(Q_0)$ applied to $\sqrt{Z_T}$ and $\sqrt{N_T - cZ_T}$, $\int \|u_T\|\, dQ_0 \le \int \sqrt{Z_T}\sqrt{N_T - cZ_T}\, dQ_0 \le \big(\int Z_T\, dQ_0\big)^{1/2}\big(\int (N_T - cZ_T)\, dQ_0\big)^{1/2} = \sqrt{\sigma^2 d - c}$. ☐

Lean remark. This route never integrates m_T or $\operatorname{tr} V_T$ over histories, only Z_T, N_T and u_T, which are parametric integrals in h (measurable by `StronglyMeasurable.integral_prod_right'`); the outline's integrability lemma $\mathbb{E}\|b_T\|^2 < \infty$ and the measurability of $h \mapsto V_T(h)$ (matrix inversion) are not needed.

16.4 The unit-ball lower bound

Lemma 337 (Lower bound on the posterior trace)**.** *Maiti2026Power.le*$_t race_p ostCov_h istS, Maiti2026Power.tr$
$\mathbb{B}_d$*, then for every history* h*,* $\operatorname{tr} V_T(h) \ge \frac{d^2}{d\sigma^{-2}+T}$*.*

Proof. lem:trace$_i nv_a mhm$tr $S_T(h) = \sum_{t<T}\|x_t\|^2 \le T$, so $\operatorname{tr}(\sigma^{-2}I + S_T(h)) \le d\sigma^{-2} + T$, and Lemma **??** applied to the positive definite matrix $\sigma^{-2}I + S_T(h)$ gives $\operatorname{tr} V_T(h) \ge d^2/\operatorname{tr}(\sigma^{-2}I + S_T(h)) \ge d^2/(d\sigma^{-2} + T)$. ☐

Lemma 338 (Expected norm of a Gaussian vector, lower bound)**.** *Maiti2026Power.le*$_i ntegral_n orm_g aussPrior,$
$\mathcal{N}(0, \sigma^2 I_d)$ *then* $\mathbb{E}\|\theta\|^2 = \sigma^2 d$ *and* $\mathbb{E}\|\theta\| \ge \sqrt{2/\pi}\,\sigma\sqrt{d}$*.*

Proof. lem:gauss$_n orm_l ower, lem : gauss_n orm_m omentsLemma$ **??**$(i)andLemma$ **??**$withtheweightsc_i =$ $1/\sqrt{d}$ and $S = \sigma^2 I_d$. ☐

Lemma 339 (Bayesian simple regret on the ball)**.** *Maiti2026Power.integral$_s$impleRegret$_b$ayesJointLaw$_g$edef B_d. Under the joint law Q of Definition **??**,*

$$\mathbb{E}[r(\hat{x},\theta)] = \mathbb{E}\|\theta\| - \mathbb{E}\langle \hat{x},\theta\rangle \;\geq\; \sqrt{2/\pi}\,\sigma\sqrt{d} - \sqrt{\sigma^2 d - \frac{d^2}{d\sigma^{-2}+T}}.$$

Proof. lem:posterior$_m$ean, $lem: posterior_second_moment, lem: pb_norm_lowerOntheunitball$, $\max_{x\in\mathbb{B}_d}\langle x,\theta\rangle = \|\theta\|$, so $r(\hat{x},\theta) = \|\theta\| - \langle \hat{x},\theta\rangle$ (Lemma **??**). The marginal of θ is π, so $\mathbb{E}\|\theta\| \geq \sqrt{2/\pi}\sigma\sqrt{d}$ (Lemma **??**), and $\mathbb{E}\langle \hat{x},\theta\rangle \leq \int \|u_T\|\,dQ_0 \leq \sqrt{\sigma^2 d - d^2/(d\sigma^{-2}+T)}$ by Lemmas **??** and **??**. ☐

Theorem 340 (Simple-regret lower bound on the unit ball)**.** *Maiti2026Power.simpleRegret$_u$nitBall$_n$onneg, Ma 1, $T \geq 1$ and* Alg *be any identification algorithm with budget T on $\mathcal{X} = \mathbb{B}_d$. With the prior $\pi = \mathcal{N}(0,\sigma^2 I_d)$, $\sigma^2 := d/(4T)$:*

(i) $\mathbb{E}[r(\hat{x},\theta)] \geq \dfrac{d}{8\sqrt{T}}$ *under the joint law Q;*

(ii) there exists $\theta \in \mathbb{R}^d$ *with* $\|\theta\| \leq \dfrac{10\,d}{\sqrt{T}}$ *and* $\mathbb{E}_\theta[r(\hat{x},\theta)] \geq \dfrac{3d}{40\sqrt{T}}$.

Proof. lem:bayes$_r$egret$_b$all, $lem: pb_joint_fubini, lem: gauss_norm_moments, lem: pb_kernel_of_likelihood (i) With \sigma^2 = d/(4T)$ we have $d\sigma^{-2} = 4T$, hence $\frac{d^2}{d\sigma^{-2}+T} = \frac{d^2}{5T}$ and $\sigma^2 d - \frac{d^2}{5T} = \frac{d^2}{4T} - \frac{d^2}{5T} = \frac{d^2}{20T}$, while $\sqrt{2/\pi}\,\sigma\sqrt{d} = \sqrt{2/\pi}\,\frac{d}{2\sqrt{T}} \geq 0.79\cdot\frac{d}{2\sqrt{T}}$ (as $\pi < 3.15$). Lemma **??** gives

$$\mathbb{E}[r(\hat{x},\theta)] \geq \frac{d}{\sqrt{T}}\Big(\frac{0.79}{2} - \frac{1}{4.47}\Big) \geq \frac{d}{\sqrt{T}}\,(0.395 - 0.2238) \geq \frac{d}{8\sqrt{T}},$$

using $4.47^2 \leq 20$.

(ii) Let $R := \frac{10d}{\sqrt{T}} = 20\sigma\sqrt{d}$. Since $0 \leq r(\hat{x},\theta) \leq 2\|\theta\|$ on the ball and $\|\theta\|\mathbf{1}\{\|\theta\| > R\} \leq \|\theta\|^2/R$,

$$\mathbb{E}\big[r(\hat{x},\theta)\mathbf{1}\{\|\theta\| > R\}\big] \leq \frac{2\,\mathbb{E}\|\theta\|^2}{R} = \frac{2\sigma^2 d}{R} = \frac{d}{20\sqrt{T}},$$

so $\mathbb{E}[r\mathbf{1}\{\|\theta\| \leq R\}] \geq \frac{d}{8\sqrt{T}} - \frac{d}{20\sqrt{T}} = \frac{3d}{40\sqrt{T}}$. By Lemma **??**, $\mathbb{E}[r\mathbf{1}\{\|\theta\| \leq R\}] = \int_{\|\theta\|\leq R} \mathbb{E}_\theta[r(\hat{x},\theta)]\,\pi(d\theta)$, where $\theta \mapsto \mathbb{E}_\theta[r(\hat{x},\theta)]$ is measurable (Lemma **??**, r being bounded by $2\|\theta\|$). If every θ with $\|\theta\| \leq R$ had $\mathbb{E}_\theta[r] < \frac{3d}{40\sqrt{T}}$, the nonnegative function $g := \mathbf{1}\{\|\theta\| \leq R\}\,(\frac{3d}{40\sqrt{T}} - \mathbb{E}_\theta[r])$ would be positive on $\{\|\theta\| \leq R\}$, a set of positive π-measure (by Markov's inequality $\pi(\|\theta\| > R) \leq \sigma^2 d/R^2 = 1/400$), so $\int g\,d\pi > 0$; but $\int g\,d\pi = \frac{3d}{40\sqrt{T}}\pi(\|\theta\| \leq R) - \mathbb{E}[r\mathbf{1}\{\|\theta\| \leq R\}] \leq 0$, a contradiction. Hence some θ with $\|\theta\| \leq R$ satisfies $\mathbb{E}_\theta[r(\hat{x},\theta)] \geq \frac{3d}{40\sqrt{T}}$. ☐

Remark 341 (Constants)*.* The outline claimed the constant 0.13 in (i), from $\frac{1}{2\sqrt{2}} - \frac{1}{\sqrt{20}} = 0.12995\ldots$, which is slightly *below* 0.13; we use 1/8 instead (the sharper Lemma **??** would even give 0.17). The truncation radius was changed from $16\sigma\sqrt{d}$ to $20\sigma\sqrt{d}$ so that Corollary **??** keeps the outline's headline constants ($\delta \le 1/500$, $T \ge d^2/(1000\varepsilon^2)$): with $R = k\sigma\sqrt{d}$ the final coefficient is $\frac{1}{8} - \frac{1}{k} - k\delta$, which for $\delta = 1/500$ is maximized near $k = 22$; $k = 20$ gives 7/200. The choice $\sigma^2 = d/(4T)$ is not optimal either but keeps the arithmetic simple.

Corollary 342 (PAC lower bound on the unit ball)**.** *$Maiti2026Power.unitBall_le_budget_of_isPAC, Maiti2026Po$* 2, $\varepsilon > 0$ *and* $\delta \le 1/500$*. Every* (ε, δ)*-PAC identification algorithm on* $\mathbb{B}_d$ *with budget* T *satisfies*

$$T \ge \frac{49\,d^2}{40000\,\varepsilon^2} \ge \frac{d^2}{1000\,\varepsilon^2}.$$

Proof. lem:pac$_e xpected_r egret, lem : pb_k ernel_o f_l ikelihood, thm : ball_s imple_r egret_l ower If$ T $= 0 thelawoftherecommendationdoesnotdependon\theta$, and the two instances $\pm 2\varepsilon e_1$ have regrets summing to 4ε at every arm, so the algorithm cannot be (ε, δ)-PAC for $\delta < 1/2$ (as in Lemma **??**). Let $T \ge 1$ and let θ be given by Theorem **??**(ii). On the ball the regret lies in $[0, 2\|\theta\|]$, so Lemma **??** applied to the law $\mathbb{P}_\theta = \kappa(\theta)$ of Lemma **??** gives $\mathbb{E}_\theta[r(\hat{x}, \theta)] \le \varepsilon + 2\delta\|\theta\| \le \varepsilon + \frac{20\delta d}{\sqrt{T}}$ (if $2\|\theta\| < \varepsilon$ this is trivial). Combining with $\mathbb{E}_\theta[r] \ge \frac{3d}{40\sqrt{T}}$,

$$\Big(\frac{3}{40} - 20\delta\Big)\frac{d}{\sqrt{T}} \le \varepsilon, \qquad \frac{3}{40} - 20\delta \ge \frac{3}{40} - \frac{1}{25} = \frac{7}{200},$$

hence $\sqrt{T} \ge \frac{7d}{200\,\varepsilon}$ and $T \ge \frac{49d^2}{40000\varepsilon^2}$; finally $40000/49 < 1000$. ☐

Lean remark. The unit ball is `unitBall = Metric.closedBall 0 1` in `EuclideanSpace` ; it is compact and spans $\mathbb{R}^d$, hence an action set. The expectation $\mathbb{E}_\theta[r(\hat{x}, \theta)]$ is `expRegret T A` , the integral of a bounded function under $\mathbb{P}_\theta^T \otimes \rho = \kappa(\theta)$ (`pairKernel A T`), and its measurability in θ is Mathlib's `StronglyMeasurable.integral_kernel_prod_right'`.

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